GUIDELINES FOR THE REGISTRATION OF PESTICIDES UNDER VARIOUS CATEGORIES AS PER THE INSECTICIDE ACT, 1968 AND RULES 1971

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INTRODUCTION

Pesticides are inherently toxic in nature as they are designed to develop to kill or combat the insects, diseases and weeds. Pesticides are being used worldwide with regulation. In India pesticides are being regulated through the Insecticide Act, 1968 and Rule 1971. The registration of pesticides is mandatory. The Act provides for use of only registered pesticides as per the label claim. Guidelines are considered as a key and fundamental document for the registration of pesticides and these guidelines are encompassing the scientific requirements and procedural aspects designed to ensure the safety, efficacy and quality of pesticides before they are marketed.

Stakeholders generating data and submitting the application for registration should ensure that the tests are conducted and data generated in accordance with established scientific procedures following the test guidelines and the principles of Good Laboratory Practices, wherever applicable. The data should be authentic, replicable, utilizable and of good quality. The complete study reports should be submitted. The requirement for registration usually includes data and information on proposed application; data on identity of the insecticide (identity, composition, analysis and quality); data to assess risk to humans and the environment; data to assess efficacy of the product; and the packaging and labelling requirements.

The data requirement for registration of insecticides varies with the type of insecticides to be registered (i.e. chemical or bio-pesticide and also type of bio-pesticide i.e. Microbial Pest Control Agent (MCPA) or Botanical or Semio-chemical/Pheromone); the type of material to be registered i.e. Technical or Formulation or Manufacturing Use Product (MUP); the type of formulation; solid (WP, granules, powder) etc. liquid (EC, EW, SC etc.) or Vapor (vaporizer, fumigants etc.); the category of registration – i.e. provisional [u/s 9(3B)]; regular [u/s 9(3)] or subsequent "Me-too" [u/s 9(4)] as per the provisions (Please refer section (9) of the Insecticides Act, 1968); and purpose of registration – domestic use or export or for both (domestic use and export); the intended use of the pesticide to be registered or its label claims etc. Hence, before starting data generation or submitting application for registration, the applicant should ensure that the requirements are being complied correctly for the type of pesticide to be registered under the desired category and for the intended purpose.

The registration of Technical Grade Pesticide is mandatory before or simultaneously in case of all registrations including Formulation Import and Formulation Indigenous Manufacturing. The data submitted by the applicant at the time of seeking registration under section 9(3b) shall not be

required to resubmit again at the time of submission of application for regular registration under section 9(3) by the same applicant, if chemical composition and other claims remain unchanged.

The registration committee has constituted an expert committee in its 466th and 468th RC meeting and committee conducted five different meetings and compiled all the guidelines approved by the registration committee in its various meetings. The overall guidelines were categorized in to five different sub categories *viz*.,

i. General guidelines
ii. Guidelines for the export and import of pesticides
iii. Guidelines for the registration of chemical pesticides for agricultural purposes
iv. Guidelines for the registration of biopesticides for agricultural purposes
v. Guidelines for the registration of pheromones, house old insecticides, public health purposes, air craft disinfectant, insecticides for the control of ectoparasites

CATEGORY: 1: GENERAL GUIDELINES

Sub category 1.1: Implementation of KYC on CROP Portal for Registration of Pesticides as per the Insecticide Act, 1968 and Rule 1971

KYC PREVIEW				
GOVERNMENT OF INDIA MINISTRY OF AGRICULTURE (DEPARTMENT OF AGRICULTURE & FARMERS WELFARE) DIRECTORATE OF PLANT PROTECTION QUARANTINE AND STORAGE NH IV, FARIDABAD (HARYANA)-121001				
l. Name of the Company/Firm/Person				
2.Full address of Registered Office				
3.PAN number & date				
4.Full address for correspondence with email & phone number				
5.GST Numbers	**Location**		**Number**	
(Full address of each manufacturing site (including the one if located at same premises as the Registered office.)	1.			
	2.			
	3.			
6.Name of the authorized person/s submitting this Information	i) Full name			
	ii) Designation			
	iii) Email i.d.			
	iv) Landline no.			
	v) Mobile no.			
7.IEC Code Number				
8.SSI Certificate Number				
9.Company URL /Website				

Information to be furnished under the following 8 headings: -

1. Policy & Practices Information.
2. Plant & Machinery Information.
3. Quality Control Equipment Information.
4. Land & Buildings Information.
5. Financial Information.
6. Compliance Information.
7. Registration Information.
8. Production Information.

a. Submit each information manufacturing site wise where applicable.

b. No question should be left blank.

c. Mark reply as Y for yes, and N for No.

d. Do mark the number indicated for each document that is attached and as indicated in the Format.

e. Do mention name of Company site on each page.

f. The KYC Form is to be updated by 30th Sept every year or earlier, based on the completion of the Company1s audit and its approval by Board of Directors/Partners/Proprietary Owner. Certain information has to be uploaded periodically, and as indicated in this performa. (SRS) for linking with CIB & RC update.

g. Each information given herein must be factual and accurate: any misleading/fake information given would be liable for prosecution.

Verification:

_____

_____

_____

_____

1. Policy & Practices Information

	Reply Y(yes)/ N (no)
1.1. ***Quality Policy***	
1.1.1. Do you have a documented Quality Policy of the company? ***If yes, then please attach a copy marked 1.1.1.***	
1.1.2. Has the Policy been communicated to all employees of the organization in the language that everyone understands? If yes, then describe	
When, How? (e.g. holding employee meeting, Notice Board)	
1.1.3. Whether the Policy has been displayed at prominent locations in the factory? If yes, where?	
Locations	
1.1.4. Do you have a clearly defined policy for handling consumer Complaints and returned product from the market? ***attach a copy marked 1.1.4***	
1.2. ***Safety Policy***	
1.2.1. Do you have a documented Safety and Environment Policy of the company? ***If yes, please attach a copy marked 1.2.1.***	
1.2.2. Has the Policy been communicated to all the employees of the organization in the language(s) that everyone understands? If yes, then please specify:	
I **When, How?**	
1.2.3. Whether the policy has been displayed at prominent location. If yes, then please specify:	
Where, How?	

1.3. ***Organization***	

1.3.1. Is the clearly defined organization chart available? *If yes, please attach a copy marked* 1.3.2. Whether the organization chart clearly defines personnel responsible for each key area? 1.3.3. Does the responsibility chart also pinpoint the person responsible for market complaints? **1.4.*Training programme*** 1.4.1. Whether employees are trained about the safety norms framed by the organization? 1.4.2. Does the company hold regular/ refresher training classes for the Personnel in the following area? a. Safety & fire fighting b. Occupational health hazards c. Spillage control d. Manufacturing operations e. On site emergency f. Offsite emergency g. Quality consciousness h. Samples handling i. Use of personal protective equipment **1.5.*Amenities*** Employee/workers Whether following amenities are available 1.5.1. Change room 15.2. Lockers for keeping personal items 1.6**. Dispensary**	

	KEY MAN POWER/EMPLOYEE Name/s of Factory lncharge..... Designation..... Production lncharge..... Designation..... Quality Control/Assurance lncharge..... Designation.....

	Safety & Environment lncharge ..... Designation.....

2. Method

2.1. Raw material and Product storage

2.1.1. Do you have a written **Standard Operating Procedures (SOP)** for Handling the following raw materials?

a. Technical

b. Solvents

c. Additives

d. Adjuvants/lnert Materials

If yes, attach a copy-marked 2.1.1.

2.1.1. Whether SOP is displayed near the workstation?

2.1.2. Whether calibration charts of storage tanks are available?

2.1.3. Whether measuring devices are calibrated at regular intervals to counter check their reliability?

2.1.4. Do you have written SOP for handling finished and packed formulations?

2.1.5. Whether the SOP for handling Packed Product is displayed near the working station? ***If yes, please attach a copy marked 2.1.6***

2.2. Manufacturing

2.2.1. Whether equipment layout and flow diagram of the plant displayed at proper place in the plant?

2.2.2. Whether each equipment/storage tank/pipeline/instrument are identified and clearly labelled?

2.2.3. Do you have written SOP for every formulation you make?

2.2.4. Check whether SOP for formulation includes Following:

a. Optimum batch size
b. Complete sequence and time of operation
c. Formula of the process

d. All the safety checks

e. Sampling procedure

f. Ventilation of the equipment involved in manufacturing/formulation

g. Decontamination after change in formulation product

2.2.5. Do you have formats for daily production report?

2.2.6. Do you have preventive maintenance plan?

2.2.7. Whether calibration of all equipment and instrument is done and calibration on shop floor?

2.2.8. Do you have SOP for reprocessing the rejected batch?

2.2.9. Do you maintain batch wise log sheet of production is recorded.

2.3. Quality Assurance

2.3.1. Do you have clearly defined system for quality assurance?

2.3.2 Do you keep records of tests done for raw materials, packaging Materials and finished products?

2.3.3. Do you have standard samples for counter testing and are recorded for traceability?

2.3.4. Do you preserve the samples till the expiry date of the batch, for testing to attend to customer complaints if received?

2.3.5. Do you have preventive and periodic maintenance plans for analytical Instruments?

2.3.6 Does reliability check plans and SOP for calibration of laboratory Equipment exist.

If yes, please attach a copy marked 2.3.12

2.4. Safety, Health & Environment

2.4.1. Whether a documented system has been prepared forthe precautions to be taken by workers before going to manufacturing area, lunch *I* dinner *I* tea and rest room etc. from manufacturing area?

If yes, please attach a copy marked 2.4.1

2.4.2. Whether this document necessarily consist of the following:

a. Where and how to keep personal belongings and eatables?

b. Where, when and how to wash hands, face etc.?

c. Where, when and how to take bath after working?

2.4.3. Whether the worker shave adequately been trained for the above?

2.4.4. Do you have safety datasheets available for the following materials:

a. For each a. i.(Technical)pesticide

b. Chemicals and intermediates

c. Solvents

d. Each and every adjuvant/additive

2.4.5. Do you have SOP for the following:

a. Handling of hazardous material?

b. Spillage control of hazardous material?

c. Decontamination of equipment?

d. Using air-breathing apparatus?

If yes, please attach a copymarked2.4.5

2.4.6. Whether SOPs are displayed at prominent places in the plant for all the persons to view?

2.4.7. Whether vents of all the equipment are connected to a suitable air treatment system?

2.4.8. Do you have waste treatment and disposal system?

Attach its flow diagram marked 2.4.8.1

2.4.9. Have you developed a safety procedure for unloading and reprocessing the rejected batch?

2.4.10 .Do you have a Manual for dealing for fire hazards?

2.4.11. Does this Manual contain the following:	
a. List of Persons responsible to deal with emergency?	☐
b. Procedure on dealing with small and big fire?	☐
c. Use of fire extinguishers?	☐
d. Responsibilities of the persons during emergency?	☐
e. List of important Telephone nos. including that of Fire brigade Police, Senior Officers of the Company.	☐
f. List of fire extinguisher and document showing location of each fire extinguisher in the factory premises?	☐
g. Document showing location of fire alarm in the factory?	☐
h. SOP for operating fire hydrant system?	☐
i. Refilling fire-fighting equipment?	☐
	☐
2.4.12. Preventive maintenance schedule of fire extinguishers and fire hydrant system has been prepared?	☐
2.4.13. Are you keeping a record of the mock drills being done for fire fighting?	☐
3. Raw material storage facilities	☐
6.1. a. Do you have safe(safety against fire and exposure) area for Storage	☐
b. Do you have a covered, adequately sized and properly ventilated shed for storage?	☐
4. *Storage*	
a. Do you have safe unloading system for solvents received in tankers?	

b. Do you have safe (in terms of stacking heights), covered, adequately sized and properly ventilated shed for storage? c. Do you have a separate area for keeping rejected raw materials?	

5. Storage and finished goods	
5.1. Do you have covered and adequately sized storage area for keeping finished product?	
5.2. Whether the storage area is provided with adequate fire proof lights?	
5.3. Whether the area is properly ventilated?	
5.4. Whether the area is provided with fire fighting equipment?	
5.5. Whether separate storage area provided for herbicides?	
5.6. Whether separate quarantine space provided for rejected/returned goods?	
6. Safety, health and environment	
6.2. Whether the worker amenity and canteen are situated sufficiently far away, as per the norms, from the manufacturing unit and storage area for raw and finished goods?	
6.3. Does the plant building been provided with lightening arrestor?	
6.4. Does the building and plant &machinery in it meant for formulation of herbicides is used exclusively for this purpose only? Is it separately located and used exclusively to avoid cross contamination.	
6.5. Does the building has separate area for storing following materials	
a. Raw materials	
b. Packing materials	
c. Rejected materials	
d. Finished products	
6.5 .Do you have proper disposal system for following:	
a. Solid waste?	
b. Rejected packing components?	
c. Rejected product?(As per CIB/Excise guidelines)	
6.6. Do you have fire-fighting equipment on each floor of the plant	
6.7. Do you have fire hydrant system?	

6.8. Do you have sufficient quantity of following personnel protective equipment:

a. Cotton hand gloves ☐
b. Hand gloves rubber ☐
c. Surgical hand gloves ☐
d. Nose filters ☐
e. Cotton masks ☐
f. Goggles ☐
g. Safety shoes ☐
h. Air breathing apparatus
i. Face shields ☐
j. Helmets ☐
k. Cotton caps ☐
l. Workers Uniform for work area ☐
m. Firefighting kit
n. Safety belt ☐

6.9. Are the workers being provided with fresh duly washed working uniform at the beginning of every shift? ☐

6.10. Do you have a collection area for used uniforms, where every employee can keep his used uniform at the end of the shift and then sent to decontamination plant for washing?

6.11. Do you have in house clothes washing machine and decontamination facility? If not, describe how the uniforms are washed after use.

6.12. Does the dispensary have following facility:

a. First aid box
b. Oxygen cylinder with its fixtures and regular
c. Stretcher
d. Wheelchair
e. Eye wash station
f. Adequate quantity of atropine injection and tablets
g. Cholinesterase kit
h. Ambulance
i. In house Doctor/paramedic
j. Doctor on call

6.13. Whether arrangements have been made with nearby specialty hospital to treat cases in emergency?

6.14. Whether plant/equipment for manufacture of herbicides/weedicides all together separate to avoid cross contamination?

6.15. Health record of workers as specified in insecticide act.

Any other remarks

Verification

I……S/D/o……..do hereby solemnly verify that to the best of my knowledge and belief, the information given in the application and the annexures attached to it are correct and complete. I clearly understand that the Certificate of Registration and Manufacturing License after issuance is liable to be cancelled, as well as any legal action against me, if any information or data submitted with the application is found incorrect or false at any stage or any condition of the Registration Certificate or Manufacturing Licence granted against this application is violated.

I hereby confirm that the equipment's installed are of the right specifications and materials of construction, and all necessary attention to safety, environmental precautions have been taken to and ensure consistent quality of the finished products manufactured by us.

I further declare that I am making this application in my capacity as……and that I am competent to make this application and verify it by virtue of…….. an attested copy of which is enclosed herewith.

Place..……..

Date……..

Signature:
Name:
Designation:
Company:

2. Plant & Machinery Information

WHICH TYPES OF FORMULATIONS

YES/NO

1.	Emulsifiable Concentrate (EC)	
2.	Suspension Concentrate (SC)	
3.	Water Dispersible Granule (WDG)	
4.	Wettable Powder (WP)	
5.	Granules Formulation	
6.	Soluble concentrate (SL)	
7.	Micro emulsion (ME)	
8.	Oil dispersion (OD)	
9.	Water soluble powder (SP)	
10.	Water dispersible powder for slurry treatment (WS)	
11.	Flowable concentrate for seed treatment (FS)	
12.	Capsule Suspension (CS)	
13.	A mix formulation of CS and SC (ZC)	
14.	A mix formulation of CS and SE (ZE)	
15.	A mix formulation of CS and EW (ZW)	
16.	Emulsion oil in water (EW)	
17.	Suspo-emulsion (SE)	
18.	Water soluble granules (SG)	
19.	Technical Grade Manufacture	

Mark Formulation wise equipment installed

YES/NO

1	Emulsion Concentrate (EC
1.1	Manufacturing equipment for EC plant

- Bulk storage tank for solvent
- SS Mixing vessel with stirrer
- Charging pump
- Fume ducting with scrubber
- Filtration equipment
- SS Storage Vessel for finished product
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking ETC
- Vent Hoods

2	Suspension Concentrate (SC)
2.1	Manufacturing equipment for SC plant

- SS pre mixing vessel with stirrer
- SS storage vessel with stirrer
- Jacketed SS Vessel for Gum solution with Stirrer
- Charging pump
- Bead mill/Colloid mill
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Chilling plant with cooling tower

3	Water Dispersible granules (WDG)
3.1	Manufacturing equipment for Extruded Granules plant

- Grinding machine pulverize/ACM/Jet Mill
- SS Pre/post mixer or blender
- Extruder
- Sieving machine
- Fluid bed or Tray Dryer
- Hot air generator
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Air compressor with receiver tank

3	Water Dispersible granules (WDG)
3.2	Manufacturing equipment for Spray Dried Granules plant

- SS pre mixing vessel with stirrer

- SS storage vessel with stirrer
- Jacketed SS Vessel for Gum solution with Stirrer
- Charging pump
- Bead mill/Colloid mill

- Spray drying plant
- Hot air generator
- Sieving machine
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking ETC
- Air compressor with receiver tank
- Chilling plant with cooling tower

4	Wettable powder (WP)
4.1	Manufacturing equipment for WP plant

- Grinding machine pulverizer/ACM/Jet Mill
- SS Pre mixer or blender
- Blender mixer
- Formulated product storage silo
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Air compressor with receiver tank
- Chilling plant with cooling tower

5	Granules Formulation
5.1	Manufacturing equipment for Granule plant

- Vibro sifter/siever for blank carrier material
- Feeding channel/Chute/buclet elevator/Screw conveyer
- Raw material feed tank
- Blender/ Rotary mixer
- Siever
- Bag filler
- Weighing machine or platform balance
- Dryer
- Sprayer
- Packing machine
- Labelling/Batch Marking etc.

6	Soluble concentrate (SL)
6.1	Manufacturing equipment for SL plant

- Bulk storage tank for solvent
- SS Mixing vessel with stirrer
- Charging pump
- Fume ducting with scrubber
- Filtration equipment
- SS Storage Vessel for finished product
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Vent Hoods

7	Micro emulsion (ME)
7.1	Manufacturing equipment for ME plant

- Bulk storage tank for water/solvent
- SS Mixing vessel with stirrer
- Charging pump
- Fume ducting with scrubber
- Filtration equipment
- SS Storage Vessel for finished product
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Vent Hoods

8	Oil dispersion (OD)
8.1	Manufacturing equipment for OD plant

- Jacketed SS high shearing mixing vessel
- SS storage vessel with stirrer
- Charging pump
- Bead mill/Colloid mill
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Chilling plant with cooling tower

9	Water soluble powder (SP)
9.1	Manufacturing equipment for SP plant

- Grinding machine pulverizer/ACM/Jet Mill
- SS Pre mixer or blender
- Blender mixer
- Formulated product storage silo
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking ETC
- Air compressor with receiver tank
- Chilling plant with cooling tower

10	Water dispersible powder for slurry treatment (WS)
10.1	Manufacturing equipment for WS plant

- Grinding machine pulverizer/ACM/Jet Mill
- SS Pre mixer or blender
- Blender mixer
- Formulated product storage silo
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Air compressor with receiver tank
-
- Chilling plant with cooling tower

11	Flowable concentrate for seed treatment (FS)
11.1	Manufacturing equipment for FS plant

- Jacketed SS high shearing mixing vessel
- SS storage vessel with stirrer
- Jacketed SS Vessel for Gum solution with Stirrer
- Charging pump
- Bead mill/Colloid mill
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Chilling plant with cooling tower

12	Capsule Suspension (CS)
12.1	Manufacturing equipment for FS plant

- Jacketed SS high shearing mixing vessel
- SS storage vessel with stirrer
- Jacketed SS Vessel for Gum solution with Stirrer
- Charging pump

- Bead mill/Colloid mill
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Steam boiler or water bath

13	A mix formulation of CS and Sc (ZC)
13.1	Manufacturing equipment for ZC plant

• SS pre mixing vessel with stirrer	
• Jacketed SS high shearing mixing vessel	
• SS storage vessel with stirrer	
• SS post mixing vessel with stirrer	
• Jacketed SS Vessel for Gum solution with Stirrer	
• Charging pump	
• Bead mill/Colloid mill	
• Dust collector with scrubber	
• Weighing machine or platform balance	
• Packing/filling machine line	
• Labelling/Batch Marking etc.	
• Chilling plant with cooling tower	
• Stearn boiler/ water bath	

14	A mix formulation of CS and SE (ZE)
14.1	Manufacturing equipment for ZE plant

- SS pre mixing vessel with stirrer
- Jacketed SS high shearing mixing vessel
- SS storage vessel with stirrerSS post mixing vessel with stirrer
- Jacketed SS Vessel for Gum solution with Stirrer
- Charging pump
- Bead mill/Colloid mill
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Chilling plant with cooling tower
- Steam boiler/ water bath

15	A mix formulation of CS and EW (ZW)
15.1	Manufacturing equipment for ZW plant

- SS pre mixing vessel with stirrer
- Jacketed SS high shearing mixing vessel
- SS storage vessel with stirrer
- SS post mixing vessel with stirrer
- Jacketed SS Vessel for Gum solution with Stirrer
- Charging pump

- Bead mill/Colloid mill
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Steam boiler/ water bath

16	Emulsion oil in water (EW)
16.1	Manufacturing equipment for FS plant

- Jacketed SS high shearing mixing vessel
- SS storage vessel with stirrer
- Jacketed SS Vessel for Gum solution with Stirrer
- Charging pump
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Chilling plant with cooling tower

17	Suspo-emulsion (SE)
17.1	Manufacturing equipment for SE plant

- Jacketed SS high shearing mixing vessel
- SS storage vessel with stirrer
- Jacketed SS Vessel for Gum solution with Stirrer
- Charging pump
- Bead mill/Colloid mill
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Chilling plant with cooling tower

18	Water soluble granules (SG)
18.1	Manufacturing equipment for Extruded Granules plant

- Grinding machine pulverizer/ACM/Jet Mill
- SS Pre/post mixer or blender
- Extruder
- Sieving machine
- Fluid bed or Tray Dryer
- Hot air generator
- Dust collector with scrubber
- Weighing machine or platform balance
- Packing/filling machine line
- Labelling/Batch Marking etc.
- Air compressor with receiver tank

Mark if manufacturing technical grade Insecticides/Pesticides

19	Technical Grade Manufacture	
19.1	Plant and Machining Installed	
	▪ Control console	
	▪ Feed tank for material	
	▪ Reactors	
	▪ Distillation towers	
	▪ Evaporators	
	▪ Condensers/Heat exchangers	
	▪ Boiler/Steam Plant	
	▪ Chilling System	
	▪ Crystallizers	
	▪ Centrifuges	
	▪ Driers	
	▪ Phase Separators	
	▪ Extractors	
	▪ Storage Tanks	
	▪ Pollution Treatment equipment	
	▪ Fire Fighting Facilities	
	▪ Packing System	
	▪ Scrubber System	
	▪ Pumps	
	▪ Filters	
	▪ Powder Transfer System	
	▪ Local Exaust Ventilation System	
	▪ Vaccum Systems (Ejectors/Vaccum Pumps)	
	▪ Utilities Facility (N2, Air, Chilled Water/Brine)	
	▪ Cooling System	

3. Quality Control Equipment Information:

QC Equipment installed in Laboratory	Y/N
• Gas Chromaography (GC)	
• GC MS	
• LC MS	
• GC Headspace	
• High performance liquid Chromatography (HPCL)	
• Sonicator	
• UV-VIS Spectrophotoometer	
• Flash point apparatus	
• Karl fisher Titrator	
• pH Meter	
• Electronic weighing Balance for Wet Lab	
• Analytical Weighing Balance	
• Heating Mantles	
• Hot Plates	
• Magnetics Stirrers	
• Wrist Shaker	
• Hot air Oven	
Analytical Instruments for Laboratory	
• Particals Size Analyzer	
• Viscometer	
• Mechanical Stirrer	
• Fume hood	
• BIS Test Sieves	
• Melting Point Apparatus	

- Stop Watch ☐
- Water Distillation Unit ☐
- Solvent Distillation Unit ☐
- Solvent Distillation apparatus ☐
- Air Conditioner ☐

4. Land & Buildings Information:

Add of the site

a) **Total land area at site:..... .....s.m.**

b) **Area used for formulation:.....s.m.**

c) **Area used for technical manufaturing.....s.m.**

d) **Covered area for QC Laboratory.....s.m.**

e) **Covered area for R&D Laboratory.....s.m.**

f) **Covered area used for waste treatment.....s.m.**

g) **Covered area used for worker facilities.....s.m.**

h) **Covered area used for medical emergency.....s.m.**

i) **Geo tagging (Log&Lot)East/West/North/South**

j) **Covered area for QC lab/Tech/Worker facility**

- **Mention what is on its:** **East.....**

  West.....

  North.....

  South.....

Other items for Laboratory

- Micro Syringe for GC and Micro Syringe for
- Internal Standards
- A Grade Glassware
- Refrigerator
- Required Laboratory regents

- Gas Cylinders of Nitrogen, Hydrogen and Zero air
- Gas Generator (Air, N2, H2) ☐
- Specification/BIS Standards of products to be tested/manufactured ☐
- Proper Storage Facilities to store lab re agents, batch wie sample of finished products ☐

5. FINANCIAL INFORMATION

Information on				
S.No.		2020-2021	2021-22	
1.	Net Sales Turnover of pesticides			
2.	Value of Plant and Machinery Gross- Net-			
3.	Value of QC Equipment Gross- Net-(After Depreciation)			
4.	Cost of Buildings			
5.	Annual Rent Paid for Building (if Rent Paid)			
6.	Annual Rent Paid for Exports			
7.	Net value of Exports			
8.	Net value of Imports			

a) The above data is required for the business of pesticides and their intermediates manufacturing only.

b) Do not include data of other products business in the above.

c) All figures are to be given in lakhs in each column.

6. COMPLIANCE INFORMATION

(To be submitted separately for each manufacturing Site).

1. Approval received from the Chief Inspector of Factories valid up to……(date)
   Marked as 2A

2. Approval received to operate from the State Pollution Control Board valid up .....**Marked as 2B**

3. Approval received from the Petroleum and Explosives Safety Orgonisation for storage of solvents etc. valid upto.....***Marked as 2C***

4. Approval/NOC from the Fire Deptt valid upto.,.....(date) ***Marked as 2D***

5. Copies of the electricity bills for immediate past 3 months**. Marked. as *2E,2F,2G***

6. Copy of insurance taken under the Public Liability Act 1961valid up to...,.....***Marked as 2H***

7. Copy of Site plan prepared for emergency response duly approved by DISH Authorities.

(Factory lnspector)

8. A map of the Factory Land showing different buildings and marked for Raw Material Storage, Production Area, Quality Control Lab, Workers Amenities and Office Premises It should be in A-4 size sheet. ***Marked as 21***

Accreditotion:

Do you have any of the following accreditation, if yes give date of accreditation of validity

GLP	
NABL	
ISO 27000	

9. Site wise option of manufacturing license.

7. Registrations Information:

A.

S.No.	Name of Product	Regn No.	Date of Registration	Section category	Type	Name of approved source of import	I/F/H/Bio	Yes/ No	Manufacturing License

*e.g. TI vs TIM)

B. Export only Registration

S.No.	Name of Product	Regn No.	Date of Registration	Section category	Type	Name of approved source of import	I/F/H/Bio	Yes/No	Manufacturing License

*e.g. TI vs TIM)

C. Registration for imports against Exports

S.No.	Name of Product	Regn No.	Date of Registration	Section category	Type	Name of approved source of import	I/F/H/Bio	Yes/No	Manufacturing License

*e.g. TI vs TIM

D. Bio Pesticide Registrations

(Type in chronological order)

S.No.	Name of Product	Regn No.	Date of Registration	Section category	Type	Name of approved source of import	I/F/H/Bio	Yes/No	Manufacturing License

E. Name and designation of the Expert Staff as given in Form lll 2(a).

S.No.	Name	Designation

F. Endorsements

Company Name change:

sub category: 1.1: Guidelines for the Inclusion of Pesticide in the Insecticide Schedule

(Approved by the Central Insecticide Board (CIB) meeting in its 22nd meeting held on 28th May 1992)

1. Common name of the proposed pesticides
2. Chemical name of the proposed pesticides
3. Literature support
    I. Chemistry
    II. Bio-efficacy
    III. Toxicity
4. Registration status in other countries

Subcategory: 1.2: Guidelines for minimum infrastructure facilities to be created by the manufactures of botanical biopesticides (Pyrethrum, Azadiractin, Cymbopogon etc.,)

(Approved by the registration committee in its 297th RC meeting held on 29.01.2009)

S. No.	Manpower Requirement
1.	Quality Control Biologist
2.	Sufficient personnel to supervise production, maintenance, stores etc.

General Requirement

1. Extraction room
2. Production room
3. Formulation unit
4. Packing and storage room
5. Quality control laboratory
6. Protective clothing
7. Respiratory devices
8. First aid measures
9. Waste disposal arrangement in compliance with Pollution Control norms

Plant Equipment/Instrument Requirement

1. Phase separation vessels
2. Decanter
3. Mixing vessels with stirrer
4. Centrifuge
5. Soxhlet apparatus
6. Electric oven with thermometer
7. Chemical measuring cylinders

8. Chemical transfer pumps
9. Filter pressure pump
10. Measuring cans
11. Electric weighing machine
12. Counter scales
13. Filtration assembly
14. Water distillation unit
15. Wrist action shaker
16. Blender
17. Vortex
18. Large air tight containers
19. Material filling trays
20. Storage tank
21. Filling machine
22. Sealing machine
23. Packing accessories

Laboratory Equipment/Instrument Requirement

1. HPLC System
2. pH meter
3. Spectrophotometer
4. Refrigerator
5. Thermometer
6. Microscope
7. Autoclave

8. U. V. Lamp
9. Abel flash point Apparatus
10. Incubator
11. Glassware *viz.* Conical Flasks, Test tubes, Beaker, Separating funnel, Funnels, Glass tubes, etc.
12. Pipette fillers
13. Pipette
14. Burette
15. TLC apparatus & accessories
16. Micro-syringes

Note: These are the general requirements of minimum infrastructure to be created by the manufacturers. However, for specific botanical biopesticides formulation(s) and their quantum of production, requirement of manpower, space, equipment /instrument may be needed.

Subcategory: 1.3: Guidelines for minimum infrastructure facilities to be created by the manufactures of Baculoviruses (Nuclear polyhedrosis virus (NPV), Granulosis virus (GV))

(Approved by the registration committee in its 297th RC meeting held on 29.01.2009)

S. No.	Manpower Requirement
1	Quality Control Biologist
2.	Sufficient personnel to supervise production, maintenance, stores etc.

General Requirement

1. Post culture production room with temperature and humidity control
2. Moth ovi-position room with temperature and humidity control
3. Production room with temperature and humidity control
4. Diet preparation room
5. Virus processing lab.
6. Quality control lab.
7. Formulation unit
8. Cold storage
9. Washing and sterilization facility
10. Stores room
11. Protective clothing
12. Respiratory devices
13. First aid measures.
14. Waste disposal arrangement in compliance with Pollution Control norms

Equipment/Instrument Requirement

1. Diet preparation machine (mixer)
2. Diet dispenser.

3 Multi channel pipette

4. Vacuum pump with an aspirator

5. Blenders

6. Multipurpose centrifuges

7. B.O.D. incubator or an incubator room or an environmental chamber

8 Scale balance

9 Digital balance

9. Haemocytometer shallow depth Counting chamber

11. Hot air oven.

12. Compound research microscope.

13. Zoom microscope.

14. Vortex mixers

15. Tally counters

16. Electric stove.

17. Autoclave

18. Shaker.

19. Water distillation unit.

20. Air conditioners

21. Humidifiers

22. Oviposition iron cage

23. Washing machine

24. Racks and cabinet

25. Refrigerators

Note: These are the general requirements of minimum infrastructure to be created by the manufacturers. However, for specific baculovirus formulation(s) and their quantum of production, requirement of manpower, space, equipment/ instrument may be needed.

Subcategory: 1.4: Guidelines for minimum infrastructure facilities to be created by the manufactures of microbial biopesticides (Antagonistic fungi, Entomopathogenic fungi, Antagonistic bacteria and Entomopathogenic bacteria.

(Approved by the registration committee in its 297th RC meeting held on 29.01.2009)

S. No.	Manpower Requirement
1	Quality Control Biologist
2.	Sufficient personnel to supervise production, maintenance, stores etc.

General Requirement

1. Production, Mixing & Drying Room and Formulation Unit for Antagonistic fungi/ Entomopathogenic fungi
2. Inoculation, Fermentor, & Sterilization Room and Formulation unit for Antagonistic bacteria/ Entomotoxic bacteria.
3. Packaging and storage room
4. Quality Control laboratory
5. Protective clothing
6. Respiratory devices
7 First Aid Measure
8. Waste disposal arrangement in compliance with Pollution Control norms

Plant Equipment Requirement

1. Plant Fermentor with all accessories / Bioreactor / Shaker
2. Steam Boiler
3. Chilling plant
4. Air compressor
5. RO/ Softener (water treatment plant)
6. Distillation Unit

7. Micro centrifuge
8. Magnetic stirrer
9. 300 □□and 160 □ Sieves
10. Electronic Weighing Balance
11. Blender/Homogenizer
12. Vortex
13. Vibro-screen
14. Autoclavable bag
15. Large air tight container
16. Pouch sealing machine
17. Box strapping machine
18. Racks and cabinet

Laboratory Equipment/ Instrument Requirement

1. Autoclave
2. Water bath
3. Shaking incubator
4. Refrigerator
5. Thermo hygrometer
6. U. V. light
7. BOD Incubator
8. Hot Air oven
9. Laminar Flow
10. pH meter
11. Balance (2-3 decimal places)

12. Vacuum pump
13. Hot plate
14. Deep freezer
15. Spirit
16. Microscope and all accessories
17. Glassware *viz.* Conical Flasks, Test tubes, Beaker etc.
18. Petri dishes
19. Titanium inoculating needles
20. Pipette fillers
21. Pipettes (0.1 ml to 20 ml)
22. Haemocytometer
23. Colony counter

Note: These are the general requirements of minimum infrastructure to be created by the manufacturers. However, for specific microbial biopesticides formulation(s) and their quantum of production, requirement of manpower, space, equipment/ instrument may be needed.

Subcategory: 1.5: Guidelines for minimum infrastructure facilities to be created by the manufactures of pesticides and their formulations to meet the requirements for issue of license by respective state governments (Licensing Authority)

(Approved by the registration committee in its 261st RC meeting held on 15.02.2006)

These are the general requirements, However, for individual chemicals, specific requirements are to be installed and additional equipment are to be supplemented to suit the reactions involved in the manufacturing process. Further, other specifications as per the Insecticides Rules viz. medical facilities, protective clothing, respiratory devices, first aid measures, training towards safety precaution and handling safety equipment, safety disposal, etc. are required to be made.

1. The minimum infrastructure with respect to manpower shall be as under

S. No.	Manpower	For Technical plant	For formulation plant
1	Production Manager	R	R
2.	Supervisor	R	R
3.	Instrument and process control personnel	R	R*/NR
4.	Maintenance Personnel (Plant and utilities)	R	R
5.	Store Keeper (Raw material and finished products)	R	R
6.	Quality Control Chemist	R	R
7.	Security personnel	R	R

*** In case of automatic plant**

2. Minimum infrastructure with respect to machinery & equipment

A. **For technical plant**

S. No	Equipment	For Technical plant
1	Control console	R
2.	Feed tank for raw material	R
3.	Reactors	R
4.	Distillation towers	R
5.	Evaporators	R
6.	Condenser / heat exchanger / boiler/ extinguisher plant / chilling system / steam plant	R
7.	Crystallizer	R

8.	Centrifuge	R
9.	Drier	R
10.	Phase separator	R
11.	Extractor	R
12	Storage tank	R
13.	Process water tank	R
14.	Pipelines with conventional colour code	R
15.	Gas plant	R

B. For Formulation plant

S. No.	Equipment	Solid, DP, WG, SP, etc	WG/G	Liquid (EC, EW, SL)
1.	Feeding channel / Chute	R	R	R
2.	Jaw Crusher	R	R	NR
3.	Raw material feed tank	R	R	R
4.	Pulveriser	R	R	NR
5.	Blender	R	R	R
6.	Siever	R	R	R
7.	Bag Filter	R	NR	NR
8.	Homogenizer	R	R	R
9.	Ball mill	R	NR	NR
10.	Weighing machine / platform balance	R	R	R
11.	Vessel with stirrer	R	R	R
12.	Drier	R	R	NR
13.	Cooling machine	NR	R	R
14.	Sprayer	NR	R	NR
15.	Water tank for liquid	NR	NR	R
16.	Filter	NR	NR	R

3. <u>Equipment for quality control laboratory</u>

1. Analytical weighing balance
2. Hot air over

3. Refrigerator
4. pH meter
5. Spectro-photometer / colourimeter
6. GLC / HPLC depending on the products analytical process as in the specification.
7. Standard glassware, chemical and general requirements for laboratory
8. Pesticide Repository
9. Specification / BIS standard of the product to be manufactured / formulated
10. Sieve shaker.
11. Fume Hood
12. Distilled water still
13. Flash point apparatus
14. Melting point apparaturs

4. Packaging plant and equipment

1. Packaging machinery.
2. Filling machine (automatic / semi-automatic).
3. Weighing machine
4. Bagging machine
5. Sealing machine
6. Labelling equipment**.**

Subcategory: 1.6: Guidelines for endorsement of additional manufacturing site

(Approved by the Registration Committee in its 325th RC meeting held on 04th January 2012)

1. The Committee decided that Toll Manufacturing and Additional Manufacturing Site are two different issues. The Committee recommends that as far Toll manufacturing is concerned, it should be regulated by the decision taken by the RC in its 296th meeting. However, endorsement of Additional manufacturing Site may be considered;
2. The Committee preliminarily recommends the following criteria/guidelines for accepting the requests for Additional Manufacturing Sites:-
   (i) The site, proposed to be additional manufacturing site, should belong to the same registrant (**it is not allowed on anyone else's premises, whatsoever**);
   (ii) the registrant should have registration for manufacture of the same insecticide/product in the country where the additional site is located;
   (iii) Letter of consent from already approved source and from new manufacturing site duly certified by the Designated National Authority (**DNA)**, which shall be verifiable by the DNA of India;
   (iv) Additional declaration from already approved source/principals as well as from new manufacturing site with respect to the formulation being made by using technical same raw material & process from the source which is already approved by the RC in India;
   (v) all operations involved in the manufacturing of the insecticide for which the additional site is being considered should be fully operated by the same registrant and have the proof of manufacturing there by submitting an undertaking to this effect, duly certified by the DNA of that country;
   (vi) the registrant shall be responsible for manufacturing the product having identical chemical composition and specifications as that of the original registered source of the product;
   (vii) five batch analytical test report shall be submitted along with the application for the five batches manufactured at the source originally registered and the five batches manufactured at the proposed additional source;
   (viii) samples of all the five-batch manufactured at the source originally registered and all the five batches manufactured at the proposed additional source shall be submitted along with the respective reference standards to the Secretariat of CIB & RC for testing at CIL for verification;
   (ix) the samples and the reference standards should be valid at least for a period of six month from the date of delivery to the Secretariat of CIB & RC;

(x) documents supporting the claim of the registrant about the additional manufacturing site, duly certified by the Designated National Authority (like, ICAMA for China), shall be submitted along with the application, which shall in turn be verified by the DNA of India before considering the endorsement;

(xi) intimation about the import of every consignment, proposed to be imported from the additional manufacturing site, shall be given to the Sectt of CIB along with the name of the port of entry into India and expected date of arrival as soon as the consignment is dispatched from the additional manufacturing site for arranging the testing of sample(s), if so considered necessary;

(xii) Justification for approval of additional manufacturing site; and

(xiii) failure to meeting any of the above-mentioned criteria shall unconditionally empower the Registration Committee to reject application for such endorsement; or endorsement, if already granted.

Subcategory: 1.7: Guidelines for enhancement of shelf life of pesticides registered under section 9(4) of the Insecticide Act, 1968

(Approved by the registration committee in its 284th RC meeting held on 13.12.2007)

i. General Information

An applicant who is registered u/s 9(4) may apply in the prescribed Performa giving his/her name, addresses of office as well as manufacturing premises, the product, for which the shelf-life enhancement is applied and the registration status of the product along with the proof for each of them in the form of notarized copies of supporting documents.

ii. Proof of Shelf-Life Enhancement

The applicant while making an application may substantiate that the shelf life of the product has already been enhanced before his application giving the registration status of the registration in respect of which the shelf life has been enhanced.

iii. Chemical Equivalence

The applicant may submit chemical composition of his product and that of the product of the registrant who has already obtained shelf-life enhancement in a comparative form to prove that the two products are identical.

iv. Equivalence of Packaging

The applicant may submit a detail of primary packaging, approved in respect of his product and compare it with the primary packaging approved based on which the shelf life of the product of another registrant was enhanced before to prove that the primary packaging of the two is identical.

v. Verification

The applicant is required to verify (declare) that the information furnished by him is correct and if the same is found to be incorrect at any point of time, the enhancement shall be liable to be withdrawn.

Application proforma for enhancement of shelf-life of pesticides,

registered under Section 9 (4) of the Insecticides Act, 1968

1. Name and Address of the applicant :
2. Address(s) of Manufacturing Premises :
3. Name of the Product for which shelf life enhancement is applied for :
4. Whether the product is registered u/s. 9(3) or 9(4) in respect of the applicant, if so enclose a notarized copy of the Certificate of Registration :
5. Name of the registrant who has got the shelf life enhanced, give particulars w.r.t Section [9(3) or 9(4) of Registration :
6. Whether the Chemical Composition of the Product of the applicant and that of the Registrant, who got enhancement earlier, is identical (give comparative detailed chemical composition) :
7. Whether the primary packaging of the product of the applicant and that of the Registrant, who got the shelf life earlier is identical (give comparative detailed Primary packaging) :

Date : Signature of the applicant with seal

Verification

I, _____, s/o _____ in my capacity as _____ do hereby solemnly verify that I am competent to make this application and that the information given in the application and the annexure(s) and statements, accompanying it, is correct and complete. I clearly understand that in the event of any information, or part thereof, being found incorrect, the enhancement shall be liable to be withdrawn without any notice.

Place :

Date : Signature of the applicant with seal

CATEGORY: 2. GUIDELINES FOR THE EXPORT AND IMPORT OF PESTICIDES

EXPORT AND IMPORT OF PESTICIDES

Report of the expert group constitutes by the Registration Committee in its 374th RC meeting under the chairmanship of Dr. Sandhya Kulshrestha, Consultant (Pharma) for review of export guidelines.
(Approved by the Registration Committee in its 380th RC meeting held on 16.11.2017).

General Remarks:

1. The registration for manufacture or imports of pesticides for export is required under the provisions of the Insecticides Act. The 'Certificate of Registration' is also a requirement by the companies/firms exporting pesticides because importing countries usually require this document to ensure that the pesticide proposed for import is legally permitted for manufacture/import in the country of export.
2. As per the provisions of the Act, the registration for manufacture or import of pesticides has to be granted after satisfying the efficacy of the product and its safety to human health and environment. However, considering the fact that the pesticide is exclusively for export, not going to be used in our country, no bio-efficacy information/data is asked from the applicant, but in order to ensure the identity of the product and prima facie its safety mainly w.r.t. the workers involved in various activities of manufacture including repacking, transport etc., minimum information on chemistry and on toxicity (only for those pesticides not registered for use in the country) is being asked so that the basic information about the pesticide to be manufactured/imported in our country for the purpose of export is available and prima facie its safety is assessed. Accordingly, precautions can be advised for safety of workers and environment. This is necessary for the Government in order to take precautions to avoid any disaster as happened in Bhopal and to act immediately in case of any untoward incidence happen.
3. Application for registration of pesticides for export usually fall under following categories:
   Category (A) : The pesticides already registered for use in the country, their higher or lower purity / concentration, any other formulation
   Category (B) : Pesticides not registered for use in the country.

   Approximately 90% of applications are under category [A] for which only basic information from chemistry to know the identity of the pesticide to be manufactured/ imported for export is being asked for. No other data/information is being asked. Only for category [B] applications information on chemistry and toxicity to prima facie assess safety to workers is being asked.
4. Applications for registration of pesticides for export are processed on priority. However, to further facilitate the export and ensure that no exports should suffer for want of requirement of certificate of registration, besides 'Star Export House Applications', a new type, i.e. 'Fast Track Applications' is proposed. These are

the applications wherein applicants have 'firm export order'. These applications will be processed on top priority with minimum information and certificate of registration will be issued in 5 working days to manufacture/import the quantity indicated in the export order. Thereafter, registrant can get the condition regarding 'quantity restriction' deleted on submission of requisite information as per the guidelines of export registration under relevant category of registration.

5. The following time frame for grant of registration has been suggested for the completion of export applications. (w.r.t all required parameters)

I	Fast Track Applications	:	05 working days
II	Star Export House Applications		
	Category (A)	:	07 working days
	Category (B)	:	10 working days
III	All other applications for Export Only		
	Category (A)	:	RC to RC basis
	Category (B)	:	03 months

6. **No separate registration is required by the company for export, if the product is already registered in his favour for use in the country.** The company can also export the said product. In order to facilitate such registrants to comply with the export requirements, the following conditions may be incorporated in all certificates of registrations sought for domestic consumption and export.

   "In case of the consignments of the product for export purpose, the primary packaging may be as per the requirement of the importing country."

7. In view of Government, 'Make in India' initiative and to prevent the diversion of unregistered material (which has not been evaluated completely for its efficacy and safety) in domestic market, it is proposed that import of technical or formulation for export of the pesticides/products already registered for use in the country, be permitted only from the source registered for use in the country. Further, if import is from unregistered source, then it will be permitted only against 'Firm Export order' for quantity specified in the order. Further, 'Firm Export Order" should be accompanied by Certificate of registration in the country of import. In such cases, the quantity restriction will not be deleted (unlike 'fast Track Registration'), however, in case of receipt of repeat order, the additional quantity may be get endorsed after submitting the authentic documents from Government Authority regarding export of total quantity permitted earlier. Any other application for import from changed source of import/import from new source for export should comply with the data requirements under guidelines for registration for import from new source or formulation import for use in the country.

8. Thousands of Certificates of Registrations for export only have been issued till date. Further, exports can also be made against the registrations granted for manufacture/import for use in the country to several thousand registrants. As export registrations are being granted on highly relaxed guidelines, only prima-facie not completely ensuring safety, without ensuring efficacy or its quality, therefore, it is suggested that statistics regarding the licenses obtained for registrations issued exclusively for export and exports undertaken pesticide-wise, company-wise may be collected to decide future direction of the policy in this regard. It is important to analyse the use and misuse of pesticides in terms of quality, health, environment and economic security.

Guidance and Requirements for Applicants for Registration under Export only Category

General Guidelines

1. Types of Applications for Export

The export applications will be considered under following three types–(Based on priority and time frame for grant of registration)

i. **Fast Track applications** with "FIRM EXPORT ORDER". These are the applications wherein applicants have "firm export order". These applications will be processed on top priority with minimum information and certificate of registration will be issued in 5 working days to manufacture / import the quantity indicated in the export order. Thereafter, registrant can get the condition regarding "quantity restriction" deleted on submission of requisite information as per the guidelines of export registration under general category.

ii. **Star Export Houses Applications**: The accreditation of Star Export Houses shall be on the same basis as that of the Ministry of Commerce and the fee shall also be charged accordingly.

iii. All other applications for export only.

2. Category of Applications

All above applications are categorized under following two categories.

A. Applications for pesticides already registered for use in the country.
B. Applications for pesticides not yet registered for use in the country.

These A and B categories are further classified into sub-categories based on whether application is for indigenous manufacture or import; for chemical pesticides or botanical pesticides or Microbial Pest Control Agents (MPCA) etc.

3. **Guidance for toxicity information:** For toxicity requirements, published information on the parameters shall be acceptable from any of the following sources:

i) From the reviews undertaken by international organization/institutions like WHO, Joint bodies of FAO/WHO, International Registry for Potential Toxic Chemicals (IRPTC), International Agency on Research on Cancer (IARC), Pesticide Manual, Merck Index etc.

ii) Any evaluation / Fact Sheet by Registration authorities of USA or EU or Canada or Japan including Material Safety Data Sheet shall be accepted. Information from other reliable sources including TOXNET, EXTONET can be provided.

If published information is not available data are to be submitted.

4. **Information on Packaging:** For packaging, labelling an undertaking with respect to type of packaging and Label and Leaflet as per Insecticides Rules, 1971 are required to be submitted for all categories and types of registration.

5. Bio-pesticides: A condition in case of Biopesticides (Botanical and microbial) shall be incorporated in the certificates of registrations for export category: -

"The product being exported is in compliance of Biological Diversity Act, 2002."

6. An Undertaking be obtained and condition on Certificate of Registration be incorporated w.r.t. compliance of provisions of PIC, CWC, WMD, and biological warfare

7. An undertaking shall be submitted by the applicant that the product manufactured / imported for export shall not be used in the country, if export could not be undertaken.

Subcategory: 2.1. Guidelines for the Registration under Indigenous Manufacture for export u/s 9(3) of the Insecticide Act, 1968.

I. For Fast Track Applications

i. Firm Export Order from the importer.

ii. The documents regarding the registration of product in the country of import.

iii. Documents for registration in any developed country like EU, USA, Australia, Japan, and Canada etc with its Decision Guidance Document. [Required only for B II, B III & B IV categories]

iv. Registration will be issued by Secretary CIB&RC for the production/import for the quantity of the export order, and subjected to ex post facto approval by the Registration Committee.

v. The applicant may then submit the requirements/ information as per the applicable category to get the condition deleted with reference to restriction on quantity.

vi. If Technical has not been registered for use in the country either under TI or TIM category and further the applicant may be allowed to procure the Technical from company who has the CR for the manufacturing of technical for exclusively for export purpose under fast-track category (**Approved by the RC in its 425th RC meeting (Vide agenda item no. 10.6) held on 25.01.2021**)

II. Type II and III applications exclusively for export category

1. Category – A I: Products registered for the use in the country by any registrant (including plant-based products, but not microbial pest control agents) for indigenous manufacture of technical with same and different impurity profile/formulations (including combination formulations) for indigenous manufacture made out of these technical and having same lower or higher concentrations, different adjuvants or new type of formulations.

i Data/ information on chemistry as per table 1 (parameter 1 & 2)

ii. An undertaking w.r.t. type of packaging & Label and leaflet as per Insecticides Rule, 1971.

2. Category – A III: In case of Microbial Pest Control Agents if application is for export of already registered strain.

i. Information on chemistry as per table 2 (except 4, 5 and 6).

ii. An undertaking w.r.t. type of packaging & Label and leaflet as per I rule, 1971.

3. Category – B I: The technical once registered exclusively for export (for indigenous manufacture or import), then subsequent registrants applying registrations for the same products (with different impurity profile their higher lower conc. or its formulations including combination formulations).

i. Information on chemistry on parameters 1, 2, 3, 4 and 6 in table 1.

ii. An undertaking w.r.t. type of packaging & Label and leaflet as per Insecticides Rule, 1971.

4. Category – B II: Indigenous manufacture/import of technical and its formulations for pesticides not registered for use in the country and is for sole purpose of export.

i. Information on chemistry on parameters 1 to 6 for both technical and formulations.
ii. Information for toxicity would be required on parameters 1 to 12 for technical and on 12 for formulation as per Table 3. In case of registration of formulation without registering technical, information on its technical is required on parameters 1 to 12 and on formulation on parameter 12.
iii. An undertaking w.r.t. type of packaging & Label and leaflet as per I rule, 1971.

5. Category B III: Indigenous manufacture of Botanical (technical/formulation) Pesticides not registered for the use in the country.

i. Information on Chemistry is required as per Table 2 (All parameters).
ii. Information for toxicity would be required on parameters 1to 6,10 and 12 as per Table
iii. An undertaking w.r.t. type of packaging & Label and leaflet as per I rule, 1971

6. Category B IV: If the Microbial Pest Control Agents (MPCA) or its strain is not registered for use in the country.

iii. Information on Chemistry is required as per Table 2 (All parameters).
iv. Information for toxicity would be required on parameters 1 to 6 and 12 as per Table 3.
v. An undertaking w.r.t. type of packaging & Label and leaflet as per I rule, 1971.

Table 1: Data requirement from the Chemistry angle for the grant of registration of chemical / plant-based pesticides exclusively for export (Category AI, AII, BI and BII)

Sl. No	Parameter	Category		
		AI	**BI**	**BII**
1.*	Chemical composition OR Botanical/common name, Scientific name, common name, if any and place of origin in the country.	R	R	R
2.	Physico-chemical properties	R	R	R
3.*	Specification in BIS format along with undertaking for product quality OR Specification of the product along with undertaking that the product is naturally occurring and not genetically engineered.	NR	R	R
4.	Method of analysis	NR	R	R
5.	Outline of process of manufacture, raw materials used / process of extraction of active ingredient	NR	NR	R
6.	Source of import/supply	NR	R	R

* Whichever is applicable as per product

Table 2: Data requirement from the Chemistry angle for the grant of registration of Microbial Pest Control Agents (MPCA) or its strain not registered for domestic use in the country exclusively for export (Category BIII) and biopesticides for export, already registered strain (Category AIII) and indigenous manufacture of MPCA and its strain not registered for use in the country (Category BIV).

Sl. No.	Parameters	Category		
		AIII	**BIII**	**BIV**
1.	Botanical/common name, Scientific name, common name, if any, and place of origin in the country.	R	R	R
2.	Specify the microbial strain/plant part for the given biological activity and its location of existence in India.	R	NR	R
3.	Composition of the product with percentage of all ingredients/chemical composition and physic chemical properties	R	R	R
4.	Specification of the product along with undertaking that the product is naturally occurring and not genetically engineered.	NR	R	R
5.	Outline of process of manufacture, raw material used/process of extraction of active ingredient.	NR	R	R
6.	**Method of analysis**			
a.	Determination of active ingredient by appropriate method	NR	R	R
b.	Determination of toxin contents using the internationally recognized scientific methods & Standard Operating Procedures in accredited and recognized laboratories only for *Bacillus sphericus* (B.s) & *Bacillus thuringiensis* / similar products.	NR	NR	R
c.	Determination of potency by bioassay	NR	NR	R
	Or			
	Determination of Colony Farming Unit (CFU), Polyhedral Occlusion Bodies (POB) counts by appropriate method	NR	NR	R
7.	Specification in BIS format* along with undertaking for the product quality	NR	R	R

* If no information on certain minor parameters is available the applicant should indicate so.

Table 3: Data requirement from the Toxicology angle for the grant of registration of pesticide exclusively for export.

Sl. No.	Parameter	CATEGORY		
		Chemical and Plant based pesticides		MPCA
		B II	B III	B IV
1.	Acute Oral rat	R	R	R*
2.	Acute dermal	R	R	R*
3.	Acute inhalation	R	R	R*
4.	Primary skin irritation	R	R	R
5.	Irritation to mucous membrane	R	R	R
6.	Allergy / Sensitization	R	R	R
7.	Effect on reproduction	R	NR	NR
8.	Developmental Toxicity	R	NR	NR
9.	Carcinogenicity	R	NR	NR
10.	Mutagenicity	R	R (Ames test)	NR
11.	Neurotoxicity	R	NR	NR
12.	Sign & Symptoms of poisoning/First Aid, treatment & other precautionary measures	R	R	R

* Single Exposure Studies.

Sub category: 2.2 Guidelines for the registration of pesticides under import for export u/s. 9(3)

I. For Fast Track Applications

i. Firm Export Order from the importer.

ii. The documents regarding the registration of product in the country of import.

iii. Registration will be issued by Secretary CIB&RC for the production/import for the quantity of the export order, and subjected to ex post facto approval by the Registration Committee.

iv. The applicant may then submit the requirements/ information as per the applicable category to get the condition deleted with reference to restriction on quantity.

v. Addition of details of product to be imported and exported and chemical composition of technical as well as formulation in certificate of registration for export certificates of import for export in Fast Track Export category u/s 9(3) export.

RC deliberated the agenda and approved the addition of following details in the Certificate of Registration and Label and Leaflet for export certificates of Import for Export in Fast Track Export category u/s 9(3) export:

1. Name and quantity of product to be imported

2. Name and quantity of product to be exported

3. Chemical composition of technical to be imported and chemical composition of formulation to be exported **(Approved by the RC in its 468th RC meeting (Vide agenda item no. 10.36) held on 12.01.2026).**

Specific requirements for different types of applications under import for export categories

Category – A II

Technical import or formulation import exclusively for export from already registered source of import for use in the country'.

i. Information on chemistry is required on parameters 1, 2 and 6 on table

ii. An undertaking w.r.t. type of packaging & Label and leaflets as per Insecticides Rules, 1971.

iii. The Committee further decided that the applicants of import for export category except Star Export category should submit the details about the Import and Exports thereof against the Certificate of Registration issued during last three years under the Import for Export category from already registered sources for the pesticides registered for use in country along with relevant authentic custom documents and/or physical verification of manufacturing/ importing sites & material imported be undertaken by a team deputed by APPA & Secretary (CIB&RC). The Certificate of Registration (CR) under this category shall be issued only after verification of documents/sites as applicable. (**Minutes of 357th Meeting of RC held on 10th August, 2015 as per Agenda Item 4.1)**

iv The Committee further decided that the applicants of import for export category shall be subjected to the decision taken by the Committee in its 357th meeting. **(Minutes of 359th Meeting of RC held on 02.11.2015, as per Agenda Item 4.1)**

Table 1: Data requirement from the Chemistry angle for the grant of registration of chemical/plant-based pesticides exclusively for export (Category All).

Sl. No	Parameter	Category
		AII
1.*	Chemical composition OR Botanical/common name, Scientific name, common name, if any and place of origin in the country.	R
2.	Physico-chemical properties	R
3.*	Specification in BIS format along with undertakingfor product quality OR Specification of the product along with undertaking that the product is naturally occurring and not genetically engineered	NR
4.	Method of analysis	NR
5.	Outline of process of manufacture, raw materials used / process of extraction of active ingredient	NR
6.	Source of import/supply	R

Subcategory 2.3: Guidelines for The Import of Sample Quantity of Pesticides for Research, Trial and Test (RTT) Purposes

(Approved by the RC in its 380th RC meeting (Vide agenda item no. 3.3, 9.8 and Annexure I) held on dated 16/11/2017 and 20.11.2017)

1. Permission for import of sample quantity of pesticides may be given by the Secretary CIB&RC for all cases except the following
   a. Pesticides falling under Red Colour toxicity triangle i.e. extremely toxic category (LD_{50} value less than 50 mg/kg body weight)
   b. Coded compounds which are not registered in any country and do not figure in pesticide manual
2. Permission issued by the Secretary CIB&RC should be brought to the RC for ex-post facto approval
3. Document requirements for seeking RTT import permit
   i. Complete FORM – C (Copy attached)
   ii. Technical bulletin from inventor / manufacturer
   iii. Information on chemistry, toxicity, bio-efficacy, along with authentic reference copy (e.g Pesticide manual)
   iv. Justification for quantity
   v. Authenticated copy of the registration certificate for manufacture of particular molecule / pesticide in that country with validity if any, issued from the Registration Authority or valid manufacturing license / permission to manufacture from the Government Department
   vi. Authorization from manufacture to the applicant to supply the product
   vii. Authorization for supplier in case of sample quantity is not supplied by the manufacturer
   viii. For Coded and non-registered pesticides authentic document from the Inventor
   ix. For import of RTT sample of Technical for the purpose of registration in India, the applicant shall submit an affidavit regarding status of registration of that product in their name under indigenous manufacturing category.

FORM NO. – C

Application for permission to import small quantity of insecticide/chemicals for the purpose of examination, analysis, tests and trials etc.

1.	Name of importer with full address	:
2.	Name of overseas manufacturer and supplier	:
3.	Name of the chemical	:
	i. Code number	:
	ii. Common accepted name	:
	iii. Chemical name	:
	(iv) Whether included in the schedule to the Insecticides Act 1968	:
4.	(a) Toxicity (LD-50 value) (Laboratory animals)	:
	i. Acute Oral	:
	ii. Acute Dermal	:
	(b) Whether manufacture, sale, distribution or use is prohibited in the country of its manufacture or any restrictions have been imported thereon. If so, please give full details	:
5.	Main active group to which the chemical belongs	:
6.	Specific purpose for which the chemical is required	:
7.	Crops/pests against which this is proposed to be tested	:
8.	(a) Name of institutions it is to be tested in the country	:
	(b) Name and Designation of the person under whose supervision tests etc. are to be conducted	:
9.	Quantity asked for and likely area to covered	
10.	Type of packing	
11.	Port of Landing/entry	
12.	Whether being imported as free sample and if so, its nominal value	
13.	Whether any examination, analysis or tests or trials are already being carried on in the country and if so please give details.	

Name of the applicant Signature of the applicant

V E R I F I C A T I O N

I, _____, S/o or D/o _____, resident of _____, do hereby verify in my capacity as _____, that the particulars given above are true and that I am competent to sign and verify the above application.

Place : **Name :**
Date : **Signature:**

Sub category: 2.4. Guidelines for the Export of Small Quantity of Pesticides Samples for Research, Trial and Test (RTT) Purposes

(Approved by the RC in its 455th RC meeting (Vide agenda item no. 10.53 A held on dated 05/12/2023; Public notice issued by the Secretariat of CIB&RC through File No. 26 26-10/2024-CIR-I(E-144492) dated 21.05.2024)

FORM C

APPLICATION FOR PERMISSION TO EXPORT SMALL QUALITY OF INSECTICIDES FOR THE PURPOSE OF RESEARCH, TRIAL AND TEST PURPOSES

1.	Name of the exporter with full address	:
2.	Name of the importing country	:
3.	Name and address of overseas importer(s)	:
4.	Name of the Insecticides (Active ingredient, formulation, etc.,) to be exported	:
5.	Name of the chemical	:
	i. Code number or common accepted name if available	:
	ii. Chemical name or chemical class	:
	iii. Whether included in the Schedule to the Insecticide Act, 1968	:
6.	Specific purpose for which the chemical is required to be exported	:
7.	Quantity to be exported	:
8.	Type of packing	:
9.	Name and address of the manufacturer and manufacturing site in India	:

Place : **Signature of the applicant**

Date : **Name of the applicant**

CATEGORY: 3. GUIDELINES FOR THE GRANT OF REGISTRATION OF CHEMICAL PESTICIDES UNDER VARIOUS CATEGORIES FOR AGRICULTURAL PURPOSES

Registration of Insecticides / Pesticides under Section 9(3B), 9(3) and 9(4) of the Insecticides Act, 1968 and the Rules 1971.

Sub category: 3.1. Division wise data requirements for the registration of pesticides under Section 9(3B), 9(3) and 9(4) of the Insecticides Act, 1968 and the Rules 1971.

1. Legal division requirements

i. Form – I duly signed/digitally signed
ii. Copy of BOD Resolution/Affidavit/Partnership deed (**Notarized**)
iii. Affidavit for Chemical composition on NJSP (**Notarized**)
iv. Certificate as per category of Industry/ Manufacturing license (**Notarized**)
v. PAN No. (**Notarized**)
vi. Incorporation Certificate (**Notarized**) and other documents as per KYC requirement
vii. Proof of ownership/lease agreement or other legal agreement of manufacturing site purported to be used as manufacturing site **(Notarized)**
viii. Technical to be used in formulation should be duly registered in India or the registration application of the Technical should be submitted in parallel to formulation registration application with necessary references (In case of import - Reference of RC meeting in which it was approved). {Only deemed registration status without issuance of Certificate of Registration shall not be considered}.
ix. Name of registrant of Formulation Import (Reference of Registration Committee Meeting in which formulation import was approved).
x. Other mandatory documents for registration u/s. 9(4) FIM/FI/TI category (Approved by the Registration Committee in its 354th RC meeting vide agenda item no. 3.4 held on 31.03.2025)

- ✓ Notarized and valid manufacturing license
- ✓ Consent letter duly authorized by the source of import and authenticated by the Consulate General of India (Indian embassy) shall be required for the followings: (a) Applications for registration of technical import of pesticides and (b) Applications for import of pesticide formulations
- ✓ List of products for which the registration has been given to the firm & Manufacturing License obtained and products actually manufactured during the previous 3 (three) years.
- ✓ Geotagging along with photographs in case of 9(4) TI/FI applications: Geotagging details and duly legalized / apostilled by the competent authority of the respective country may be considered as a proof along with a location map indicating the place where the manufacturing activities are carried out in that country. (Approved by the Registration Committee in its 470th RC meeting vide agenda item no. 8.2 held on 23.03.2026)

Note: The RC has decided to implement the KYC as per decision taken in its 424th meeting. As per requirement the module has been brought to working condition. The applicant are supposed to feed their documents are per requirement. If, the "KYC" is activated, then the applicant needs not to submit the following documents repeatedly:

1. Copy of BOD Resolution/Affidavit/Partnership deed (**Notarized**)
2. Certificate as per category of Industry/ Manufacturing license (**Notarized**)
3. PAN No. (**Notarized**)
4. Incorporation Certificate (**Notarized**) and other documents as per KYC requirement
5. Proof of ownership/lease agreement of manufacturing site purported to be used as manufacturing site**(Notarized)**
6. List of products for which the registration has been given to the firm and Manufacturing License obtained and products actually manufactured during the previous 3 (three) years.

2. Chemistry - Data requirement from for the registration of pesticides in various categories

Sl. No.	Parameter	9(3B)			9(3)				9(4)			
		TI	TIM	FIM	TI	TIM	FI	FIM	TI	TIM	FI	FIM
1	2	3	4	5	6	7	8	9	10	11	12	13
1.	Details of source of supply of Technical	R	NR	R	R	NR	R	R	R	NR	R	R
2.	Chemical Composition (clearly showing claims of purity of active ingredient, impurities or adjuvants, as the case may be) in Form-I and L/L	R	R	R	R	R	R	R	R	R	R	R
3.	Physical and Chemical Properties of the active ingredient in Technical and adjuvant in case of a formulation	R	R	R	R	R	R	R	NR	R	NR	NR
4.	Technical Bulletin	R	NR	NR	R	NR	R	NR	NR	NR	NR	NR
5.	Product Specification in BIS format/BIS No. if published	R	R	R	R	R	R	R	NR	R*	NR	NR
6.	Method of Analysis	R	R	R	R	R	R	R	NR	R*	NR	NR
7.	Analytical Test Report (ATR) from GLP/NABL accredited laboratory	R	R	R	R	R	R	R	NR	R	NR	NR
8.	Characterization (Identity Test) of active ingredient by UV-VIS, IR, MS and NMR spectra)	R	R	NR	R	R	NR	NR	NR	R	NR	NR
9.	Identification& Quantification of Impurities	R	R	NR	R	R	NR	NR	NR	R	NR	NR
10.	Shelf-life claim	R	R	R	R	R	R	R	R	R	R	R
11.	Storage Stability Data (samples stored in three varied agro-climatic conditions) for six months in excess of claimed shelf-life along with meteorological data for corresponding period	R	R	R	R	R	R	R	NR	NR	NR	NR
12.	Establishment of Chemical Equivalence	NR	NR	NR	NR	R*/ NR*	R*/ NR	R*/ NR*	R	R	R	R

13.	Detailed stepwise manufacturing process (provide chemical reactions explained with structural formulae, all applicable reaction other conditions in case of technical grade insecticides including flow sheet diagram).	R	R	R	R	R	R	R	NR	R	NR	NR
14.	Information about Raw Materials Used along with their source of supply	R	R	R	R	R	R	R	NR	R	NR	NR
15.	Effluent Treatment method with complete details	R	R	R	R	R	R	R	NR	R	NR	NR
16.	Legalized letter of consent that the manufacturer is registered the Technical Grade Pesticide/Insecticide/Formulation and that the consents supplying the Technical Grade Pesticide/Insecticide/Formulation to the applicant.	R	NR	R	R	NR	R	**R**	R	NR	R	NR
17.	Registration Certificate from Designated National Authority (DNA) of the pesticides from source country.	R	NR	NR	R	NR	R	NR	R	NR	R	NR
18.	In case the supplies are to be made through a supplier, a duly legalized certificate from the exporting manufacturer that the supplier is his authorized agent and that the invoice would originate from the approved source of import (actual manufacturer). Or Principal company or Subsidiary of registrant of pesticide in India or the supplier authorized by manufacturer and mention the full details of the source of import in the invoice	R	NR	NR	R	NR	R	NR	R	NR	R	NR
19.	An In-Process sample to be drawn from the & D/manufacturing facility of applicant in case of technical indigenous manufacture of the insecticide/sample to be submitted in case of technical	R	R	R	R	R	R	R	NR	R	NR	NR

	import[$]; along with certified reference material (CRM) and standard impurities along with purity certificate in case of technical grade insecticides for pre-registration verification in the Central Insecticides Laboratory, Faridabad (CIL). In case of FIM and FI category sample and CRM to be submitted in CIL Faridabad.											
20.	Methodology for residue estimation in BIS format.	R	R	R	R	R	R	R	NR	NR	NR	NR

Remarks on chemistry parameters:

1. S. No. (6) & (7) R* wherever If BIS published; than not required, if not, published Registered product specification is required.
2. S. No. 12. Not required (NR)- For first Registration, Required (R)- for subsequent registration after the first registration.
3. Adjuvant(s) shall be mentioned by their common names(s) and not by code names or numbers and their complete chemical identity shall be provided.
4. Sample shall be drawn in case of 9(3) TIM and 9(4) TIM as per approved procedure in 442ndRC or as and when amended by RC. In case of 9(3)TI5 batch GLP analysis report to be submitted.
5. In case of the insecticides for Seed Treatment, 'Adhesion to Seed Test' shall be invariably provided.
6. Same method of analysis should be used in generating ATR and shelf-life data.
7. Accelerated Storage Data can be considered for grant of provisional shelf-life. However, in such cases the Certificate of Registration (CR) shall be issued with a validity of two years. Shelf-life claim of up to 2-years or as the case may be (provisionally) be granted to the insecticides with a condition that applicant is required to submit real time / actual storage stability study data in the proposed construct and container of sale for duration of minimum 30 months, within two and half years of submission of application for granting the registration, failing which Registration Certificate shall stand invalid.
8. In reference to the sample drawn / submitted for pre-registration verification, another chance may be given to the willing applicant for re-analysis of the product. Further it was decided that a team of officers be constituted by APPA & Secretary (CIB&RC) from the officers of Secretariat of CIB&RC to draw the in-process sample, in case of indigenously manufacture of Technical/Formulation as per procedure and/or submission of a fresh sample to the Secretariat of CIB&RC, in case of import of Technical/Formulation (Approved by the RC in its 460th RC meeting vide agenda item no. 9.7)
9. 442nd RC minutes modality for the sample drawling for five batch analysis
10. Standard Operating Procedure (SOP) for pre-registration verification and sample drawl through crop portal enclosed as Annexure I. **11.** In case of 9(3b) TI, 9(3) TI/FI & 9(4) TI/FI category, a condition in CR to be incorporated as an analytical test report by the manufacturer (exporting to India) about the quality of the insecticide/Pesticide from a NABL or ISO 17025:2017 compliance /GLP certified laboratory (Such analytical test report in respect of the batch(s) shall accompany each & every consignment exported to India)
12. In case of formulation of pre-mix combination product having three active ingredients, preferably a single method of analysis must be used, otherwise applicant should be submitted technical justification for using more than one method of analysis.

3. Bio-efficacy - Data requirement for the registration of pesticides under various categories

Sr. No.	Parameters	9(3b)			9(3)					9(4)
		TI	TIM	FIM	TI	TIM	FI	FIM	Label expansion	TI/TIM/FI/FIM or endorsement of already approved label expansion.
1.	Bio-effectiveness	R$	R$	R*	R$	R$	R**	R**	R**	No data requirement. The claim will be granted as per approved formulation u/s 9(3)/ approval claims of label expansion.
2.	Phytotoxicity	R$	R$	R*	R$	R$	R**	R**	R**	
3.	Effect on germination of Seed (in case of seed treatment of fungicide and insecticide)	R$	R$	R*	R$	R$	R**	R**	R**	
4.	Effect on natural enemies (parasitoids and/or predators) (Insecticides)	NR	NR	R*	NR	NR	R**	R**	R**	
5.	Effect on beneficial soil micro-organisms &physico-chemical properties (Herbicide only). Effect on beneficial soil micro-organisms in case of Seed treatment and on soil applied pesticides (Fungicides, Insecticide and PGRs).	NR	NR	R*	NR	NR	R**	R**	R**	
6.	Translocation in Plants	R	R	NR	R	R	NR	NR	NR	
7.	Metabolism in Soil	R	R	NR	R	R	NR	NR	NR	
8.	Metabolism in Water	R	R	NR	R	R	NR	NR	NR	
9.	Metabolism in Plant	R	R	NR	R	R	NR	NR	NR	
10.	Persistence in Soil	R	R	R	R	R	R	R	NR	
11.	Persistence in Water	R	R	R	R	R	R	R	NR	
12.	Persistence in Plant	NR	R	R	NR	R	R	R	R	
13.	Compatibility with other chemicals, if claimed	NR	NR	R	NR	NR	R	R	R	
14.	Residues in Plant	NR	NR	R#	NR	NR	R#	R#	R	
15.	Residues in Soil	NR	NR	R#	NR	NR	R#	R#	NR	

Sr. No.	Parameters	9(3b)			9(3)					9(4)
		TI	TIM	FIM	TI	TIM	FI	FIM	Label expansion	TI/TIM/FI/FIM or endorsement of already approved label expansion.
16.	Residue tolerance limits fixed by foreign countries	NR	NR	R	NR	NR	R	R	R	
17.	Cost Benefit Ratio or Per Rupee return (per ha)	NR	NR	NR	NR	NR	R	R	R	
18.	Registration status in foreign countries	NR	NR	R	R	NR	R	R	R	
19.	MRL Performa to submitted in the CROP portal at the time for submission of application.	NR	NR	R	NR	NR	R	R	R	
20.	Label and Leaflets (as per Insecticides Rule 18 and 19)	R	R	R	R	R	R	R	R	

Remarks on Bio-efficacy parameters:

a. **At S. No. 5-** Applicant may submit position paper and published data related to toxicological effects on beneficial insects in case FI/FIM**.**

b. **R***: Two seasons/years data generated at minimum two different agro-climatic conditions.

c. **R**:** Two seasons/years data generated at minimum three different agro-climatic conditions

d. **R#:** One season/year data generated at minimum four different agro-climatic conditions for fungicides, insecticides and PGRs. **Whereas, in case of herbicides, two seasons/" two" year data generated at minimum two agro-climatic zones in case of 9(3b) and two seasons/two-year data generated at minimum three agro-climatic conditions in case of 9(3).** In case of seed treatment one season / year data generated at minimum three different agro-climatic conditions will be required. In case of soil applied Pesticides (Insecticide/ Fungicide/ Herbicide), if residue at harvest is above LOQ then 2nd season data will be required or else one season data will suffice.

e. For commercial non-edible crops (like jute, jatropha, rubber, etc) only, the data on residue and persistent on plant is not required.

f. **R$:** If Technical Import & Technical Indigenous Manufacture application which are submitted together with Formulation Import & Formulation Indigenous Manufacture applications, no bio-efficacy data required.

Note: In case of herbicides data on effect on soil Physico- chemical and biological properties and effect on normally cultivated three succeeding crops is required along with residue studies in the same plots of the field.

***Example*:** for a herbicide intended to be registered for use in wheat crop data on effect on succeeding crops of maize at location one, green gram at location two and sesame at location three may be generated along with residue studies. However, this is only an example and data on any other normally cultivated succeeding crop may be generated.

1. For registration of Combination products of two / three registered pesticides (All sort of combination should be addressed in this)**:**data is required as per the guidelines of Formulation Import / Formulation Indigenous Manufacture. However, data on Sr. No. 1, 2 (if any individual component is not registered as formulations, the two seasons comparable data are required to be generated against target insects/ diseases/ weeds using stable / viable formulations. Already registered formulations of individual component of the formulation at approved dosage shall also be included as check), 4, (required only for insecticides {Natural enemy data not applicable for PGRs})10, 11, 12, 13, 14, 15,16,17, 18 & 19 (Table – 1) if the a.i. content of either component of the combination product is higher than the already registered same formulations combinations, the applicant is required to submit the data.

   Note: S. No. 14 & 15: Data on residue required for as many seasons and locations as required in case of existing FIM u/s 9(3) guidelines. In case of herbicides data on effect on succeeding crops and soil physico-chemical and biological properties are also required along with residue studies as detailed in existing FIM under section 9(3) bio-efficacy guidelines for herbicides.

   S. No. 10,11,12,14 & 15: These data shall only be required when the concentration / a.i. dose are higher than already registered and/or in case new compound is formed (Waiver on residue/persistence studies can be accepted vide 313rd RC Meeting Decisions) and/or any adjuvants/diluents (other than water)/carrier is different that are in the registered formulations of the individual components. Data on 12, 14, & 15 are also required for the additional crop label claims to common of the individual registered.

2. **For use of Surfactant with registered pesticide Formulations,** data on two seasons Sr. No. 1 & 2 (Table – 1) from minimum three different agro-climatic zones are to be generated with surfactant (tank mix) and without surfactant. The data on Sr. No. 10, 11, 12, 14 & 15 (Table – 1) are to be generated with surfactant as per the requirement of the general guidelines u/s 9(3) of the Insecticides Act, 1968.

3. For registration of twin pack of two registered Herbicides and Surfactant**,** data on Sr. No. 1, 2 & 13 (Table – 1) if proposed to mix, Sr. No. 10, 11 & 12, 14, 15, 18 & 19 (Table – 1) is required.

   Note: Two season data on Sr. No. 1 & 2 (Table – 1) are to be generated on combination of the two herbicides with & without surfactant and individual herbicide from minimum three different agro climatic conditions. The bio-efficacy data on other parameters are to be generated on the combination of the two herbicides with surfactant as per the requirement of general guidelines u/s 9(3) of the Insecticide Act, 1968.

4. MRL fixation may be required for seed treatment products on those crops (such as leafy vegetables, etc.) which are being consumed within one month of sowing.

   Note: For Fungicides the efficacy trials should be conducted in areas where the claimed crop is the main crop of that particular area and the area should be hot spot of the claimed disease i.e disease pressure should be in the range of 15-20% (on disease basis) during the trial period and the fungicide should show minimum 60-70% control. This must be certified by the SAU/ICAR institute where the study has been done. In case of insecticide reduction over control should not be less than 70% and in case of herbicides Weed Control Efficiency (WCE) of individual weed species should be minimum 70%, which is duly authenticated by ICAR/SAUs.

4. Toxicity-Data requirement for the registration of pesticides under various categories

Sl. No.	Parameters	Technical		Formulation		Technical/ Formulation	
		9(3b)	9(3)	9(3b)	9(3)	9(4)	9(4)
		TI/TIM	TI/TIM	FIM	FIM/FI	TIM	TI/FIM/FI
1.	Acute oral Rat	R	R	R	R	NR	NR
2.	Acute Dermal- Rat / Rabbit	R	R	R	R	NR	NR
3.	Acute inhalation-Rat	R	R	R	R	NR	NR
4.	Primary Skin Irritation-Rabbit	R	R	R	R	NR	NR
5.	Acute Eye Irritation- Rabbit	R	R	R	R	NR	NR
6.	Skin Sensitization Test	R	R	R	R	NR	NR
7.	Repeated dose range finding oral toxicity study (28 days)*-Rat	R	R	NR	NR	NR	NR
8.	Repeated dose 90 days oral (Rat)	R	R	NR	NR	NR	NR
9.	Repeated dose 90 days oral toxicity-Dog***	R	R	NR	NR	NR	NR
10.	Repeated dose dermal toxicity-Rat/Rabbit	R	R	NR	NR	NR	NR
11.	Repeated dose inhalation toxicity*#- Rat	R	R	NR	NR	NR	NR
12.	Acute Neuro-toxicity-Rodent*	R	R	NR	NR	NR	NR
13.	Repeated dose Neurotoxicity- Rodent*	R	R	NR	NR	NR	NR
14.	Delayed Neurotoxicity- OP compound- Acute exposure-Laying Hen	R	R	NR	NR	NR	NR
15.	Delayed neurotoxicity- OP compound- Repeated Administration- Laying Hen	R	R	NR	NR	NR	NR
16.	Developmental Neurotoxicity*- Rodent	R	R	NR	NR	NR	NR
17.	Combined carcinogenicity & chronic toxicity-Rat	R	R	NR	NR	NR	NR
18.	Carcinogenicity-Mice	R	R	NR	NR	NR	NR
19.	Developmental toxicity study- a) Rat & b) Rabbit	NR	R	NR	NR	NR	NR

20.	Two generation reproduction toxicity in Rat*	NR	R	NR	NR	NR	NR
21.	Mutagenicity **	NR	R	NR	NR	R	NR
22.	Pharmacokinetics and Metabolism in Rat.	R	R	R NR	R NR	NR	NR
23.	Feeding study including Metabolism in Livestock (Goat, Cow) / Poultry (Hen)	R	R	NR	NR	NR	NR
24.	Pharmacokinetics and Metabolism in other mammals and its similarities or differences from humans.	NR	NR	NR	NR	NR	NR
25.	Immuno-toxicity study*	R	R	NR	NR	NR	NR
26.	Acute Avian toxicity *	R	R	R	R	NR	NR
27.	Repeated dose Avian toxicity (One species)	R	R	NR	NR	NR	NR
28.	Avian Reproduction toxicity (One species)	R	R	NR	NR	NR	NR
29.	Acute toxicity to Fresh Water Fish (One species)	R	R	R	R	NR	NR
30.	Acute Toxicity- Honey Bee (Oral & Contact)	R	R	R	R	NR	NR
31.	Acute Toxicity- Earthworm	R	R	R	R	NR	NR
32.	Medical data	R	R	R	R	R	NR
33.	Human toxicity information from foreign countries	NR	R	NR	R	NR	NR
34.	Health recordsof Industrial workers	NR	R	NR	R	NR	NR
35.	International report on Carcinogenicity & Genotoxicity	NR	R	NR	NR	NR	NR

* Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.

** In Mutagenicity test; an Ames test, any two *in-vitro* and one *in-vivo* Mutagenicity tests are required.

Repeated Dose Inhalation Toxicity Study for 90 days' exposure would be required if there is likelihood of significant repeated inhalation exposure as in case of gas, vapors, aerosols, fumigants or likely duration of human exposure via inhalation is long viz. Mosquito coils; sprays used repeatedly.

Footnote:

1. Health records of industry workers shall be submitted in accordance with the provisions of the Insecticides Rules, 1971, as specified in Form-12.
2. Foreign country toxicity records and International carcinogenic reports shall be submitted to provide robust and relevant information sourced from credible public and research domains.
3. Based on the toxicological profile, mode of action, and potential risks associated with the active ingredient, its metabolites, relevant impurities, adjuvants, or other formulation components, the regulatory authority may recommend submission of additional toxicological, Ecotoxicological, or mechanistic data to facilitate a comprehensive and science-based risk assessment.
4. Pharmacokinetics and Metabolism in other mammals and its similarities or differences from humans are required in case of new molecule on case-to-case basis.

1. Technical (TI and TIM) u/s 9(3b), 9(3) and 9(4):-

- In case of Technical (both TI and TIM), name and Percentage of relevant and toxic impurities and/or metabolites should be indicated. Also data/information from peer reviewed journals or fact sheet should be provided about their toxicity.

- **Toxicity study No. 9**- ***Repeated dose 90 days oral toxicity study (Dog): Studies, as per the guidelines, will be required for new molecules. However, peer reviewed internationally published data/ literature is also acceptable for such studies for applications of other categories.

- **Toxicity study no. 17,18**

  i. The Combined Carcinogenicity and Chronic Toxicity study in Rat and

  ii. Separately carcinogenicity study in mice is required.

- **Toxicity study no. 23-**Feeding and metabolism study can be done together.

  Some metabolites can be more harmful than parent compound even in very small quantity due to bio- activation so it can be conditional requirement, depending on case to case basis.

A. TI u/s 9(3):-

a. In case of Technical Import (NEW MOLECULE), data from Sl. No. 1 to 35 are required.

b. In case of Technical Import (NEW SOURCE), data from Sl. No. 1 to 11; 21; 26; 27; 29-34 are required. (in alignment with the 470th RC decision)

c. In case of Technical Import (NEW SOURCE), the name and percentage of relevant and toxic impurities and/or metabolites must be identical or lower compared to the registered source. Any deviation will require submission of supporting data. Additionally, toxicity information should be provided for adjuvants, impurities, or metabolites if identified as known toxicants.

B. TIM u/s 9(3):-

a. In case of Technical Indigenous Manufacture (New molecule), data from Sl. No. 1 to 35 are required.

b. In case same technical is registered for import (**TIM Vs TI**) or formulation made from the same technical is registered for import or indigenous manufacture **(TIM vs FI WRT/ TIM vs FIM WRT):**

I. If impurities are identified, quantified and impurities are within limits (i.e. within maximum of already registered technical), then data on parameters 1-6, 21 and 32 are required. Additionally, health records of industrial workers are to be submitted if manufacturing unit is already existing. (359th RC decision)

II. In case impurities which are not toxic and not relevant and are not within maximum limits of registered technical however are within +3%) and no new impurity is there, then same data as above (Parameters 1 to 6, 21 and 32) are required. Additionally, health records of industrial workers are to be submitted if manufacturing unit is already existing. (359th RC decision)

III. If impurities are identified, quantified and are not within limits of registered technical or if impurities are not within +3%) and/or any additional impurity is present then in addition to the tests as indicated in (4) above, additional tests based on the nature and quantity of impurity and QSAR alert will be taken on case-to-case basis.

c. Technical indigenous manufacture u/s 9(3) in case same technical is registered for import or formulation made from the same technical is registered for import by the same applicant with same composition, process of manufacture etc., (TIM vs FI Original Registrant), then data on Ames Test only is required. (389th RC decision)

C. TI u/s 9(4):-

- In case of technical import u/s 9(4), no studies are required.

D. TIM u/s 9(4):-

- In case of Technical Indigenous Manufacture u/s 9(4), data on Sl. No. 21 data only AMES test is required. In case found to be mutagenic then additional data on one *in vivo* and two *in vitro* are required.
- The Ames Test shall be generated using the technical grade active ingredient (TGAI) from the manufacturing source under consideration for registration.

2. Formulation (FI and FIM) u/s 9(3b) and 9(3) :-

- ✓ In case of formulation, name and properties of adjuvant, along with data/information should be provided about their toxicity and possible interaction with active ingredient (AI). Also, data/information from peer reviewed journals or fact sheet should be provided about their toxicity.
- ✓ Additional toxicological studies may be required in either of the following circumstances:

(i) Where the formulation exhibits greater toxicity than the corresponding technical grade active ingredient, additional studies on sub-acute oral, sub-acute dermal, sub-acute inhalation, neurotoxicity, and developmental toxicity shall be submitted on the formulation.

(ii) For applications under Section 9(3) and Section 9(3B) pertaining to FI/FIM products containing more than one active ingredient, studies to evaluate potential synergistic, potentiating, additive, or antagonistic effects among the active ingredients may be required, as deemed necessary.

A. FIM u/s 9(3):-

a. In case of Formulation Indigenous Manufacture, data from Sl. No. 1 to 6, 26, 29-35 are required.

b. In case same formulation is registered for import (FIM vs FIT)

i. For indigenous manufacture of a pesticide formulation identical in composition to an already registered imported formulation under Section 9(3):
No data required, provided the technical is duly registered and label claims remain unchanged.

ii. For indigenous manufacture of a pesticide formulation with a different composition than an already registered imported formulation under Section 9(3):
Full data must be submitted as per existing FIM guidelines under Section 9(3), provided the technical is duly registered.

c. Herbicides in twin pack with no other herbicide or with surfactant: MSDS and Acute toxicity information on surfactant should be submitted along with data on herbicide formulation.

B. FI u/s 9(3):-

a. In case of Formulation Indigenous Manufacture, data from Sl. No. 1 to 6, 26, 29-35 are required.

C. FI/FIM u/s 9(4):-

In case of Technical import u/s 9(4), no studies are required.

Note: The general recommendations as mentioned in the Guidance Document on Toxicology for registration of Chemical Pesticides in India will be applicable on the basis of merit case to case basis.

'Waiver would be considered when existing information provides robust and full scientifically sound weight of evidence approach and based on read across/bridging from structurally and /or biologically related similar pesticides, physico-chemical/biological properties and use and exposure patterns case to case basis.

The replacement alternatives not involving experiments on animals would be considered and accepted in those cases where data from valid 3Rs (Replacement, Reduction and Refinement) methods recognized by OECD and other renowned Regulatory Authorities is provided wherever applicable, case to case basis.

5. Packaging-Data requirement for the registration of pesticides under various categories

(Chapter V of the Insecticides Rules 1971 in the Insecticides Act, 1968, the rule 16 to 20 of the said chapter deals with the Packaging and Labelling)

Sl. No.	Parameter	Section 9(3B)			Section 9(3)				Section 9(4)			
		TIM	FIM	TI	TIM	FIM	TI	FI	TIM	FIM	TI	FI
1.	Labels and Leaflets per IR-1971, all fields (as applicable) a n d as amended from time to time	R	R	R	R	R	R	R	NR	NR	NR	NR
2.	Manner of labeling and Leaflet	R	R	R	R	R	R	R	R	NR	NR	NR
3.	Type of packaging (Ultra small, small or Bigb whichever is applicable)	R	R	R	R	R	R	R	R	NR	NR	NR
4.	Manner of packaging	R	R	R	R	R	R	R	R	NR	NR	NR
5.	Specification for primary, Secondary and Transport packages (whichever is applicable)	R	R	R	R	R	R	R	R	NR	NR	NR
6.	Details of packaging material and its compatibility with content	R	R	R	R	R	R	R	R	NR	NR	NR
7.	Performance of container with content during storage stability test(Shelf life Study)	R*	R*	R*	R	R	R	R	NR	NR	NR	NR
8.	Transport worthiness test	R*	R*	R*	R	R	R	R	NR	NR	NR	NR

R*- Before Commercialization the data will be required.

Note:

1. In case of additional packaging endorsement applications (already approved packaging), the data at Sl. No. 05, 06, 07& 08, are not required if similar packaging (material)and manner of packaging is being sought by the applicant as has been granted to earlier 9(3) registrant.
2. Specification of Bureau of Indian Standard (BIS) must be followed for all the packaging requirements (Wherever available and applicable).
3. All Packaging tests must be carried out with the product of same batch and in its commercial package preferably in Indian condition.
4. The duration of the test and the conditions including geographical conditions must be mentioned.
5. Storage stability data should be generated keeping at least the following parameters in test protocol such as test temperature, test duration, test packaging material, content of active ingredient (a.i.) in the product during and after storage, test humidity, exposure to light, physical and chemical properties of the product during and after storage etc.
6. The testing protocols must have their basis in the WHO/FAO/ CIPAC/ASTM recommendations or other validated methodology of GLP/ NABL accredited laboratory having packaging testing (chemical / mechanical as applicable etc.) in the scope.
7. The Accelerated Storage Study (ASS) test must be conducted at $54^0C \pm 2^oC$ (wherever applicable) containing corrosiveness study of the product in reference to packaging material for 14 days as per FAO/ WHO manual for claiming appropriate shelf life of the product which can be maximum two years, subject to the condition of providing the ambient storage stability study data of thirty months or as the case may be within thirty months from the date of application for the registration.

Sub category: 3.2. Guidelines for the registration of post-harvest treatment of crop produce

a) Data requirements for Registration of Technical

Note: The data should be submitted on Parameters for Bio-efficacy as per the guidelines approved by the RC from time to time for registration of Chemical pesticides under TIM / TI U/s 9(3) category, as the case may be. The specific data requirements / information for registration of technical (TC/TK) are as under: -

i. Fate and behavior in treated crop produce

ii. Fate and behavior in air (for gases and fumigants)

iii. Fate and behavior in water (for solid and liquid pesticides)

iv. Fate and behavior in soil (for solid and liquid pesticides) and data on Sr. No. 16 & 18 (Table – 1) is required.

v. If the product technical is not in gas form and/or also to be used to make formulation(s) for use for other than fumigation of post-harvest crop produce (e.g. for use in field crops, household purposes etc.), the additional data requirements on technical to be submitted as per the guidelines approved by the 427thRC for registration under TIM, TI category U/s 9(3), as the case may be.

b) Date requirement for registration of MUP (Manufacturing Use Product) (Not for ripening agent)

Note: -

1) If technical grade of the pesticide is not registered, complete data with respect to product technical on Bio-efficacy as per above guidelines under I. above for technical indigenous manufacture (TIM) / technical Import (TI) U/S 9(3), as the case may be, to be submitted along with data on MUP as listed below.

2) If the technical grade and source for import is duly registered as per guidelines of the Registration Committee, data on MUP to be submitted as listed below:

i. Registration status of MUP in other countries.

ii. If the product MUP is not in gas form and/or also to be used to make formulation(s) for use other than fumigation of post-harvest crop produce (e.g. for use in field crops, household purposes etc.) the additional data requirements for registration of MUP to be submitted as per the guidelines approved by the RC from time to time for registration of Chemical Pesticides U/s 9(3).

c) Data requirements for Registration of formulation –

Note:

1. For registration of a formulation under FIM / FI U/s 9(3) category, the technical (TC/TK) and / or MUP from which formulation to be manufactured, should be registered under the Insecticides Act, 1968.
2. An applicant seeking registration of a formulation without registering technical for import/indigenous manufacture U/s. 9(3), required to subunit complete data as per the guidelines for registration of formulation listed below, along with complete data as per above guidelines for registration of technical for import. Indigenous manufacture U/s. 9(3) category, as the case may be.
3. The data parameters for Bio-efficacy shall be as per the guidelines approved by the RC from time to time for registration of Chemical Pesticides under FIM/ FI u/s 9(3) category, as the case may be.

The specific data requirements / information for registration of formulation on post-harvest crop produce (PGR) (**except for Ethephon (formulation) for use as ripening agent as per directions of Hon'ble Delhi High court in W.P. (C) No. 1987 of 2022)** are as under**:**

Table – 2

1.	Data on bio-effectiveness Adverse effects on postharvest crop produce, if any (phyto-toxicity, change in appearance and flavour etc.)	For label claim of controlled atmospheric conditions data on 1. And 2. To be submitted for three repeated trials (same temperature and relative humidity) (or) For label claim of ambient atmospheric conditions data to be submitted for minimum three repeated trials, regardless of season.
2.	Residue and persistence at different interval (immediately after treatment, start from 0 hours till BDL or acceptable level) in post-harvest crop produce.	For label claim of controlled atmospheric conditions (same temperature and relative humidity) data to be submitted for three repeated trials. (or) For label claim of ambient atmospheric conditions (same temperature and relative humidity) data to be submitted for minimum three repeated trials. (or) The requirement of one season four locations will be applicable as per insecticide, fungicide and PGR guidelines.
3.	Registration status of formulation in other countries	Required
4.	MRL Performa and label leaflets	Required
5.	Label claims as per product Bio-efficacy information as per gazette notifications issued by Govt. of India from time to time.	

Guidelines for import and indigenous manufacture of Ethephon (formulation) for use as ripening agent. (As approved in 447RC meeting vide agenda item no. 10.51)

Sl. No.	Parameter		9(3)	
			FI	FIM
1	2		3	4
A.	**CHEMISTRY**			
1	Chemical composition of formulation.		R	R
2	Chemical identity of technical and adjuvants		R	R
3	Physico-chemical properties of technical and adjuvants		R	R
4	Specification of the product and Method of analysis		R	R
5	Analytical Test report		R	R
6	Data on rate of release/application		R	R
7	Shelf life claim & data		R	R
8	Process of manufacture, Step-wise manufacturing process and flow sheet diagram of process of manufacture		R	R
9	Information about raw material used and their source of supply		R	R
10	The applicant should provide sample along with ref./analytical standard for pre-registration verification along with Specification, Method of analysis, Chromatograms etc.		R	R
11	Methodology for residue estimation as per BIS format		R	R
12	Source of supply of technical: The technical should be registered in India for use in country and proof of supply that the formulated product has been manufactured using the particular technical only for consent provided		R	R
13	Form-I, Labels/Leaflets and copy of RTT permit, if relevant.		R	R
14	ETP certificate, if applicable		R	R
B.	**BIO-EFFICACY & RESIDUES**		**FI**	**FIM**
1	Data on bio-effectiveness.	For label claim of controlled atmospheric conditions data on 1 and 2 to be submitted for three repeated trials (same temperature and relative humidity). Or For label claim of ambient atmospheric condition data to be submitted form three repeated trials Of different locations.	R	R
2	Adverse effect on post-harvest crop produce, if any(phytotoxicity, change in appearance and flavor etc.)		R	R

3	Residue and persistence at different interval (immediately after treatment start from 0 hours till BDL or acceptable level) in post-harvest crop produce	For label claim of Controlled atmospheric conditions (same temperature and relative humidity) data to be submitted for three repeated trails. Or For label claim of ambient atmospheric condition data to be submitted form three repeated trials of different locations.	R	R
4	Registration status of formulation in other countries		R	R
5	MRL Performa and label leaflets		R	R
6	Label claims as per product Bio-efficacy : Information as per latest Gazette Notification issued by the Government of India		R	R
C.	**TOXICITY**		**FI**	**FIM**
1	Acute Oral toxicity in rat		R	R
2	Acute dermal toxicity		R	R
3	Acute inhalation toxicity		R	R
4	Primary skin irritation in rabbit		R	R
5	Irritation to mucous membrane in Rabbit		R	R
6	Skin sensitization test		R	R
7	Observation in man if the concentration is lower and WP in place of EC etc., data may not be required (NR type of formulation is such i.e. as per decision of 318th RC held on 27.04.2011)		R	R
8	Toxicity to birds (One species only)		R	R
9	Medical data		R	R
	Note: If the formulation is more toxic than its technical, then the toxicity data on sub-acute oral rat, sub-acute dermal, sub-acute inhalation, neurotoxicity, sub-acute oral dog (Any peer reviewed published data/information for sub-acute oral dog only as approved in 343 and 344 RC meeting) will be acceptable.			
D.	**PACKAGING**		**FI**	**FIM**
1	**Labels and leaflets as per-1971 existing norms**			
a	For size 250 ml/gm & below		R	R
b	For 500 ml/gm & above		R	R
2	**Label to contents**			
a	Detailed Chemical composition		R	R
b	Purpose for import/manufacture		R	R
c	Antidote		R	R
d	Toxicity triangle		R	R
e	Cautionary statement		R	R
f	Brief direction concerning usages		R	R
g	Restriction if any		R	R
3	**Leaflets to contain:**			
a	Detailed chemical composition leaflets accompanying small labels (up to 250mg/ml size container)		R	R

b	Introductory para about the pesticide	R	R
c	Detailed directions concerning usages	R	R
d	Time of application	R	R
e	Application equipment	R	R
f	Waiting period	R	R
g	Symptoms of Poisoning	R	R
h	First-aid measures	R	R
i	Antidote and treatment	R	R
j	Restriction, if any	R	R
k	Instruction for storage	R	R
l	Information regarding disposal of used packages	R	R
4	**Type of packaging (packing material + compatibility with content)**	R	R
5	**Manner of packing**		
a	Specification for primary package	R	R
b	Specification for secondary packaging	R	R
c	Specification for transport packaging	R	R
d	Manner of labeling	R	R
e	Performance of container during storage stability test	R	R
f	Transport worthiness Test	R	R
E	**LEGAL**	**FI**	**FIM**
1	Copy of BOD Resolution/ Affidavit / partnership deed.{notarized)	R	R
2	Affidavit for Chemical composition on NJSP (Notarized)	R	R
3	Certificate as per category of industry/ Manufacturing license (notarized)	R	R
4	PAN No. (notarized)	R	R
5	Incorporation Certificate (Notarized)	R	R
6	Proof of ownership of industry plot	R	R
7	Proof of Source of Technical to be used in formulation is duly registered. (ln case of indigenous manufacture/ import- Reference of RC meeting in which it was approved). {Only deemed registration status without issuance of Certificate of registration shall not be considered].	R	R
8	Letter of consent, duly legalized from Indian Embassy / High Commission / Consulate in the Country of origin (Applicable in case of 9(4) TI/FI Applications)	R	R
9	Copy of 9(3b) Registration certificate, if relevant	R	R

Remarks- Bio-efficacy data on target pests (diseases, insects, nematodes and weeds etc.) generated by ICAR/SAUs and institutes under National Agriculture Research System and other institutes approved by registration committee will only be acceptable.

Sub category 3.3: Guidelines for the registration of Petroleum derived products like spray oil – natural mineral oil products (Approved by the registration Committee in its 326th RC meeting held on 15.02.2012)

S.No	Parameter	Provisional Registration U/s 9 (3B)	Regular Registration U/s 9 (3)
1	2	3	4
A.	**Chemistry**		
1	Source of Supply	R	R
2	Chemical Composition	R	R
3	Chemical Identity	R	R
4	Physico - Chemical Properties	R	R
5	Technical Bulletin	R	R
6	Specification	R	R
7	Method of Analysis	R	R
8	Analytical Test Report	R	R
9	Identification& Quantification of identifiable Impurities	R	R
10	Shelf-life	R	R
10a.	Shelf-life Data	NR	R
11	Process of Manufacture	R	R
a.	Information about Raw Materials Used	R	R
b.	Source of Supply of the raw materials	R	R
c.	Stepwise Manufacturing Process	R	R
d.	Formula of the product	R	R
e.	Flow sheet diagram of process of manufacture	R	R
12	Documents such as registration certificate / manufacturing license or any other approval under any Government regulation will be acceptable to support that manufacturer is actual producer	R	R
13	Certificate from manufacturer that the dealer/ trader is an authorized dealer/ trader of the manufacturer	R	R
14	A test report about the quality of the product from a laboratory as per GLP scheme or from a company with ISO-9000 certification. This requirement will be provided along with first consignment. Thereafter, each consignment should have proper analytical test report of the manufacturer	R	R
15	The applicant should provide sample along- with standards technical sample from the principals/ authorized dealers for chemical verification. In case of technical grade pesticides, samples of standard impurities are also to be provided for chemical verification. In process sample is to be provided with appropriate undertaking	R	R
16	Methodology for residue estimation as per BIS format	R	R
B.	**Bio-efficacy & residue**		
17	Bio-effectiveness *	R	R
18	Phytotoxicity*	R	R

19	Effect on parasite & predators*	R	R
20	Compatibility with other chemicals if claimed	R	R
21	Residues in plant	R	R
22	Residues in soil	R	R
23	Residue tolerance limits fixed by foreign countries	R	R
24	Registration status in foreign countries	R	R
C.	**Toxicity**		
25	Acute oral Rat	R	R
26	Acute Dermal- Rat / Rabbit	R	R
27	Acute inhalation-Rat	R	R
28	PrimarySkin Irritation- Rabbit	R	R
29	Acute Eye Irritation- Rabbit	R	R
30	Skin Sensitization Test		
31	Neurotoxicity	NR	R
32	Teratogenicity	NR	R
33	Effect on reproduction	NR	R
34	Carcinogenicity	NR	R
35	Metabolism	NR	R
36	Mutagenicity	NR	R
37	Toxicity to birds	R	R
38	Toxicity to fish	R	R
39	Toxicity to honeybees	R	R
40	Medical data	R	R
41	Human toxicity information from foreign countries	R	R
D.	**Packaging & labeling**		
42	Labels and leaflets as per IR-1971 existing norms (i) for size 250 ml & below (ii) for 500 & above	R	R
43	Labels to contents	R	R
a.	Detailed Chemical composition	R	R
b.	Purpose for import / manufacture	R	R
c.	Antidote	R	R
d.	Toxicity triangle	R	R
e.	Cautionary statement	R	R
f.	Brief direction concerning usages	R	R
g.	Restriction, if any	R	R
44	Leaflets to contain	R	R
a.	Detailed Chemical composition on leaflets accompanying small labels (upto 250 ml size container)	R	R
b.	Introductory paragraph about the pesticide	R	R
c.	Detailed directions concerning usages	R	R
d.	Time of application	R	R
e.	Application equipment	R	R
f.	Waiting Period	R	R

g.	Symptoms of poisoning	R	R
h.	First-aid measures	R	R
i.	Antidote & treatment	R	R
j.	Restriction, if any	R	R
k.	Instruction for storage	R	R
l.	Information regarding disposal of used packages.	R	R
45	Type of packaging (packaging material + compatibility with content)	R	R
46	Manner of packaging	R	R
47	Specification for primary package	R	R
48	Specification for secondary packaging	R	R
49	Specification for transport packaging	R	R
50	Manner of labeling	R	R
51	Performance of container during storage stability test	NR	R

**** Bio-efficacy data on target pests (diseases, insects, nematodes etc.) be generated by ICAR/CSIR Institutes/SAUs based on multilocation (minimum three locations for 9(3) & two locations for 9 (3b) for two season is required.***

Refer to 'Toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.'

Sub Category 3.4: Guidelines for The Registration of Combination Products Of Two Registered Herbicides

(Approved by the registration committee in its 291st RC meeting held on 28.07.2008)

A.	**CHEMISTRY**	
1.	Detailed chemical composition.	R
2.	Chemical identity of individual a. i content	R
3.	Physico-chemical properties of the combination product	R
4.	Specifications	R
5.	Method of analysis	R
6.	Analytical test report	R
7.	Shelf-life claim / data	R
8.	Manufacturing process	R
B.	**BIO-EFFICACY**	
1.	Bio-effectiveness	R
2.	Phytotoxicity.	R
3.	Persistence in soil*	R
4.	Persistence in plant*	R
5.	Persistence in water*	R
6.	Compatibility with other chemicals**	R
7.	Residues in plant*	R
8.	Residues in soil*	R
	* "if the a.i. content of either component of the combination product is higher than the already registered same formulations combination, the applicant is required to submit the data". ** If the product is proposed to be mixed with other chemicals, the data on compatibility with these chemicals are required to be submitted. **Note:** Data on Bioeffectiveness, phytotoxicity and Residues*(with physico-chemical & biological properties of soil & effect on succeeding crops) to be generated as per the requirement of general guidelines for Registration of Herbicides U/s 9(3).	
C.	**TOXICOLOGY- Revised in 296th RC held on 24-12-2008**	
1.	The technical product of individual pesticides should be registered U/s. 9(3) of Insecticides Act 1968. Or Data on technical is to be submitted as per existing guidelines in respect of that	R

	individual component/(s) which is/are not registered U/s.9(3) of Insecticides Act 1968. This data submission would be irrespective of applicant desire for registration of individual product(s) for registration of technical compounds.	
2.	Data on individual components of formulation is not required.	R
3.	The following data will be required (observed value*) on combination formulation to be registered:- a) Acute oral Rat b) Acute Dermal- Rat / Rabbit c) Acute inhalation-Rat d) Primary Skin Irritation- Rabbit e) Acute Eye Irritation- Rabbit f) Skin Sensitization Test g) Acute Avian toxicity * h) Toxicity to fish i) Toxicity to honeybees j) Toxicity to earthworm k) Medical data l) Human toxicity information from foreign countries m) Health records of Industrial workers **Note :** The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein. Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.	R
4.	The predicted LD_{50}/LC_{50} value* are to be calculated as per WHO formula and should be submitted along with calculation of the raw data and statement on observed* Vs predicted values to be submitted in tabular form.	NR
D.	**PACKAGING AND LABELLING**	
1.	Label and Leaflets as per Insecticides Rules, 1971 existing norms.	R
2.	Type of packing	R
3.	Manner of Packing (Packaging material plus compatibility with content)	R
4.	Specifications for primary packing	R
5.	Specifications for secondary packing	R
6.	Specifications for transport packing	R
7.	Manner of Labelling	R
8.	Instructions for storage and use, etc.	R
9.	Information regarding disposal of used package.	R

R – Required

Sub category 3.5: Guidelines for Registration of Combination of Pesticides (not for herbicides) for Indigenous Manufacture u/s 9(3)

(Approved by the Registration Committee in its 211th RC meeting and revised in its 296th RC meeting)

S. No	Parameter	CBN (IM)
1	2	3
A. CHEMISTRY		
1.	Source of Supply of Technical	R
2.	Chemical Composition	R
3.	Chemical Identity of technical	R
4.	Physico - Chemical Properties of adjuvants	R
5. 5(a)	Technical Bulletin Copy of RTT permit	NR NR
6.	Specification	R
7.	Method of Analysis	R
8.	Analytical Test Report	R
9.	Identification& Quantification of identifiable Impurities	NR
10a.	Shelf-life claim	R
10b.	Shelf-life Data	R
11.	Establishment of Chemical Equivalence	NR
12a.	Process of Manufacture	R
12b.	Information about Raw Materials Used	R
12c.	Their Source of Supply.	R
12d.	Step-wise Manufacturing Process.	R
12e.	Chemical Equation	NR
12f.	Formula	NR
12g.	Flow sheet diagram of process of manufacture	R
12h.	Effluent Treatment method	R

13.	Documents such as registration certificate / Certificate of DNA/manufacturing licence or any other approval under any Govt. regulation will be acceptable to support that manufacturer is actual producer	NR
14.	Certificate from manufacturer that the dealer/ trader is an authorised dealer/ trader of the manufa - cturer.	NR
15.	A test report about the quality of the product from a laboratory as per GLP scheme or from a company of ISO-9000. This requirement will be provided along with first consignment. There after, each consignment should have proper analytical test report of the manufacturer.	NR
16.	The applicant should provide sample along- with standards technical sample from the principals/ authorized dealers for chemical verification. In case of technical grade pesticides u/s 9(3), samples of std. impurities are also to be provided for chemical verification. In process sample to be provided in case of indigenous manufacture of technical u/s 9(3) TIM & 9(4) TIM with undertaking	R
17.	Methodology for residue estimation as per BIS format.	R
B. Bioefficacy		
18a.	Bio-effectiveness	R*
18b.	Phytotoxicity	R*
18c.	Effect on parasites & predators	R*
19.	Translocation in plants	NR
20.	Metabolism in soil	NR
21.	Metabolism in water	NR
22.	Metabolism in plant	NR
23.	Persistence in soil	R
24.	Persistence in water	R
25.	Persistence in plant	R
26.	Compatibility with other chemicals, if claimed	R
27.	Residues in plant	R#
28.	Residues in soil	R#
29.	Residue tolerance limits fixed by foreign countries	NR

30.	Cost benefit ratio	R
31.	Registration status in foreign countries	NR

R* **:** Two seasons/years data generated at minimum three agro-climatic conditions.

R #: One season / year data generated at minimum four agro-climatic conditions.

NOTE:

(i) Data on effect on parasites and predators is not required in case of fungicides.

(ii) If the a.i. content in the combination product is higher than the a.i. content of the registered formulation under the Insecticides Act, the applicant is required to submit the data on persistence (soil, water & Plant) & Residues (plant & soil).

C. TOXICITY (Revised in 296th RC held on 24-12-2008)

I. The technical product of individual pesticides should be registered U/s. 9(3) of Insecticides Act 1968.

or

Data on technical is to be submitted as per existing guidelines in respect of that individual component/(s) which is/are not registered U/s.9(3) of Insecticides Act 1968. This data submission would be irrespective of applicant desire for registration of individual product(s) for registration of Technical component(s).

II. Data on individual components of formulation is not required.

III. The following data will be required (observed value*) on combination formulation to be registered:-

32.	Acute oral Rat	R
33.	Acute Dermal- Rat / Rabbit	R
34.	Acute inhalation-Rat	R
35.	Primary Skin Irritation- Rabbit	R
36.	Acute Eye Irritation- Rabbit	R
37.	Skin Sensitization Test	R
38.	Acute Avian toxicity *	R
39.	Toxicity to fish	R
40.	Toxicity to honeybees	R
41.	Toxicity to earthworm	R

53.	Medical data	R
42.	Human toxicity information from foreign countries	R
43.	Health records of Industrial workers **Note :** The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein. Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.	R
D. Packaging		
44.	Labels and leaflets as per IR-1971 existing norms (i) for size 250 ml & below (ii) for 500 & above.	R
45	Labels to contents	R
a.	Detailed Chemical composition	R
b.	Purpose for import / manufacture.	R
c.	Antidote	R
d.	Toxicity triangle	R
e.	Cautionary statement	R
f.	Brief direction concerning usages	R
g.	Restriction if any	R
46.	Leaflets to contain	
a.	Detailed Chemical composition on leaflets accompanying small labels (up to 250 ml size container)	R
b.	Introductory para about the pesticide	R
c.	Detailed directions concerning usages	R
d.	Time of application	R
e.	Application equipment	R
f.	Waiting Period	R
g.	Symptoms of poisoning	R
h.	First aid measures	R

i.	Antidote & treatment	R
j.	Restriction, if any	R
k.	Instruction for storage	R
l.	Information regarding disposal of used packages.	R
47.	Type of packaging (pkg material + compatibility with content)	R
48.	Manner of packaging	R
48.1.	Specification for primary package	R
48.2	Specification for secondary packaging.	R
48.3.	Specification for transport packaging.	R
49.	Manner of labelling	R
50.	Performance of container during storage stability test	R
51.	Transport worthiness test	R

Sub category 3.6: Guidelines for Registration of Combination Product of One Registered Insecticide and One Registered Fungicide for foliar spray u/s 9(3)

(Approved by the Registration Committee in its 284th RC meeting held on 13.12.2007)

A. CHEMISTRY

1.	Detailed chemical composition	R
2.	Chemical identity of individual a.i. content	R
3.	Physico-chemical properties of the combination product	R
4.	Specifications	R
5.	Method of analysis	R
6.	Analytical test report	R
7.	Shelf-life claim/data	R
8.	Manufacturing process	

B. BIO-EFFICACY

1.	Bio-effectiveness*	R
2.	Phytotoxicity	R
3.	Persistence in soil**	R
4.	Persistence in plant**	R
5.	Persistence in water**	R
6.	Compatibility with other chemicals***	R
7.	Residues in plant**	R
8.	Residues in soil**	R

*Bio-efficacy evaluation of insecticide and fungicide alone and in combination is to be evaluated against insect pest(s) and disease(s) as per label claim(s) under different agro-climatic conditions for two seasons.

**If the a.i. content in the combination product is higher than the a.i. content of the registered formulation under the Insecticides Act, the applicant is required to submit the data.

***If the product is proposed to be mixed with other chemicals, the data on compatibility with these chemicals are required to be submitted.

C. TOXICOLOGY

a) Acute oral Rat
b) Acute Dermal- Rat / Rabbit
c) Acute inhalation-Rat
d) Primary Skin Irritation- Rabbit
e) Acute Eye Irritation- Rabbit
f) Skin Sensitization Test
g) Acute Avian toxicity *
h) Toxicity to fish
i) Toxicity to honeybees
j) Toxicity to earthworm

k) Medical data
l) Human toxicity information from foreign countries
m) Health records of Industrial workers

Note : The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein.

Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.

D. PACKAGING AND LABELLING

1.	Label and Leaflets as per Insecticides Rules, 1971 existing norms	R
2.	Type of packing	R
3.	Manner of Packing (Packaging material plus compatibility with content)	R
4.	Specifications for primary packing	R
5.	Specifications for Secondary packing	R
6.	Specifications for transport packing	R
7.	Manner of Labelling	R
8.	Instructions for storage and use, etc.	R
9.	Information regarding disposal of used package.	R

R-Required.

Sub category: 3.7: Guidelines/Data Requirement for Registration of New Formulation for Seed Treatment U/s.9(3).

(Approved by the Registration Committee in its 296th RC meeting held on 24.12.2008)

A. CHEMISTRY

i) Source of supply of technical
ii) Chemical composition
iii) Chemical identity of technical
iv) Physico-chemical properties of technical and adjuvants
v) specification method of analysis
vi) Analytical test report
vii) Shelf life claim & data
viii) (a) Process of manufacture
(b) Information about raw material used
(c) Their source of supply
(d) Step-wise manufacturing process
(e) Flow-sheet diagram of process of manufacture
(ix) The applicant should provide sample along with ref./analytical standard for pre-registration verification.
(x) (1) Methodology for residue estimation as per BIS format
(2) Copy of Form-I
(3) Labels/Leaflets
(4) Copy of RTT permit, if relevant

B. BIOEFFICACY AND RESIDUES

1. **Bio-effectiveness**

Bio-efficacy evaluation of the product for seed treatment is to be evaluated against insects pest (s), diseases (s) as per label claim (s) under different agro-climatic zones/ multi-location field trial for two season/ year.

2. Phytotoxicity

Data required as for bio-effectiveness above.

3. Effect on parasites & predators

Data on effect on parasites and predators is not required in case of fungicides, but data on beneficial flora and fauna in soil is required.

4. Persistence in soil, water and plant- Data as per existing guidelines required

5. Compatibility with other chemicals, if claimed

If the product is proposed to be mixed with other chemicals, the data on compatibility with other pesticides/bio-pesticides and bio-fertilizers used for seed dressing are required to be submitted.

6. Residue in plant and soil : Residue data as per label claim for four locations, one season is required to be submitted.
7. Registration status in foreign countries- required
8. Copy of Form-I- required
9. Chemical composition- required
10. Copy of RTT permit, if relevant - required
11. Labels/Leaflets- required

C. TOXICITY

a. Acute oral Rat
b. Acute Dermal- Rat / Rabbit
c. Acute inhalation-Rat
d. Primary Skin Irritation- Rabbit
e. Acute Eye Irritation- Rabbit
f. Skin Sensitization Test
g. Acute Avian toxicity *
h. Toxicity to fish
i. Toxicity to honeybees
j. Toxicity to earthworm
k. Medical data
l. Human toxicity information from foreign countries
m. Health records of Industrial workers

Note : The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein.

Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.

D. Packaging and Labeling

1. Labels and leaflets as per-1971 existing norms (i) for size 250 ml & below (ii) for 500 & above.
2. Label to contents
   - (a) Detailed Chemical composition
   - (b) Purpose for import/manufacture
   - (c) Antidote
   - (d) Toxicity triangle
   - (e) Cautionary statement
   - (f) Brief direction concerning usages
   - (g) Restriction if any

3. Leaflets to contain

(a) Detailed Chemical composition leaflets accompanying small labels (up to 250 ml size container)

(b) Introductory para about the pesticide

(c) Detailed directions concerning usages

(d) Time of application

(e) Application equipment

(f) Waiting period

(g) Symptoms of poisoning

(h) First-aid measures

(i) Antidote and treatment

(j) Restriction, if any

(k) Instruction for storage

(i) Information regarding disposal of used packages

4. Type of packaging (packing material + compatibility with content)
5. Manner of packing
6. Specification for primary package
7. Specification for secondary packaging
8. Specification for transport packaging
9. Manner of labelling

10. Performance of container during storage stability test

11. Copy of Form-I

Sub category : 3.8: Guidelines/data Requirements for Registration of Combination product of One Registered Insecticide and One Registered Fungicide for Seed Treatment u/s 9(3)

(Approved by the Registration Committee in its 296th RC meeting held on 24.12.2008)

A. Chemistry

1. Detailed chemical composition
2. Chemical identity of individual a.i. content
3. Physico-chemical properties of the combination product
4. Specifications
5. Method of analysis
6. Analytical test report
7. Shelf-life claim/data
8. Manufacturing process

B. Bioefficacy

1. Bio-effectiveness

Bioefficacy evaluation of insecticide and fungicide alone and in combination is to be evaluated against insect pest(s) and disease(s) as per label claim(s) under different agro-climatic conditions for two seasons.

2. Phytotoxicity- data required as for Bio-effectiveness above**.**

3. Persistence in soil

If the a.i. content in the combination product is higher than the a.i. content of the registered formulations under the Insecticides Act, the applicant is required to submit the data.

1. Persistence in plant

If the a.i. content in the combination product is higher than the a.i. content of the registered formulation under the Insecticides Act, the applicant is required to submit the data.

2. Persistence in water

If the a.i. content in the combination product is higher than the a.i. content of the registered formulation under the Insecticides Act, the applicant is required to submit the data.

3. Compatibility with other chemicals

If the product is proposed to be mixed with other chemicals, the data on compatibility with these chemicals are required to be submitted.

4. Residues in plant

If the a.i. content in the combination product is higher than the a.i. content of the registered formulation under the Insecticides Act, the applicant is required to submit the data.

C. TOXICOLOGY

a. Acute oral Rat
b. Acute Dermal- Rat / Rabbit
c. Acute inhalation-Rat
d. Primary Skin Irritation- Rabbit
e. Acute Eye Irritation- Rabbit
f. Skin Sensitization Test
g. Acute Avian toxicity *
h. Toxicity to fish
i. Toxicity to honeybees
j. Toxicity to earthworm
k. Medical data
l. Human toxicity information from foreign countries
m. Health records of Industrial workers

Note : The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein.

Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.

D. PACKAGING AND LABELLING

1. Label and Leaflets as per Insecticides Rules, 1971
2. Type of packing
3. Manner of Packing (Packaging material plus compatibility with content)
4. Specifications for primary packing
5. Specifications for transport packing
6. Manner of Labelling
7. Instructions for storage and use, etc.
8. Information regarding disposal of used package.

Note:- All data related to Chemistry, Bio-efficacy, Toxicology, Packaging & Labeling are required individually and on combined product.

Sub category 3.9: Guidelines for the Registration of New Formulation (of same a.i.) where another formulation (of same a.i.) is already registered for seed treatment u/s 9(3).

(Approved by the Registration Committee in its 296th RC meeting held on 24.12.2008)

A. CHEMISTRY

1. Chemical composition
2. Physico-chemical properties of adjuvants
3. Specification & Method of Analysis
4. Analytical Test Report
5. Shelf-life claim and data
6. (a) Process of manufacture

   (b) Information about raw materials used

   (c) Their source of supply

   (d)Stepwise manufacturing process

   (e) Flow-sheet diagram of process of manufacture
7. The applicant should provide sample of formulation along with reference/analytical standard.
8. Method of residue estimation as per BIS specification
9. Copy of Form-I
10. Labels/Leaflets

B. BIO-EFFICACY & RESIDUES

1. **(a) Bio-effectiveness**
   Data under different agro-climatic zones/ multi-location field trials for two seasons/ year is required if the pest is different while for the same pest one season under three to four different locations / agro-climatic zones is required to be submitted.
2. **Phytotoxicity**- data required as for bio-effectiveness above.
3. **Residue in plant**
   (i) Required for additional crop as per label claim
   (ii) Required for vegetable crops and crops where young seedlings are consumed as vegetables.
4. **Persistence in plant, soil & water:**
   (i) Required for additional crop as per label claim
   (ii) Required for vegetable crops and crops where young seedlings are consumed as vegetables.
5. **Registration status in foreign countries- required**
6. **Effect on parasites and predators (applicable in case of insecticides only)**- Data required as per existing guidelines.

TOXICOLOGY

a. Acute oral Rat
b. Acute Dermal- Rat / Rabbit
c. Acute inhalation-Rat
d. Primary Skin Irritation- Rabbit
e. Acute Eye Irritation- Rabbit
f. Skin Sensitization Test
g. Acute Avian toxicity *
h. Toxicity to fish
i. Toxicity to honeybees
j. Toxicity to earthworm
k. Medical data
l. Human toxicity information from foreign countries
m. Health records of Industrial workers

Note : The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein.

Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.

Subcategory. 3.10: Guidelines for the registration of Pesticide Formulation FIM vs FIT u/s 9(3) for indigenous manufacture having the identical chemical composition that of formulation already registered for import u/s 9(3) for the same label claims.

(Approved by the Registration Committee in its 317th RC meeting held on 31.03.2011)

Name of the Applicant :- M/S ----------

Product :- ----------

Comments: The above highlighted information has no relevance and suggested to be deleted.

1. Index
2. Page Numbering
3. Authentication of pages by authorized signatory
4. Form – I duly signed
5. Demand draft as registration Fee
6. Requisite number of stamped envelops
7. Copy of BOD Resolution/Affidavit/Partnership deed (**Notarized**)
8. Affidavit for Chemical composition on NJSP (**Notarized**)
9. Certificate as per category of Industry/ Manufacturing license (**Notarized**)
10. PAN No. (**Notarized**)
11. Incorporation Certificate **Notarized**)
12. Labels/leaflets
13. Proof of Source of Technical to be used in formulation is duly registered. (In case of indigenous manufacture / import - Reference of RC meeting in which it was approved). {Only deemed registration status without issuance of Certificate of registration shall not be considered}.

14. Name of registrant of Formulation Import (Reference of Registration Committee Meeting in which formulation import was approved).

15. ATR from independent laboratory

16. A Sample for Analysis by CIL

In view of above, application is

i) Accepted for further scrutiny

ii) Not accepted

Sub category 3.11: Guidelines for Registration of Pesticide Formulation of Pre-mix Combination having Three Active Ingredients for Indigenous Manufacture for Use in Agriculture

(Approved by the Registration committee in its 336th RC meeting held on 01.03.2013)

GENERAL REQUIREMENTS

1. All pesticide active ingredients constituents in the combination must be manufactured from the technical grade pesticides registered under the Insecticides Act, 1968.
2. In case of any active ingredient(s) not yet registered in the country, the entire data / separate application on its technical has to be submitted as per guidelines for TIM U/s 9(3) / TI U/s 9(3) for evaluation of the combination product.
3. If source of supply of technical of any a.i. component(s) of the combination is not yet registered, separate application to be submitted as per relevant guidelines of RC.
4. There should be single clear analytical separation method of active ingredients.
5. The proposed effective a.i. dosages / concentration in combination constituent should not be generally higher than already registered individual a.i. dosages. However, if there is a need, the rationality of such higher dose/ concentration has to be justified.
6. There should not be the chemical Interaction among the a.i. in the combination product.
7. Ingredients of mixture could be from any group of pesticides (e.g. I / F / H etc) depending on target pests, provided rationality is there.
8. All toxicity data should be generated as per RC approved protocols.

Abbreviations: FIM: Formulation Indigenous Manufacture
TIM: Technical Indigenous Manufacture
TI : Technical Import

A.	**CHEMISTRY**
1.	Source of Supply of Technical
2.	Chemical Composition
3.	Physico – Chemical Properties of adjuvants
4.	Technical Bulletin/ scientific information of the combination formulation, If available
5.	Draft Specification to be submitted for future consideration of BIS
6.	Method of Analysis (A Clear analytical separation method of active ingredients)

7.	Analytical Test Report
8.	Shelf-life claim supported by Shelf-life Data – Required as per the requirement of FIM u/s 9(3) guidelines
9.	Information on outline of Process of formulation as per the requirement of FIM u/s 9(3) guidelines
10.	Effluent Treatment method
11.	A sample of the product for pre registration verification
12.	Method of Analysis for residue estimation as per BIS format
13.	Data on Chemical Interaction among the a.i. in the combination product. Protocol attached at **Appendix -I**.
14.	Traceability record of samples used in data generation.
B.	**Bioefficacy**
1	**Rationale** for registering the combination of more than two active ingredients
2	**Bio-effectiveness** : Two seasons/years data on combination formulation from minimum three different agro climatic conditions. If any individual component is not registered as formulation, the two seasons comparable data are required to be generated against the targeted insects / diseases/ weeds using stable/ viable formulation. Already registered formulation of individual component of the formulation at approved dosage shall also be included as check**.**
3	**Phytotoxicity** : Two seasons data on combination formulation from minimum three different agro climatic conditions
4	**Effect on major parasitoids & predators:** Required only for Insecticides (field data to be recorded along with the bioefficacy trial for same number of seasons and locations)
5	*Persistence in soil
6	*Persistence in water
7	*Persistence in plant
8	Compatibility with other chemicals, if claimed
9	*Residues in plant**

10	*Residues in soil**
11	Residue tolerance limits fixed by foreign countries
12	Registration status of combination product in foreign countries
13	Cost-benefit ratio

**** 9&10**: Data on Residue required for as many seasons & locations as required in case of existing FIM u/s 9(3) guidelines. In case of herbicides data on effect on succeeding crops, and soil physico-chemical & biological properties are also required along with residue studies as detailed in existing FIM u/s 9(3) bioefficacy guidelines for herbicides.

*** 5,6,7,9&10**: These data shall only be required when the concentration / a. i. dose are higher than already registered and /or in case new compound is formed and / or any adjuvant / diluents(other than water) / carrier is different that are in the registered formulation of the individual components. Data on **7,9&10** are also required for the additional crop(s) label claims to common of the individual registered.

C.	**TOXICITY**
1	Acute oral Rat
2	Acute Dermal- Rat / Rabbit
3	Acute inhalation-Rat
4	PrimarySkin Irritation- Rabbit
5	Acute Eye Irritation- Rabbit
6	Skin Sensitization Test
7	Mutagenicity
8	Acute Avian toxicity *
9	Toxicity to fish
10	Toxicity to honeybees
11	Toxicity to earthworm
12	Medical data
13	Human toxicity information from foreign countries
14.	Health records of Industrial workers

Note : The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various

	Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein. Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.
D.	**Packaging**
1	Labels and leaflets as per IR-1971existing norms (i) for size 250 ml & below (ii) for 500 & above
2	Labels to contents
2a	Detailed Chemical composition
2b	Purpose for import / manufacture.
2c	Antidote
2d	Toxicity triangle
2e	Cautionary statement
2f	Brief direction concerning usages
2g	Restriction if any
3.	Leaflets to contain
3a	Detailed Chemical composition on leaflets accompanying small labels (upto 250 ml size container)
3b	Introductory para about the pesticide
3c	Detailed directions concerning usages
3d	Time of application
3e	Application equipment
3f	Waiting Period
3g	Symptoms of poisoning
3h	First aid measures
3i	Antidote & treatment
3j	Restriction, if any

3k	Instruction for storage
3l	Information regarding disposal of used packages.
4a	Type of packaging (Packaging material)
4b	Data on Container Content compatibility
5	Manner of packaging
5a	Specification for primary package
5b	Specification for secondary packaging.
5c	Specification for transport packaging.
6	Manner of labeling
7	Performance of container during storage stability test
8	Transport Worthiness Test

Sub category 3.12: Guidelines / data requirements for registration for import of pesticide formulation of pre-mix combination having three active ingredients for use in Agriculture

(Approved by the Registration committee in its 336th RC meeting held on 01.03.2013)

GENERAL REQUIREMENTS

1. All pesticide active ingredients constituents in the combination must be manufactured from the technical grade pesticides registered under the Insecticides Act, 1968.
2. In case of any active ingredient(s) not yet registered in the country, the entire data / separate application on its technical has to be submitted as per guidelines for TI U/s 9(3) for evaluation of the combination product.
3. If source of supply of technical of any a.i. component(s) of the combination is not yet registered, separate application to be submitted as per relevant guidelines of RC.
4. There should be single clear analytical separation method of active ingredients.
5. The proposed effective a.i. dosages of combination constituent should not be higher than already registered individual a.i. dosages for the same uses. However, if there is a need, the rationality of such higher dose/concentration has to be justified.
6. There should not be the chemical Interaction among the a.i. in the combination product.
7. Ingredients of mixture could be from any group of pesticides (e.g. I / F / H etc) depending on target pests, provided rationality is there.
8. All toxicity data should be generated as per RC approved protocols.

Abbreviations: FI: Formulation Import
TI: Technical Import

A.	**CHEMISTRY**
1.	Source of Supply of Technical
2.	Chemical Composition
3.	Physico - Chemical Properties of adjuvants
4.	Technical Bulletin/ scientific information of the combination formulation, If available
5.	Copy of RTT Permit
6.	Draft Specification to be submitted for future consideration of BIS

7.	Method of Analysis (A Clear analytical separation method of active ingredients)
8.	Analytical Test Report
9.	Shelf-life claim supported by Shelf-life Data – Required as per the requirement of FI u/s 9(3) guidelines
10.	A sample of the product for pre-registration verification
11.	Letter of consent from source of import along with Certificate of Registration issued by and duly authenticated by the registration authority verifiable from DNA.
12.	Certificate from Manufacturer that Dealer/Trader is an authorized dealer/trader of the manufacturer
13.	A test report about the quality of the product from laboratory as per GLP scheme or from a company of ISO-9000. This requirement will be provided along with first consignment. Thereafter, each consignment should have proper analytical test report of the manufacturer.
14.	Method of Analysis for residue estimation as per BIS format
15.	Data on Chemical Interaction among the a.i. in the combination product. Protocol attached at **Appendix-I**
16a. 16b.	Copy of RTT permit issued by RC Traceability record of samples used in data generation.
B.	**Bioefficacy**
1	**Rationale** for registering the combination of more than two active ingredients
2	**Bio-effectiveness** : Two seasons/years data on combination formulation from minimum three different agro climatic conditions. If any individual component is not registered as formulation, the two seasons comparable data are required to be generated against the targeted insects / diseases/ weeds using stable/ viable formulation. Already registered formulation of individual component of the formulation at approved dosage shall also be included as check**.**
3	**Phytotoxicity** : Two seasons data on combination formulation from minimum three different agro climatic conditions
4	**Effect on major parasitoids & predators:**

	Required only for Insecticides (field data to be recorded along with the bioefficacy trial for same number of seasons and locations)
5	*Persistence in soil
6	*Persistence in water
7	*Persistence in plant
8	Compatibility with other chemicals, if claimed
9	*Residues in plant**
10	*Residues in soil**
11	Residue tolerance limits fixed by foreign countries
12	Registration status of combination product in foreign countries
13	Cost-benefit ratio

**** 9&10**: Data on Residue required for as many seasons & locations as required in case of existing FI u/s 9(3) guidelines. In case of herbicides data on effect on succeeding crops, and soil physico-chemical & biological properties are also required along with residue studies as detailed in existing FI u/s 9(3) bioefficacy guidelines for herbicides.

*** 5,6,7,9&10**: These data shall only be required when the concentration / a. i. dose are higher than already registered and /or in case new compound is formed and / or any adjuvant / diluents(other than water) / carrier is different that are in the registered formulation of the individual components. Data on **7,9&10** are also required for the additional crop(s) label claims to common of the individual registered.

C.	**TOXICITY**
1	Acute oral Rat
2	Acute Dermal- Rat / Rabbit
3	Acute inhalation-Rat
4	PrimarySkin Irritation- Rabbit
5	Acute Eye Irritation- Rabbit
6	Skin Sensitization Test
7	Mutagenicity
8	Acute Avian toxicity *

9	Toxicity to fish
10	Toxicity to honeybees
11.	Toxicity to earthworm
12.	Medical data
13.	Human toxicity information from foreign countries
14.	Health records of Industrial workers
Note : The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein. Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.	
D.	**Packaging**
1	Labels and leaflets as per IR-1971 existing norms (i) for size 250 ml & below (ii) for 500 & above.
2	Labels to contents
2a	Detailed Chemical composition
2b	Purpose for import / manufacture.
2c	Antidote
2d	Toxicity triangle
2e	Cautionary statement
2f	Brief direction concerning usages
2g	Restriction if any
3.	Leaflets to contain
3a	Detailed Chemical composition on leaflets accompanying small labels (upto 250 ml size container)
3b	Introductory para about the pesticide
3c	Detailed directions concerning usages

3d	Time of application
3e	Application equipment
3f	Waiting Period
3g	Symptoms of poisoning
3h	First aid measures
3i	Antidote & treatment
3j	Restriction, if any
3k	Instruction for storage
3l	Information regarding disposal of used packages.
4a	Type of packaging (Packaging material)
4b	Data on Container Content compatibility
5	Manner of packaging
5a	Specification for primary package
5b	Specification for secondary packaging.
5c	Specification for transport packaging.
6	Manner of labeling
7	Performance of container during storage stability test
8	Transport Worthiness Test

Sub category 3.13: Guidelines for Registration of Technical for Indigenous Manufacturing (TIM) where Pesticides Formulation is registered for Import without registering its Technical:

(Approved by the Registration committee in its 359th RC meeting held on 02.11.2015)

In view of the para 47(b) of the interim Order dated 05.09.2013 and 28.03.2014 passed by Hon'ble High Court of Gujarat in SCA No. 2530 of 2011 and Miscellaneous Civil Application No. 2483 of 2013, the following draft Guidelines are proposed:

1. An application may be submitted for registering of a pesticides Formulation for Import under section 9(3) of the Insecticides Act, 1968, without registering its Technical.
2. The data requirement for registration for "Technical for Indigenous Manufacture" (TIM) contained in a 'Formulation registered for Import' (FI) without registering the "Technical" will be enumerated by the "Registration Committee" (RC) under a new category for registration called "TIM vs FI". Any applicant desirous of registration of the "Technical for Indigenous Manufacture (TIM)" with respect to a registered 'Formulation for Import (FI)' without registration of its Technical, will be required to submit bridging data on shelf-life and packaging and also any other data which the RC may deem fit, and may include data on bio-efficacy and toxicity, based on trial in India and chemical comparison of the Technical grade pesticides.
3. A certificate of Registration of the Technical for Indigenous Manufacture (TIM) will be considered for approval immediately upon evaluation of the data required under the 'TIM vs FI' category by the Registration Committee.
4. In compliance with interim orders of the Hon'ble High Court of Gujarat, dated , 05.09.2013 in WP No. 2530 of 2011 & others, no technical grade pesticide in the formulation registered under the FI category shall be considered as ' deemed to be registered'.
5. In compliance with the orders of the Hon'ble High Court of Gujarat, dated , 28.3.2014 in MCA No. 2483 of 2013 and others , pre-registration verification of the chemical composition of the ' Technical ' in the pesticide Formulation sought to be registered will be mandatory prior to grant of any such registration. As such, the applicant desirous of registration for 'Formulation for Import' will mandatorily be required to submit a sample of the technical grade material for evaluation in the Central Insecticides Laboratory (CIL) , Faridabad.
6. The Registration Committee will be frame and adopt Guidelines on similar lines for registration of 'Technical for Import' where 'Formulation Indigenous Manufacture' has been registered without registering the 'Technical'.
7. All pending cases for registration of 'Technical for Indigenous Manufacture' in respect of already registered FIs will be subjected to the same procedures in accordance with the Guidelines for "TIM vs FIs'.

S. No.	Parameter	TIM vs FI-WRT*/ FIM-WRT*	TI vs FIM*
A. CHEMISTRY			
1	Source of Supply of Technical	NR	R
2	Chemical Composition	R	R
2a.	Chemical Identity of technical	R	R
2b.	Physico - Chemical Properties of pesticides/AI	R	R
2c.	Physico - Chemical Properties of adjuvant	R	R
3.	Technical Bulletin	NR	R
4.	Specification of the product	R	R
5.	Method of Analysis of Active ingredients and its impurities	R	R
6.	Analytical Test Report as per specification	R	R
7.	Identification & Quantification of identifiable Impurities with relevant chromatogram, spectra etc.	R	R
8.	Shelf-life claim	R	R
9.	Shelf-life Data	R	R
10.	Establishment of Chemical Equivalence with relevant chromatogram, spectra etc.	R	R
11.	Process of Manufacture (1st step onward)	R	R
12.	Information about Raw Materials Used and their name, formula,	R	R
12.a.	Source of Supply of raw material	R	R
12.b.	Chemical Equation	R	R
12.c.	Formula of impurities etc	R	R
12.d.	Flow sheet diagram of process of manufacture	R	R
12.e.	Effluent Treatment method	R	NR
13.	Documents such as registration certificate, notarized photocopy containing registered Chemical composition (OR any other legal documents containing Chemical composition of Technical) in the country of origin or	NR	R

	authentic proof of Chemical composition approved by the Govt. agency		
13.a.	Certificate from DNA of respective country	NR	R
13.b.	Certificate of manufacturing license if issued or any other approval under any Govt. regulation to support that applicant is a manufacturer.	R	R
14.	Certificate from manufacturer that the dealer/ trader is an authorized dealer/ trader of the manufacturer.	NR	R
15.	A test report about the quality of the product from a laboratory as per GLP scheme or from a company of ISO-9000. This requirement will be provided along with first consignment. Thereafter, each consignment should have proper analytical test report of the manufacturer.	NR	R
16.	The applicant should provide sample along with standards technical sample from the principals/ authorized dealers for chemical verification. In case of technical grade pesticides u/s 9(3), samples of std. impurities are also to be provided for chemical verification. In process sample to be provided in case of indigenous manufacture of technical u/s 9(3) TIM & 9(4) TIM with undertaking.	R	R
17.	Methodology for residue estimation as per BIS format.	R	R
	FI* Registration of Formulation Import (FI) without registration of its Technical in to India. FIM* Registration of Formulation Indigenous Manufacture (FIM) without registration of its Technical in to India. TIM= Technical Indigenous Manufacture TI= Technical import		
B. BIOEFFICACY			
18.a.	Bio-effectiveness	R+	R+
18.b.	Phytotoxicity	R+	R+
19.	Residues in plant	R^	R^
C.	**TOXICITY**		
20.	Acute oral in rat	R	R
21.	Acute dermal	R	R

22.	Acute inhalation	R	R
23.	Primary skin irritation	R	R
24.	Skin Sensitization Test	R	R
25.	Irritation to mucous membrane	R	R
26.	Mutagenicity	R	R
27.	Medical data	R	R
28.	Health records of Industrial workers.	NR* /R**	NR*/ R**
NR* : If Manufacturing unit is new ; R**: If manufacturing unit is already existing/manufacturing			
Note : The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein. Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.			
D.	**PACKAGING**		
28.	Labels and leaflets as per IR-1971 existing norms (i) for size 250 ml & below (ii) for 500 & above.	R	R
29.	Labels to contains	R	R
29.a.	Detailed Chemical composition	R	R
29.b.	Purpose for import / manufacture.	R	R
29.c.	Antidote	R	R
29.d.	Toxicity triangle	R	R
29.e.	Cautionary statement	R	R
29.f.	Brief direction concerning usages	R	R
29.g.	Restriction if any	R	R
30.	Leaflets to contains	R	R
30.a.	Detailed Chemical composition on leaflets accompanying small labels (up to 250 ml size container)	R	R
30.b.	Introductory para about the pesticide	R	R
30.c.	Detailed directions concerning usages	R	R
30.d.	Symptoms of poisoning	R	R
30.e.	First aid measures	R	R

30.f.	Antidote & treatment	R	R
30.g.	Restriction, if any	R	R
30.h.	Instruction for storage	R	R
30.i.	Information regarding disposal of used packages.	R	R
31.	Type of packaging (pkg material)	R	R
32.	Manner of packaging	R	R
33.	Specification for primary package	R	R
33.a.	Specification for secondary packaging.	R	R
33.b.	Specification for transport packaging.	R	R
34.	Manner of labelling	R	R
35.	Container Content compatibility data	R	R
36.	Performance of container during storage stability tests.	R	R
37.	Transport worthiness test data	R	R

Sub category 3.14: Guidelines for registration of TIM (by the original registrant under Formulation Import (FI category) u/s 9(3)

(Approved by the Registration committee in its 389th RC meeting held on 15.06.2018)

Discipline	Data Requirement
(A) General/Legal;	
	(i)Statutory Form-1
	(ii)Agreement of Technology transfer in India
	(iii)Letter of consent from original source/Sponsor of the supplier for utilization of data submitted for registration under FI category.
	(iv)Letter of confirmation/undertaking from original FI registrant for product manufactured at new manufacturing site is of same specifications & quality as that of the registered source for which the data was submitted.
	(v)Affidavit/Declaration confirming the same registered chemical composition.
	(vi)Undertaking/Agreement for adhering to the same manufacturing process provided by original manufacturer under technology transfer.
	(vii)Rational for approval of additional manufacturing site in India.
(B) Chemistry;	
	(i) Same as "A" above
	(ii) Drawl of in-process sample manufactured at the proposed additional source in line with 9(3) & (4) registration drawn sample should be submitted along with the respective reference standards to the CIL for testing & verification.
	(iii) Analytical test report from GLP/NABL accredited Laboratory for identification & Quantification of its identifiable impurities (with relevant chromatogram, spectra etc.) along with other relevant analytical parameters
	(iv) Submission of Technical Sample along with standards for technical and impurities from original source for chemical verification at CIL.
	(v) The samples and the reference standards from original source should be valid at least for a period of six months from the date of delivery to the Sectt. of CIL;
	(vi) Establishment of chemical Equivalence with the date of original source submitted at the time of registration of product under FI registered for import.
	(vii) Information about Raw material used and their source of supply
	(viii) Step-wise Manufacturing Process
	(ix) Shelf Life data (by ASS test).
(C) Toxicity;	
	(i) Same as "A" above
	(ii) Information about Raw material and their source of supply.
	(iii) AMES Test
(D) Packaging;	
	(i) Same as "A" above
	(ii) Type and manner of packaging.
	(iii) Container content capabilities report.
	(iv) Performance of container during storage stability test.
	(v) Transport worthiness test.
	(vi) manner of labelling strictly in compliance with relevant provision of Insecticide Act. 1968 & rule 1971.
	(vii) Shelf-Life data (by ASS test).

Abbreviations:

TIM: Technical Indigenous
Manufacture; FI: Formulation Import;

Sub category 3.15: Guidelines for The Registration of Manufacture Use Product (MUP) for Indigenous Manufacturing / Import of Pesticide U/S 9(3)

(Approved by the Registration Committee in its 399th RC meeting Vide agenda item no. 9.3.1)

Note:-

1. If the technical grade pesticide from which MUP is to be prepared is not registered, complete data with respect to product Chemistry (along with sample and reference standard of technical grade pesticide and impurities), Bio-efficacy, Toxicity and Packaging as per the applicable guidelines for registration of technical should be submitted.
2. If the technical grade and source for import is duly registered as per guidelines of the Registration Committee, data on MUP to be submitted as listed below.
3. Rationale for Indigenous manufacture / import of the MUP to be submitted in Chemistry, Bio-efficacy, Toxicity and Packaging folders.

Note: 2. For Manufacturing Use Product (MUP) of pesticide u/s 9(3) (Other than fruit ripening agent):

a. If the technical grade pesticide is not registered, complete data with respect to product technical on Bio-efficacy as per guidelines for technical import u/s 9(3) to be submitted along with data on MUP as listed below.
b. If the technical grade and source for import is duly registered as per guidelines of the registration committee, no data on Bio-efficacy will be required except Registration status of MUP in foreign countries.
c. Rationale for import and registering the MUP to be submitted in Bio-efficacy.

S. No	Parameter
A.	**Chemistry**
1.	Source of Supply of Technical
2.	Chemical Composition of MUP
3.	Chemical Identity
4.	Physico - Chemical Properties
5.	Technical Bulletin (applicable in case of Import)
6.	Copy of RTT permit (applicable in case of Import)
7.	Specification
8.	Method of Analysis
9.	Analytical Test Report
10.	Shelf-life claim
11.	Shelf-life Data

12.	Documents such as registration certificate / Certificate of DNA /manufacturing licence or any other approval under any Govt. regulation will be acceptable to support that manufacturer is actual producer (in case of import).
13.	Certificate from manufacturer that the dealer/ trader is an authorized dealer/ trader of the manufacturer (in case of import).
14.	A test report about the quality of the product from a laboratory as per GLP scheme. This requirement will be provided along with first consignment. Thereafter, each consignment should have proper analytical test report of the manufacturer (in case of import).
15.	The applicant should provide sample along- with standard technical sample from the principals/ authorized dealers for chemical verification.
B.	**Bio-efficacy**
1.	Persistence in soil
2.	Persistence in water
3.	Registration status of Manufacturing use product (MUP) in foreign countries
C.	**Toxicity**
1.	Acute oral Rat
2.	Acute Dermal- Rat / Rabbit
3.	Acute inhalation-Rat
4.	PrimarySkin Irritation- Rabbit
5.	Acute Eye Irritation- Rabbit
6.	Skin Sensitization Test
7.	Mutagenicity
8.	Medical data
9.	Health records of Industrial workers
NR*: If Manufacturing unit is new ; R: If manufacturing unit is already existing/manufacturing**	
Note : The corresponding footnotes of the "Guidelines for Data Requirements for Registration of Chemical Pesticides under Various Categories for Agricultural Purposes" shall apply to this guideline, unless otherwise specified herein. Refer to 'toxicity data requirements for chemical pesticides' of 'guidance document on toxicology for registration of chemical pesticides in India.	
D.	**Packaging**
1.	Labels / Leaflets (Ls/Ls) as per the Insecticides Rules, 1971.
2.	Type of packaging
i	Information about packaging material proposed of the product.

ii	Container Content Compatibility data.
3.	Manner of packaging
i	Information regarding filling, sealing of closure system, pilfer proofness, leak proofness on primary, secondary and transport packaging.
ii	Specification of Primary, secondary and transport packaging.
4.	Transportation: undertaking for safe storage handling and transportation of the product.
5.	Manner of labeling
i	Information as per provisions of section 18 & 19 of Insecticides Rules, 1971
ii	Instructions for storage and use including precautionary measures.
iii	Instruction regarding disposal of used material and washing of Insecticides should be proposed.
iv	Information on disposal of empty used containers
6.	Transport Worthiness Test report for the proposed packaging system.
7.	Performance of containers report during the storage stability tests.

Sub category: 3.16. Guidelines for registration of pesticides for Technical Import from New Source (TI-New Source) v/s against the molecule registered under FIM-WRT, FI-WRT and TI U/S 9(3).

(Approved by the Registration Committee in its 470th RC meeting held on 23.03.2026)

S. No.	Parameter	Requirement
1	2	3
A.	**Chemistry**	
1.	Name of source of supply of Technical	R
a.	Office address of the Source of Manufacturer with documentary evidence with geo tagging details.	R
b.	Manufacturing/factory address of the Manufacturer with documentary evidence with geo tagging details.	R
c.	Copy of registration certificate in the exporting country (Where technical is not granted regn. Proof of registered chemical composition is required) *	R
d.	Documentary evidence of manufacturing address to prove that the manufacturing address belongs to the manufacturer with geo tagging details.	R
e.	Copy of manufacturing license of the manufacturer, (In case license of manufacturing of technical is not issued in the exporting country then a notification from the exporting country that the manufacturing license is not issued to the manufacturer in the country as per law).	R
2.	Detailed Chemical Composition	R
a.	Chemical Identity & Physico - Chemical Properties of technical (Molecule).	R
b.	Physico - Chemical Properties of all adjuvants	R
c.	IUPAC name, CAS name & CAS no of Raw materials/adjuvants.	R
d.	Technical Bulletin of the product.	R
3.	Specification of the product in BIS format.	R
a.	Method of Analysis of active ingredients, impurities and adjuvants separately.	R
b.	Analytical Test Report	R
c.	Identification & Quantification of identifiable Impurities from GLP lab (Current 5 Batch)	R
4.	Shelf-life claim	R
5.	Shelf-life Data from GLP lab	R
6.	Establishment of Chemical Equivalence	NR
7.	Process of Manufacture in details	R
7.1	Information about Raw Materials Used & their Source of Supply	R
7.2	Step-wise Manufacturing Process, conditions etc.	R
7.3	Route of formation of impurities, chemical reaction and flow chart diagram.	R
7.4	IUPAC name, CAS name, CAS no., chemical structure, molecular weight of known/detected Impurities.	R

7.5	Manufacturing Chemical Equation in detailed with molecular balancing.	R
7.6	Chemical Formula of all raw material, stepwise products formed etc. with molecular mass and recovery of resultant.	R
7.7	Detailed flow sheet diagram of process of manufacture with conditions such as temp. duration etc.	R
7.8	Effluent Treatment method	R
8.	Documents such as registration certificate / Certificate of DNA giving detailed CC, Manufacturing license or any other approval under any Govt. regulation will be acceptable to support that manufacturer is actual producer.	R
9.	Certificate from manufacturer that the supplier is an authorized supplier of the manufacturer, wherever applicable.	R
10.	A test report about the quality of the product from a laboratory as per GLP scheme. This requirement will be provided along with first consignment. Thereafter, each consignment should have proper analytical test report of the manufacturer from GLP lab.	R
11.	The applicant should provide sample through drawl of sample along-with standards sample from the principals/ authorized supplier for competency & chemical verification. In case of technical grade pesticides u/s 9(3), samples of std. impurities are also to be provided for chemical verification. An undertaking is also to be submitted by the importer for readiness for drawl of in-process sample from the manufacturer.	R
12.	Methodology for residue estimation as per BIS format.	R
B.	**Bioefficacy**	
13.	Bio-effectiveness*	R (2 seasons data on each crop mentioned in labels / leaflets at least at 2 climatic zones)
14.	Phytotoxicity*	
15.	Effect on parasites & predators	NR
16.	Translocation in plants	NR
17.	Metabolism in soil	NR
18.	Metabolism in water	NR
19.	Metabolism in plant	NR
20.	Persistence in soil	NR
21.	Persistence in water	NR
22.	Persistence in plant	NR
23.	Compatibility with other chemicals, if claimed	NR

24.	Residues in plant	R (on representative crop of each group on which pesticide is approved, data for 2 years / seasons from one location for herbicides and repeatedly used fungicides /insecticides at same location). For insecticides and fungicides data for one season from two agro climatic locations.
25.	Residues in soil	R [same as for residue in plant]
26.	Residue tolerance limits fixed by foreign countries	NR
27.	Cost benefit ratio	NR
28.	Registration status in foreign countries	NR
	NOTE: Bio-effectiveness & Phyto-toxicity data to be submitted on all registered formulations of same technical as per RC guidelines on all approved crops at the time of issue of import permit provided the application for registration is received within 4 years of issue of import Permit (313th RC decision). No exemption is allowed in any condition. For herbicides data on effect on Physico-chemical and biological properties of soil and effect on normally cultivated succeeding crops are to be submitted along with residue studies.	
C.	**Toxicity**	
29.	Acute oral in rat	R
30.	Acute dermal	R
31.	Acute inhalation	R
32.	Primary skin irritation	R
33.	Irritation to mucous membrane	R

34.	Skin sensitization	R
35.	Sub-acute oral rat 28 day (Dose range finding)	R
36.	Sub-acute oral rat 90 days	R
37.	Sub-acute dermal	R
38.	Sub-acute inhalation	R
39.	Neuro-toxicity	NR
40.	Synergism & potentiation	NR
41.	Teratogenicity	NR
42.	Effect on reproduction	NR
43.	Carcinogenicity	NR
44.	Metabolism in Rat	NR
45.	Mutagenicity (AMES test, any two In Vitro and one In Vivo Test)	R
46.	Toxicity to birds	R
47.	Repeated dose Avian toxicity (One species)	R
48.	Acute Toxicity- Earthworm	R
49.	Toxicity to fish	R
50.	Toxicity to honeybees (Oral & Contact)	R
51.	Metabolism to live stock	NR
52.	Medical data of industrial workers	R
53.	Human toxicity information from foreign countries	R
54.	Observation in man (Health records of spray operators)	NR
55.	Health records of Industrial workers (a certificate from exporter)	R
56.	Toxicity to live stock (Field trial & observation)	NR
57.	International report on carcinogenicity & genotoxicity status	NR
D.	**Packaging**	
58.	Labels and leaflets as per IR-1971 existing norms (i) For size 250 ml & below For 500 & above.	R
59.	Labels to contents	R
a.	Detailed Chemical composition	R
b.	Purpose for import / manufacture.	R
c.	Antidote	R
d.	Toxicity triangle	R
e.	Cautionary statement	R

f.	Brief direction concerning usages	R
g.	Restriction if any	R
60.	Leaflets to contain	
a.	Detailed Chemical composition on accompanying leaflets small labels (up to 250 ml size container)	NR
b.	Introductory para about the pesticide	R
c.	Detailed directions concerning usages	NR
d.	Time of application	NR
e.	Application equipment	NR
f.	Waiting Period	NR
g.	Symptoms of poisoning	R
h.	First aid measures	R
i.	Antidote & treatment	R
j.	Restriction, if any	R
k.	Instruction for storage	R
l.	Information regarding disposal of used packages.	R
61.	Type of packaging (pkg material + compatibility with content)	R
62.	Manner of packaging	R
a.	Specification for primary package	R
b.	Specification for secondary packaging.	R
c.	Specification for transport packaging.	R
63.	Manner of labelling	R
64.	Performance of container during storage stability test	R
65.	Transport worthiness test	R

*For proof of Chemical Composition (Where Chemical Composition is not published on the certificate of registration) the documents may be accepted: Such as apostille/notarized copy of application form containg Chemical Composition submitted to the regulatory authority in the exporting country with a undertaking by the source that their source can be cancelled without any notice, if their Chemical Composition is found different from the registered one at the time of registration.

Note: Data requirements of chemical pesticides are also applicable for registration of chemical plant growth regulators.
Abbreviations: R: Required; NR: Not Required; TI: Technical Import
FI-WRT: Formulation Import Without Registering Technical; FIM-WRT: Formulation indigenous manufacturing without Registering Technical\

CATEGORY: 4. GUIDELINES FOR THE GRANT OF REGISTRATION OF BIOPESTICIDES UNDER VARIOUS CATEGORIES FOR AGRICULTURAL PURPOSES

Sub category: 4.1. Guidelines for the registration of microbial based pest control agents

General requirements for Bio-pesticides/ Microbial Pest Control Agents (MPCA)

1. The applicant needs to submit MOU/license agreement between the applicant and the inventor (either own R&D Laboratory or outsourced Research Institute/Facility) or Authorization letter from the inventor of strain OR undertaking by the applicant about the name of inventor/source of strain as per **Annexure-I.**
2. Undertaking declaring that the product is free from Chemical pesticides/Botanicals pesticides/Other Agro-Chemicals as **Annexure-II.**
3. Undertaking on bio-pesticides composition as **Annexure-III.**
4. Updated stakeholder list for all members in Association / Organization claiming for MOU/authorization for data/technology utilization for mass multiplication/commercialization of the strain, **if applicable**.
5. Form-I dully filled and signed giving complete details along with requisite fee as applicable.
6. Notarized copy of BOD Resolution/ affidavit in case of proprietor/ partnership deed in case of partnership firms.
7. Correct composition as per earlier 9(3) / 9(3B) registrant of bio-pesticide strain.
8. The applicant should also, submit notarized copy of the Permanent Account No. (PAN), allotted by the Income Tax Department.
9. In case of company, the Certificate of Incorporation granted by the Registrar of Companies.
10. Fee receipt issued by NBAIM, Maunath Bhanjan, UP after submission of sample for DNA verification.
11. Undertaking that the product is free from GMO as **Annexure-IV**

The Data Requirements for Microbial Pest Control Agents (MPCA) Pesticides

i) Entomo-pathogenic/ Entomo-toxic Bacteria

ii) Antagonistic Bacteria

iii) Entomo-pathogenic Fungi

iv) Antagonistic Fungi

v) Nuclear Polyhedrosis Virus (NPV) & Granulosis Virus (GV)

A. Chemistry:

Sl. No.	Characteristics	Microbial (Antagonistic bacteria, Entomo-pathogenic/Entomo-toxic bacteria, Entomo-pathogenic fungi, Antagonistic fungi, and Baculovirus			
		Primary/Mother Culture#		Formulated product	
		9(3B)	**9(3)**	**9(3B)**	**9(3)**
1.	Systematic name (Genus and species)	R	R	R	R
2.	Strain name	R	R	R	R
3.	Common name, if any	R	R	R	R
4.	Source of origin	R	R	R	R
5.	Specification of the product containing Habitat, Physical appearance and morphological description, pH, particle size, suspensibility, miscibility etc parameters	R	R	R	R
6.	Isolation of strain & Manufacturing process	R	R	R	R
7.	Methods of analysis including Quantitative analysis	R*	R*	R*	R*
8.	Shelf life claims	R	R	R	R
9.	Data on storage stability as per shelf Life claims	NR	R	NR	R
10.	Composition of the product	R	R	R	R
11.	Potency of product by bioassay method (LC 50 (Beta, Delta, Cry toxin endotoxin content, classification(delta endotoxin)	NR#	NR#	NR#	NR#
12.	CFU/g or ml	R	R	R	R
13.	POB/Capsule count per ml/g of the product	R%	R%	R%	R%
14.	Adjuvants	NR	NR	R	R
15.	Human pathogens (culture method)	R	R	R	R
16.	Percent content of the Bio-control mass/organism in the formulation & nature of biomass.	R	R	R	R

17.	Percentage of carrier/filler, wetting/ dispending agent, stabilizers/ emulsifiers, contaminants/ impurities etc.	R	R	R	R
18.	Moisture content	R	R	R	R
19.	Contaminants: Pathogenic contaminants such as Salmonella, Shigella, Vibrio and such other microbial, not to exceed 1 x 10^4 count per ml or per g of formulation.	R	R	R	R
20.	Undertaking for free from Chemical and botanical pesticide contaminants	R	R	R	R
21.	Natural occurrence of the organism	R	R	R	R
22.	PCR / Immunology assays ELISA Test	NR$^{\$}$	NR$^{\$}$	NR$^{\$}$	NR$^{\$}$
23.	Separation and purification of crystals	NR	NR	NR	NR
24.	A sample for verification (500$^{@}$ g or 500$^{@}$ mL as the case may be)				
	a. DNA fingerprinting for the strain verification from NBAIM, Maunath Bhanjan.	R	R	R	R
	b. Pre-registration verification at Central Insecticide laboratory (CIL)	R	R	R	R

Note: In case of first-time registration of any strain if not proposed to be used in the field, Chemistry data is not required.

*Test procedure and criteria used for identification – morphology, biochemistry, serology/ Immunology for Entomo-toxic-bacteria.

- # and $ these parameters are required for Entomo-toxic bacteria.
- % these parameters are required for Virus.
- @ Two samples of same batch of 500 gm/ml each along with copy of the Fee receipt shall be submitted to Central Insecticides Laboratory, Faridabad for PRV purpose by the applicant for Entomo-toxic bacteria.
- POB/Capsule count per ml/g of the product only for NPV.
- Viral unit: NPVs $1x10^9$ POB/ml or gm. minimum, GVs: $5x10^9$ capsules /ml or gm minimum.
- Dual culture to attain at least 50% reduction in target organism (35% for antagonistic bacteria). Bioassay: based on diseased severity and root colonization.
- Natural occurrence of the organism, Immunology assays: Elisa and Separation and purification of crystals are required for Entomo-toxic bacteria.
- Test procedure and criteria used for identification by DNA test (Restriction enzymes analysis test).
- Biological assays for determining the LC_{50} / LD_{50} of the formulation for Entomo- pathogenic Viruses. Production of Entomo-pathogenic Viruses at commercial- scale was done exclusively *in-vivo* by culturing large number of larvae of host insect and subsequently feeding them with semi-synthetic diet contaminated with virus inoculums in laboratory. Viruses' production *in-vitro* by culturing insect cells in bioreactors was a substitute for labor intensive maintenance of the massive host-insect colony.
- Manufacturing process including type of fermentation and biological end products. The microbial cultures are multiplied by liquid solid fermentation. Information pertaining to use of entire mycelia mats with spores separated must be provided in terms of biomass.

Documents to be mandatorily furnished by applicant applying u/s 9 (3)/ 9(3b) for all categories of bio pesticides

1. Verification of the Authorization letter submitted by the applicant via e mail by Secretariat from the original inventor/source of the strain for data utilization for mass multiplication.
2. MOU/license agreement between the applicant and the inventor (either own R&D Laboratory or outsourced Research Institute/Facility) or Authorization letter from the inventor of strain OR undertaking by the applicant about the name of inventor/source of strain as per **Annexure-I**
3. Updated Stakeholder list for all members in Association/ Organization claiming for MOU/authorization for data /technology utilization for mass multiplication/commercialization of the strain.
4. Relevant Affidavit/Undertakings:-
   - **a)** Affidavit on bio-pesticide composition on NJSP duly notarized.
   - **b)** Notarized copy of depositing microbial bio-pesticides strain sample in National Repository with reference code number.
   - **c)** Undertaking on NJSP duly notarized that product do not contain any genetically modified organism in the prescribed format.
   - **d)** Undertaking on NJSP duly notarized that product is free from chemical/botanical pesticides / other agro-chemicals.
   - **e)** Affidavit of strain Innovator or applicant
   - **f)** Copy of 9(3B) Registration certificate, if relevant.

Note:

1. Bt products should be labeled with bio potency and (or) toxin content. In addition, the labels will have to contain a measurement of toxin protein as percent protein, referring to the Lepidopteran-active toxin(s) present in thecrystal.
2. The presently used Bt var. kurstaki standard is HD-1-S-1980 and its potency was calculated at 16,000 IUs per milligram of powder (Beegle et al. 1986. Standardization of HD-1-S-1980: US Standard for Lepidopterous-active *Bacillus thuringiensis*. Bulletin Ent. Soc. America 32: 44-45.). This standard strain is now available with PDBC, Bangalore and DOR,Hyderabad.
3. Defined potency and toxin concentration – Bioassay would require the use of an insect species. Normally manufacturers could select *Trichoplusiani*/ *Helicoverpaarmigera* for Lepidopteran specific Bt formulations. *Spodoptera* Units (SPU), *Leptinotarsa* Units (LTUs) or International Toxin Units (ITUs) are to be used for denoting a specific insect.
4. No test for beta exotoxin is required for *Bacillus sphaericus*, because this species is not known to produce exotoxins.
5. The bio-potency of products based on *B. thuringiensis* subsp. *Israelensis* (*Bti*) is compared

against a reference strain IPS82, 1884 using early fourth-instar larvae of *Aedes aegypti* (strain Bora Bora). The toxicity of IPS82 has an arbitrarily assigned toxicity of 15,000 ITU/mg powder.

6. The biopotency of products based on *B. sphaericus* (*Bsh*) is determined against a reference standard SPH88, strain 2362 using early fourth-instar larvae of Culex pippins (strain Montpellier). The toxicity of SPH88 has an arbitrarily assigned toxicity of 1,700 ITU/mg of the powder (Guidelines for laboratory and field testing of mosquito larvicides, WHO 2005 pp45).
7. The use of alternative bacterial reference powders and / or strains must be approached cautiously. Such alternatives must be the subject of careful cross- calibration against the reference powders and should be conducted by recognized laboratories and should be made available to anyone who wishes to use, or check, the test with the alternative powders/strains.
8. Water content should not exceed 8 %, (12% in *Pseudomonas spp*) to preclude premature degradation of the product.

Sub category 4.1.1: Guideline for already registered formulation/Strain u/s 9(3) under the IA, 1968

Applicant shall submit only one folder containing the following documents:

I. Form-I dully filled and signed giving complete details along with requisite fee as applicable.

II. Notarized copy of BOD Resolution/ affidavit in case of proprietor/ partnership deed in case of partnership firms.

III. Correct composition as per earlier 9(3) registrant of bio-pesticide strain.

IV. The applicant should also, submit notarized copy of the Permanent Account No. (PAN), allotted by the Income Tax Department.

V. In case of company, the Certificate of Incorporation granted by the Registrar of Companies.

VI. Authorization letter from the inventor of strain OR undertaking by the applicant about the name of inventor/source of strain as per **Annexure-I.**

VII. Requisite number of stamp and envelopes.

VIII. Copies of Label Leaflets of the product as approved by RC of already registered strain.

IX. Copy of letter of Accession No. of strain or information on Accession number of strain.

X. Undertaking declaring that the product is free from Chemical pesticides/Botanicals pesticides/Other Agro-Chemicals as **Annexure-II**

XI. One sample of 500 gm/ml quantity shall be deposited to NBAIM, Maunath Bhanjan for test relating to DNA fingerprinting particularly partial gene code sequencing of desired strain and a fee may be paid directly to NBAIM through DD or online.

XII. Original fee receipt issued by NBAIM, Maunath Bhanjan, UP.

XIII. Undertaking that the product is free from GMO as **Annexure-IV**

XIV. Undertaking on bio-pesticides composition as **Annexure-III**

XV. Toxicology data shall be accepted from Non-GLP laboratory also for encouraging of new strain registration/any new or repeat studies for old strain. This decision shall be applicable to all categories of bio-pesticide registration, henceforth. The applications under scrutiny in the Secretariat of CIB&RC are also covered under this decision.

XVI. A sample of 500 gm/ml shall be deposited in the Secretariat of CIB&RC along with File/documents for PRV purposes.

XVII. The above folder shall be scrutinized by the Chemistry division of the Secretariat of CIB&RC.

XVIII. No preliminary scrutiny is required for already registered strain of bio- pesticides.

XIX. A letter may also be written by the Secretariat of CIB&RC to the Director, NBAIM, Mau Nath Bhanjan, UP for submitting the DNA finger print report directly to the Secretariat of CIB&RC, certifying that the DNA of strain submitted by the applicant (Strain No.) matches with original Strain or otherwise.

XX. RC also decided that any government laboratory willing to undertake such studies on the terms and conditions as approved by the committee may request Secretariat of CIB&RC so as to seek approval from RC.

Annexure-I

AFFIDAVIT ON BIO-PESTICIDE STRAIN BY INVENTOR OR APPLICANT

I, …….. S/o ……, aged …….years, resident of

……..and Proprietor/Authorized person of the firm M/s ……..., having its office at … .....do hereby declare and solemnly affirms as under:

That I am in the capacity of …….. of firm M/s …… do hereby declare that the information furnished with respect to composition in Form-I, Label/Leaflet and bonafide verification of the application fir registration of (Name of the product)……..., CFU/PBO…….per gm or ml min; Strain No. (Name and number of registered strain)……. invented by M/s ..... is registered under section 9(3b) or 9(3) of the Insecticides Act, 1968.

1. That a shelf life of the product shall be twelve/Six/four months.
2. The product shall be packed as per IS: 8190 (Part-I) 1988 for Solid Pesticide (Second Revision).
3. That there will be no change in chemical composition, shelf-life, packaging requirement and the product will have the quality and packaging as per the relevant IS or as per specification approved by Registration Committee for 9(3b) & 9(3)registrant.

I, shall be responsible for adhering to the above composition and strain while manufacturing and marketing the product for distribution or sale. In case of any violation of the above declaration and also the conditions laid down on the Certificate of Registration of the said Bio-Pesticide, interalia, Product Quality Speciation submitted by us and also to the specification as and when the same are formulated and published by BIS amendments thereof, I am liable to be prosecuted/rejection of application under the provisions of Insecticide Act, 1968 and the Rules 1971 and amendments thereof.

Deponent

VERIFICATION

I,....., the above deponent do hereby verify that what has been declared above is true to the best of my knowledge and belief and nothing has been concealed there from.

Deponent

Annexure-II

UNDERTAKING FOR ABSENCES OF CHEMICAL/ BOTANICAL PESTICIDES/ CONTAMINATS/OTHER AGRO-CHEMICALS;

I, …….. S/o ……, aged..... years, resident of ……..and Proprietor/Authorized person of the firm M/s ……..., having its office at ……. do hereby declare and solemnly affirms asunder:

That the product, (Name of the product)…….. formulation containing CFU/PBO/Delta endotoxin content……. per gm or ml min, Strain code, (if any)……..Strain No: ……. manufactured by (Name of the applicant)…… does not contain any Chemical/Botanical Pesticide/Contaminants/other Agro-Chemicals.

That I/we shall provide the samples of our product (Name of the product) … ..... as and when desired by the competent Authorities of Government of India for verification.

That my/our above undertaking is true, and no portion is false and I have concealed nothing relevant to the above matter.

Place & date: Deponent

Signature: Name Designation:

Company Seal:

Annexure-III

AFFIDAVIT ON BIO-PESTICIDE COMPOSITION

I, …….. S/o ……, aged …….years, resident of.....and Proprietor/Authorized person of the firm M/s

……..., having its office at ……. do hereby declare and solemnly affirms asunder:

That I am in the capacity of …….. of firm M/s …… do hereby declare that the information furnished with respect to composition in Form-I, Label/Leaflet and bonafide verification of the application for registration of (Name of the product) …….., CFU/PBO…….per gm or ml min; Strain No. (Name and number of registered strain)…..... under section 9(3b) or 9(3) of the Insecticides

Act, 1968 is as under:-

1. COMPOSITION: (SPECIMEN FOR Pseudomonas fluorescence WP) Components:

A.	Quantity	:	(%w/w)
a)	Pseudomonas fluorescence	:	CFU 1x108 CFU/gm min1.0%
b)	Carboxy methyl cellulose	:	1.0%
c)	TalcPowder	:	98.0%
	Total	:	100.0%

2. That a shelf life of the product shall be twelve/Six/fourmonths.

3. The product shall be packed as per IS:8190 (Part-I) 1988 for Solid Pesticide (Second Revision).

4. That there will be no change in chemical composition, shelf-life, packaging requirement and the product will have the quality and packaging as per the relevant IS or as per specification approved by Registration Committee for 9(3b)registrant.

5. Bonafide declare that M/s....., manufacturing premises proposed/located at .....having Registration Certificate total no., if any or Nil and manufacturing license no. if any orNil.

I shall be responsible for adhering to the above composition while manufacturing and marketing the product for distribution or sale. In case of any violation of the above declaration and also the conditions laid down on the Certificate of Registration of the said Bio-Pesticide, interalia,Product

Quality Speciation submitted by us and also to the specification as and when the same are formulated and published by BIS amendments thereof, I am liable to be prosecuted/rejection of application under the provisions of Insecticide Act, 1968 and the Rules 1971 and amendments thereof.

Deponent

VERIFICATION

I,....., the above deponent do hereby verify that what has been declared above is true to the best of my knowledge and belief and nothing has been concealed therefrom.

Deponent

Annexure-IV

UNDERTAKING BY MANUFACTURERS OF MICROBIAL PESTICIDES

I,------,aged -years, s/o --------, R/o---------an----------of M/s.-------- ---------- Registered Office at do hereby undertake as follows:

That the product----------based on------Strain No.-------, manufactured by M/s.------ and /or imported by M/s…....does not contains any genetically modified organism (GMO) .

a) That I/We shall abide by the provisions contained in the International Plant Protection Convention with regard to the import of this product.

b) That I/We shall abide by the provisions in context of International Standards for Phyto-Sanitary Measures-Code of Conduct for the import and release of exotic bio- pesticidesoftheInternationalPlantProtectionConvention(IPPC),FAO,Rome/Plant Quarantine (regulation of Import into India) order,2003.

c) That I/We shall provide the samples of our product as and when desired by the competent authorities of Government of India for verification.

d) That I/We further undertake that in the event of the above product having proved otherwise by any competent authority and resulting in environmental damage, I/We shall inform to Plant Protection Adviser, Directorate of PPQ&S, Sectt. of Central Insecticides Board and Registration Committee, and other relevant authorities for Manufacturing Licensing, Pollution Control and of appropriate District/State/National Level and shall comply with the directions from them.

e) That my/our above undertaking is true, and no portion is false and I have concealed nothing relevant to the above matter.

Signature: ----------

Date------- Name----------

Place:---------- Designation-&Seal of the Company-----

B. Bio-efficacy

Sr. No.	Particulars	Primary culture/mother culture		Formulated product	
		9(3B)	9(3)	9(3B)	9(3)
1	**Field studies:** Data on bio-effectiveness and phyto-toxicity generated at ICAR, SAUs, CSIR / ICMR institutes. The data should be certified either by the Director or Head of the Institute.	NR	NR	R**	R***
2	**Laboratory studies:** The product should be tested at a laboratory under ICAR/ SAU/ CSIR/ICMR. 2.1) LC50 values for each insect species under laboratory conditions should be generated at least at two institutes of ICAR, SAUs, CSIR andICMR. 2.2) Data on LC50 values for each target insect species should be generated at a laboratory under ICAR/ SAU/CSIR/ICMR	R	R	R	R
3	Data on non-target organism: One season/one year on effect on product against natural parasites/ predators.	NR	NR	R	R

- **R = Required, NR = Not Required**
- **R** - Two seasons/years data on bio-effectiveness from two agro-climatic Zones**
- **R *** - Two seasons/years data on bio-effectiveness from minimum three agro climatic Zones.**

- **2.1) - Applicable for Entomo-toxic Bacteria**
- **– Applicable for NPV & GV.**
- **Sr. No. 3 - Required in case of Entomo-pathogenic fungi, Entomo-pathogenic Bacteria.**
  Note: No bio-efficacy data required for already registered strains of Bio-pesticides. Certificate of Registration will be granted as per approved formulation u/s 9(3)

C. Toxicity

S. No.	Parameters	Microbial (Antagonistic bacteria, Entomo-pathogenic/-Entomo-toxic bacteria, Entomo-pathogenic fungi, Antagonistic fungi and Baculo-virus			
		Primary culture/mother culture*		Formulated product	
		9(3b)	9(3)	9(3b)	9(3)
1.	**Single Dose Oral – Rat** (Toxicity/Infectivity/Pathogenicity)	R	R	R	R
2.	**Single Dose Dermal – Rabbit** (Toxicity/Infectivity/Pathogenicity)	R	R	R	R
3.	**Acute Inhalation (a)**	R	R	R	R
4.	**Single Dose Pulmonary –Rat (b)** (Toxicity/Infectivity/Pathogenicity)	R	R	R	R
5.	**Single Dose Intra-peritoneal–Rat (c)** (Toxicity/Infectivity/Pathogenicity)	R	R	R	R
6.	**Single dose intravenous (d)**	R	R	R	R
7.	**Primary Skin Irritation - Rabbit**	R	R	R	R
8.	**Primary Eye Irritation - Rabbit**	R	R	R	R
9.	**Skin Sensitization - Guinea pig**	R	R	R	R
10.	**Cell culture (d)**	R	R	R	R
11.	**Human Safety Records** (Effect/Lack of effects)	NR	R	NR	R
12.	**Toxicity to bird (1 species)** (Toxicity/Infectivity/Pathogenicity)	NR	NR	NR	R (Only 1 Species)
13.	**Toxicity to Fresh water Fish** (Toxicity/Infectivity/Pathogenicity)	NR	NR	NR	R
14.	**Toxicity to Honey bees (e)**	NR	NR	NR	R
15.	**Toxicity to Silkworm (f)**	NR	NR	NR	R
16.	**Toxicity to Earthworm (g)**	NR	NR	NR	R

*** Note: In case of first-time registration of any strain if not proposed to be used in the field, tox data is not required.**

Note

- **a.** Inhalation toxicity study required for registration of entomo-pathogenic/entomo-toxic bacteria
- **b.** Pulmonary toxicity study required for registration of antagonistic bacteria, antagonistic fungi, entomo-pathogenic fungi, baculo-virus
- **c.** Intra-peritoneal toxicity study required for registration of antagonistic fungi, entomo- pathogenic fungi, antagonistic bacteria
- **d.** Cell culture and Intravenous study required for registration of baculo-virus. e and f – required for all except antagonistic fungi
- **e.** required for all except entomo-pathogenic/entomo-toxic bacteria

- ✓ No data required for already registered strain from the same source with same strain designation and accession number.
- ✓ If genome sequence of conserved region of the microbial strains/microbes used as microbial pest control agent is identical (minimum 98%) with already registered strain then data is not required from toxicity angle.
- ✓ Formulations developed from similar already registered mother culture using similar ingredient and process of manufacture then no data is required from toxicity.
- ✓ The general recommendations as mentioned in the Guidance document on Toxicology for registration of chemical pesticides in India will also be applicable for registration of Microbial Pest Control Agents (MPCA) Pesticides on merit case to case basis.
- ✓ Waiver would be considered when existing information provides robust and full scientifically sound weight of evidence approach and based on read across/bridging from structurally and /or biologically related similar pesticides, physicochemical/biological properties and use and exposure patterns case to case basis
- ✓ The replacement alternatives not involving experiments on animals would be considered and accepted in those cases where data from valid 3Rs (Replacement, Reduction and Refinement) methods recognized by OECD and other renowned Regulatory Authorities is provided wherever applicable, case to case basis.

D. PACKAGING

Sl. No.	Parameter	Primary culture/mother culture		Formulated product	
		9(3B)	**9(3)**	**9(3B)**	**9(3)**
1.	Labels and Leaflets as per IR-1971, all fields (as applicable) and as amended from time to time	R	R	R	R
2.	Manner of labeling and Leaflet	R	R	R	R
3.	Type of packaging (Ultra small, small or Big whichever is applicable)	R	R	R	R
4.	Manner of packaging	R	R	R	R
5.	Specification for primary, Secondary and Transport packages (whichever is applicable)	R	R	R	R
6.	Details of packaging material & its compatibility with content	R	R	R	R
7.	Performance of container with content during storage stability test(Shelf life Study)	R*	R	R*	R
8.	Transport worthiness test	R*	R	R*	R

*** Note: In case of first-time registration of any strain if not proposed to be used in the field, Packaging data is not required.** R*- Before Commercialization the data will be required.

1. In case of **additional packaging endorsement** applications, the data at Sl. No. 05, 06, 07& 08, are not required if similar packaging (material) and manner of packaging is being sought by the applicant as has been granted to earlier 9(3) registrant.
2. Specification of Bureau of Indian Standard (BIS) must be followed for all the packaging requirements (Wherever available and applicable).
3. All Packaging tests must be carried out with the product of same batch and in its commercial package preferably in Indian condition.
4. The duration of the test and the conditions including geographical conditions must be mentioned.
5. Storage stability data must be generated keeping at least the following parameters in test protocol viz., temperature, duration, test packaging material, content of active ingredient in the product, test humidity, exposure to light (if applicable), physical and chemical properties of the product during and after storage etc.
6. The testing protocols must have their basis in the WHO/FAO/ CIPAC/ASTM recommendations or other validated methodology of GLP/ NABL accredited laboratory having packaging testing chemical / mechanical as applicable etc.) in the scope.
7. The storage stability data for microorganisms can vary depending on the type of microbes. For Fungi, maximum storage stability study data will be 12 months at ambient temperature. For Gram negative bacteria like *Pseudomonas fluorescens* or other *Pseudomonas* species the maximum storage stability study data should be 8 months at ambient temperature. For spore forming gram positive bacteria like *Bacillus* species the maximum storage stability study data should be 18 months at ambient temperature.

   Note: Additional two months' data for six months self-life claim / three months additional data for one year and six months additional data for 18 months shelf-life claim at two/three different agro climatic locations at ambient temperature along with meteorological data should be submitted.

Sub category: 4.1.2. Guidelines for the registration of Consortium of Bio-pesticides

Efficiency of bio-control agents could be increased by the development of mixture of compatible strains of different bio-control organisms by considering the following norms. While developing a consortia formulation, the following needs to be addressed:

1. Compatible strains combination that differs in pattern of plant/site of colonization.
2. Compatible strains combination is broad spectrum of action against different plant pathogens
3. Compatible strains combination with different modes of action under similar conditions.
4. Compatible strains combination of genetically diverse group to adapt to different pH, moisture, temperature and relative humidity.

- ✓ The guidelines of Chemistry, Bio-efficacy, packaging for registration of consortia of Bio-pesticides are similar with the guidelines of Bio-pesticides except the following points.
- ✓ Guidelines of mother culture/Primary culture of already registered bio-pesticides u/s 9(3) category are not required for registration of consortium Bio-pesticides. Only the guidelines of formulated product (Consortium) will be required. **Ratio of each strain in the formulation is required.**

1. Following toxicology guidelines for consortia of Bio –pesticides are required.

Note:

- **a.** Inhalation toxicity study required for registration of entomo-pathogenic/entomo-toxic bacteria
- **b.** Pulmonary toxicity study required for registration of antagonistic bacteria, antagonistic fungi, entomo-pathogenic fungi, baculo-virus
- **c.** Intraperitoneal toxicity study required for registration of antagonistic fungi, entomo-pathogenic fungi, antagonistic bacteria
- **d.** Cell culture and Intravenous study required for registration of baculovirus. e and f – required for all except antagonistic fungi
- **e.** Required for all except entomo-pathogenic/entomo-toxic bacteria

Note:-

a. If genome sequence of conserved region of the microbial strains/microbes which are used in consortia of Bio-pesticides to be used as microbial pest control agent is identical (minimum 98%) with already registered strain, then data is not required for mother culture but data is required for combination/consortia from toxicity angle.

b. If any new formulation of microbes is made by using new ingredients with different processes of manufacture than data is required for the formulation.

c. If any new combination/consortia /Mixture of microbial strains/microbe developed from already registered microbial strain than data is required only for the mixture and not for mother cultures from toxicity angle.

List of Fungi, Bacteria and Viruses for Consortia (Any new MPCA {Bio-pesticides} other than the list could be developed as Consortia after inclusion in Schedule)

S. No.	Name of the bio-pesticides
1.	*Trichoderma* spp.*(T. viride /T. asperellum, T. harzianum, T. virens etc.)*
2.	*Gliocladium virens T.virens*
3.	*Ampyliomyces quisqualis*
4.	*Coniotyrium minitans*
5.	*Chaetomium globosum*
6.	*Aspergillus niger* (non-pathogenic/biotype)
7.	*Bacillus subtilis*
8.	*Pseudomonas fluorescens, P. protegens, P. entomophila,*
9.	*Streptomyces griseoviridis*
10.	*Streptomyces lidicus*
11.	*Agrobacterium radiobacter*K84
12.	*Beauveria bassiana*
13.	*Metarrhizium anisopliae*
14.	*Nomuraearileyi*, (New name: *Metarhiziumrileyi*)
15.	*Verticillium lecanii,* (New name: *Lecanicilliumlecanii*)
16.	*Verticilium chlamydosporium*, (New name: *Pochonia chlamydosporium*)
17.	*Paecilomyces lilacinus,* (New name: *Purpureocilliumlilacinum*)
18.	*Myrothecium verrucaria-nematicide*
19.	*Bacillus species* (includes *Bacillus sphaericus (syn: Lysinibacillus, Bacillus thuringiensis var. galleriae, Bacillus thuringiensis var. israelensis, Bacillus thuringiensis var. kurstaki, Bacillus thuringiensis var. tenebrionsis, Bacillus thuringiensis var. sandiego, Bacillus thuringiensis var. tolworthi, Bacillus firmus and Bacillus albus.*
20.	Granulosis Viruses (GV)
21.	Nuclear Polyhedrosis Viruses (NPV) (includes *Spodoptera litura* NPV, *Spodoptera frugiperda* NPV, *Heicoverpa armigera* NPV, *Spodoptera mauritia* NPV, *Mythimna separata* NPV,

A. INDIAN STANDARDS ANTAGONISTIC FUNGI -DRAFT SPECIFICATIONS

1. Form and appearance
2. pH
3. Composition
   a. CFU/g of the product
   b. Percent loading content of the Bio-control organism in the formulation & nature of biomass.
   c. Percentage of carrier/filler, wetting/ dispending agent,
   d. Stabilizers/ emulsifiers, contaminants/ impurities etc.
   e. Moisturecontent
   f. CFU counts: *Trichoderma* $2x10^6$CFU/ml or gm min. (Stability at 30^oC and 65%RH).

4. Contaminants**:**
   a. Biological Contaminants:
   b. Pathogenic Contaminants: such as gram negative bacteria *Salmonella, Shigella, Vibrio* etc.:**absent**
   c. Other contaminants should not exceed $1x10^4$/ml org
   d. Chemical/ botanical pesticides contaminants: **absent.**
5. Method of analysis**:**
   a. CFU counts by serial dilution and examination under regular compound research microscope with bright field optics.
   b. Plating for contaminants on specific media
   c. Antagonistic mycolytic capability on target organism by bioassay on plants (Laboratory test).
   d. Bioassay procedure based on diseased severity and root colonization as detailed in Appendix-I

Bioassay for plant disease antagonists based on disease severity and root colonization.

- ✓ The target pathogen to be tested against has to be grown in Sand maize medium. The Sand-maize medium is prepared by adding sand 90g, maize 10g. and water 10ml in a saline or any glass bottle of 300ml capacity and then autoclaved twice. Then 5 mycelial discs of the test pathogen are transferred into the bottle and left for incubation for 15 days. Once the culture has grown well, the sand maize medium is mixed along with the fungal growth and 1g from this preparation is used as the inoculum after adjusting the CFU to 1 x10/g by addition of sand.
- ✓ The plastic cups (5-6 cm diameter) filled with soil and FYM (3:1) have to be used. In each cup the filling should be done up to ¾th level. The pathogen inoculum is mixed with sand has to be applied up to 2cm depth in the plastic cups.
- ✓ The bio-efficacy of the bio-agent shall be tested by both seed treatment and soil application. For seed treatment, the recommended dose of the formulation has to be used (5 to 10g.). For soil application, the bio-agent is added at the rate of 1g of formulation (minimum CFU should be the $2x10^6$). The germination percentage, disease intensity and seedling vigour are to be recorded.
- ✓ Another set of plastic cups filled with sterile soil and sterile FYM has to be used to confirm whether the bio-efficacy was due to the isolate of the bio-agent tested or due to the native isolates of the bio-agent present in the soil.
- ✓ The keys for grading the efficacy mentioned below shall be used (Srivastava et al., 2002). However, for the registration purpose, the bio-agents that are Highly Efficient, Efficient or Moderately Efficient in the plastic cup test under glass house condition (in the presence of pathogen) can be allowed (i.e.) germination percentage of 70% or above, disease incidence of 30% or less can be considered for registration.
- ✓ For the bacterial antagonists, the above bioassay procedure has to be followed where only the % root colonization will be considered and other parameters are not required. The % root colonization required is 80%.

Disease Grading Key

Disease incidence (%)	Description	Rating of bio- efficacy of bioagents
0	Germination>90%, no seed rotting, seedling healthy, root and shoot portions well developed	Highly Efficient (HE)
1-15	Germination 80-90%, infection on main as well as lateral roots, seedlings are well developed	Efficient (E)
16-30	Germination 70-80%, development of roots restricted and growth is less compared to Score 1. Infection occurred on roots. Shoot portions developed but growth retarded compared to Score 1.	Moderately Efficient (ME)
31-45	Germination 60-70%, length of roots and shoots short compared to Score 1. Germination of seeds inhibited. 50% of root area infected. Shoot portions also showed infection	Moderately Inefficient (MI)
46-60	Seed germination 50-60%. Development of roots and shoots greatly retarded. Shoot portions also showed infection.	Efficient (I)
Above 60	Less than 50% germination and seed rotting	Highly Inefficient (HI)

For the root colonization assay, the rhizosphere region of the plants tested above have to be collected and the soil adhering to the root surface has to be removed by gently tapping the roots. The root bits have to be cut into 1 cm bits and randomly 25 bits should be selected for each treatment. They have to be plated on (TSM) and the percentage of root bits colonized has to be recorded. This has to be performed in the sterile soil and non-sterile soil. One control treatment without the Biocontrol agent, being tested, should be kept for both the sterile and non-sterile soil to rule out of the possibility of interference of native micro flora in the bio- efficacy assay.

B. INDIAN STANDARDS ANTAGONISTIC BACTERI DRAFT SPECIFICATIONS

1. Form and appearance
2. pH
3. Composition
   a. Percent content of the Bio-control organism in the formulation & nature of biomass
   b. CFU/g or ml of the product.
   c. Percentage of other components: carrier /filler, wetting/ dispersing agent, stabilizers/emulsifiers, contaminants/impurities etc.
   d. Moisture content.
   e. CFU counts: Minimum $1x10^8$ CFU/ml or gm. (Stability at 30^oC and65%RH).
4. Contaminants:
   a. Biological Contaminants:
   b. Pathogenic Contaminants: such as gram-negative bacteria *Salmonella, Shigella, Vibrio* etc.:**absent**
   c. Other contaminants should not exceed $1x10^4$/ml or g
   d. Chemical/botanical pesticides contaminants: **absent.**
5. Method of analysis:
   a. CFU counts on specific medium.
   b. Plating for contaminants on specific media
   c. Antagonistic capability on target organism by bioassay.
   d. Bioassay procedure based on diseased severity and root colonization as detailed in Appendix-I

c. INDIAN STANDARDS ENTOMOPATHOGENIC FUNGI - DRAFT SPECIFICATIONS

1. Form and appearance
2. pH
3. Composition
   - **a.** CFU/g of the product
   - **b.** Percent content of the Bio-control organism in the formulation & nature of biomass.
   - **c.** Percentage of carrier/filler, wetting/ dispending agent &stabilizers/ emulsifiers, contaminants/ impurities etc.
   - **d.** Moisture content
4. CFU counts: Minimum $1x10^8$ CFU/ml or gm. (Stability at 30^oC and65%RH).
5. Contaminants:
   - **a.** Biological Contaminants:
   - **b.** Pathogenic Contaminants: such as gram negative bacteria Salmonella, *Shigella, Vibrio* etc: **absent**
   - **c.** Other contaminants should not exceed $1x10^4$/ml org
   - **d.** Chemical/botanical pesticides contaminants: **absent.**
6. Method of analysis:
   - **a.** CFU counts by serial dilution and examination under regular compound research microscope with bright field optics.
   - **b.** Plating for contaminants on specific media
   - **c.** Entomo-pathogenic capability on target insects by bioassay.

Laboratory bioassay procedures for screening fungal pathogens on *Spodoptera litura*and *Helicoverpa armigera*

Insect pathogens:

Beauveria bassiana, Metarhizium anisopliae, Nomuraea rileyi

Preparation of Fungal inoculum for bioassays:

The fungus is grown on SDAY/SMAY medium for 10 days in slants and aqueous spore suspensions of various concentrations are prepared using sterile water. The spore count is estimated by Haemo cytometer. (10^4 - 10^{10} spores/ml). Tween-80 is added @ 0.01% to get uniform spore suspension.

Rearing insects:

H.armigera, S.litura - Artificial diet(Semi-

syntheticdiet) Stage of insect for bioassay

H.armigera, S.litura- II instar larvae to be used for bioassay protocols for lepidopteron pests

Method of inoculation

S. litura

1. Cut castor leaf discs of 3.0cm diameter, rinse in sterile distilled water and place each leaf disc in a sterile Petri plate and allow it air dry in a laminar flow system

2. Apply ten micro liters of the spore suspension of each concentration on the leaf disc and spread it uniformly on the leaf surface and allow it air dry in a laminar flow system. Treat the other side of the disc similarly.

3. Release ten numbers of second instar larvae of *S. litura* on the leaf surface and incubate the discs in an incubator at 25^0C and 90%RH

4. After 24hours, shift the larvae to the poly-pots containing the semi-synthetic diet and incubate in an incubator at 25^0C and 90%RH

5. After 5 days of incubation, mortality of the larvae are recorded in each concentration tested

6. Lc-50 can be calculated using SPSS package

Standard for LC-50: Not more than $2.00X10^6$ spores/ml ($3.0X10^3$ spores/mm^2)

H. amigera:

Instead of castor leaves, soybean leaves can be used for ***H. amigera*** and the procedure is same as above.

Standard for LC50: Not more than $4.00X10^6$ spores/ml (6.0X103 spores/mm^2)

Appendix-II

Bioassay procedure for *Plutella xylostella*

Various concentrations of *Beauveria bassiana* formulation ranging from 6×10^8 to 2×10^{10} are to be screened to assess the mortality.

Fresh undamaged radish leaves free from pesticide application are to be collected and washed thoroughly in sterile distilled water and air-dried. Individual leaves are dipped in respective concentrations for 30 seconds. After complete drying of leaves ten late 2^{nd}instar larvae of *Plutella xylostella*are released per treatment. A water dipped radish leaf is maintained simultaneously as control.

To prevent desiccation of leaves, the petiole is covered with a moist cotton swab. Each treated leaves are placed in a plastic container of dimension 12.5 x 10 cm containing moist filter paper, Whatman No.41 to provide humidity.

Each treatment has to be replicated thrice. Fresh radish leaves were provided as feed at 24 hours interval. This set up has to be maintained at $25+1^0$C and 70-80% RH for 7 days. Observations on larval mortality are to be made at 3, 5 and 7 days aftertreatment.

Standard for LC50 = Not more than 3×10^9 cfu/g

D. **INDIAN STANDARDS, ENTOMO-TOXIC BACTERIATECHNICAL /FORMULATION - DRAFT SPECIFICATIONS**

S.No.	Details
1.SCOPE	
1.1	This Indian Standard prescribes the requirements and the method of sampling and test for Entomo-toxic bacteria technical and formulation. The product is a bio-pesticide active against target insects. The product is not for human consumption.
2.REQUIREMENTS	
2.1	Common name: i.e., *Bacillus thuringiensis or B. sphaericus etc.*
2.2	Systematic name (Genus, species, serotype, strain and Cry-toxin* along with cry gene)
2.3	Physical specification 2.3.1 Form and appearance 2.3.2 Moisture content 2.3.3 pH
2.4	Composition 2.4.1 Delta endo-toxin content (Minimum 2.0%) – estimation as per Appendix-V 2.4.2 Adjuvant 2.4.3 Beta Exo-toxin content – Negative through housefly bioassay test as per Appendix-IV 2.4.4 Human pathogens (gram negative bacteria Salmonella, shigella &vibrio etc) - Absent 2.4.5 Other microorganisms (not more than 10^4 /g) 2.4.6 Chemical/botanical pesticide contamination –Absent
2.5	Natural occurrence of the organism
	2.5.1 Its relationship of the organisms 2.5.2 History (exotic or indigenous strain) 2.5.3 The isolate should not be genetically modified organism(GMO).
3.SAMPLING	
3.1	Representative samples of the material shall be drawn in accordance with IS 10946:1984
4.TESTS	

4.1	An appropriate test procedure and criteria used for identification, such as morphology, biochemistry and / or serology / immunology 4.1.1 Morphology description, particle size 4.1.2 Immunology assays: ELISA / Dot blot assay test or any other sensitive standard immunology test. 4.1.3 Method of analysis 4.1.4 Level of beta exo-toxins to be identified if expressed by Housefly bio- assay method. 4.1.5 Potency of product by bioassay method(**Appendix-II**) 4.1.5.1 Bio-assay method a) LC 50 on target larvae and potency against a reference using artificial diet or leaf disc method or in the water for mosquito larvae(Appendix-I) b) Housefly Bioassay method for Beta-exotoxin content (Appendix-IV) c) Determination of toxin content by ELISA / Dot Blot Assay Method(Appendix-V) 4.1.5.2 A technique for the separation and purification of the crystals (Appendix III) is to be used by the manufacturer and the antisera to be raised using solubilized toxin. Toxin content (3.5 %) to be standardized in the formulation using this antisera (ELISA /Dot blot assay).

2.2 Crytoxin* If H-Serotype is not known, it is mandatory to provide the details of Cry toxin to confirm that it is *Bacillus thuringiensis.*

Appendix-I

Bioassay Method Diet incorporation

The following protocol is used for diet incorporation of oral toxicants to test their toxicity on target insects. The example presented here is to bioassay Cry I Ac on *H. armigera* (First instar larva of other test insects are used for similar bioassay).

1. Pipette out 3 ml of the solution into a 40 ml plastic cup.
2. Pour lukewarm diet, approx. 60^0 C, into the cup to a total volume of 30 ml. Place the lid and shake the cup vigorously for a minute to mix properly.
3. Pour the diet to 0.5 cm height, into wells of a 24-cell insect- rearing tray. Allow the diet to cool in laminar airflow under UV lamps for I h to surface sterilize the diet.
4. If concentration of the toxicant in the stock solution was 2 µg/ml, the final concentration in the diet would now be 0.2µg/ml diet. Thus the final concentration of toxin in diet was diluted10-fold.
5. Release first instars into the diet rearing trays at the rate of one per well. Cover the diet tray with semi-permeable wrap and close the lid.
6. It is recommended that the lid be tightly secured to the tray with rubber bands, to prevent the larvae from escaping. Because the diet is unsuitable, larvae try constantly to escape from the diet rearing trays.
7. Keep controls with larvae released on untreated diet, for all the experiments.
8. The unused rearing trays with diet can be stored in a refrigerator for a week.
9. Change the diet for the larvae every two or three days.
10. Record mortality observations at 8 hourly intervals until the end of seven days, for median lethal time LT_{50} calculations. LT_{50} is the time at which 50 % of' the test population is killed with the specific dose tested. A simple linear regression equation can be worked out to calculate the LT_{50}
11. Otherwise, record mortality at alternate days until the end of seven days, for median lethal concentration LC_{50} calculations. LC_{50} is the concentration that kills half the test population.
12. Record weights of surviving larvae at the end of seven days, for median effective concentration EC_{50} and LC_{50} is the concentration that prevents half the test population from reaching 50% of the weight attained by control larvae. For example, if the average weight of larvae on the control diet (without toxin) was 80 mg, EC_{50} represents the concentration at which 50% of the test population is unable to gain a weight more than 40 mg. LC_{50} is the concentration that inhibits half the test population from reaching the third instar.

Diet incorporation for filter paper bioassays

1. For bioassays with bollworms, 10 ml toxin incorporated diet is poured over a16 sq cm piece of filter paper. The filter papers layered with diet are cooled and cut into smaller squares of 2 x 2 cm, and 10 first instar larvae are released in small plastic cups 3 x 3 cm (d x h) cups containing a square. Change the strips every alternate day.

2. Record mortality observations until the seventh day.

Surface coating of semi-synthetic diet

1. Prepare the diet and pour it into the trays or the rearing plastic cups. Generally, 10 µl of the toxin can be used to coal 1 sq cm surface area. Gently swirl the diet surface to ensure uniform and complete spread of the solution over the diet surface.
2. Allow the surface to dry in a laminar airflow under UV light for 2-3 hours to surface sterilize.
3. Release one first instar *H. armigera* larva per well. Always maintain proper controls with un treated diet.
4. Change the diet on alternate days and record mortality until the seventh day. Then, weight of surviving larvae should be recorded on the final day of the bioassay.

The method has the advantage of obtaining constantly reliable results because the toxin is unlikely to be affected by either improper mixing or heat as can occur in the diet-incorporation method. Moreover, less amount of the toxin is required for the assay, compared to the diet- incorporation method.

Calculation of results:

The potency of the sample (International Units – IUs)

$$\textbf{IU/mg sample} = \frac{\text{LC50 Standard}}{\text{LC50 Sample}} \times \text{IU/mg Standard}$$

(IU/mg Standard, i.e., HD-1-S-1980 is 16,000 IUs/mg; the US standard is available with PDBC, Bangalore; each registrant should prepare a “self reference” and should deposit it with the Registering Authority. Each self reference will be expressed as IU/mg using International standard)

Exo-toxin determination by PCR studies

Methodology:

Some *Bacillus thuringiensis* strains secrete type I β- exotoxin, which is a non-specific insecticidal and thermo-stable adenine nucleoside oligosaccharide. Toxicity bioassays and HPLC are traditional methods for detecting β-exotoxin. For rapid approach for prediction of type I β-exotoxin production, PCR-based method can be followed as per Diego H Sauko*et al*. (2014). One of these ORFs encodes the Exo-protein that was proved to be responsible for the phosphorylation of a β-exotoxin precursor at the last step of their biosynthesis process (Liu et al., 2010). Primers BEF (forward; 50- CGGCAGCCGTTTATTCAAA-30) and BER (reverse; 50-CCCCTTCCCATGGAGAAACA-30) amplify a 406-bp DNA fragment of thu E between nucleotides 373 and 778. All *B. thuringiensis* strains are grown on nutrient agar plates for 16h. A loopful of cells is transferred to 100 µl H2O and boiled for 10 min to make DNA accessible for PCR amplification. The lysate is centrifuged briefly (5 s at 20,000g), and 5 µl supernatant is used as a DNA template in each polymerase chain reaction. This is performed with a final volume of 25 µl containing 2.5 µl 10x reaction buffer, 0.5 µl 50 mM MgCl2, 0.5 µl 100 mM deoxy nucleoside triphosphate mixture, 8 pmol each primer, and 1 U of Taq polymerase (In-vitro-gen). The PCR amplification consisted of DNA denaturation at 94°C for 2 min followed by 25 cycles of amplification with a gradient thermo-cycler. Each cycle consisted of a de-naturation step at 94°C for 1 min, an annealing step at 54°C for 1 min, and a chain elongation step at 72°C for 1 min. The final elongation step was extended for an additional 5 min. Subsequently, 10 µl PCR product is analysed by 1.0% agarose gel electrophoresis. A positive control can be also used for better results.

Appendix-II

Dot Blot assay of *Bacillus thuringiensis* (Bt) toxin protein as alternate of Bioassay

1) Bt grown till sporulation in shake flask or in fermenter vessel and let the cells lyse and release spore/crystals into the medium

2) Cells are harvested by centrifugation at 10k for 15mins.

3) Wash the pellet with 1M NaCl to remove the B.t. associated seine/metallo proteases and washed twice with sterile distilled water.

4) Pellet suspended in 50MM NaOH to solublize the toxin protein for 2 hours at R.T. with slow shaking and centrifuged again at 10K for 15Mins.

5) Supernatant was adjusted to pH 8.0 with Tris HCL pH8.8

6) Protein contents estimated by Lowry'sprotocol.

7) Two fold serial dilutions of test protein were made in PBS and known amount at protein applied on NCP using S&S or Biorad Dot Blot manifold apparatus and applying water vacuum for 30mins.

8) NCP was carefully removed from Dot Blot set and soaked in excess of 3% Skim milk in PBS for blocking the remaining acetic sites on NCP for 2-3 hours at R.T/O/N at4^{o}C.

9) Wash the NCP with excess PBS with 0.01% Tween 20, 3-4 times and then finally with PBS

10) Polyclonal antiserum raised against total crystal protein was suitably diluted in PBS and added to the 'seal a meal' containing NCP and incubated for 1-2 hours withshaking.

11) Remove the NCP from the bag and was several times (as mentioned instep.No.9)

12) Anti-rabbit antibodies conjugated with HRPO/alkaline Phosphate was diluted as per the suppliers instruction and incubated NCP (as in step10)

13) Was as in step11

14) ForHRPO:

    a) Diaminobenzene (4mg/10ml PBS)/4-Chloro-1-Napthol (4mg/10ml 20% Alcohol) were dissolved and 10ml of 30% of H2O2 per 10 ul substrate solution was added and colour reaction developed in dark for 5-10 mins (DAB gives brick red colour. 40N gives blue colour).
    b) For alkaline Phosphatase:

Alkaline Phosphatase Buffer:

1M TrispH8.8	-10ml/
4MNaCI	- 2.5ml/ make up to100ml
1MMgc12	-0.5ml/

For 10ml of above buffer add NBT-66 µl and BCIP-33 µl and developed and colour reaction

15. Stop the reaction by removing the substrate and washing with PBS.
16. Keep on filter paper anddry.

DIFFERENT PROTEIN CONCENTRATION

10µg 5µg 2.5µg 1.25µg 512.5ng 256.25ng 128ng 64ng 32ng 16ng 8ng 4ng

Determination if cell dry weight

Take a known volume of Bacterial culture spin down at 4R

formin. # Wash the pellet in minimal distilled water

Transfer to a pre weighed container

Incubate at 80^0C for 16-18 hours till become dry and weight becomes constant.

Appendix-III

PURIFICATION OF CRYSTALS BY GELATIN METHOD

Centrifuge the sporulated material and wash pallet twice with 1M Nacl. Add 200ml. of 0.5% Gelatin, stir and remove all froth completely. Dilute with sterile water and centrifuge. Take debris and stir with 20ml. of 1.5M sucrose. Further add 50 ml of 1.5M sucrose, stir and centrifuge at 3000 RPM for 2 hours. Remove supernatant and purified crystals are harvested.

E. INDIAN STANDARDS BACULORIVUS DRAFTSPECIFICATIONS

1. Form and composition of the product
1.1 Viral Unit: POB/Capsule count per ml/g of the product
1.2 Percent content of the bio-control organism in the formulation and nature of biomass
1.3 Percent of carrier/filler, wetting/dispersing agent, stabilizers/ emulsifiers, containments/ impurities etc.
2. Moisture content
3. pH
4. Viral Unit

NPVs (*Helicoverpa& Spodoptera*) - $1x10^9$ POB/ml or gm (minimum) (POB –Polyhedral Occlusion Body)

GV (*Chilo, Plutella & Acheae*) - $5x10^9$ Capsules/ml or g. (minimum).

5. Contaminants**:**

5.1 Biological contaminants:

5.1.1 Pathogenic contaminants: Pathogenic contaminants such as gram negative bacteria *Salmonella, Shigella, Vibrio* etc. should be **absent:**

5.1.2 **Other microbial contaminants:** Other microbial contaminants should not exceed $1x10^4$ /ml or g

5.2 Chemical/botanical pesticides contaminants should be absent**.**
6. Identification of Baculovirus by DNA test (Restriction enzyme analysis test).
7. An undertaking should be submitted that the strain is indigenous, naturally occurring and not exotic and not genetically modified as perAnnexure-1.1
8. Method of analysis:

Viral Unit:

NPVs (*Helicoverpa and Spodoptera*) =$1x10^9$POB/mlor gm. minimum

GVs = $5x10^9$ Capsules/ml or gm .minimum.

8.1 In case of NPVs/, POB/Capsule count should be taken with Haemocytometer using shallow depth counting chamber as detailed in Appendix – I

8.2 Biological assay for determining the LC_{50} or LD_{50} of the formulation:
8.2.1 Bioassay for NPV by the Diet Surface Contamination Method as detailed in Appendix-II OR

8.2.2 Bioassay for GV against *Chiloinfuscatellus* as detailed in Appendix-IIIOR

8.2.3 Bioassay for GV against *Plutellaxylostella* as detailed in Appendix-IV.

8.2.4 Bioassay for GV against *Acheaejanta*as detailed in Appendix-V.

8.3 Plating for contaminants on specified media.

Appendix-I

COUNTING OF NPV/GV (POB/CAPSULE) USING IMPROVED NEUBAUER HAEMOCYTOMETER COUNTINGCHAMBER.

A haemocytometer is used for estimating of NPVs/GVs in a unit volume of the product. The Improved Neubauer Haemocytometer comprised a thick glass slide with a shallow depression in the central section divided into two halves (figure-1). Each side, the base of the depression has a fine ruled grid of squares (figure-2) which is visible under a microscope. The dimensions of this grid are defined. Place a standard cover slip placed over the depression and a one half halves of the slide chamber using a micro pipette. The particles require 2-5 minutes to sediment to the chamber floor.

Either dark field or a phase contrast microscope is used to identify and count polyhedral occlusion bodies (POB) or capsule. With the counting chamber under the microscope, the number of Polyhedra/capsule in a given number of grid squares can be counted. Each count consists of a tally of the number of polyhedra completely contained within a big square plus the number of touching the top and left sides. Polyhedra touching the bottom and right sides are not counted. Since both the depth of the chamber and the grid dimensions are known. It is then a straight forward calculation to determine the number of polyhedra /capsule per ml of test suspension.

$$\text{Number of NPV (POB) per ml/gm} = \frac{D \times X}{N \times K}$$

Where:

D = Dilution factor

X = Total number of polyhedra

counted N = Number of squares

counted

K = Volume above one small square in cm^3 =($2.5 \times 10^{-7} cm^3$)

Area of each small square is 1/400 mm^2 = 0.0025 mm^2. Depth of chamber is o.1mm. Volume of liquid above a single small square is 0.0025 mm^2 x 0.1mm= 0.00025 mm^3. To covert to cm^3 multiply by 1/1000 to get a volume of 2.5 x 10^{-7} cm^3 above 1 small square. Hence,K=$2.5 \times 10^{-7} cm^3$

Worked example:

Suppose in a sample diluted by a factor of 1000 we count 535 polyhedra in 160 small squares then:

D = 1000

X = 535

N = 160

K = 2.5×10^{-7} cm^3

$$\text{Thus, POB count} = \frac{1000 \times 535}{160 \times 2.5 \times 10^{-7}} = 1.34 \times 10^{10} \text{ POB/ml of test sample}$$

Note: (i) Usually, this procedure is repeated 3times and an average taken to get a more accurate estimate.

(ii)Same procedure will be used for GV also for counting the number of capsule per unit volume of the product**.**

Appendix-II

PROCEDURE FOR ESTIMATION OF LC50 OF NPV BY THE DIET SURFACE CONTAMINATION METHOD.

i) **Diet to be used**: The standard chickpea-based diet without formalin.

ii) **Bioassay bottles**: 5ml. vials with a diameter of 18 mm (255 mm^2 surface area)

iii) Doses of NPV to be tested**:**

Helicoverpa armigera		***Spodoptera litura***	
POB/ml	POB/mm^2	POB/ml	POB/mm^2
a)$5x10^4$	1.96	$1x10^6$	39.21
b)$1x10^4$	0.39	$2x10^5$	7.84
c) $2x10^3$	0.078	$4x10^4$	1.57
d)$4x10^2$	0.016	$8x10^3$	0.31
e)$0.8x10^2$	0.003	$16x10^2$	0.062
f)1.6x10	0.0006	$3.2x10^2$	0.013

iv) **Method of dosing**: Dispense 10 Micro litre aliquots into each vial and spread uniformly over the entire diet surface using a polished rounded lip of 4 mm glass rod and allow to dry off under flow laminar hood for 10minutes.

v) **No. of larvae/dose**:50 (Maintain 50 healthy larvae without virus inoculation for control)

vi) **Stages of larvae**: II instar larvae (Preferably 4daysold) Release one larva/vial and plug mouth with sterile absorbent cotton. Incubate at 25 ± 1^oC for 7days.

vii) Record mortality in different doses on the 7^{th}day.

viii) Apply Abott's formula for correction of mortality in control treatment.

ix) Subject the dose – mortality response to probit analysis using relevant statistical software.

x) Express LC 50 as POB/mm^2 of diet surface.

Expected standards for NPV for II instar larvae

Species	LC 50POB//mm^2
1. *Heliocoverpa armigera*	<0.5
2. *Spodoptera litura*	<20.0

Appendix-III

Bioassay for GV against *Chiloinfuscatellus*

Determination of LD50:

To determine the LD50 of the GVs, third instar larvae should be used. The larvae are to be microfed (one micro litre per larva) with six different doses, viz. 1.1 x 10^1, 10^2, 10^3, 10^4, 10^5, and 10^6 IBs/larva. One hundred freshly moulted larvae have to be used for each treatment. Larvae fed with equal quantity of distilled water serve as control. The mortality has to be recorded daily. The LD50 of the virus is determined following the probit analysis method (Finney,1962).

LD50 =< 1x10^3 OB for third instar larvae by micro-feeding.

Appendix-IV

Laboratory bioassay procedures for estimation of LC50 of *Plutellaxylostella* (PxGV) by leaf disc method:

1. Cut leaf discs of cauliflower (3.2cm). Soak it in 0.1N NaOCl for 5 min. and wash thoroughly in distilled water. Air dry these leaf discs for 2-3 minutes. (Fifth leaf from top to be used)
2. PxGV (containing 0.01per cent Triton X 100) of different concentrations 28000, 2800, 280, 28, 2.8 OB/mm^2 on the leaf disc) is prepared
3. Aliquots of 12ul of each concentrating of GV is dispensed on the upper surface of the leaf disc and spread uniformly with a blunt end glass rod (use separate tips and glass rods for each treatment)
4. Air dry these leaf discs for 2-3minutes
5. Repeat the same on the lower surface of the leaf disc.
6. Control discs were treated with distilled water containing 0.01 per cent Triton X 100 only.
7. The leaf discs are placed in Petri dishes lined with wet filter paper discs and 35-second instar larvae of *P. xylostella*(starved for 6 hours) are released on each leaf disc starting
   from control treatment to highest concentration. This is replicated three times.
8. Incubate these larvae at25^oC
9. After 24 hours remove the treated leaves (partially eaten) and provide the larvae with fresh cauliflower rleaves.
10. The leaves are changed daily and mortality data recorded everyday.
11. The dosage and time mortality responses are subjected to probit analysis.
12. If the mortality in the control excess 10% repeat the

experiment. LC50 = < 0.15 OB/mm^2 for second instar larvae by

disc method.

Laboratory bioassay procedures for estimation of LC50 of *Achaea janata* Granulosis virus (AjGV) by leaf disc method:

1. Cut leaf discs of castor (8cm dia) and wash in distilled water. Air dry these leaf discs for 5minutes.
2. Treat the leaf disc on both the upper and lower surfaces with 200 µl suspension of AjGV (containing 0.02% Tween-80) of different concentrations ($5x10^8$, $5x10^7$, $5x10^6$, $5x10^5$, $5x10^4$ corresponding to 19884, 1988, 198, 19, 1.9 OB per mm^2 on the leaf disc)
3. Aliquots of 100 µl of each concentration of GV is first dispensed on the upper surface of the leaf disc and spread uniformly with a blunt end of glass rod (use separate tips and glass rods for each treatment)
4. Air dry these leaf discs for 5minutes
5. Repeat the same on the lower surface of leaf disc
6. Control leaf discs were treated with distilled water containing 0.02% Tween- 80only
7. The leaf discs are placed in Petri dishes (9.0cm dia) line on wet filter paper discs and 35 second instar larvae (third day after hatching) of *A. Janata* are released on each leaf disc starting from control treatment to highest concentration. This is replicated three times.
8. Incubate these larvae at25^oC.
9. After 24-48 hours remove the treated leaves (partially eaten) and provide the larvae with fresh castor leaves
10. The leaves are change daily and mortality data recorded everyday
11. The dosage and time mortality responses are subjected to probit analysis
12. If the mortality in the control exceeds 10% repeat the experiment.

Recommended LC50 GV (*Achaea janata) – LC50* <4 OB/mm^2 for second instar larvae by the leaf disc method.

SUB CATEGORY 4.2: GUIDELINES FOR THE REGISTRATION OF BOTANICAL / PLANT ORIGIN PESTICIDES

Plant Based Products

Botanical Pesticides/Pesticides of any Plant Origin like Eucalyptol, Cymbopogon formulation, Neem Based Pesticides, Pyrethrum Extract & Rotenone for Pisci- culture (Formulation), Essential Oils (formulations only) and other similar categories.

General Guidance

1. It is clarified that traditional remedies or products prepared by the individuals for their own-self use do not need registration if the produce is not sold. If the product prepared or the treated produce is to be sold in the market, then the registration is required under the provisions of the Act.
2. Sometimes the active substances to be used as Plant origin or botanical bio-pesticides, are vastly studied and the information of good quality which is consistent with current methodology, information/data is available, then waivers for submission of certain data, if justified, can be granted. Further, in some cases, based on the information and characteristics or the results of studies, additional data may be required.
3. For registration of extract u/s 9(4): It shall be obtained from the same part of the plant with same process of extraction and should have identical chemical composition as of 9(3) registrant.
4. Formulation to be registered u/s 9(4) should have identical chemical composition and required to be prepared from the extract already registered under the Act for use in the country.

Chemistry

Sr. No.	Parameters	9(3B)			9(3)				
		TI	TIM	FIM	TI	TIM	FI	FIM	Label expansion
1	Name of the part of the Plant(s) to be used for extraction of the active ingredients	R	R	R	R	R	R	R	-
2	Source of supply of the technical grade material	R	R	R	R	R	R	R	-
3	Outline of Process of formulation	R	R	R	R	R	R	R	-
4	Extract Contents active ingredient	R	R	R	R	R	R	R	-
5	Chemical composition of the formulations	R	R	R	R	R	R	R	-
6	Physico-chemical properties of a.i.& adjuvants	R	R	R	R	R	R	R	-
7	Product Specifications as BIS format	R	R	R	R	R	R	R	-
8	Method of analysis (a.i.)	R	R	R	R	R	R	R	-
9	An undertaking that the product does not contain any other chemical pesticide except plant extract	R	R	R	R	R	R	R	-
10	Analytical test report (generated by the NABL / GLP certified lab	R	R	R	R	R	R	R	-
	from the source of manufacturing country)								
11	Shelf-life claim (In the case of)	R	R	R	R	R	R	R	-

12	Shelf-life data **(# in the case of Eucalyptus based products containing 'Eucalyptol', Rotenone for pisci-culture, Cymbopogan spp.)**	R/# NR	R/#NR	R/#NR	R	R	R	R	-
13	Process of mfg in step wise manner along with the list of RAW material with impurity	R	R	R	R	R	R	R	-
14	Submission of sample to CIL along with slandered	R	R	R	R	R	R	R	-
15	Drawl of in process of sample to submission of CIL.	NR	R	NR	NR	R	NR	NR	-
16	Copy of RTT permit	R	NR	NR	R	NR	R	NR	-
17	DNA from the source country along with copy of Registration (CR) issued from importing country	R	NR	NR	R	NR	R	NR	-
18	In case of import consent letter from source of supply	R	NR	NR	R	NR	R	NR	-

Bio-efficacy

Sr. No.	Parameters	9(3b)			9(3)					9(4)
		TI	TIM	FIM	TI	TIM	FI	FIM	Label expansion	TI/TIM/FI/FIM or endorsement of already approved label expansion
1	Bio-effectiveness against target pest and disease in specified crops	R	R	R*	R	R	R	R**	R**	No data requirement
2	Phytotoxicity (as per standard tests)	R	R	R*	R	R	R	R**	R**	
3	Compatibility with other agro- chemicals, if claimed	N R	NR	R	NR	NR	R	R	R	
4	Stability of formulation (Photo, thermal etc.) in aqueous dilution (acidic, neutral & basic)	N R	NR	R	NR	NR	R	R	R	
5	Direction concerning dosage for each target pest and disease	N R	NR	R	NR	NR	R	R	R	
6	Stage of crop for use and stage of target pest and disease	N R	NR	R	NR	NR	R	R	R	
7	Waiting period	N R	NR	NR	NR	NR	N R	NR	NR	
8	Application equipment	N R	NR	R	NR	NR	R	R	R	
9	Information regarding registration status in other countries, if any.	N R	NR	R	R	NR	R	R	R	
10	Time of Application	N R	NR	R	NR	NR	R	R	R	
11	Purpose of Manufacture	R	R	R	R	R	R	R	R	
12	Residue in plant	N R	NR	NR	NR	NR	N R	NR	NR	
13	Residue in soil	N R	NR	NR	NR	NR	N R	NR	NR	
14	Registration status in foreign countries	N R	NR	R	R	NR	R	R	R	
15	Residue tolerance limit fixed by foreign countries	N R	NR	NR	NR	NR	N R	NR	NR	

R[*]: Two seasons /year data generated at minimum two different agro climatic zones **R**[**]**:** Two seasons / year data generated at minimum three different agro-climatic zones NR: Not required

Toxicology

Sr. No.	Parameters	9 (3b)			9(3)					9(4)
		TI	TIM	FIM	TI	TIM	FI	FIM	Label expansion	TI/TIM/FI/FIM or endorsement of already approved label expansion
1	Acute Oral Toxicity-Rat	R	R	R	R	R	R	R	-	NR
2	Acute dermal Toxicity-Rabbit	R	R	R	R	R	R	R	-	NR
3	Acute Inhalation-rat	R	R	R	R	R	R	R	-	NR
4	Primary Skin Irritation-Rabbit	R	R	R	R	R	R	R	-	NR
5	Primary Eye Irritation-Rabbit	R	R	R	R	R	R	R	-	NR
6	Skin sensitization-Guinea pig	R	R	R	R	R	R	R	-	NR
7	Sub-acute oral-rat	NR	NR	NR	NR/R*	NR/R*	NR	NR	-	NR
8	Sub-acute oral-dog**	NR	NR	NR	R	R	NR	NR	-	NR
9	Sub-acute dermal	NR	NR	NR	R	R	NR	NR	-	NR
10	Sub-acute inhalation	NR	NR	NR	R	R	NR	NR	-	NR
11	Neuro-behavioral toxicity	NR	NR	NR	R	R	NR	NR	-	NR
12	Teratogenicity	NR	NR	NR	R	R	NR	NR	-	NR
13	Effect on Reproduction	NR	NR	NR	R	R	NR	NR	-	NR
14	Carcinogenicity/chronic	NR	NR	NR	R	R	NR	NR	-	NR
	toxicity/combined toxicity study									
15	Metabolism	NR	NR	NR	R	R	NR	NR	-	NR
16	Mutagenicity* (AMES+2in-vitro+1 in-vivo)	NR	NR	NR	R	R	NR	NR	-	NR

17	Toxicity to birds	R (Two species)	R (Two species)	R (one species)	R (Two species)	R (Two species)	R(one species)	R (one species)	-	NR
18	Toxicity to fresh water fish	R	R	R	R	R	R	R	-	NR
19	Toxicity to honey bees	R	R	R	R	R	R	R	-	NR
20	Toxicity to earthworm	R	R	R	R	R	R	R	-	NR
21	Medical data	R	R	R	R	R	R	R	-	NR
22	Human toxicity information	NR/ R*	NR/ R*	NR/R *	NR/ R*	NR/ R*	NR/ R*	NR/R *	-	NR
23	Health record of industrial workers	NR	NR	R	NR	NR	R	R	-	NR
24	International report on carcinogenicity & genotoxicity study	NR/ R	NR/ R	NR/R	NR/ R	NR/ R	NR/ R	NR/R	-	NR

Note:

1. The requirement for sub-acute studies shall be determined on the basis of results of other toxic study reports.
2. * Please refer to guidance document on toxicology for registration of chemical pesticides in India

Toxicity Study No. 8: Sub-acute oral toxicity study (Dog): ** Peer reviewed international published literature is also required for dog study.

Note:

1. The general recommendations as mentioned in the guidance document on toxicology for registration of chemical pesticides in India will also be applicable for registration of botanical pesticides on merit case to case basis.

2. Waiver would be considered when existing information provides robust and full scientifically sound weight of evidence approach and based on read across/bridging from structurally and /or biologically related similar pesticides, physicochemical/biological properties and use and exposure patterns case to case basis

3. The replacement alternatives not involving experiments on animals would be considered and accepted in those cases where data from valid 3Rs (Replacement, Reduction and Refinement) methods recognized by OECD and other renowned Regulatory Authorities is provided wherever applicable, case to case basis.

Packaging

Sr. No.	Parameters	9 (3b)			9(3)					9(4)
		TI	TIM	FIM	TI	TIM	FI	FIM	Label expansion	TI/TIM/FI/FIM or endorsement of already approved Packaging
1	Labels and leaflets per IR- 1971, all fields (as applicable) and as amended from time to time	R	R	R	R	R	R	R	NA	R*
2	Manner of labeling and leaflet	R	R	R	R	R	R	R	NA	R*
3	Type of packaging (Ultra small, small or Big whichever is applicable	R	R	R	R	R	R	R	NA	R*
4	Manner of packaging	R	R	R	R	R	R	R	NA	R*
5	Specification for primary, secondary and transport packages (whichever is applicable)	R	R	R	R	R	R	R	NA	R*
6	Details of packaging material and its	R	R	R	R	R	R	R	NA	R*
	compatibility with content									
7	Performance of container with content during storage stability test (Shelf life study)	R**	R	R	R	R	R	R	N.A	R*
8	Transport worthiness test	R**	R	R	R	R	R	R	N.A	R*

Note: R*- Required only if manner of packaging is different then the 9(3) registrant. R**- Before commercialization the data will be required. N.A-Not Applicable

Note:

1. In case of additional packaging endorsement applications, the data at Sl. No. 05, 06, 07& 08, are not required if similar packaging (material)and manner of packaging is being sought by the applicant as has been granted to earlier 9(3) registrant.
2. Specification of Bureau of Indian Standard (BIS) must be followed for all the packaging requirements (Wherever available and applicable).
3. All Packaging tests must be carried out with the product of same batch and in its commercial package preferably in Indian condition.
4. The duration of the test and the conditions including geographical conditions must be mentioned.
5. Storage stability data should be generated in consonance of BIS or product specification.
6. The testing protocols must have their basis in the WHO/FAO/ CIPAC/ASTM recommendations or other validated methodology of GLP/ NABL accredited laboratory having packaging testing (chemical / mechanical as applicable etc.) in the scope.
7. The Accelerated storage study (ASS) test must be conducted at 54± 2°C (wherever applicable) for 14 days as per FAO/ WHO manual for claiming appropriate shelf life of the product which can be maximum two years, subject to the condition of providing the ambient storage stability study data of thirty months or as the case may be within thirty months from the date of application for the registration.

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Sub category 4.2.1: Guidelines for the Registration of Indigenous Manufacture of Eucalyptus Based Products Containing "Eucalyptol"

(Approved by the Registration Committee in its 293rd RC meeting held on 26.09.2008)

	PARAMETER	Extract / Concentrate		Formulation	
		9(3B)	9(3)	9(3B)	9(3)
A.	**Biological characteristic and Chemistry**				
1	Name of the Part of the Plant(s) to be used for extraction of the active ingredients / components.	R	R	R	R
2	Outline process of extraction of active ingredients.	R	R	R	R
3	Outline of process of manufacture of formulation clearly identifying the chemicals as indicated in point (4) below.	R	R	R	R
4	(a) The active ingredient (a.i.) content of the 'Eucalyptol' in the extract (concentrate) /formulation. (b) When the insecticidal a.i. is other than Eucalyptol then the applicant has to indicate the name, quality and quantity of that particular a.i. (s).	R	R	R	R
5	Chemical identity of the ingredient as stated at point (4) above.	R	R	R	R
	(a) Structure				
	(b) Chemical formula				
	(c) Isomers, if any				
	(d) Conditions of isomerism				
6	Physico-chemical properties including stability parameters	R	R	R	R
	(e.g. Photo, Thermo etc.)				
7	Specifications of ingredient including adjuvants as indicated at point (4) above.	R	R	R	R
8	Method of analysis of Eucalyptol as well as other components / ingredients (as at point 7)	R	R	R	R
9	Analytical test report	R	R	R	R
10	Shelf-life claim.	R	R	R	R
11	Shelf-life data	NR	R	NR	R
B.	**BIO-EFFICACY**				
1	Bio-effectiveness against target pest species in specified crops	R* (2 seasons data from minimum 2 agro-climatic	R* (2 seasons data from minimum 3 agro-climatic	R (2 seasons data from minimum 2 agro-climatic	R (2 seasons data from minimum 3 agro-climatic

		condit ions)	condit ions)	condit ions)	condit ions)
2	Phytotoxicity (as per standard tests)	R* (same as above at 1.	R* (same as above at1.	R (same as above at 1.	R (same as above at 1.
3	Compatibility with other agro-chemicals, if claimed	R*	R*	R	R
4	Stability of formulation (Photo, Thermal etc.) in aqueous dilution (acidic, neutral & basic)	R*	R*	R	R
5	Direction concerning dosage for each target pest species	R*	R*	R	R
6	Stage of crop for use and stage of target pest	R*	R*	R	R
7	Waiting period	R*	R*	R	R
8	Application equipment	R*	R*	R	R
9	Information regarding registration status in other countries, if any.	R	R	R	R
C.	**TOXICITY (As per data requirement of chemical Pesticides)**				
1.	Acute oral in rat & mice	R	R	R	R
2.	Acute dermal	R	R	R	R
3.	Acute inhalation	R	R	R	R
4.	Primary skin irritation	R	R	R	R
5.	Irritation to mucous membrane	R	R	R	R
6.	Sub-acute oral rat	R	R	NR/R	NR/R
7.	Sub-acute oral dog	R	R	NR/R	NR/R
8.	Sub-acute dermal	R	R	NR/R	NR/R
9.	Sub-acute inhalation	R	R	NR/R	NR/R
10.	Neuro-toxicity	NR	R	NR	NR/R
11.	Synergism & potentiation	NR	R	NR	NR/R
12.	Teratogenicity	NR	R	NR	NR
13.	Effect on reproduction	NR	R	NR	NR
14.	Carcinogenicity	NR	R	NR	NR
15.	Metabolism	NR	R	NR	NR
16.	Mutagenicity	NR	R	NR	NR

17.	Toxicity to birds	R	R	R	R
18.	Toxicity to fish	R	R	R	R
19	Toxicity to honeybees	R	R	R	R
20.	Toxicity to live stock	R	R	NR	NR
21.	Medical data	R	R	R	R
22	Human toxicity information	R	NR	R	NR
23.	Observation in man (Health records of spray operators)	NR	NR	NR	R
24.	Health records of Industrial workers.	NR	NR	NR	NR
25.	Toxicity to live stock (Field trial & observation)	NR	NR	NR	R
26.	International report on carcinogenicity & mutagenicity/genotoxicity status	NR	R/NR	NR	NR
D.	**PACKAGING AND LABELLING**				
1	Labels and Leaflets	R	R	R	R
2	Type of packing	R	R	R	R
3	Manner of packing	R	R	R	R
4	Content container compatibility data shall be generated in an independent reputed laboratory.	NR	R	NR	R
5	Manner of labelling	R	R	R	R
6	Specifications of packing	R	R	R	R
	i. Primary packing	R	R	R	R
	ii. Secondary packing	R	R	R	R
	iii. Transport packing	R	R	R	R
7	Information for safe storage and use	R	R	R	R
8	Information for disposal of empty containers/ used packages.	R	R	R	R
9	Performance test of containers during storage conditions for the shelf life of the product as per relevant IS.	NR	R	NR	R
	Note: The applicant shall submit complete data for registration of indigenous manufacture of botanical extract/ concentrate along with complete data sets in respect of registration of indigenous manufacture of formulation u/s 9(3)				
	R = Required, NR = Not required, R* = Required, if used in concentrated form				

Sub category 4.2.2: Guidelines For Registration of Rotenone For Pisci-Culture

(Approved by the registration committee in its 280th RC meeting held on 06.09.2007)

A.	CHEMISTRY	9(3B)	9(3)
1	Name of the Part of the Plant(s) to be used for extraction of the active ingredients.	R	R
2	Outline of Process of Formulation.	R	R
3	Extract Contains Rotenone as active constituent.	R	R
4	Chemical composition of the formulations	R	R
5	Physico-chemical properties of a.i. & adjuvants	R	R
6	Product Specifications as BIS format	R	R
7	Method of analysis for Rotenone a.i.	R	R
8	An undertaking that the product does not contain any other chemical except plant extract	R	R
9	Analytical test report (a) Independent laboratory (b) Central Insecticides laboratory	R	R
10	Shelf-life claim	R	R
11	Shelf-life data	NR	R
B.	**BIO-EFFICACY**		
1	Bioeffectiveness data on weed fish (Puntius spp.) and predatory fish (Murrel/Channa spp.)based on two season and two locations for the registration u/s 9(3b) and based on two season and three locations for the registration u/s 9(3) generated by ICAR/CSIR/SAUs	R	R
2	Effect on natural fish food organisms i.e. phytoplankton and zooplankton	R	R
3	Effect on water quality parameters i.e. – physico–chemical properties with reference to major nutrients	R	R
4	Persistence toxicity against cultured species – carp /catfish /prawn /shrimp (depending on fresh water/brackish water application)	R	R
5	Compatibility with other chemicals if claimed	R	R
6	Restocking period*	R	R
7	Direction concerning dosage – dose of application per hectare metre of water	R	R
8	Method of application	R	R
9	Time of application	R	R
10	Purpose of manufacture	R	R
11	Information regarding registration status in other countries, if any	R	R
C.	**TOXICITY**		
1	Acute Oral rat and mice	R	R
2	Acute dermal	R	R
3	Acute inhalation	R	R
4	Primary skin irritation	R	R
5	Irritation to mucous membrane	R	R
6	Sub-acute oral rat & dog	R	R
7	Sub-acute dermal	R	R

8	Sub-acute inhalation	R	R
9	Neuro-toxicity	R	R
10	Synergism & potentiation	R	R
11	Teratogenicity	R	R
12	Effect on reproduction	R	R
13	Carcinogenicity	R	R
14	Metabolism	R	R
15	Mutagenicity	R	R
16	Toxicity to birds	R	R
17	Toxicity to fish	R	R
18	Toxicity to honeybees	R	R
19	Toxicity to live stock	R	R
20	Medical data	R	R
21	Health monitoring study on applicators	R	R
22	Health records of Industrial workers.	R	R
D.	**PACKAGING AND LABELLING**		
1	Label and Leaflets as per IR 1971 existing norm	R	R
2	Type of package	R	R
3	Manner of Package (Container - content compatibility)	R	R
4	Specification for primary package	R	R
5	Specification for secondary package	R	R
6	Specification for transport package	R	R
7	manner of labelling	R	R
8	Instruction for storage and use	R	R
9	Information regarding disposal of used package	R	R
10	Process of manufacturing / indicating material balance generation of wasted.	R	R
	R- Required NR- Not required. *Time from the application of rotenone to introduction of fish seed.		

Sub category 4.2.3: Guidelines for the registration of formulations Cymbopogan spp. Based plant extract

(Approved by the registration committee in its 202nd RC meeting held on 17.08.2000)

A.	CHEMISTRY	9(3B)	9(3)
1	Name of the Part of the Plant(s) to be used for extraction of the active ingredients.	R	R
2	Outline of Process of Formulation.	R	R
3	(a) Extract Contents "Geraniol" as one of the major active constituents. The concentration of a.i. i.e. Geraniol (Cymbopogom sp.) in the formulated extract should contain not less than ppm of Geraniol (Cymbopogom sp.) a.i.	R	R
4	Chemical composition of the formulations	R	R
5	Physico-chemical properties of a.i. & adjuvants	R	R
6	Product Specifications as BIS format	R	R
7	Method of analysis for Geraniol (Cymbopogom sp.) a.i.	R	R
8	An undertaking that the product does not contain any other chemical except plant extract	R	R
9	Analytical test report	R	R
10	Shelf-life claim	R	R
11	Shelf-life data	NR	R
B.	**BIO-EFFICACY**		
1	Bio-effectiveness	R	R
2	Phytotoxicity	R	R
3	Compatibility with other chemicals	R	R
4	Purpose of manufacture	R	R
5	Direction of concerning dosage	R	R
6	Time of application	R	R
7	Waiting period	R	R
8	Application equipment	R	R
9	Information regarding registration status in other countries, if any	R	R
C.	**TOXICITY**		
1	Acute Oral rat and mice	R	R
2	Acute dermal	R	R
3	Primary skin irritation	R	R
4	Primary eye irritation	R	R
5	Allergy/Sensitization	R	R
D.	**PACKAGING AND LABELLING**		
1	Label and Leaflets as per IR 1971 existing norm	R	R
2	Type of packaging	R	R
3	Manner of Packaging (Container - content compatibility)	R	R
4	Specification for primary package	R	R
5	Specification for secondary package	R	R
6	Specification for transport package	R	R

7	manner of labelling	R	R
8	Instruction for storage and use	R	R
9	Information regarding disposal of used package	R	R
10	Process of manufacturing / indicating material balance generation of wasted.	R	R
	R- Required		
	NR- Not required.		

Sub category 4.2.4: Guidelines for the Import of Concentrated Pyrethrum Extract

(Approved by the registration Committee in its 148th RC meeting held on 11.11.1994)

A.	If the higher concentration of Pyrethrum extract is imported from the already approved source (which is approved for import of Pyrethrum extract (2%) i.e., Pyrethrum market board, Nakura, Kenya in that case the following simplified data in all disciplines is required to be submitted.	
	CHEMISTRY	
1	Detailed chemical composition.	R
2	Physico-chemical properties	R
3	Specification and method of analysis alongwith undertaking for product quality.	R
4	Analytical test report from principal	R
5	Identification and quantification of identifiable impurities in addition to the presence of active constituents i.e. Pyrethrum 1 & 2, Cinerin 1 & 2 and Jasemolin 1 & 2.	R
6	Shelf-life data as per the relevant guidelines is required to be submitted.	R
	BIO-EFFICACY	
	The higher concentrate is proposed to be imported to make the approved formulation. As such data on approved formulations is required w.r.t. bioefficacy	R
	TOXICITY (Toxicity data Requirements from alternate source are available below this table as approved in 303rd meeting of RC held on 05-08-2009)	
1	Acute oral toxicity in rat and mice	R
2	Acute dermal toxicity	R
3	Acute inhalation toxicity	R
4	Primary skin irritation	R
5	Irritation to mucous membrane.	R
	PACKAGING	
	If kerosene is used as a solvent then container content compatibility data is not required. If the solvent is other than kerosene then container content compatibility data is required.	
B.	In case higher concentration of Pyrethrum is to be imported from any other source then the approved one i.e. Pyrethrum Marketing Board, Nakura, Kenya then the applicant has to generate complete data as per the relevant guidelines u/s 9(3) of the Act.	
1	The individual component(s) is / are higher than registered formulations strength one acute toxicity data on higher strength has to be generated.	
2	Data / information on toxicity of registered strength / formulation have to be provided for comparision.	
3	The components of combination should preferably be limited to two only.	
	PACKAGING AND LABELLING	
1	Label and Leaflets as per Insecticides Rules, 1971 existing norms.	R
2	Type of packing	R
3	Manner of Packing (Packaging material plus compatibility with content)	R
4	Specifications for primary packaging	R
5	Specifications for secondary package	R
6	Specifications for transport package	R
7	Manner of Labelling	R
8	Instructions for storage and use, etc.	R

9	Information regarding disposal of used package.	R
	R- Required	

Sub category 4.2.5: Guidelines for the registration of herbal / botanical plant regulators under section 9(3) of Insecticides Act, 1968

(Approved by the registration Committee in its 161st RC meeting held on 06.03.1996)

A.	CHEMISTRY	
1	Source	R
2	Composition	R
3	Identity of the a. i. and Physico-chemical properties	R
4	Method of extraction in details (Manufacturing process)	R
5	Specifications	R
6	Chemical method of analysis only.	R
7	Shelf-life	R
8	Test Report	R
B.	**BIO-EFFICACY**	
1	Two seasons multi-location data on bio-effectiveness and phytotoxicity.	R
2	Compatibility - If the product is proposed to be mixed with other chemicals, data on compatibility are required to be submitted.	R
3	Registration status in other countries	R
4	One season data on persistence in soil and plants.	R
5	Bio-efficacy claims on the labels & leaflets as per guidelines approved by Registration Committee.	R
C.	**TOXICITY**	
1	Acute oral toxicology study in rats and mice.	R
2	Acute dermal toxicology study in rats and rabbits.	R
3	Primary skin irritation	R
4	Irritation to mucous membrane	R
5	Actual oral toxicity to birds (Chicken & Pigeons)	R
6	Toxicity to Honeybees	R
7	Toxicity to fish	R
	Data / information on following parameters are also required to be submitted.	
	Long Term Toxicity	R
i.	Neuro-behavior toxicity	R
ii.	Mutagenicity	R
iii.	Carcinogenicity	R
iv.	Effect on reproduction	R
v.	Health records of workers / spray operators as per the protocol.	R
D.	**PACKAGING AND LABELLING**	
1	Label and Leaflets	R
2	Type of packing	R
3	Manner of Packing	R
4	Content container compatibility	R
5	Manner of Labelling	R
6	Specifications of packing	R
	i. Primary packing	R

	ii. Secondary packing	R
	iii. Transport packing	R
7	Instructions for storage and use	R
8	Disposal of empty containers	R
	R- Required ; NR- Not required.	

Sub category 4.2.6: Guidelines for the registration of neem-based products containing Azadiractin to be used as household insecticides

(Approved by the registration committee in its 186th RC meeting)

	PARAMETER	9(3B)	9(3)
A.	**CHEMISTRY**		
	The common name of the household insecticides which is to be printed on the labels and leaflets should also contain minimum percentage of Azadirachtin content.	R	R
1	Chemical composition and allied data requirements.		
	Source of supply of the technical grade material / neem extract concentrate with minimum purity to be used in the formulation should be given.	R	R
2	Chemical Composition		
2.1	Complete chemical composition of the formulation should be submitted.	R	R
2.2	Complete identity and physico-chemical characteristics of various ingredients used in the formulation should be submitted.	R	R
2.3	The chemical composition for the formulation should include the kind, name and percentage of all the ingredients that is to be used in the formulation.	R	R
2.4	The active ingredients in the house hold insecticides should be only Azadirachtin and other constituents associated during extraction of neem. Seed Kernel free friom aflatoxin.	R	R
3	Specifications		
	Specification depending on the coding of the product as per BIS format alongwith undertaking for quality should be submitted.	R	R
4	Analytical test report	R	R
	The analytical test report giving the analysis of a particular batch of all the parameters included in the specifications should be submitted from an independent reputed national / international laboratory.	R	R
5	Shelf-life claim		
	An underting regarding shelf life claim should be submitted	R	R
6	Shelf-life data		
	Shelf life data as per existing guidelines in support of shelf life claim should be submitted.	NR	R
7	Method of analysis		
	Method of analysis should be submitted (whereever BIS method is available, the same should be submitted). The sample of the product and reference standard should be submitted. Detailed method of extraction of the active ingredient and process of formulation should be submitted.	R	R
B.	**BIO-EFFICACY**		
1	Bio-effectiveness		
	i. Bio-efficacy data generated at National Laboratories	R (Two National Labs)	R (Three National Labs)
2	Residual Toxicity	R	R

	i. Bio-efficacy tests on the proposed formulations should be conducted under Indian conditions at National Laboratories	R (Two National Labs)	R (Three National Labs)
	ii. Data on persistance on residual toxicity of insecticides on different type of surfaces should be generated.	R (Two National Labs)	R (Three National Labs)
3	General Information on label / leaflets		
	i. A brief direction concerning the major uses of the insecticide should be given on the labels.	R	R
	ii. Instruction regarding Insecticide : Not to be used on any food crops" to be given both on label and leaflets	R	R
	iii. Detailed information on the uses of Insecticide indicating the name of insects, method of application, dosage, places of treatments, Plant Protection Equipments to be used etc. should be given in the leaflets.	R	R
C.	**TOXICITY**		
1	Acute oral toxicity rat.	R	R
2	Acute oral toxicity mice	R	R
3	Acute dermal toxicity	R	R
4	Acute inhalation toxicity*	R	R
5	Primary skin irritation	R	R
6	Irritation to mucus membrane	R	R
7	Neuro-behavioral toxicity	NR	R
8	Reproduction toxicity	NR	R
9	Carcinogenicity	NR	R
10	Mutagenicity	NR	R
11	Health monitoring study of the users and /or spray operators by using the household pesticide in actual use. These studies shall be conducted as per the protocols approved by the Registration Committee.	NR	R
	* To be submitted for only category III household insecticides which emit vapours e.g. mats and coils etc.		
D.	**PACKAGING AND LABELLING**		
I.	Label and Leaflets		
	i. 7 copies of labels and leaflets prescribed for small size container upto 250 ml. capacity as per provisions of Insecticides Rules, 1971.	R	R
	ii. 7 copies of labels and leaflets prescribed for large size container above 250 ml. capacity as per provisions of Insecticides Rules, 1971.	R	R
II.	Type of Packaging		
	i. Information about the packaging materials proposed packaging of the product in retail or bulk.	R	R
	ii. Data on compatibility of the packaging materials with the content as per the method prescribed in relevent IS standard.	R	R
	iii. Information / data on suitability of the proposed packaging system for protecting the content through out the shelf life.	NR	R

	iv. Information on the proposed packaging materials regarding its incorporation in IS standard / Environment friendly notification by the Ministry of Environment.	R	R
III.	Manner of Packaging		
	i. Information regarding filling, sealing of closure system, pilfer proofness, leakproofness on primary packaging, secondary packaging and transport packaging.	R	R
	ii. Undertaking for the use of pilfer proof, leak proof and easily disposable container for packing the product	R	R
IV.	Transportation		
	Undertaking for safe storage, handling and transportation of the product.	NR	R
V.	Manner of Labelling		
	i. Information as per provisions 18 & 19 of Insecticides Rules, 1971 should be furnished	R	R
	ii. Instruction for storage and use including precautionary measures should be proposed.	R	R
	iii. Instructions regarding disposal of used material and washing of insecticides should be proposed.	R	R
	R- Required		
	NR- Not required.		

Sub Category 4.3: Guidelines for the Registration of Pheromones / Semio Chemicals

Definitions as per OECD:

- Semio-chemicals are chemicals emitted by plants, animals, and other organisms, and synthetic analogues of such substances, that evoke a behavioural or physiological response in individuals of the same or other species. They include pheromones and allele-chemicals. Here, Semio-chemicals refers to only those that affect the behavior of arthropods.
- Pheromones are Semio-chemicals produced by individuals of a species that modify the behavior of other individuals of the same species (i.e. an intra-specific effect).
- Straight-chained lepidopteran pheromones (SCLPs) are a group of pheromones consisting of un-branched aliphatic having a chain of nine to eighteen carbons, containing up to three double bonds, ending in an alcohol, acetate or aldehyde functional group.

Data requirements for the registration of pheromones / Semio Chemicals

A. Chemistry

Technical for import/indigenous manufacture:

S No	Parameters	9 (3b)	9(3)	9(4)
1.	Active ingredient/Tech conc.	R	R	R
2.	Chemical composition (If impurities in SCLP are also SCLPs, they should be summed-up)	R	R	R
3.	Chemical name (s)	R	R	R
4.	Common name	R	R	R
5.	Physical chemical properties	R	R	R
6.	Manufacturing process	R	R	R
7.	Analytical Test Report from GLP/ NABL accredited laboratory	R	R	R
8.	Storage condition with special reference to temperature	R	R	R
9.	Source of import (In case of import only)	R	R	R
10.	Sample for PRV purpose	R	R	R
11.	Purpose of import/indigenous manufacture	R	R	R

TECHNICAL FOR IMPORT/INDIGENOUS MANUFACTURE:

1. Active ingredient

2. Laboratory test: Lure manufactured from the particular ingredient will be tested by using wind tunnel and should demonstrate minimum 50% attractancy.

LURE/DISPENSER FOR IMPORT/MANUFACTURE:

1. Laboratory test; The lure/dispenser should demonstrate at least 50% attractacy using the wind tunnel.
2. Field test: The data on bio-efficacy based on two seasons field trials from two different agro-climatic conditions in the form of authentic/published report.
3. Compatibility: No data on compatibility are required unless the product is recommended for use in combination with pesticides or other agrochemicals.
4. Time and method of application: Information on timing, disruption is to be furnished.
5. Intended uses.
6. Mode of action and degree of specificity.
7. Target pest (s) and crops or premises to be protected.
8. Application rate.
9. Manner, rate and frequency of application.
10. Limitations of use.

Note: As per decision of 248th meeting of R.C. Pheromones used for monitoring and mass trapping are not covered under the various provisions of the Insecticides Act, 1968.

B. Bio-efficacy

S. No.	Particulars	9 (3b)	9(3)	9(4)
1.	**Active ingredients**	**R**	**R**	No bio-efficacy data required. Claim will be granted as per approved formulation u/s 9(3).
2.	**Laboratory test (**attractancy**):** Lure manufactured from the particular ingredient will be tested by using wind tunnel and should demonstrate minimum 50% attractancy	**R**	**R**	
LURE/DISPENSER FOR IMPORT/MANUFACTURE:				
3.	**Laboratory test:** The lure/dispenser should demonstrate at least 50% attractantcy using the wind tunnel.	R	R	
4.	Field test	The data on bio- efficacy based on two seasons field trials from one agro-climatic conditions in the form of authentic/published report.	The data on bio-efficacy based on two seasons field trials from two different agro-climatic conditions in the form of authentic/published report.	
5.	Compatibility: No data on compatibility are required unless the product is recommended for use in combination with pesticides or other agrochemicals.	**R**	**R**	
6.	Time and method of application: Information on timing, disruption is to be furnished.	**R**	**R**	
7.	Intended uses.	**R**	**R**	
8.	Mode of action and degree of specificity.	**R**	**R**	
9.	Target pest (s) and crops or premises to be protected.	**R**	**R**	
10.	Application rate.	**R**	**R**	
11.	Manner, rate and frequency of application.	**R**	**R**	
12.	Limitations of use.	**R**	**R**	

Note: As per decision of 248th meeting of R.C. Pheromones used for monitoring and mass trapping are not covered under the various provisions of the Insecticides Act, 1968.

SPECIFICATIONS FOR PHEROMONE TRAPS AND LURES

The committee after detailed discussions on the existing specification of various models of pheromones traps decided not to make any change in these specifications. However, these specifications may further be examined by SAU and SDAS of various States:

Funnel shaped trap: This trap is generally used for trapping the moths of *Helicovarpa armigera, Spodoptera litura, Earisa spp*. etc.

Colour: Any colour other than black.

Structure: The funnel trap may have three parts (1) canopy (2) funnel shapped "trap base" and (3) a collection device.

Canopy: Dia : 120-160mm

Thickness : 1.0-3.0mm (There should be a provision for fixing the canopy to the "trap base" and also the (pheromone lure)

Trap base:

Dia of the mouth : 75- 20mm ; Height of funnel : 45-190mm ; Dia of the bottom hole :20-30 mm

Should possess a "L" or "T" shaped handle or any other device by which the other device by which the "trap" may be fixed to the support. The "Trap base" may be provided with 2 to 4 stalks for fixing the canopy to the "trap base". The canopy should be firmly rest on stalks so that the canopy is not dislodged due to wind.

Collection device: It should be made of polythene or other suitable material. It should withstand wind, temperature and rain water. Should be fixed to the "trap base" in such a way that the device remains attached to the trap under field conditions.

II. **Sticky trap (for pink boll worm etc.):**

* Corrugated DVC, Plastic laminated card board, tin or any other suitable material that should be water-proof.
* The sticky glue should be non- drying.
* The outer surface of trap should be water proof.
* The colour may be except black.
* There should be provision for fixing the trap for support.

III. **Fly trap (For fruit/vegetable flies):**

* Material construction as described in sticky/funnel trap.
* Any colour except black.
* Should withstand rainfall, heat/temperature and wind.

* Should be structured in such a way that the trap is escape proof.

Specification of Lures:

1. Lure made of sulphur free rubber/polypropylene/PVC, Impregnated with specific pheromone blends.
2. Field efficacy should be minimum for 15 days after application.
3. Impregnated lures should be packed singly in individual tri laminated pouches with 30 M1 Aluminum foil.
4. Shelf-life of Lure in original pack should be minimum 6 months at room temperature.
5. Lures should attract insect species only, with 50% insect attractancy by pheromone/lure/dispenser by using wind tunnel method.

C. Toxicology

S. No	Parameters	Semio-chemicals / pheromones		
		9(3b)	9(3)	9(4)
1.	Acute oral toxicity- rat	R	R	No data required if chemical equivalence is established with 9(3) registrant
2.	Acute dermal toxicity- rat	R	R	
3.	Acute Inhalation – rat	R	R	
4.	Primary Skin Irritation - Rabbit	R	R	
5.	Primary Eye Irritation - Rabbit	R	R	
6.	Skin Sensitization - Guinea pig	R	R	
7.	Sub- acute oral- rat	NR	NR/R	
8.	Sub-acute dermal	NR	NR/R	
9.	Sub- acute inhalation	NR	NR/R	
10.	Neurotoxicity	NR	NR/R	
11.	Teratogenicity	R	NR/R	
12.	Effect on Reproduction	NR	NR/R	
13.	Carcinogenicity/chronic toxicity/combined toxicity study*	NR	NR/R	
14.	Metabolism	NR	NR/R	

15.	Mutagenicity (AMES + 2 in-vitro +1 in-vivo)*	R	NR/R	
16.	Cellular immune response	R	R	
17.	Toxicity to bird (2species)*	R	NR/R	
18.	Toxicity to fresh water fish	R	NR/R	
19.	Toxicity to Honey bees	NR	NR/R	

Note:

1. Lepidopterean Pheromones that are naturally occurring compound designated by an unbranched aliphatic chain (between 9 and 18 carbons) ending in an alcohol, aldehyde or acetate functional group and containing up to 3 double bonds in aliphatic backbones can be exempted from the sub-acute toxicity carcinogenicity, effect on reproduction and metabolism if their use rates do not exceed 150 gm/acre/year with Good Agricultural Practices (GAP) and used in solid matrix dispensers.
2. For formulation products no toxicity data are required unless it is added with some other pesticides.
3. Data for 9(3) will be conditionally required on case to case basis.

***Please refer to guidance document on toxicology for registration of chemical pesticides in india.**

Note:

1) The general recommendations as mentioned in the Guidance document on Toxicology for registration of chemical pesticides in India will also be applicable for registration of Semio-chemicals / Pheromones on merit case to case basis.
2) Waiver would be considered when existing information provides robust and full scientifically sound weight of evidence approach and based on read across/bridging from structurally and /or biologically related similar pesticides, physicochemical/biological properties and use and exposure patterns case to case basis
3) The replacement alternatives not involving experiments on animals would be considered and accepted in those cases where data from valid 3Rs (Replacement, Reduction and Refinement) methods recognized by OECD and other renowned Regulatory Authorities is provided wherever applicable, case to case basis.

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D. Packaging

Sl. No.	Parameter	Semio-chemicals (pheromones)		
		9(3b)	9(3)	9(4)

1.	Labels and Leaflets per IR-1971, all fields (as applicable) and as amended from time to time	R	R	R
2.	Manner of labeling and Leaflet	R	R	R
3.	Type of packaging (Ultra small, small or Big whichever is applicable)	R	R	R
4.	Manner of packaging	R	R	R
5.	Specification for primary, Secondary and Transport packages (whichever is applicable)	R	R	R
6.	Details of packaging material and its compatibility with content	R	R	R
7.	Performance of container with content during storage stability test(Shelf life Study)	R*	R	R
8.	Transport worthiness test	R*	R	R

R*- Before Commercialization the data will be required.

Note:

1. In case of additional packaging endorsement applications, the data at Sl. No. 05, 06, 07& 08, are not required if similar packaging (material) is being sought by the applicant as has been granted to earlier 9(3) registrant.
2. Specification of Bureau of Indian Standard (BIS) must be followed for all the packaging requirements (Wherever available and applicable).
3. All Packaging tests must be carried out with the product of same batch and in its commercial package preferably in Indian condition.
4. The duration of the test and the conditions including geographical conditions must be mentioned.
5. Storage stability data should be generated keeping at least the following parameters in test protocol such as test temperature, test duration, test packaging material, content of active ingredient (a.i.) and relevant impurities in the product during and after storage, test humidity, exposure to light, physical and chemical properties of the product during and after storage etc.
6. The testing protocols must have their basis in the WHO/FAO/ CIPAC/ASTM recommendations or other validated methodology of GLP/ NABL accredited laboratory having packaging testing (chemical / mechanical as applicable etc.) in the scope.
7. The Accelerated storage study (ASS) test must be conducted at 54± 2°C (wherever applicable) for 14 days as per FAO/ WHO manual for claiming appropriate shelf life of the product which can be maximum two years, subject to the condition of providing the ambient storage stability study data of thirty months or as the case may be within thirty months from the date of application for the registration.

Foot Note: ***As per 397th RC meeting (Agenda 3.12), repeat registration for same strain/compound of Bio pesticides/botanical products and pheromones should be granted u/s 9(3) instead of 9 (3b) considering the fact that these have already been introduced in the country as per relevant provision of the Act and accepted the above suggestion in principle.***

CATEGORY: 5. GUIDELINES FOR THE REGISTRATION OF PESTICIDES FOR OTHER PURPOSES (i.e. HOUSEHOLD, PUBLIC HEALTH, AIR CRAFT DISINFECTANT, ANY OTHER)

SUBCATEGORY: 5.1. GUIDELINES FOR THE REGISTRATION OF HOUSE HOLD INSECTICIDES

(*Reference of 211th meeting of RC held on 30-05-2001 based on approval of 93rd meeting of 28-08-1987 and additions based on approval in 266th meeting of 20-07-2006*).

Household pesticides are commonly used indoors to control pests such as ants, cockroaches, houseflies, mosquitoes, fleas, ticks, bedbugs, termites, rodents, mites and microbes. ... Household pesticides may contain one or a combination of active ingredients of synthetic (chemical) or natural (plant or microorganism) origin.

GENERAL REQUIREMENT:

1. All household pesticides as defined must be manufactured only from the technical grade pesticides which are registered under the Insecticides Act, 1968
2. Data on household pesticides formulation shall be considered along with the data on technical grade pesticides and not in isolation.
3.

CHEMISTRY:

The common name of the household insecticides which is to be printed on the labels and leaflets should be written as household pesticides containing 'x'...% w/w 'y'....% w/w and 'ź.....% w/w etc.

1. **Source of Supply of the Technical grade** material with minimum purity to be used in the formulation.
2. **Chemical Composition**

2.1 Complete Chemical composition of the formulation

2.2 Complete data on identity and Physico - Chemical Properties of various ingredients of **the formulations**.

2.3 The Chemical composition for the formulation should include the kind, Name & Percentage of all the ingredients that is to be used in the formulation.

2.4 The number of active ingredients in household combination product should not be more than two. However, if any applicant is able to submit satisfactory method of analysis for more than two a.i. in the combination product, the committee may consider such request on its merit.

3. **Specification**

3.1 Wherever no specifications are available, the registration secretariat may harmonize specification evolving its own procedure.

4. **Analytical Test Report**

4.1 The ATR giving analysis of a particular batch of all the parameters included in the specifications shall be submitted from an independent reputed National / International laboratory.

5. **Shelf-life Data**

5.1 The expected shelf-life claim for the product along with the supporting data, be submitted. This data should conform to the guidelines prescribed by the central Insecticides Board.

6. **Method of Analysis**

6.1 Wherever applicable ISI methods of analysis are followed.

6.2 The method for the analysis of various a.i. used in a formulation be provided. As far as possible, single multiple pesticides analysis methods be suggested.

6.3 For identification of a.i. in the formulation, suitable quantitative methods may be developed and provided followed by detailed analysis as per 6.2 above.

7. **Process of Manufacture**

7a. Information about Raw Materials Used & Their Source of Supply.

7b. Step-wise Manufacturing Process.

7c. Flow sheet diagram of process of manufacture

8. Effluent Treatment method
9. The applicant should provide sample along- with standards technical sample from the principals/ authorized dealers for chemical verification.
10. Methodology for residue estimation as per BIS format.

(Above 7-10 are added, based on approval of 266th meeting of RC held on 20-07-2006)

DATA REQUIREMENTS FROM BIOEFFICACY AND RESIDUE PARAMETERS:

A. Bio-efficacy claims in case of formulation

(1) Bio-efficacy claims to be given on the labels

(i) A brief direction concerning the major usages of the pesticides should be given on the labels.

(ii) Whenever the Registration Committee has approved the product for restricted use, this should be indicated very clearly on the labels in capital letters. 'For use only'.

(iii) Instruction regarding Insecticide 'Not to be used on any food crop to be given'.

(2) Bio-efficacy claims to be given on the leaflets

(i) Detailed information on the usages of insecticide indicating the name of insects, method of application, dosage, places of treatment, PP equipment to be used etc. should be given in paragraphs forms. Common name of the insects should be given.

(ii) Whenever the registration committee has approved the product for restricted use, this fact should be indicated very clearly on the leaflets in bold letters.

(iii) Instructions regarding Insecticides 'Not to be used on any food crop to be given'.

B. Bio-efficacy claims to be given on the labels and leaflets in case of technical grade material.

1. The purpose of import/manufacture of technical grade material is required to be given on the labels and leaflets.

C. Data requirements on Bio-efficacy and Residues for formulation of pesticides for provisional registration of U/s 9(3B).

(I) The applicant should submit published / cited Indian data on bio-effectiveness in support of the claims indicated on the labels / leaflets.

The data should be produced from 2 National laboratories based on minimum 2 repeated trials. This should be further supported with any published information available from elsewhere (overseas data).

(II) Information on secondary pests outbreaks particularly of ticks and mites should be given where residual pyrethroids are being used.

(III) Data on Residues:

Data on persistence, of the pesticides which should be on different types of surfaces should be submitted / generated obtained under foreign / Indian conditions from 2 laboratories. This may also be supported by data generated elsewhere.

(IV) Data on concentration of a.i. in Air - for Aerosols (e.g. Coil, mats, liquid vaporizer etc.). (Added based on approval in 266th meeting of RC held on 20-07-2006)

(V) Registration Status in foreign countries. (Added based on approval in 266th RC meeting)

D. Data requirements on Bio-efficacy & Residues data requirement for regular registration of formulations of pesticides U/s 9(3):

(i) Bio-efficacy test on the proposed formulations should be conducted under Indian conditions minimum on two trials in each three national laboratories.

(ii) Data on persistence of pesticides on different types of surfaces should be generated in three national laboratories wherever applicable.

(iii) Information on secondary pests outbreaks particularly of ticks and mites should be given where residual Pyrethroids are being used.

(iv) Data on concentration of a.i. in Air - for Aerosols (e.g. Coil, mats, liquid vaporizer etc.). (Added based on approval in 266th meeting of RC held on 20-07-2006)

(v) Registration Status in foreign countries. (Added based on approval in 266th RC meeting)

E. Data requirements for the registration of new formulations of the approved pesticides

Data on bio-effectiveness and persistence on different surfaces as required in case of Registration of formulations for regular registration under section 9(3).

F. Data required for combination products

Bio-efficacy data on combination products v/s individual products. Data on persistence on different types of surfaces should also be submitted. All above data should be generated as per the requirement indicated in case of pesticides required for regular registration under section 9(3).

G. Methodology:

Flying/crawling insects

Residual films of insecticides prepared by spraying insecticides on different types of surfaces, such as Glass, Wood, Mud & Cement surfaces. Insects to be exposed for 30 minutes and then shifted to recovery chambers for 24 hours after which the mortality count should be made and the satisfactory mortality of insects would be more than 90%. The residual toxicity of insecticides should also be studied at different intervals., Evaluation of space spray against flying insects should be conducted in PEET GRADY Chamber as per standard ISI specification 1824, mats/coils could also be evaluated inside the Peet Grady Chambers against caged mosquitoes and the knock down effect is to be recorded at different intervals. Aerosols are to be evaluated inside a standard room. The tests are to be conducted as per WHO technical reports series No. 206.

PACKAGING & LABELLING DATA REQUIREMENTS

Guidelines on the data requirements for the packaging & Labelling for the grant of registration of Household products of Pesticides formulations.

Type of formulation	**Physical state of formulation**
	Solid - coil, cake-mat, liquid, paste, aerosol and others.
1. Proposed labels	As per Insecticides Rules, 1971 / Existing norms of the Registration Committee.
2. Proposed leaflets	As per Insecticides Rules, 1971 / Existing norms of the Registration Committee.

3. Manner of Packaging

(1) In general, the requirements for packaging of household pesticides are governed according to IS: 8190 (part-III)-1979 and amendment thereof, if any.

(2) Glass bottles of glass used as primary packaging material is not permitted for household pesticides

(3) Details of points (a) to (d) as under should be submitted. If the packaging material dose not conform to IS specifications.

(a) Specification for primary packages.

(b) Specification for secondary package

(c) Specification for transportation packaging.

(d) History of conventional usage of packaging.

4. Compatibility: if the compatibility of container with pesticides is already established the relevant Indian Standards number for said packaging material may be quoted. In case compatibility is not established detailed specifications for primary package is required to be furnished indicating dimension, material construction design etc. along with test reports with reference to physico-chemical requirements for field trial/studies for transport worthiness. The analysis report should be according to relevant Indian standards specifications on packaging and

quality control or as per specifications/test methodology, approved by the Registration Committee.

5. Instructions for storage and use including first aid and precautionary measures proposed for labelling

6. Information regarding disposal of used package, surplus material and washing of insecticides proposed for labelling.

TOXICITY - TOXICOLOGICAL DATA REQUIREMENTS

Category - 1 (Solid)

1. Acute oral Rat
2. Acute Dermal- Rat / Rabbit
3. Acute inhalation-Rat
4. Primary Skin Irritation- Rabbit
5. Acute Eye Irritation- Rabbit
6. Skin Sensitization Test
7. Medical data
8. Human toxicity information from foreign countries

Category - II (Liquid)

1. Acute oral Rat
2. Acute Dermal- Rat / Rabbit
3. Acute inhalation-Rat
4. Primary Skin Irritation- Rabbit
5. Acute Eye Irritation- Rabbit
6. Skin Sensitization Test
7. Medical data
8. Human toxicity information from foreign countries

Category - III (Vapours)

1. Same as category II

Health monitoring study of the user by using the household pesticides in its actual use. The symptoms and signs as a result of sub-acute chronic or delayed effects on man from single or multiple exposure to the product shall be observed and submitted to the RC.

There would be two types of House-hold products from the view point of safety viz:

I. Ready- to - use formulations for house- hold purposes;

II. Concentrated forms of formulations to be used only by the pest control operators, commercial establishments, Government agencies, for large scale disinfestations etc. However, in case of specific requirements, the type I for use in concentrated form could be considered by the committee on merit basis.

REFER TO 'TOXICITY DATA REQUIREMENTS FOR CHEMICAL PESTICIDES' OF 'GUIDANCE DOCUMENT ON TOXICOLOGY FOR REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.'

Note: The application may be submitted according to the Checklist for submission of application in 5 folders for Registration of Pesticides U/s (3) / 9(3B).

-Reference icon 'Checklists' on the website.

Sub category: 5.2. Guidelines for Registration of Insecticides for Use in Public Health Programme u/s 9(3)

1. Bio-efficacy data generated by ICMR / MOH&FW Institutes based on multi-centric three years / seasons as per their protocol (Bio-efficacy data must be generated as per the Common Protocol for Evaluation of Public Health Pesticides for use in Vector Control published by Indian Council of Medical Research (ICMR), 2023)
2. For WHOPES recommended insecticides, Phase I data is exempted (As per 449th RC decision, Vide agenda item no. 10.32 and Annexure 10.32.1)
3. Other data w.r.t Chemistry, Bio-efficacy, Toxicity & Packaging shall be as per the requirements of General Guidelines for Registration of Insecticides.
4. Data requirement from Toxicity angle

   1. Acute oral Rat
   2. Acute Dermal- Rat / Rabbit
   3. Acute inhalation-Rat
   4. Primary Skin Irritation- Rabbit
   5. Acute Eye Irritation- Rabbit
   6. Skin Sensitization Test
   7. Acute Avian toxicity *
   8. Acute toxicity to Fresh Water Fish (One species)

   Recommendations from National Vector Disease Control Program, M/o Health & Family Welfare are required.

   For LLIN: Acute toxicity studies (Six pack) i.e. S.No. 1 to 6 are required with Premix and health monitoring studies in users as per approved protocol is required with final product.

REFER TO 'TOXICITY DATA REQUIREMENTS FOR CHEMICAL PESTICIDES' OF 'GUIDANCE DOCUMENT ON TOXICOLOGY FOR REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.'

Sub category : 5.3. Guidelines for the registration of insecticidal formulation for use in air craft disinfection

(Approved by the Registration Committee in its 280th RC meeting held on 6th September 2007)

General requirement

All insecticides for aircraft disinsection must be manufactured only from the technical grade insecticides which are registered under the Insecticides Act, 1968

Data on insecticides formulation shall be considered along with the data on technical grade insecticides and not in isolation.

Data requirement from Chemistry parameters The common name of the aircraft disinsection insecticides which is to be printed on the labels and leaflets should be written as aircraft disinsection insecticide containing 'X'……. %w/w and 'Z'……%w/w etc.

1 Chemical composition and allied data requirement

1.1 Source of supply of the technical grade material with minimum purity to be used in the formulation should be given.

2 Chemical composition

Sample analysis to be carried out in the Central Insecticide Laboratory (CIL), DPPQS, Faridabad.

2.1 Complete chemical composition of the formulation should be submitted.

2 Complete data on the identity and physico-chemical characters of various ingredients of the formulation should be submitted.

2.3 The chemical composition for the formulation should include the kind, name and percentage of all the ingredients that is to be used in the formulation.

2.4 "The number of active ingredients in the aircraft disinsection combination product should not be more than two. However, if any applicant is able to submit satisfactory method of analysis for more than two active ingredients in the combination product, the Committee may consider such request on its merit".

Specification

Wherever no specifications are available, the Registration Secretariat may harmonize specification evolving its own procedure.

Analytical Test Report

The analytical test report giving the analysis of a particular batch of all the parameters included in the specifications shall be submitted from an independent reputed National/International laboratory.

Shelf life data

The expected shelf life claim for the product along with the supporting data be submitted. This data should confirm to the guidelines prescribed by the Central Insecticides Board.

Method of analysis

Wherever available, an BIS method for analysis is followed.

The methods for the analysis of various active ingredients used in a formulation be provided. As far as possible, single, multiple insecticides analysis method be suggested.

For identification of active ingredient in the formulation, suitable qualitative methods may be developed and provided followed by detailed analysis as for quantification as stated above.

Bio-efficacy claims to be given on the labels

i A brief direction concerning the major usage of the insecticide should be given on the labels.

ii Wherever the Registration Committee has approved the product for restricted use, this should be indicated very clearly on the labels in capital letters. 'For _____ use only"

iii Instructions regarding insecticide 'Note to be used on any food as well as crops to be given'.

Bioefficacy claims to be given on the leaflets

i Detailed information on the usage of insecticide indicating the names of insects, method of application, dosage, places of treatments, appliances to be used etc. should be given in paragraph forms. Common name of the insects should be given.

ii Wherever the Registration Committee has approved the product for restricted use, this fact should be indicated very clearly on the leaflets in bold letters.

iii Instructions regarding insecticide 'Not to be used on any food crop to be given'

Bio-efficacy and residue data

i Bio-efficacy test on the proposed formulation should be conducted in Indian conditions minimum 2 trials in each of 3 National Laboratories / NABL accredited laboratories, recognized by Government of India.

ii Data on persistence of pesticides on commonly used surfaces in the aircrafts and concentration in air, as applicable, should be generated in three National Laboratories/ NABL accredited laboratories.

Toxicology

1. Acute oral rat
2. Acute dermal Rat / Rabbit
3. Acute inhalation Rat
4. Primary skin irritation Rabbit
5. Irritation to mucous membrane Rabbit
6. Skin sensitization test

7. Health monitoring studies on spray men and/or volunteers on board as per the protocol to be approved by the Registration Committee

Packaging and labelling

A Type of packaging

1 Specification of primary packaging system.

2 Specification of secondary packaging system.

3 Specification of transport packaging system.

B **Manner of packing**

Detailed information with reference to method by which inner most pack shall be further packed in secondary and transport packing.

C **Container content compatibility study**

As per relevant B.I.S. method

D **Manner of labelling**

Method of labelling as per requirements of Insecticides Rule, 1971 along with direction of use.

E **Labels and leaflets**

7 copies of labels and leaflets as per requirements of Insecticides Rule, 1971 for

1 Small size packing up to 250ml capacity and below

2 Large size packing up to 500ml capacity and above.

Note: The above guidelines are for registration of those insecticidal formulations whose technical grade material is registered u/s 9(3) of the Insecticides Act, 1968.

Subcategory 5.4. Guidelines for the registration of Insecticides for the control of Ectoparasites (Mites, Bedbugs Ticks etc.)

(As approved by the registration committee in its 301st RC meeting held on 05.06.2009)

S. No.	Parameter	Technical		Formulation	
		9(3B)	9(3)	9(3B)	9(3)
A.	**CHEMISTRY**				
1	Source of Supply	R	R	R	R
2	Chemical Composition	R	R	R	R
3	Chemical Identity	R	R	R	R
4	Physico - Chemical Properties	R	R	R	R
5	Technical Bulletin	R	R		
6	Specification	R	R	R	R
7	Method of Analysis	R	R	R	R
8	Analytical Test Report	R	R	R	R
9	Identification& Quantification of identifiable Impurities	R	R		
10a.	Shelf-life claim	R	R	R	R
10b.	Shelf-life Data	NR	R	R	R
11a.	Process of Manufacture	R	R	R	R
11b.	Information about Raw Materials Used	R	R	R	R
11c.	Their Source of Supply.	R	R	R	R
11d.	Step-wise Manufacturing Process.	R	R	R	R
11e.	Chemical Equation	R	R	NR	NR
11f.	Formula	R	R	R	R
11g.	Flow sheet diagram of process of manufacture	R	R	R	R
11h.	Effluent Treatment method	R	R	R	R
12	Documents such as registration certificate / manufacturing licence or any other approval under any Govt. regulation will be acceptable to support that manufacturer is actual producer (source in case of import)	R	R	NR	NR

13	Certificate from manufacturer that the dealer/ trader is an authorized dealer/ trader of the manufacturer (Authorization of supplier by the source).	R	R	R	R
14	A test report about the quality of the product from a laboratory as per GLP scheme or from a company of ISO-9000. This requirement will be provided along with first consignment. Thereafter, each consignment should have proper analytical test report of the manufacturer.	R	R	R	R
15	The applicant should provide sample along- with standards, technical sample from the principals/ authorized dealers for verification of the claim and the method of analysis for quality monitoring. In case of technical grade pesticides, samples of std. impurities are also to be provided for chemical verification. In process sample to be provided with undertaking	R	R	R	R
16	Methodology for residue estimation as per BIS format.	R	R	R	R
B.	**BIOEFFICACY & RESIDUES**				
17	Bioeffectiveness – data generated in National/ ICAR/SAU Laboratories [three for 9(3) and 2 for 9(3B)] against the claimed pest based on repeated trials.	NR	NR	R	R
18	Effect on layers (duration 3 months, study to commence preferably at the age of 40 weeks) - data generated in National/ ICAR/SAU Laboratories [three for 9(3) and 2 for 9(3B)] on	NR	NR	R	R
	(i) change in body weight,				
	(ii) feed in take, water intake, Feed Conversion Ratio (FCR)				
	(iii) mortality and morbidity pattern,				
	(iv) * Clinical symptoms/morphological changes in organs,				

	(v) * Blood profile and Enzymology, (vi) egg production records for 15 days				
19	Metabolism in soil	R	R	NR	NR
20	Metabolism in water	R	R	NR	NR
21	Persistence in soil	R	R	NR	NR
22	Persistence in water	R	R	NR	NR
23	Persistence and residue on treated surface	NR	NR	R	R
24	Compatibility with other chemicals, if claimed	NR	NR	R	R
25(a).	Residues in various organs of birds and edible products	NR	NR	R	R
	Residue in birds excreta	NR	NR	R	R
25(b).	Residue tolerance limits fixed by foreign countries	R	R	R	R
.	Registration status in foreign countries	R	R	R	R
C.	**TOXICITY**	R	R	R	R
26	Acute oral in rat	R	R	R	R
27	Acute dermal	R	R	R	R
28	Acute inhalation	R	R	R	R
29	Primary skin irritation	R	R	R	R
30	Irritation to mucous membrane	R	R	R	R
31	Skin sensitization	R	R	R	R
32	Sub-acute oral rat	R	R	NR/R	NR/R
33	Sub-acute oral dog	R	R	NR/R	NR/R
34	Sub-acute dermal	R	R	NR/R	NR/R
35	Sub-acute inhalation	R	R	NR/R	NR/R
36	Neuro-toxicity	NR	R	NR	NR
37	Synergism & potentiation	NR	R	NR	NR/R
38	Teratogenicity	NR	R	NR	NR
39	Effect on reproduction	NR	R	NR	NR
40	Carcinogenicity	NR	R	NR	NR
41	Metabolism	NR	R	NR	NR
42	Mutagenicity	NR	R	NR	NR
43	Toxicity to birds (two species other than poultry)	R	R	R	R

44	Toxicity to fish	R	R	R	R
45	Toxicity to earthworm	R	R	R	R
46	Toxicity to live stock	R	R	NR	NR
47	Medical data	R	R	R	R
48	Human toxicity information from foreign countries	R	R	NR/R	NR/R
49	Observation in man (Health records of spray operators)	NR	NR	NR	R
50	Health records of Industrial workers.	NR	R	NR	R
51	Toxicity to live stock (Field trial & observation)	NR	R	NR	NR
52.	International report on carcinogenicity & genotoxicity status	NR/R	R	NR/R	NR/R
D.	**PACKAGING AND LABELLING**				
53	Labels and leaflets as per IR-1971 existing norms (i) for size 250 ml & below (ii) for 500 & above.	R	R	R	R
54	**Labels to contents**				
a.	Detailed Chemical composition	R	R	R	R
b.	Purpose for import / manufacture.	R	R	R	R
c.	Antidote	R	R	R	R
d.	Toxicity triangle	R	R	R	R
e.	Cautionary statement	R	R	R	R
f.	Brief direction concerning usages	R	R	R	R
g.	Restriction if any	R	R	R	R
55	**Leaflets to contain**				
a.	Detailed Chemical composition on leaflets accompanying small labels (upto 250 ml size container)	NR	NR	R	R
b.	Introductory para about the pesticide	R	R	R	R
c.	Detailed directions concerning usages	R	R	R	R
d.	Time of application	NR	NR	R	R
e.	Application equipment	NR	NR	R	R
f.	Waiting Period	NR	NR	R	R
g.	Symptoms of poisoning	R	R	R	R
h.	First aid measures	R	R	R	R
i.	Antidote & treatment	R	R	R	R

j.	Restriction, if any	R	R	R	R
k.	Instruction for storage	R	R	R	R
l.	Information regarding disposal of used packages.	R	R	R	R
56	Type of packaging (pkg material + compatibility with content)	R	R	R	R
57	Manner of packaging	R	R	R	R
58	Specification for primary package	R	R	R	R
59	Specification for secondary packaging.	R	R	R	R
60	Specification for transport packaging, including UN Number	R	R	R	R
61	Manner of labelling	R	R	R	R
62	Performance of container during storage stability test	NR	R	NR	R

* To be scrutinized by the Toxicity expert; R : Required; NR-Not required

Note: In case of Import of formulation without registering technical, whole set of data on technical shall be submitted along with the application

Sub category 5.5: Guidelines for the registration of Biocide and Biocide Products (Manufacturing Use Product)

(Approved by the registration Committee in its 389th RC meeting held on 13.06.2018 and 15.06.2018)

Biocidal products are necessary for the control of organisms that are harmful to human or animal health and are required for the control of organisms that cause damage to natural or manufactured materials. These substances usually contains pesticides and can pose risks to humans, animals and the environment due to their intrinsic properties and associated use patterns and hence are required to be regulated under the Insecticide Act, 1968 and the Insecticide Rule, 1971, in order to ensure safety to human, animals and environment.

The purpose of these guidelines is to ensure safety of workers and the general public at large while handling and use of Biocides and products containing biocides and to provide guidance on its regulatory requirements under the Insecticides Act, 1968.

When the sole purpose of incorporation of biocide in the product (paint) is to protect the product (paint) itself, and no label claims are made for inhibiting the growth of micro organisms which may cause odours or to inhibit the growth of mould and mildews or repelling or killing insects, then such products (paints) do not require registration under the Insecticides Act, 1968.

Guidelines for use of biocide in paint by the paint industry as preservatives to enhance the shelf life of the paints

The biocide (whether technical or formulated biocide product) used in paint by paint industry as preservatives to enhance the shelf-life of the paints and also as dry film preservative are required to be registered under the Insecticides Act, 1968. Data required for registration of technical and formulated biocide product are annexed (Annexure-I).

1. The TGAIs used in biocide products must be registered under the Insecticides Act.

2. In case of registration of biocide formulated product without registering technical grade pesticide, every biocide product manufacturer / importer is required to declare the source(s) of Biocide-TGAI/s in such products. Complete data of TGAI w.r.t. chemistry, bio efficacy and toxicity must be submitted to CIBRC in case of import/ or indigenous manufacture of biocide product without registering its TGAI.
3. Only CIBRC registered Biocides-TGAIs should be used for manufacture of biocide products. In case the candidate TGAIs also qualify for direct/ indirect usages in agriculture, bio efficacy studies are mandatorily required to be conducted as defined for agricultural pesticides and data submitted for registration to CIBRC. Otherwise bio-efficacy study on TGAI Biocide/s may be done from the view point of relevant use / application needs of expected antimicrobial-performance in Industrial end-products. As a general principle, tests shall be conducted according to the methods described / recommended by international bodies / CIB&RC. In the event of, lack of an appropriate protocol, appropriately justified methods endorsed/ approved by CIBRC may be used. All toxicological tests should be GLP compliant. Tests on physico-chemical properties and

safety-relevant data should be performed according to international standards. Effect of biocides, on environment and surface under protection, bio-efficacy, etc., must be carried out as per the internationally accepted protocols (like ASTM 5589, 5590) / protocols approved by CIB&RC.

4. Paint Industry (as also other industries) must use the CIB&RC registered biocide products at the recommended dosages as per the defined maximum limit of usage of TGAI in the end product/s (especially Diuron and Carbendazim). At present, the set-limit on Diuron is 0.5% w/w of final composition of Paint, and for carbendazim it is 0.1% -0.5% (w/w). Products containing more than 1000 ppm (0.1%) carbendazim will attract "skull and crossbones" and "dead fish plus dead tree" labels together with the phrases "Toxic and Dangerous for the Environment", "Mutagen Category 2" and "Reprotox Category 2". Paint products with over 0.1% ppm of carbendazim will be restricted for sale to professionals only. However, for other TGAIs if any, concentrations will be specified as per defined maximum limits of usage (documented in the literature/ or as determined based on data generation and approved by CIBRC).

5. Where in no dosage limit has been defined the applicant must provide valid data to evaluate possible effects at species, population or community and ecosystem level. Studies must be carried out in systems representative to habitats to which the product is applied. Important aspects to consider are the use of reference areas, history of the (treated and non-treated) areas, climatic conditions, timing, duration of exposure, frequency, dosage and concentration, distribution in time and location, etc.

6. Biocide Product Manufacturers would be required to provide

a. Bio-efficacy data of the antimicrobial products on the performance of a paint film to inhibit the microbial growth by generating the data on the representative sample of commercial paint at the dosage of the Biocide Product at its minimum required level, w.r.t. TGAI contents in the dosed Biocide Product.

b. Data to be provided on the actual dosage being recommended as safe dosage to ensure the inhibition of microbial growth on the film and from that what is retained in the paint film w.r.t. TGAI contents after leaching of such paint film by flow of water. Different leaching rates may be required, for example in relation to leaching during the washing of freshly preserved film, leaching from treated film outdoors with a risk of wetting, leaching from the treated film when washed indoors or otherwise in contact with water during its service life, and volatilization from the treated film in contact with indoor or outdoor air.

7. No change will be permitted thereafter in the composition of biocide products so- registered (unless under exceptional conditions in accordance with Government policy).

8. Data related to Product-stability and performance under actual usage- have to be generated under Indian Climatic conditions.

9. Biocide (Technical) / Biocide formulated products should follow the labelling requirements as provided under the Insecticides Act, 1968.

10. Labelling on the containers with respect to its composition and specific end use is required to be incorporated on Biocide products.

11. Labelling is mandatorily required on the container when the Registered Biocide Products are present in dosage in excess to the prescribed dosage (Diuron - 0.5% w/w of final composition of Paint, Carbendazim - 0.1% to 0.5% w/w and 0.1 (%)??? for others until the limit is amended based on the generated data/ scientifically documented literature) in exterior and interior decorative Paints/similar products and meant as formulations of Insecticides.

12. No labelling is required if these biocide products are being added in the product as antimicrobials to ensure sufficient surface protection from the likely microbial attack and as per the dosage not exceeding the maximum permissible content of Active ingredients coming from the said biocide products.

13. As the Biocide Industry falls under the realms of the Insecticide Act, 1968, records on the import/ manufacture /consumption of TGAIs in biocide products and formulated must be submitted to CIB&RC monthly and annual basis.

II. Guidelines for registration of biocides OR pesticides to be used in paints for killing or repelling insects with label claim

Following are required for Paint Industry who proposes to manufacture Insect Repellent /Killing Paints (pesticide formulation):

a. Insect Repellent /Killing Paints (pesticide formulation) are required to be registered under the Insecticides Act as Pesticide Formulation **(Annexure II)**.

Annexure–I

S. No.	Parameter	TI	TIM	FI	FIM	TI (New Source)	Comments
A.	**Chemistry**						
1	Source of Supply of Technical	R	NR	R	R	R	
2	Chemical Composition	R	R	R	R	R	
3	Chemical Identity of technical	R	R	R	R	R	
4	Physico – Chemical Properties of adjuvants	R	R	R	R	R	
5	Technical Bulletin	R	R	R	NR	R	
6	Specification	R	R	R	R	R	
7	Method of Analysis	R	R	R	R	R	
8	Analytical Test Report	R	R	R	R	R	
9	Identification & Quantification of identifiable Impurities	R	R	NR	NR	R	
10a.	Shelf-life claim	R	R	R	R	R	
10b.	Shelf-life Data	R	R	R	R	R	
11a.	Process of Manufacture	R	R	R	R	R	
11b.	Information about Raw Materials Used	R	R	R	R	R	
11c.	Their Source of Supply.	R	R	R	R	R	
11d.	Chemical Equation	R	R	R	R	R	
11e.	Formula	R	R	R	R	R	
11f.	Flow sheet diagram of process of manufacture	R	R	R	R	R	
11g.	Effluent Treatment method	NR	R	NR	R	NR	
12	Documents such as registration certificate / Certificate of DNA/manufacturing license or any other approval under any Govt. regulation will be acceptable to support that manufacturer is actual producer	R	NR	R	NR	R	

13	Certificate from manufacturer that the dealer/ trader are an authorized dealer/trader of the manufacturer.	R	NR	R	NR	R	
14	A test report about the quality of the product from a laboratory as per GLP scheme or from a company of ISO-9000. This requirement will be provided along with first consignment. Thereafter, each consignment should have proper analytical test report of the manufacturer.	R	NR	R	NR	R	
15	The applicant should provide sample along with standards technical sample from the principals/ authorized dealers for chemical verification. In case of technical grade pesticides u/s 9(3), samples of std. impurities are also to be provided for chemical verification. In process sample to be provided in case of indigenous manufacture of technical u/s 9(3) TIM & 9(4) TIM with undertaking	R	R	R	R	R	
B.	**BIOEFFICACY-**						
16a.	Bio-effectiveness	R /NR	R /NR	R	R	R/NR	Required if TGAI goes directly into end use product
		R /NR	R /NR	R	R	R	Effectiveness against target organism using a representative end use product
17	Metabolism in soil	R/ NR	R/ NR	R/ NR	R/ NR	R/ NR	
18	Metabolism in water	NR/ R	NR/R	NR/ R	NR/ R	NR/ R	
19	Persistence in soil	NR	NR	NR	NR	NR	
20	Persistence in water	NR	NR	NR	NR	NR	
21	Residues in soil	NR	NR	NR	NR	NR	
22	Cost benefit ratio	NR	NR	NR	NR	NR	

23	Registration status in foreign countries	R	R	R	R	R	
C.	**TOXICITY**						
24	Acute oral in rat	R	R	R	R	R	
25	Acute dermal	R	R	R	R	R	
26	Acute inhalation	R	R	R	R	R	
27	Primary skin irritation	R	R	R	R	R	
28	Irritation to mucous membrane	R	R	R	R	R	
29	Allergy/Sensitization	R	R	R	R	R	
30	Sub-acute oral rat	R	R	R/ NR	R/ NR	R	
31	Sub-acute oral dog*	R/ NR	R/NR	R/ NR	R/ NR	R/ NR	*In case no data is available, information maybe submitted
32	Sub-acute dermal	R	R	R/ NR	R/ NR	R	
33	Sub-acute inhalation	R	R	R/ NR	R/ NR	R	*if inhalation exposure is major route and pesticides does not have use in agriculture
34	Neuro-toxicity	R	R	NR	NR	NR	
35	Teratogenicity (rat & rabbit)	R	R	NR	NR	NR	
36	Effect on reproduction	R	R	NR	NR	NR	
37	Carcinogenicity (rat & mice)	R	R	NR	NR	NR	
38	Metabolism (rat)	R	R	NR	NR	NR	
39	Mutagenicity	R	R	NR	NR	R	
40	Toxicity to birds (two)	R	R	R	R	R	
41	Toxicity to fish (freshwater) a) Fresh water fish b) Marine water fish (When the claim is for marine use)	R R/ NR	R R/ NR	R R/ NR	R R/ NR	R R/ NR	

42	Toxicity to live stock	NR/ R*	NR/ R*	NR	NR	NR/R*	*based on the end use of products made out from the technical
43	Medical data	R	R	R	R	R	
44	Human toxicity information from foreign countries	R	NR/ R	NR	NR	R	
45	Health records of Industrial workers.	R/ NR	R/ NR	R/ NR	R/ NR	R/ NR	
46	International report on carcinogenicity & genotoxicity status	R	R/NR	R	NR/ R	R	
D.	**PACKAGING**						
47	Labels and leaflets as per IR-1971 existing norms (i) for size 250 ml & below (ii) for 500 & above.	R	R	R	R	R	
48	Labels to contents	R	R	R	R	R	
a.	Detailed Chemical composition	R	R	R	R	R	
b.	Purpose for import / manufacture.	R	R	R	R	R	
c.	Antidote	R	R	R	R	R	
d.	Toxicity triangle	R	R	R	R	R	
e.	Cautionary statement	R	R	R	R	R	
f.	Brief direction concerning usages	R	R	R	R	R	
g.	Restriction if any	R	R	R	R	R	
49	Leaflets to contain						
a.	Detailed Chemical composition on leaflets accompanying small labels (upto 250 ml size container)	R	R	R	R	R	
b.	Introductory para about the pesticide	R	R	R	R	R	
c.	Detailed directions concerning usages	R	R	R	R	R	
g.	Symptoms of poisoning	R	R	R	R	R	
h.	First Rd measures	R	R	R	R	R	
i.	Antidote & treatment	R	R	R	R	R	
j.	Restriction, if any	R	R	R	R	R	
k.	Instruction for storage	R	R	R	R	R	
l.	Information regarding disposal of used packages.	R	R	R	R	R	
50	Type of packaging (pkg material + compatibility with content)	R	R	R	R	R	
51	Manner of packaging	R	R	R	R	R	

51.1	Specification for primary package	R	R	R	R	R	
51.2	Specification for secondary packaging.	R	R	R	R	R	
51.3	Specification for transport packaging.	R	R	R	R	R	
52	Manner of labelling	R	R	R	R	R	
53	Performance of container during storage stability test	R	R	R	R	R	
54	Transport worthiness test	R	R	R	R	R	

TI: Technical Import

TIM: Technical Indigenous manufacture

FI: Biocide Formulated product for Import

FIM: Biocide Formulated product for Indigenous manufacture

Annexure–II

Guidelines for registration of biocides or pesticide to be used in paint for killing or repelling insects with label claim.

1. Manufactured Paints with claims to repel or kill insects, fungus, weeds or any other pesticide claims are considered as pesticide formulations of paint. The pesticides used in such pesticide formulations of paint must be obtained from the source registered under the Insecticides Act, 1968.
2. In case the paint industry wants to import / manufacture indigenously the pesticide for use in paint which are not registered for use in India then they have to take registration for import / indigenous manufacture of technical by submitting data as per guidelines for Technical for import (TI) / TIM u/s 9(3), as the case may be.
3. The data requirement remains the same as that of TI/ TIM u/s 9(3)and under FI/ FIM for registration for use in the country with following special conditions:

S.No	Parameter	Regular Registration U/s 9 (3)
A.	**Chemistry**	
1	Source of Supply of technical	R
2	Chemical Composition	R
3	Chemical Identity	R
4	Physico - Chemical Properties	R
5	Specification	R
6	Method of Analysis	R
7	Analytical test report about the quality of the product from a laboratory having NABL/ GLP accredited laboratory	R
8	Shelf-life claim	R
9	Shelf-life Data	R
10	Process of Manufacture	R
11	Information about raw materials used	R
12	Source of Supply of the raw materials	R
13	Stepwise Manufacturing Process	R
14	Flow sheet diagram of process of manufacture	R
15	Effluent Treatment	NR
16	Documents such as registration certificate /manufacturing license or any other approval under any Government regulation will be acceptable to support that manufacturer is actual producer of the product of the claimed chemical composition	R

(B) Bio-efficacy:-

1. Data on evaluation of efficacy against designated pest.
2. Persistence on treated painted surface.

(C) Packaging

S.No.	Parameter	Regular Registration U/s 9 (3)
1	Labels and leaflets as per IR-1971 existing norms (i) for size 250 ml &below (ii) for 500 & above.	R
I.	Labels to contents	R
II.	Detailed Chemical composition	R
III.	Purpose for import / manufacture.	R
IV.	Antidote	R
V.	Toxicity triangle	R
VI.	Cautionary statement	R
VII.	Brief direction concerning usages	R
VIII.	Restriction if any	R
IX.	Leaflets to contents	R
2	Detailed Chemical composition on leaflets accompanying small labels (up to 250 ml size container)	R
I.	Introductory para about the pesticide	R
II.	Detailed directions concerning usages	R
III.	Applications	R
IV.	Application equipment	R
V.	Persistence after application	R
VI.	Symptoms of poisoning	R
VII.	First Aid measures	R
VIII.	Antidote & treatment	R
IX.	Restriction, if any	R
X.	Instruction for storage	R
XI.	Information regarding disposal of used packages.	R
XII.	Type of packaging (packing material + compatibility with content)	R
XIII.	Manner of packaging	R
XIV.	Specification for primary package	R
XV.	Specification for secondary packaging.	R
XVI.	Specification for transport packaging.	R
VII.	Manner of labeling	R
VIII.	Performance of container during storage stability test	R
XIX.	Transport worthiness test	R

D. Toxicology

1. The insecticides registered with label claim in public health will not be allowed for use in paint formulations. Insecticides will be allowed on case to case basis in consultation with NVBDCP. Therefore, studies are to be initiated with prior permission of Registration Committee.

2. The technical source of the insecticides should be registered with CIBRC. Otherwise, data must be provided on the technical counterpart of the active ingredient intended for use in the formulations.

3. All acute mammalian studies as prescribed in "Guidance Document of Toxicity for the Registration of Pesticides in India" i.e. acute oral – rat, acute dermal, acute inhalation, Primary skin irritation, Mucous Membrane irritation, skin sensitization.

4. Eco-toxicity studies depending on use pattern.

5. The chemical composition of the formulation (Paint) to be disclosed under the affidavit and Material Safety Data Sheet (MSDS)/Safety Data Sheet (SDS) must be provided for each component along with prevailing literature.

6. The applicant must undertake?? a statement?? or certify that no component present in formulation (Paint) are known to cause Carcinogenicity, Teratogenicity (Developmental toxicity) and effect on reproduction and other chronic disorders to human beings at the dosages/ concentration in use.

7. The persistence of the pesticides in the paint, rate of degradation during use along with residue data in paint scrap, for the shelf life period

8. Use pattern of the paint with intended insecticide across the globe and its registration status with label claims.

9. The air concentration of insecticide in the air near the painted surface at different time interval.

10. Human and environmental toxicity information from foreign countries with use of insecticidal paint.

11. Specify the use pattern of these paints in the country and mention how publicity will be done if, allowed to use.

12. LabelandleafletsasperInsecticidesRules1971supportedbySafetyrelated issues to be provided.

ADDITIONAL INFORMATION

Requirements for the submission of application in five folders for registration of pesticides u/s 9(3)/ 9(3b)
(Approved by the registration committee in its 266th RC meeting held on 20th July, 2006)

Name of the applicant: M/s……

Name of the product: …….

Type of the pesticide & its use: …….

Folder A. Administration / Legal:

1) Index
2) Page Numbering
3) Authentication of pages by authorized signatory
4) Form – I duly signed
5) Demand draft as registration Fee
6) Requisite number of stamped envelops
7) BOD Resolution/Affidavit
8) Copy of 9(3B) Registration certificate, if relevant

Complete / Deficient

Folder B. Chemistry:

1) Copy of Form-I
2) Chemistry data
3) Labels/leaflets
4) Copy of RTT permit, if relevant
5) MRL proforma
6) Copy of 9(3B) Registration certificate, if relevant

Complete / Deficient

Folder C. Bioefficacy:

1) Copy of Form-I
2) Bioefficacy data
3) Chemical composition
4) MRL proforma
5) Labels/leaflets
6) Copy of RTT permit, if relevant
7) Copy of 9(3B) Registration certificate, if relevant
8) Summary of bioefficacy and residue data (field data) in duplicate

(Added as per decision of 279th RC held on 23-08- 2007)

Complete / Deficient

Folder D. Toxicity:

1) Copy of Form-I
2) Toxicity data
3) Chemical Composition
4) Copy of specification
5) Details of adjuvant/solvents, as submitted in Chemistry Folder
6) MRL proforma
7) Labels/leaflets
8) Copy of RTT permit, if relevant
9) Copy of 9(3B) Registration certificate, if relevant

Complete / Deficient

Folder E. Packaging and labeling:

1) Copy of Form-I
2) Packaging & Labelling data
3) Copy of 9(3b) Registration certificate, if relevant

Complete / Deficient

Requirements for submission of application in five folders for registration of biopesticides u/s 9(3)/ 9(3b)
(Approved by the Registration Committee in its 323rd RC held on 29th Sept, 2011)

Name of the applicant: M/s……

Name of the product: ……..

Type of the biopesticide & its use: ……..

General Requirements for all folders

1) Index
2) Page Numbering
3) Authentication of pages by authorized signatory

Complete / Deficient

Folder A. Administration /Legal:

1) Form – I duly signed
2) Demand draft as registration Fee
3) Requisite number of stamped envelops
4) Notarized copy of BOD Resolution/ Partnership deed / Affidavit
5) Affidavit on bio-pesticide composition on NJSP duly notarized (Original)
6) Notarized copy of depositing microbial bio-pesticides strain sample in National Repository with reference code number
7) Undertaking on NJSP duly notarized that product do not contain any genetically modified organism in the prescribed format (in original).
8) Undertaking on NJSP duly notarized that product is free from chemical / botanical pesticides / other agro-chemicals (in Original).
9) Copy of 9(3B) Registration certificate, if relevant

Complete / Deficient

Folder B. Chemistry:

1) Copy of Form-I
2) Data on biological characteristics & chemistry as per the guidelines of RC
3) Labels/leaflets
4) Copy of Affidavit on bio-pesticide composition on NJSP (Same as in legal folder)
5) Copy of depositing microbial bio-pesticides strain sample in National Repository with reference code number (same as in legal folder)
6) Copy of Undertaking on NJSP that product do not contain any genetically modified organism in the prescribed format (same as in legal folder)
7) Copy of Undertaking on NJSP that product is free from chemical / botanical pesticides / other agro-chemicals (same as in legal folder)

8) Copy of RTT permit, if relevant
9) Copy of 9(3B) Registration certificate, if relevant

Complete / Deficient

Folder C. Bioefficacy:

1) Copy of Form-I
2) Bioefficacy data as per the guidelines of RC
3) Copy of biopesticide composition (Same as in legal folder)
4) Labels/leaflets
5) Copy of depositing microbial bio-pesticides strain sample in National Repository with reference code number (same as in legal folder)
6) Copy of RTT permit, if relevant
7) Copy of 9(3B) Registration certificate, if relevant
8) Summary of bioefficacy data in duplicate

Complete / Deficient

Folder D. Toxicity:

1) Copy of Form-I
2) Toxicity data as per the guidelines of RC
3) Copy of biopesticide composition (Same as in legal folder)
4) Copy of specification
5) Details of adjuvant/solvents, as submitted in Chemistry Folder
6) Labels/leaflets
7) Copy of RTT permit, if relevant
8) Copy of 9(3B) Registration certificate, if relevant

Complete / Deficient

Folder E. Packaging and labeling:

1) Copy of Form-I
2) Packaging & Labelling data as per the guidelines of RC
3) Copy of 9(3B) Registration certificate, if relevant

Complete / Deficient

In view of above, the application is

1) Accepted for further scrutiny
2) Not accepted

Requirements for Extension of Provisional Registration Under Section 9(3b)
(For Chemical pesticides & Biopesticides both)
(Approved by the Registration Committee in its 263rd RC held on 16th May, 2006)

Name of the applicant: M/s……

Name of the product: ……..

A. Administration/legal:

i) Index
ii) Page numbering
iii) Authentication of pages by authorized signatory
iv) Authorization for the signatory (Notarized copy of BOD resolution/affidavit/partnership deed)
v) Form-I duly signed
vi) Demand draft as registration fee
vii) Requisite number of stamped envelops
viii) Justification for extension
ix) Copy of all earlier CRs for the product
x) Details of data generated to fulfill the conditions of provisional registration.
   a. Chemistry
   b. Bioefficacy
   c. Toxicity
   d. Packaging

Requirements for the registration of pesticide formulation "FIM vs FIT u/s 9(3)" for indigenous manufacture having the identical chemical composition that of formulation already registered for import u/s 9(3) for the same label claims

(Approved by the Registration Committee in its 317th RC held on 31st March, 2011)

Name of the applicant: M/s……

Name of the product: ……..

1) Index
2) Page Numbering
3) Authentication of pages by authorized signatory
4) Form – I duly signed
5) Demand draft as registration Fee
6) Requisite number of stamped envelops
7) Copy of BOD Resolution/Affidavit/Partnership deed (**Notarized**)
8) Affidavit for Chemical composition on NJSP (**Notarized**)
9) Certificate as per category of Industry/ Manufacturing license (**Notarized**)
10) PAN No. (**Notarized**)
11) Incorporation Certificate (**Notarized**)
12) Labels/leaflets
13) Proof of Source of Technical to be used in formulation is duly registered. (In case of indigenous manufacture / import - Reference of RC meeting in which it was approved). {Only deemed registration status without issuance of Certificate of registration shall not be considered}.
14) Name of registrant of Formulation Import (Reference of Registration Committee Meeting in which formulation import was approved).
15) ATR from independent laboratory
16) A Sample for Analysis by CIL

In view of above, the application is

i) Accepted for further scrutiny
ii) Not accepted

Requirement for 9(4) Tim Cases Under Insecticides Act, 1968

(Approved by the Registration Committee in its 256th RC held on 12th September, 2005)

Name of the applicant: M/s……

Name of the product: ……..

A. Administration/Legal:

1) Index
2) Page numbering
3) Authentication of pages by authorized signatory.
4) Demand draft.
5) Requisite number of stamped envelopes
6) Form-I duly signed
7) Authorization for the signatory (Notarized copy of BOD resolution/affidavit/partnership deed)

B. Chemistry:

1) Chemical Composition
2) Chemical Identity/Structural formula
3) Physico-Chemical properties
4) Specification
5) Method of analysis
6) Analytical Test Report from an independent laboratory
7) Identification and quantification of associated standard identifiable
8) Shelf- life claim and Products Quality undertakings
9) Establishment of chemical equivalence
10) Process of manufacture
11) Information about raw material
12) Source of supply
13) Step-wise manufacturing process
14) Step-wise chemical equations
15) Flow –sheet diagram of manufacturing process
16) Effluent treatment method/flow – sheet diagram, and
17) Drawl of in-process sample and its Analysis Test Report (ATR) from CIL

C. Toxicology:

1) Acute oral in rat & mice
2) Acute dermal
3) Primary skin irritation

4) Irritation to mucous membrane

D. Packaging:

1) Seven copies Label/leaflets as per IR 1971 existing norms
2) Manner of packaging
3) Manner of Labeling

Requirement Under Section 9(4) Fim Category Of The Insecticides Act, 1968

(Approved by the Registration Committee in its 256th RC held on 12th September, 2005)

Name of the applicant: M/s……

Name of the product: ……..

1) Index
2) Page numbering
3) Authentication of pages by authorized signatory.
4) Demand draft.
5) Requisite number of stamped envelopes
6) Form-I duly signed

Notarized copies of the following:

1) BOD Resolution/Partnership Deed/Affidavit
2) Certificate as per category of Industry / manufacturing license
   (Modified as per the decision of 276th meeting of RC held on 31st May, 2007)
3) PAN NO.
4) Incorporation Certificate
5) List of products for which the registration has been given to the firm and Manufacturing License obtained; and products actually manufactured during the previous 2 (two) years.
   (Added as per decision of 315th meeting of RC held on 22-02-2011)

Requirement under Section 9(4) Ti/Fi Category Of The Insecticides Act, 1968
(Approved by the Registration Committee in its 256th RC held on 12th September, 2005)

Name of the applicant: M/s……

Name of the product: ……..

1) Index
2) Page numbering
3) Authentication of pages by authorized signatory.
4) Demand draft.
5) Requisite number of stamped envelopes
6) Form-I duly signed

Notarized copies of the following:

1) BOD Resolution/Partnership Deed/Affidavit
2) Certificate as per category of Industry / manufacturing license
   (Modified as per the decision of 276th meeting of RC held on 31st May, 2007)
3) PAN NO.
4) Incorporation Certificate
5) Letter of consent, duly legalized from Indian Embassy/High Commission/ Consulate in the Country of origin.
6) List of products for which the registration has been given to the firm and Manufacturing License obtained; and products actually manufactured during the previous 2 (two) years. (Added as per decision of 315th meeting of RC held on 22-02-2011)

Requirement for the registration of pesticides for export under section 9(3)
(Approved by the Registration Committee in its 256th RC held on 12th September, 2005)

Name of the applicant: M/s……

Name of the product & category: ……..

A. Administration / Legal:

1) Index
2) Page Numbering
3) Authentication of pages by authorized signatory
4) Demand draft as registration Fee
5) Requisite number of stamped envelops
6) Authorization for the signatory (Notarized copy of BOD Resolution/Partnership Deed/Affidavit)

B. Chemistry:

Already registered molecule

1) Chemexcil certificate
2) Form – I duly signed
   6(i)-Chemical composition
   10(i)- BIS no/specification & method of analysis.
3) 7 copies of Labels/leaflets
4) For category I and VI- "Source of Import."
5) "Process of manufacture" (Indigenous manufacture) for category I and II

New molecule, not registered for use in the country

a) Chemistry-information as in (B) above.
b) Toxicity information
   i) Acute oral in rat
   ii) Acute dermal
   iii) Acute inhalation
   iv) Primary skin irritation
   v) Irritation to mucous membrane

<u>For Technical Additional Requirement</u>

vi) Effect on reproduction
vii) Carcinogenicity
viii) Mutagenicity

Bt. & Neem: As per guidelines for export- Chemistry + Toxicity information

Requirement for the label expansion U/S 9(3)

(Approved by the Registration Committee in its 263rd RC held on 16th May, 2007)

Name of the applicant: M/s……

Name of the product: ……..

1) Index
2) Page Numbering
3) Authentication of pages by authorized signatory
4) Requisite number of envelop
5) Copy of Registration Certificate
6) Authorization for the signatory
7) Affidavit duly notarized regarding Label expansion made earlier in respect of Certificate for which the present request is being made

Bioefficacy:

1) Bioeffectiveness
2) Phytotoxicity
3) Residue in plant
4) Persistence in plant
5) Residue tolerance limits fixed by foreign countries
6) MRL proforma in duplicate
7) Registration status in foreign countries
8) Effect on parasites and predators (applicable in case of insecticides only)
9) Seven copies of revised labels/leaflets in English & Hindi
10) Summary of bioefficacy and residue data (field data) in duplicate
(Added as per decision of 279th RC held on 23.08.2007)

Draft/Specimen

A F F I D A V I T

(As approved by RC in its 314th meeting held on 24th & 27th Jan, 2011)

(To be furnished by the Applicant or Authorised person, duly Notarized)

I.....s/o ..... aged about ..... years, working as..... in ....., having registered office at .....do hereby confirm, declare and solemnly affirm as under:

1. That following Endorsement regarding change in Company name and / or transfer of Certificate of Registration were applied / made previously in respect of Certificate(s) of Registration for the product (Name of the product with Registration No.) ..... from …… to …….. (In case more than one Certificate please enclose a list of Certificates (**Product wise**) along with their specific **Registration Numbers** allotted by the CIB&RC.

Note: *(In case no Endorsement regarding change in company name / transfer of certificate was applied / made previously, then clearly state that no Endorsement of the said category was made earlier).*

2. That following Endorsement regarding factory/office address, duplicate certificate, packaging, shelf life, source of Import, or any other category etc. were applied / made previously in respect of Certificate(s) of Registration for the product (Name of the product with Registration No.) …… (In case more than one Certificate please enclose a list of Certificates (**Product wise**) along with their specific **Registration Numbers** allotted by the CIB&RC.

Note: *(In case no Endorsement regarding above categories were applied / made previously, then clearly state that no Endorsement of the said category was made earlier).*

Deponent
(Name & Designation)

V E R I F I C A T I O N

I ....., the above named deponent do solemnly verify that the information given above is true to the best of my knowledge & belief and nothing has been concealed there from.
I, further declare that I am making this declaration in my capacity as ..... (Designation) and that I am competent to make this declaration.

Deponent

Requirement for the label expansion of insecticide for household use u/s 9(3)
(Approved by the Registration Committee in its 263rd RC held on 16th May, 2006)

Name of the applicant: M/s……

Name of the product: ……..

1) Index
2) Page Numbering.
3) Authentication of pages by authorized signatory
4) Requisite no. of stamped envelopes
5) Authorization for the signatory (Notarized copy of BOD resolution/ affidavit/ partnership deed)
6) Bioeffectiveness data of minimum two repeated trials from three National Laboratories
7) Information on secondary pest outbreaks particularly of ticks & mites where residual pyrethroids are being used
8) Data on persistence on different type of surfaces generated under Indian conditions from three laboratories (except coil/mats/liquid vaporizer)
9) Data on concentration in air in case of coil/mats/liquid vaporizer
10) Seven copies of Label/leaflets in Hindi and English
11) Affidavit duly notarized regarding label expansion made earlier in respect of certificate for which the present request is being made

Requirement for the endorsement of bio-efficacy claim on label/leaflet of registration certificate u/s 9(4) of the insecticides act, 1968

(Approved by the Registration Committee in its 263rd RC held on 16th May, 2006)

Name of the applicant: M/s……

Name of the product: ……..

1) Index
2) Page numbering
3) Authentication of pages by authorized signatory
4) Notarized copy of BOD Resolution/Affidavit/Partnership Deed
5) Notarized copy of Registration Certificate of the product with approved label/leaflet
6) If Registration Certificate is more than 2 years old, a certificate from the concerned State/UT Govt. as a proof of commencement of actual production of the pesticide
7) Affidavit duly notarized regarding endorsement made earlier in respect of certificate for which the present request is being made

Requirement for the endorsement of new/alternate packaging
(Approved by the Registration Committee in its 256th RC held on 12th September, 2005)

Name of the applicant: M/s……

1) Index
2) Page Numbering.
3) Authentication of pages by authorized signatory
4) Authorization for the signatory (Notarized copy of BOD resolution/ affidavit/ partnership deed)
5) Justification for change
6) Copy of Certificate of Registration
7) Details of system of packaging w.r.t. Primary, secondary and transport packaging
8) Affidavit duly notarized regarding endorsement made earlier in respect of Certificate for which the present request is being made.
9) Data supporting the suitability of the content with the co
10) Container
*(for new packaging material)

*10) Data on performance of container during storage stability (for new packaging material)

*11) Data on transport worthiness test (If packaging material is new and/or if the packaging material is same but pack sizes are different than BIS specified pack size)

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*10) & *11): Added as per decisions of 299th & 307th meeting of RC held on 28-04-2009 and 24-02-2010, respectively.

Requirement for the endorsement of change of name of the company/firm
(Approved by the Registration Committee in its 256th RC held on 12th September, 2005)

Name of the Applicant: M/s…….

1. Index
2. Page Numbering.
3. Authentication of pages by authorized signatory
4. Authorization for the signatory (Notarized copy of BOD resolution/affidavit/partnership deed)
5. Affidavit duly notarized regarding endorsement made earlier in respect of certificates for which present request is being made.
6. Copy of Certificate of registration.
7. In case of shifting of factory premises, a No Objection Certificate from the Director of Agriculture of the State/UT for setting of the factory at the newly proposed site. In case of shifting of office premises, full and complete address of the Head Office including the branch office.
8. In case of firm, copy of the Partnership Deed/ in case of Company, (a) Memorandum of Articles & Association (b) fresh Certificate of Incorporation consequent on change of Name, approved by Registrar of the Company, Ministry of Law, Justice & Company Affairs.
9. An affidavit regarding no court case is pending under any provisions of the Insecticides Act and Rules framed there under.

Requirement for the endorsement of factory address for the first time/shifting of factory and establishment of second factory address

(Approved by the Registration Committee in its 256th RC held on 12th September, 2005)

Name of the Applicant: M/s ……..

I. Factory Address for the first time:

1. Index
2. Page numbering
3. Authentication of pages by authorized signatory.
4. Authorization for the signatory (Notarized copy of BOD resolution/ affidavit/ partnership deed)
5. Affidavit, duly notarized, regarding endorsement made earlier in respect of Certificate for which present request is made.
6. Copy of Certificate of Registration
7. Copy of approved labels and leaflets or 7 copies of labels and leaflets (If not approved earlier)

II. Shifting of Factory and endorsement of second Factory Address:

A. All above from 1-6

B. 'No Objection Certificate' from the State Director of Agriculture for shifting of the factory / endorsement of second address.

Requirement for the transfer of certificate of registration of one person/ undertaking in the name of another person/ undertaking

(Approved by the Registration Committee in its 256th RC held on 12th September, 2005)

Name of the Applicant: M/s.....

1. Index
2. Page Numbering.
3. Authentication of pages by authorized signatory
4. Authorization for the signatory (Notarized copy of BOD resolution/affidavit/partnership deed)
5. Affidavit duly notarized regarding endorsement made earlier in respect of certificates for which present request is being made
6. Joint application signed by both the parties in respect of each registration certificate on a non-judicial stamp paper duly Notarized with the following details:
    i. Date of commissioning the unit/Accepted date of commissioning of unit
    ii. Complete address of the manufacturing unit
    iii. Details of production of different products for last one year
7. Attested copy of the manufacturing licence issued by the State Licensing Authority containing the name of products for which a request for transfer of certificate have been made.
8. Certificate of incorporation issued by the Registrar of Companies in case of Private/Public Ltd. Companies together with copies of Memorandum and articles of association.
9. Attested copies of the Deed for dissolution of partnership or deed of partnership as the case may be.
10. Attested copies of the registered sale deed/gift/inheritance
11. No Objection Certificate from the competent court towards its transfer, in case of any Court case is pending under any provisions of the Insecticides Act and Rules framed there under against the transfer.

Requirement for the endorsement of change of name of source of import of chemicals due to disinvestment
(Approved by the Registration Committee in its 261th RC held on 151th Feb, 2006)

Name of the applicant: M/s……

Name of the product: …….

i) Index
ii) Page Numbering
iii) Authentication of pages by authorized signatory
iv) Authorization for the signatory (Notarized copy of BOD resolution/affidavit/partnership deed)
v) Formal request letter by the dis-investor
vi) Copy of Certificate of Registration
vii) Certificate with reference to disinvestments along with other relevant supporting documents
viii) Undertaking from the buyer with reference to maintenance of same quality, specification, process of manufacture of technical product
ix) Analytical test report of the technical product being manufactured by the buyer with reference to quality specification as well as purity, impurity profile of these batch samples
x) Consent letter to supply the product to India
xi) Affidavit with reference to earlier endorsement being made

Requirement for the change of name of source of import

(Approved by the Registration Committee in its 263rd RC held on 16th May, 2006)

Name of the applicant: M/s……

Name of the Product……

Administration/Legal

1. Index
2. Page numbering
3. Notarized copy of BOD Resolution/Affidavit/Partnership Deed
4. Authentication of pages by authorized signatory.
5. Affidavit duly notarized regarding endorsement made earlier in respect of Certificate for which the present request is being made
6. Documentary proof of change of name applicable in respect of country duly authenticated by Consulate/Embassy/High Commission
7. Documentary proof as per change of manufacturing premises duly authenticated by Consulate/Embassy/High Commission
8. Copy of certificate of Registration

Chemistry

1. Reason/justification for the change in name clarifying whether it is a case of merger/taking over of company/internal re-arrangement/shifting of manufacturing unit
2. Analytical test reports from the approved source as well as from the newly named source along with undertaking w.r.t. quality specification and process of manufacture
3. Supportive document from the designated National Authority/Registration Document
4. Letter of consent, duly legalized from Indian Embassy/High Commission/Consulate in the country of origin
5. Any other additional relevant document/information.

Requirement for the endorsement for inclusion of already approved additional source of import in the registration certificate

(Approved by the Registration Committee in its 266th RC held on 20th July, 2006)

Name of the applicant: M/s……

Name of the Product……

Administration/Legal

1. Index
2. Page numbering
3. Notarized copy of BOD Resolution/Affidavit/Partnership Deed
4. Authentication of pages by authorized signatory
5. Affidavit duly notarized regarding endorsement made earlier in respect of Certificate for which the present request is being made
6. Letter of consent, duly legalized from Indian Embassy/High Commission/ Consulate in the Country of origin
7. Reference of Registration Committee meeting in which additional source of Import is approved
8. Copy of Registration Certificates along with label/leaflets

Requirement for the endorsement of enhancement of shelf life of pesticides u/s 9(4) of the insecticides act, 1968

(Approved by the Registration Committee in its 305th RC held on 20th Oct, 2009)

Name of the applicant: M/s……

Name of the Product……

1. Index
2. Page Numbering
3. Authentication of pages by authorized signatory
4. Notarized copy of BOD resolution/Partnership Deed/Affidavit
5. Notarized copy of manufacturing license
6. Affidavit regarding previous Endorsement(s) made on the Certificate of Registration for which present request has been made.
7. Application Proforma for enhancement of shelf life of insecticides registered u/s 9(4) (approved by the Registration Committee in its 284th meeting held on 13.12.2007)
8. Name of 9(3) registrant and reference of CIB meeting in which shelf life enhancement has been approved

Requirement For Free Sale Certificate

(Approved by the Registration Committee in its 315th RC held on 22th Feb, 2011)

Note: Regular registrants may submit separate application for sole purpose of Export or for Use in India as well as Export.

Name of the applicant: M/s……

1. Index
2. Page Numbering
3. Authentication of pages by authorized signatory
4. Authorization in favour of authorized signatory (notarized copy of BOD resolution / Partnership deed / Affidavit)
5. Notarized copy of Certificate of Registration(s) along with relevant Endorsements i.e. Transfer of Certificate of Registration, Change of Name of Registrant, Change of Address of Registrant etc.
6. Notarized copy of valid Manufacturing License
7. Justification for the requirement of F.S.C.

Requirement for the endorsement of change in office address with respect to certificates of registration
(Approved by the Registration Committee in its 316th RC held on 08th March, 2011)

1. Index
2. Page Numbering
3. Authentication of Pages by authorized signatory
4. Affidavit duly notarized regarding endorsement made earlier in respect to Certificate for which the present request is made
5. Copy of Certificate(s) of Registration with Ls/Ls
6. (a) Notarized copy of proof of change in company office address from Registrar of Companies under the Companies Act, 1956, in case of Company /

   (b) Affidavit with details of previous address along with reasons for change in address duly notarized, in case of Proprietorship/firm
7. Affidavit regarding no court case along with list of products duly notarized

Requirement for the submission of application for attestation of copy of already issued certificate(s) of registration

(Approved by the Registration Committee in its 318th RC held on 27th Apr, 2011)

1) Index
2) Page numbering
3) Authentication of pages by authorized signatory
4) Notarized copy of BOD resolution/affidavit/partnership deed
5) Notarized copy of Certificate(s) of Registration along with labels & leaflets in duplicate (Second copy may be the photocopy of notarized copy of certificates of Registration duly authenticated by the applicant. Submission of each certificate copy should be indexed with page no.)
6) Affidavit by applicant in the format approved by RC in its 314th meeting held on 24th & 27th Jan., 2011 (attached)
7) Original Certificate(s) of Registration are to be produced before the designated officer / committee for verification purpose

Draft/Specimen

A F F I D A V I T

(To be furnished by the Applicant or Authorised person, duly Notarized)

I.....s/o ..... aged about ..... years, working as..... in ....., having registered office at .....do hereby confirm, declare and solemnly affirm as under:

1. That I/we are the registrant for the product(s) ..... (Name of the product(s) with Registration No.), duly notarized copy/copies ***(IN DUPLICATE)*** of the Certificate of Registration is/are enclosed.

2. That copy/copies of the above the mentioned certificate(s) is/are being submitted to the Secretariat of Central Insecticide Board & Registration Committee, Faridabad (CIB&RC) for attestation as per the requirement of the importing countries and shall not be used/misused for any other purpose than for the export of the above mentioned product(s) only.

3. That a duly notarized copy of authorization in favour of deponent is also enclosed.

Deponent
(Name & Designation)

V E R I F I C A T I O N

I ....., the above named deponent do solemnly verify that the information given above is true to the best of my knowledge & belief and nothing has been concealed there from.

I, further declare that I am making this declaration in my capacity as ..... (Designation) and that I am competent to make this declaration.

Deponent

Requirement for the endorsement of name of supplier of already approved source of import in the certificate of registration

(Approved by the Registration Committee in its 318th RC held on 27th April, 2011)

1. Index
2. Page numbering
3. Notarized copy of BOD Resolution/Affidavit/Partnership Deed
4. Authentication of pages by authorized signatory
5. Affidavit duly notarized regarding endorsement made earlier in respect of Certificate for which the present request is being made
6. Letter of consent, from approved source of import duly legalized from Indian embassy/ high Commission/ Consulate in the Country of origin
7. Reference of Registration Committee meeting in which additional source of Import is approved
8. Copy of Registration Certificates along with label/leaflets.

Requirement for the submission of application for small quantity import of pesticides for research, test and trial (RTT) purposes

(Approved in 320th meeting of RC held on 30-06-2011 and modified as per decisions of 329th meeting held on 08-06-2012)

1) Index
2) Page Numbering
3) Authentication of pages by the authorized signatory.
4) Notarized copy of BOD Resolution/ Partnership deed/ Affidavit;
5) Complete Form-C duly signed.
6) Justification for the quantity for import
7) Technical bulletin from inventor / manufacturer.
8) Information on Chemistry, toxicity and bio-efficacy of product along with its reference.
9) In case of microbial Biopesticides- a notarized affidavit by the applicant that product does not contain any genetically modified organism. Also, a document from the manufacturer in this regard.
10) Authenticated copy of the Registration Certificate for manufacture of particular molecule/residue that country, issued from the Registration Authority.
11) Authorization for supplier in case of sample quantity is not supplied by the Manufacturer.
12) For coded & non-registered pesticides authentic document from the Inventor.

Requirement for the issue of duplicate registration certificate
(Approved in 320th RC held on 30th June, 2011)

Name of the applicant: M/s……

Name of the product: ……..

1) Index
2) Page Numbering.
3) Authentication of pages by authorized signatory
4) Required fees
5) Affidavit duly notarized regarding endorsement made earlier in respect of certificates for which present request is being made along with details of Certificate of Registration.
6) Copy of the F.I.R. in respect of loss of Certificate of Registration
7) Cutting of the leading Newspaper advertisement regarding loss of the said certificate after a fortnight of the advertisement
8) An affidavit to the effect that as when original certificates are found in future, the same shall be returned to the Registration Committee immediately and shall not be misused.
9) Copy of the letter written to the State Licensing authority for intimation of the loss of certificate(s).
10) Photocopy of Certificate(s) and Label/Leaflets lost, duly notarized.

Requirement for extension of shelf life of pesticides u/s 9(3b)/ 9(3) for data generated in continuation
(Approved by the Registration Committee in its 320th RC held on 30th June, 2011)

Note: 1. For enhancement of shelf-life of Biopesticides, one of the two locations shall be Central Insecticides Laboratory (Decisions of 333rd RC held on 30-10-2012)

2. For Chemical Pesticides, additional location for generation of shelf life data shall be the Central Insecticides Laboratory, Faridabad (Decisions of 333rd RC held on 30-10-2012)

1. Index
2. Page numbering
3. Authentication of pages by authorized signatory
4. Notarized copy of BOD resolution/ Partnership deed/ affidavit
5. Notarized copy of Registration Certificate with label & leaflets
6. Copy of Form-1, submitted with the application for registration of the product mentioned at Sl. No. 5
7. Shelf life claim
8. Shelf life data along with Meteorological data

Requirement for the endorsement of extension of validity/ revalidation of registration certificates issued under section 9(4) of the insecticides act, 1968

(Approved by the Registration Committee in its 320th RC held on 30th June, 2011)

Note: The above endorsement shall not be applicable for the certificates of registration for 9(4) formulations (decision of 333rd RC held on 30-10-2012)

1. Index
2. Page numbering
3. Authentication of pages by authorized signatory
4. Notarized copy of BOD resolution/partnership deed/affidavit
5. Copy of SSI / other category of Industry
6. Copy of valid manufacturing license w.r.t. other pesticides.
7. Copy of Registration Certificate(s) along with labels & leaflets.
8. Affidavit regarding details of endorsements made earlier with product name and Certificate No., duly notarized.
9. Justification for not obtaining the manufacturing license for the pesticides for which Registration Certificate(s) u/s 9(4) was/were obtained, supported by documentary evidence, duly notarized.
10. NOC from Pollution Control Board, if already taken.

As per 442 RC; Agenda item 10.36; 9(3)

1. Covering Letter
2. Index
3. Notarized copy of BOD resolution/partnership deed/affidavit
4. Endorsement Affidavit
5. Copy of Certificate of Registration from source/manufacturer/ICAMA
6. Original hard copy of ICAMA/ Certificate of Registration
7. Legalized Consent Letter
8. NOC from Pollution Control Board

Requirements for the change of name of already approved source of import for office address only

Name of the applicant: M/s……

Name of the Product……

Administration/Legal

1. Index
2. Page numbering
3. Notarized copy of BOD Resolution/Affidavit/Partnership Deed
4. Authentication of pages by authorized signatory
5. Affidavit duly notarized regarding endorsement made earlier in respect of Certificate for which the present request is being made
6. Documentary proof of change of name applicable in respect of country duly authenticated by Consulate/Embassy/High Commission.
7. Copy of certificate of Registration.

Chemistry

1. Reason/justification for the change in name clarifying whether it is a case of merger/taking over of company/internal rearrangement/shifting of manufacturing unit
2. Analytical test report from the newly, named source along with undertaking w.r.t. quality specification and process of manufacture
3. Supportive document from the designated National Authority/Registration Document
4. Letter of consent, duly legalized from Indian Embassy/High Commission /Consulate in the country of origin and documentary evidence to prove the same
5. Any other additional relevant document/information

Requirements for the change of address of manufacturing premises of import in the country of origin in case of closure of old unit

Name of the applicant: M/s……

Name of the Product……

Administration/Legal

1. Index
2. Page numbering
3. Notarized copy of BOD Resolution/Affidavit/Partnership Deed
4. Authentication of pages by authorized signatory
5. Affidavit duly notarized regarding endorsement made earlier in respect of Certificate for which the present request is being made
6. Documentary proof of change of address applicable in respect of country duly authenticated by Consulate/Embassy/High Commission
7. Copy of certificate of Registration

Chemistry

1. Reason/justification for the change in address clarifying whether it is a case of merger/taking over of company/internal rearrangement/shifting of manufacturing unit
2. 5-batch analysis report from the changed source along with undertaking w.r.t. quality specification and process of manufacture
3. Supportive document from the designated National Authority/Registration Document
4. Letter of consent, duly legalized from Indian Embassy/High Commission /Consulate in the country of origin
5. Any other additional relevant document/information

Requirements for the change of name of source of import or change of manufacturing premises in the country of origin

Name of the applicant: M/s……

Name of the Product……

Administration/Legal

1. Index
2. Page numbering
3. Notarized copy of BOD Resolution/Affidavit/Partnership Deed
4. Authentication of pages by authorized signatory
5. Affidavit duly notarized regarding endorsement made earlier in respect of Certificate for which the present request is being made
6. Documentary proof of change of name applicable in respect of country duly authenticated by Consulate/Embassy/High Commission

   OR

   Documentary proof as per change of manufacturing premises duly authenticated by Consulate/Embassy/High Commission
7. Copy of certificate of Registration along with label / leaflet

Chemistry

1. Reason/justification for the change in name clarifying whether it is a case of merger/taking over of company/internal rearrangement/shifting of manufacturing unit
2. Analytical test report from the new promises along with undertaking w.r.t. quality specification and process of manufacture
3. Supportive document from the designated National Authority/ Registration Document
4. Letter of consent duly legalized from Indian Embassy/ High Commission/ Consulate in the country of origin
5. Any other additional relevant document/information.

Agricultural Chemicals for which no Maximum Residue Limit (MRL) applies

The Scientific Panel on Pesticide Residue (SPPR), FSSAI has discussed the subject on "Agricultural Chemicals for which no maximum residue limit (MRL) applies" in its 88th SPPR meeting and decided to seek the comments from the secretariat of CIB&RC and RC deliberated the matter in its 464th RC meeting (Vide agenda item no. 10.25) and agreed the decision taken by the SPPR. RC decided to seek the relaxation for the fixation of MRL on seed treatment products, Petroleum derived spray oils, and pesticides to be applied under nursery stage.

Agricultural Chemicals for which No Maximum Residue Level Applies

No	Pesticide	Condition
1.	1-Triacontanol	When used as a plant growth regulator on pasture
2.	1,4-Dimethylnaphthalene	When used for maintenance of sprout inhibition in stored potatoes
3.	9,10-Anthraquinone	Used as a bird repellent for grapes
4.	Ammonium sulfamate	General purpose herbicide
5.	Ammonium thiosulphate	Applied during flowering for fruit reduction in fruit plants
6.	Extract of Azadirachta indica (Neem) (containing azadirachtin)	Where the primary mode of action derives from the presence of azadirachtin, and; When used as an insecticide for food producing plant species
7.	Bacillus thuringiensis	Used as an insecticide
8.	*Banda de Lupinus albus doce*	Used as fungicide
9.	Barium carbonate	Used as a rodent poison
10.	Benzalkonium chloride	Bactericidal and fungicidal agent.
11.	Biocyclopyrone	Herbicide
12.	Boric acid	When applied as a fungicide for pruning wound treatment of fruit and control of canker in apple orchards
13.	Bromochlorodimethylhydantoin	When applied as a biocide to fruits and vegetables
14.	C9 – C16 Alkanes	When used as an agricultural chemical. Includes any structural isomer of linear alkanes, branched alkanes

		and cycloalkanes within the specified carbon number. The alkanes must contain no heteroatoms
15.	Calcium polysulphide (lime sulphur)	Used as a fungicide or insecticide on food producing plant species
16.	Camphor	Acting as a natural insect repellent, antifungal agent, and growth stimulant for certain crops.
17.	Carbon disulphide	Used as fungicides, biocides, and soil fumigants
18.	Chitosan	No condition of use applies
19.	Cholecalciferol (Vitamin D3)	Used as a rodenticide and food fortifier.
20.	Chlorine dioxide	When applied as an agricultural chemical to fruit and vegetables at a concentration not exceeding 10 ppm
21.	*Chromo bacterium subtsugae* PRAA4-1T and its metabolites, including violacein. Excludes metabolites that have been isolated as independent compounds.	When used as an agricultural chemical
22.	Cinnamaldehyde	Used as fungicide, insecticide, and post-harvest preservation agent due to its antifungal, antibacterial, and insecticidal properties.
23.	Copper and its salts	When used as an agricultural chemical
24.	3-Decane-2-one	A natural plant growth regulator used post-harvest on stored potatoes to inhibit sprouting. Also used as a flavouring ingredient.
25.	Elemental iron, iron complexes, and iron salts	When used in pellet form as a molluscicide
26.	Ethyl formate	Used as a post-harvest fumigant on cereal grains, fruit, oilseeds, and vegetables
27.	Ethylene	When used for ripening and de-greening of fruits
28.	Eugenol	When used as an insect attractant and fungicide on grapevines

29.	Fatty acids of 8 carbons or more in their chains, and their salts	When used as an agricultural chemical
30.	Geraniol	When used as a fungicide on grapevines
31.	Gibberellic acid (gibberillins GA3, GA4 and GA7 and potassium gibberellate)	Used as a plant regulator applied at <200 g a.i./ha/year
32.	Harpin αβ protein	The source(s) of the harpin αβ protein must be non-pathogenic or non-toxic to humans. The extracted harpin αβ protein must be purified and be free of any bacterial cells
33.	Hydrogen peroxide	When used as a spray-on fungicide and bactericide treatment for fruits and vegetables
34.	Laminarin	Laminarin, a polysaccharide from brown seaweed, is used in agriculture as a plant defence stimulant, triggering plant's natural defence mechanisms against diseases like blights, rusts, and leaf spots.
35.	Methyl anthranilate	Used as a bird repellent
36.	Microbial Active Ingredients (any organism classified as a microorganism including but not limited to bacteria, protozoa, fungi and viruses, or the genetically modified or naturally occurring mutants of any of these microorganisms. This includes whole organisms (either viable or non-viable), organism organelles, organism metabolites, organism spores, or occlusion bodies.) This entry	When used as the active ingredient in an agricultural as registered pesticide.

	applies when the Microbial Active Ingredient leaves no residues of the parent organism, its metabolites, or toxins on the treated crop at levels that are considered a dietary intake concern. This entry does not include metabolites produced by a microorganism that have been isolated as an independent active ingredient.	
37.	Mixtures of chito-oligosaccharides and oligogalactonurans	No condition of use applies
38.	6-Benzyladenine	Used as a plant growth regulator
39.	Pimaricin (Natamycin)	Fungicide
40.	Pelargonic acid	Used as a non-selective, contact herbicide to control weeds, and as a blossom thinner for fruit trees.
41.	*Neotyphodium uncinatum* strain AR1006 (containing the Loline alkaloids: N-acetylloline, Nacetylnorloline, N-formylloline)	Where the primary mode of action derives from the presence of Loline alkaloids, and; When used as an insecticide for food producing plant species
42.	Ozone	When used as an agricultural chemical
43.	Phosphorous acid	When directly used as an agricultural compound, or when representative of the use of fosetyl aluminium as an agricultural compound
44.	Pine oil	When used as an herbicide
45.	Piperin	Insecticidal and antimicrobial properties

46.	Plant extracts (unrefined)	Except where listed in Schedule 1 of this Notice: Where the extract is in a product registered under the Agricultural Compounds and Veterinary Medicines Act 1997 and intended for use as an agricultural chemical, and; Where the extract is derived from plants of the following species: *Apocynum Venetum* *Beta vulgaris* (Sugar beet, beetroot, and chard) *Camellia sinesis* (Tea) *Clitorea tematea* (Butterfly Pea) *Fallopia sachalinensis* (Giant knotweed) *Melaleuca alternifolia* (Tea Tree) *Optuntia linheimeri* (Texas prickly pear) *Quercus falcate* (Southern red oak) *Rhizophoria mangle* (Red mangrove) *Rhus aromatica* (Fragrant sumac) *Ribes nigrum* (Blackcurrant) *Swinglea glutinosa* *Undaria pinnatifida* (Wakame)
47.	Polysaccharides	Used as an agricultural chemical
48.	Polyoxin D Zinc Salt	When used as an agricultural chemical
49.	Potassium bicarbonate	When used as an agricultural chemical
50.	Prohydrojasmon	When used as a colour enhancer on apples
51.	S-Abscisic acid	When used as a plant growth regulator
52.	Salicylic acid	When used on any fruit
53.	Scilliroside	Rodenticide
54.	Sulphur	When used as an agricultural chemical
55.	Synthetic latex	Used as an anti-pod shatter
56.	Tar oils (Tar Acid)	Used as pest control, seed protection, and soil stabilization

57.	Thymol	When used a fungicide on grapevines
58.	Piperin	Piperine is a major component of black pepper. It is used as repellent for various animals and effective in killing mosquito larvae.
59.	Methyl chavicol	Methyl chavicol, also known as estragole, is a natural compound found in plants like basil, tarragon, and fennel. This biopesticide is effective in controlling Fall armyworm.

Standard Operating Procedure (SOP) for use of Drone application with pesticides for crop protection in agricultural, forestry, non-cropped areas, etc.

(Approved by the RC in its 434th RC meeting held on 30.11.2021)



सत्यमेव जयते

Government of India

Ministry of Agriculture and Farmers Welfare
Department of Agriculture and Farmers Welfare
Krishi Bhawan

New Delhi

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Endorsement

This Standard Operating Procedure (SOP) for use of Drone application with pesticides for crop protection in agricultural, forestry, non-cropped areas, is prepared by the committee constituted vide O.M. No 13-15/2020- M&T (I&P), dated 31st of March, 2021 under the chairmanship of Plant Protection Adviser, Directorate of Plant Protection, Quarantine & Storage, Faridabad, for undertaking safe and effective control of Pest and diseases by Drone application with pesticides through aerial spraying. This SOP will render guidance to the stakeholders/ pilot/operators/ users/regulators while undertaking safe and effective control of Pest and diseases by Drone based Pesticide application.

(Dr Ravi Prakash)

Plant Protection Adviser
Directorate of Plant Protection, Quarantine & Storage,
NH-IV, Faridabad-121001 India

1. Objective:

Indian Agriculture has gone through rapid advancements and has benefited from research and adoption of new technologies by farmers. Technologies like drip irrigation, mechanized farming for planting, harvesting and grading are being successfully used for sustainable agriculture in India.

In recent years, use of drone in agriculture has gained prominence and some states are actively engaged in checking the suitability of this new technology in Indian agriculture. Application of pesticides using drones has a great potential as we move towards commercialization and achieving precision in agricultural crops.

Drones are going to be important for increasing efficiency of application of crop protection chemicals by reducing manpower requirement, reducing time of application, reducing volume of water, quantity of chemicals and saving drift to environment along with reducing exposure to human being to hazardous chemicals.

In conventional agricultural practices, pesticides are sprayed either manually or with the help of tractor-mounted sprayers where high quantity of pesticides and water are used and where a sizable portion of spray goes waste in environment. However, drone-based spray requires less amount of water, as well as pesticides, due to better application and bio-efficiency.

To achieve the above stated objectives, Drones as a future technology for pesticide spraying, custom hiring and cooperative use will be encouraged and facilitated by Ministry of Agriculture and Farmers Welfare so that they are widely available and easily accessible to the farmers of the country. Drone as future technology for pesticide spraying are also expected to be linked with Agriculture insurance system for tackling any kind of damage and loss.

The SOP for drone regulation for pesticide application therefore covers important aspects like statutory provisions, flying permissions, area distance restrictions, weight classification, overcrowded areas restriction, drone registration, safety insurance, piloting certification, operation plan, air flight zones, weather conditions, SOPs for pre, post and during operation, emergency handling plan etc.

2. Statutory Provisions:

As per the provisions of the Insecticides Rules1971, Under the Insecticides Act, 1968 one of the functions of the board (Central Insecticides Board) constituted under section 4 of the Act, is to specify the uses of the classification of insecticides on the basis of their toxicity as well as their being suitable for aerial application [Rule 3 (b)].

Further, as per Insecticide Rules 1971, Chapter VIII Rule 43 on Aerial Spraying Operations, the aerial application of Insecticide shall be applicable for pesticide application through drones subject to the following provisions:

a. marking of the area shall be the responsibility of the operators;
b. the operators shall use only approved insecticides and their formulations at approved concentration and height;
c. washing decontamination and first-aid facilities shall be provided by the operators;
d. All aerial operations shall be notified to the public not less than twenty-four hours in advance through competent authorities (written intimation to be sent to the Executive Officer of the respective Gram Panchayat & Panchayat Samiti, as well as the concerned Agriculture Officer of the area, at least 24 hours in advance).
e. Animals and persons not connected with the operations shall be prevented from entering such areas for a specific period; and
f. The pilots shall undergo specialization training including clinical effects of the insecticides (details on Pilot Training are given at point no.5.4 of this SOP).

3. The drone operations are being permitted by Ministry of Civil Aviation (M/o CA) and Director General of Civil Aviation (DGCA) through the conditional exemption route. New drone directorates have been established in DGCA recently to speed up the approval process. They have published a detailed DGCA Requirements for Operation of Civil Remotely Piloted Aircraft System (RPAS) Guidance Manual with an objective to acquaint the public and the industry with the procedures being followed for processing all matter pertaining to issue of unique identification number, unmanned aircraft operator permit, and related activities. It will help understand the flow of various processes involved and understand the intricacies of the system.

Further, Requirements for Operation of Civil Remotely Piloted Aircraft System (RPAS) mentions about the Civil Aviation Requirements (CAR) issued under the provisions of Rule 15A and Rule 133A of the Aircraft Rules, 1937 under the Aircraft Act, 1934 (22 of 1934)and lays down requirements for obtaining Unique Identification Number (UIN), Unmanned Aircraft Operator Permit (UAOP) and other operational requirements for civil Remotely Piloted Aircraft System (RPAS).These operations of drones are regulated by Unmanned Aircraft System (UAS) Rules, 2021, as published by Ministry of Civil Aviation vide G.S.R. 589(E) dated 25th August, 2021. Operators have the flexibility to utilize any RPAS/drone in case of agricultural pesticide spraying, irrespective of weight category or use case, provided the discharge of substance is cleared and mentioned in the Unmanned Aircraft Operator Permit(UAOP) issued by the DGCA.

4. Details, precautions and prerequisites etc. for Drone based Pesticide application

The following details, precautions, prerequisites etc. shall be adhered to before, during and post-operation:

4.1. Pre-application:

i. Confirm not to fly in the drone-forbidden area (airport or electronic station).No prior permission is required for operating an unmanned aircraft system in a green zone.

ii. Ensure your Drone is Digital Sky Compliant with „No Permission – No Takeoff" hardware and firmware;

iii. Obtain Unique Identification Number (UIN) from DGCA for operating in controlled airspace and affix it on your drone.

iv. Obtain Unmanned Aircraft Operator Permit (UAOP), if applicable from DGCA for commercial operations and keep it handy

v. Ensure drone is in good condition (not damaged) and fit for flying safely.

vi. Keep an eye on interference: Interference can be from mobile devices or blockage of signals, do watch out when flying your drone.

vii. Fly in visual line of sight (VLOS): Always be within visual range of your drone.

viii. Ensure the operators are trained on both drone operation and safe use pesticide.

ix. No alcoholic drinks should be consumed within 8 hours preceding operation.

x. Calibrate drone spray system to ensure nozzle output and accurate application of labelled rates.

xi. Check if drone is in good condition and that there is no leak in the spraying system.

xii. Confirm a place for take-off, landing and tank mix operations.

xiii. Check and mark proposed treated area, boundary, the obstacles (walls, trees) around the field for safe operation.

xiv. Set up the least buffer zone (as specified by DGCA/ CIB&RC) between drone treatment and the non-target crop.

xv. Confirm presence of water sources - Do not spray pesticides near water sources (less than 100 m from target area) to avoid polluting water sources.

xvi. Check if the public has been notified/intimated at least twenty-four hours in advance through a written intimation to the competent authorities, that is the Executive Officer of the respective Gram Panchayat & Panchayat Samiti, as well asthe concerned Agriculture Officer of the area.

xvii. Concerned territory in-charge/operators to ensure that animals and persons not connected with the operations shall be prevented from entering such areas for a specific period.

xviii. Do log your flights and intimate concerned authorities (like DGCA, local police etc.)of any incidents/ accidents

xix. Don‟t fly drone over groups of people, public events, or stadiums full of people without permission

xx. Don‟t fly drone over government facilities/military bases or over/ near any no-drone zones.

xxi. Don‟t fly drone over private property unless permission is given.

xxii. Don't fly drone in controlled airspace near airports without filing flight plan or AAI/ADC permission (at least 24 hours before actual operation).

xxiii. Don‟t drop or carry hazardous material

xxiv. Don‟t fly drone from a moving vehicle, ship or aircraft.

xxv. Don‟t fly the Drone in contraventions to theUnmanned Aircraft System Rules (UAS), 2021 rules as published by Ministry of Civil Aviation vide G.S.R. 589(E) dated 25thAugust, 2021and as amended from time to time.

4.2. During Application:

i. Read labels carefully to understand safety guidance.
ii. Wear Personal Protect Equipment (PPE).
iii. Do not eat, drink or smoke while spraying.
iv. Confirm the flying route was reasonable to minimize turn around.
v. Operation team shall always stay at the downwind end of the field and backlight direction.
vi. Spray with pure water first to test operation for at least 5 minutes wherever feasible and required.
vii. Ensure two-step dilutions to fully dissolve the pesticide.
viii. Adopt proper pressure for optimized droplet spectrum (>100μm).
ix. Check weather conditions for:
    a. Appropriate Wind speed,
    b. Appropriate Temperature,
    c. Appropriate Humidity
x. Ensure appropriate flying height above target crop.
xi. Ensure appropriate water volume.
xii. Ensure appropriate flying speed.
xiii. Avoid having to walk through crop which has been contaminated by drifting spray.
xiv. Do not spray during active bee foraging period of the day. Avoid spray drift to flowering nectar crop.
xv. When spraying pesticides that are toxic to non-target organisms such as fish, birds and silkworm, strictly abide by the product label requirements and take effective measures to avoid risks.
xvi. Use anti-drift nozzle to decrease drift to human and environment.

4.3. Post Application:

i. Timely evacuation and transfer to fresh air.
ii. Triple rinse of empty container is mandatory.
iii. Ensure waste generated is kept to a minimum.
iv. The disposal of waste must conform to the local laws.
v. Never burn or bury hazardous waste.

vi. Never leave empty containers in the field. It should be disposed of as per the Insecticides Rule 1971.

vii. Set up warning signs in the spray area for reminding people.

viii. Take a shower and put on clean clothes.

ix. To prevent leakage of plant protection products in the process of transport and waiting to use.

x. Securely store plant protection products away from unauthorized people, animals and food when transporting and storing PPP. Safely dispose all spills immediately**.**

xi. Follow the maintenance schedule as prescribed by the Drone manufactures.

5. Critical parameters to be considered for Drone based pesticide application:

5.1. Drone related:

i. Only Director General of Civil Aviation (DGCA) certified/approved drone shall be permitted to carry-out agriculture spray. The reliability of the drone is assured through DGCA certification process.

ii. The drone must have capability to handle variable payload (depleting tank). The nozzle system should be attached in a manner that the spray swath is continuous when sprayed from the minimum permitted height above the uniformly distributed crop (e.g. paddy/sugarcane).

iii. The drone must be fitted with accurate altitude sensor to ensure desired height above the crop is maintained throughout the spraying mission.

iv. The GPS accuracy of the drone and accuracy of the map shall be characterized and the same shall be utilized to define the safety/buffer margin while creating the geo-fencing around the field or obstacles.

v. The drone spray system must support variable flow control to ensure uniform dispensing of the payload.

vi. The drone must have necessary fail-safes including Return to Home (RTH) on empty tank and auto mission restart from the point RTH was engaged.

vii. The drone spray system should be leak proof and dripping of Pesticides/Insecticides should be avoided during the application. (Check before flight)

5.2. Pesticides/Insecticides

i. Only Central Insecticides Board and Registration Committee (CIB&RC) approved Pesticides/Insecticides shall be used.

ii. The dose has to remain within the range approved by CIB&RC.

iii. The Pesticides/Insecticides (liquid/solid) compatibility with the drone spray system shall be established prior to the mission for the desired dilution. This is to ensure Pesticides/Insecticides solubility formulation stability and ability to spray with the type of

nozzles provided in the drone. In case of mixing of more than one Pesticides / Insecticides, CIB&RC specified guidelines must be adhered to.

iv. The minimum dilution shall be decided based on the fulfilment of the above-mentioned requirements and ensuring satisfactory coverage of sprayed input both horizontally and vertically.
v. Pesticides/Insecticides should be diluted only with clean water wherever applicable or with other suitable ingredient as has been approved by the CIB&RC.

5.3. Environment Limitations

The drone based spray should be undertaken and may be permitted under the conducive weather conditions to get the best results in terms of appropriate Wind speed, Temperature& Relative Humidity etc.

5.4. Pilot Training

i. Only DGCA certified pilots shall be permitted to fly the agri drones.
ii. A training module as devised by NIPHM, Hyderabad encompassing pesticide/ insecticide handling, agri-mission specific operational protocols, and relevant crop protection guidelines shall be made mandatory for pilots operating pesticide/ insecticide spray drones.

5.5. Drift Management -Critical Operational Parameters

Majority of land holdings are small and there are possibilities of spray drift to the nearby crop fields while spraying with drones. To minimize the spray drift, apart from wind limitation, the following are important and to be suitably adopted:

i. Spray height above crop canopy
ii. Speed of the drone
iii. Suitable nozzles and Droplet size
iv. A buffer zone can be demarcated (geo fenced)
v. Spray schedule should be at appropriate time gap either before rainfall or after rainfall.
vi. Others, as specified in guidelines issued by CIB&RC from time to time.

5.6. Safe guarding the non-targets

The non-targets shall be safe guarded by adhering to the following operational protocols:

i. Buffer zone as per CIB&RC approved guidelines to cater to Pesticides/Insecticides drift, shall be maintained between the adjacent farms or different crops to avoid spray on non-targets.
ii. During the spray operation operator should always maintain the distance approved by CIB&RC from the drone and avoid windward direction as much as possible.

iii. No human or animal movement shall be permitted within the farm during and immediately after the spray operations.

iv. Drone based pesticides spray operations should be conducted at a distance approved by CIB&RC from water bodies, residential areas, fodder crops, public utilities, dairy, poultry etc. and as per DGCA guidelines.

6. Registration requirements of pesticides for drone application:

The Registration requirements of pesticides for drone application and the modalities are dynamic in nature and will be considered based upon the safety, efficacy, statutory and legal requirements and published by the Central Insecticides Board and Registration committee (CIB&RC) from time to time. Drone users shall use only Central Insecticides Board and Registration Committee approved pesticides.

For registration of insecticides/pesticides and for use with drone, the applicant shall apply before Secretariat of CIB&RC, in the manner, as prescribed by CIB&RC under the Insecticides Act, 1968.

7. Spray Monitoring Form& Data Submission:

Spray monitoring data dully filled by the operator/ service provider shall be submitted through email(as mentioned in the guidelines issued by CIB&RC)/online portal created for the purpose, with-in seven days of application of pesticides through drone, in the form as prescribed in Annexure-I below.

SPRAY MONITORING FORM

Annexure-I

1	**LOCATION of Trial Details**	1		2		3		4		5		6	
1-1	Date												
1-2	Name of the location												
2	**VEGETATION DATA**												
2-1	Vegetation type (Crop, Grass, Bushes, Trees,)	G B T C		G B T C		G B T C		G B T C		G B T C		G B T C	
2-2	Height from ground/Crop Canopy (m)												
2-3	Crop names and pest/disease/weed infection/infestation/intensity (%)												
3	**PESTICIDEDATA**												
3-1	Trade name& Common name												
3-2	Concentration (g a.i./l or %)												
3-3	Formulation (EC, ULV, Dust)	E U D		E U D		E U D		E U D		E U D		E U D	
3-4	expiry date												
3-5	Is the insecticide mixed with water or solvent?	Y N		Y N		Y N		Y N		Y N		Y N	
3-6	If yes, what solvent and mixing ratio												
4	**WEATHER CONDITIONS**												
	Start and end of control operations	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End
4-1	Date &time												
4-2	Temperature (°C)												

4-3	Relative humidity (%)												
4-4	Wind speed (m/s)												
4-5	Wind direction (degrees from N)												
4-6	Spray direction (degrees from N)												
5	**SPRAY APPLICATION**												
5-1	Sprayer type (Rotary, Airblast, ENS, Hydraulic, Other)	R A E H O		R A E H O		R A E H O		R A E H O		R A E H O		R A E H O	
5-2	Sprayer operator (Pilot, Driver, Hired, Other)	P D L H O		P D L H O		P D L H O		P D L H O		P D L H O		P D L H O	
5-3	Sprayer manufacturer												
5-4	Sprayer model												
5-5	Sprayer platform (Aerial, Vehicle, Handheld)	A V H		A V H		A V H		A V H		A V H		A V H	
5-6	Date of last calibration												
5-7	Atomizer height above ground (m)												
5-8	ROTARY SPRAYERS: speed setting (blade angle, pulley setting, no. batteries)												
5-9	Speed of atomizer (rpm)												
5-10	Flow rate setting (which nozzle or restrictor used)												
5-11	Flow rate/atomizer (l/min)												
5-12	Number of atomizers												
5-13	Track spacing (m)												
5-14	BARRIERS ONLY: width and spacing (m)												
5-15	Forward speed (km/h)												
5-16	AERIAL SPRAYING: support supplied	GP = ground party available RC = radio communication with aircraft TG = DGPS track guidance											
		GP RC TG		GP RC TG		GP RC TG		GP RC TG		GP RC TG		GP RC TG	
5-17	Ground marking (GPS, Flag, Mirror, Smoke, Vehicle, None)	G F M S V N		G F M S V N		G F M S V N		G F M S V N		G F M S V N		G F M S V N	
6	**CONTROL EFFICACY**												
6-1	Effectiveness/mortality (%)												
6-2	Days after treatment/ time after treatment (hours)												
6-3	Yield of Crop/method of Effectiveness /mortality estimation (Quadrats, Target size, Visual, Cages, Other)	Q T V C O		Q T V C O		Q T V C O		Q T V C O		Q T V C O		Q T V C O	
7	**SAFETY AND ENVIRONMENT**												
7-1	Protective clothing: what did the operator wear?	G = goggles M = mask L = gloves O = overalls B = boots											
		G M L O B		G M L O B		G M L O B		G M L O B		G M L O B		G M L O B	
7-2	Was soap and water available?	Y N		Y N		Y N		Y N		Y N		Y N	
7-3	Who was informed of spraying? (Farmers, Nomads, Villagers, Officials, Beekeeper, others, etc.)	F N V O B		F N V O B		F N V O B		F N V O B		F N V O B		F N V O B	
7-4	Effect on non-target organisms	Y N		Y N		Y N		Y N		Y N		Y N	
7-5	If yes, what type of effect?												
7-6	Details of anyone who felt unwell or if other problems were encountered												
7-7	Details about nearby water-bodies and water channels												

***

SPRAY MONITORING FORM

Annexure-II

1	**LOCATION of Trial Details**	1		2		3		4		5		6	
1-1	Date												
1-2	Name of the location												
2	**VEGETATION DATA**												
2-1	Vegetation type (Crop, Grass, Bushes, Trees,)	G B T C		G B T C		G B T C		G B T C		G B T C		G B T C	
2-2	Height from ground/Crop Canopy (m)												
2-3	Crop names and pest/disease/weed infection/infestation/intensity (%)												
3	**PESTICIDEDATA**												
3-1	Trade name& Common name												
3-2	Concentration (g a.i./l or %)												
3-3	Formulation (EC, ULV, Dust)	E U D		E U D		E U D		E U D		E U D		E U D	
3-4	expiry date												
3-5	Is the insecticide mixed with water or solvent?	Y N		Y N		Y N		Y N		Y N		Y N	
3-6	If yes, what solvent and mixing ratio												
4	**WEATHER CONDITIONS**												
	Start and end of control operations	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End
4-1	Date &time												
4-2	Temperature (°C)												
4-3	Relative humidity (%)												
4-4	Wind speed (m/s)												
4-5	Wind direction (degrees from N)												
4-6	Spray direction (degrees from N)												
5	**SPRAY APPLICATION**												
5-1	Sprayer type (Rotary, Airblast, ENS, Hydraulic, Other)	R A E H O		R A E H O		R A E H O		R A E H O		R A E H O		R A E H O	
5-2	Sprayer operator (Pilot, Driver, Hired, Other)	P D L H O		P D L H O		P D L H O		P D L H O		P D L H O		P D L H O	
5-3	Sprayer manufacturer												
5-4	Sprayer model												
5-5	Sprayer platform (Aerial, Vehicle, Handheld)	A V H		A V H		A V H		A V H		A V H		A V H	
5-6	Date of last calibration												
5-7	Atomizer height above ground (m)												
5-8	ROTARY SPRAYERS: speed setting (blade angle, pulley setting, no. batteries)												
5-9	Speed of atomizer (rpm)												
5-10	Flow rate setting (which nozzle or restrictor used)												
5-11	Flow rate/atomizer (l/min)												
5-12	Number of atomizers												
5-13	Track spacing (m)												
5-14	BARRIERS ONLY: width and spacing (m)												
5-15	Forward speed (km/h)												
5-16	AERIAL SPRAYING: support supplied	GP = ground party available RC = radio communication with aircraft TG = DGPS track guidance											
		GP RC TG		GP RC TG		GP RC TG		GP RC TG		GP RC TG		GP RC TG	
5-17	Ground marking (GPS, Flag, Mirror,	G F M S V		G F M S V		G F M S V		G F M S V		G F M S V		G F M S V	

	Smoke, Vehicle, None)	N	N	N	N	N	N
6	**CONTROL EFFICACY**						
6-1	Effectiveness/mortality (%)						
6-2	Days after treatment/ time after treatment (hours)						
6-3	Yield of Crop/method of Effectiveness /mortality estimation (Quadrats, Target size, Visual, Cages, Other)	Q T V C O	Q T V C O	Q T V C O	Q T V C O	Q T V C O	Q T V C O
7	**SAFETY AND ENVIRONMENT**						
7-1	Protective clothing: what did the operator wear?	G = goggles M = mask L = gloves O = overalls B = boots					
		G M L O B	G M L O B	G M L O B	G M L O B	G M L O B	G M L O B
7-2	Was soap and water available?	Y N	Y N	Y N	Y N	Y N	Y N
7-3	Who was informed of spraying? (Farmers, Nomads, Villagers, Officials, Beekeeper, others, etc.)	F N V O B	F N V O B	F N V O B	F N V O B	F N V O B	F N V O B
7-4	Effect on non-target organisms	Y N	Y N	Y N	Y N	Y N	Y N
7-5	If yes, what type of effect?						
7-6	Details of anyone who felt unwell or if other problems were encountered						
7-7	Details about nearby water-bodies and water channels						

Standard Operating Procedures (SOP) on aerial spraying using aircraft/ helicopter/drone for control of Desert Locust

(Approved by the RC in its 434th RC meeting held on 30.11.2021)



Government of India

Ministry of Agriculture and Farmers Welfare
Department of Agriculture, Cooperation and Farmers Welfare

Directorate of Plant Protection, Quarantine & Storage
N.H.IV., Faridabad-121001 (ppa@nic.in)

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Endorsement

This Standard Operating Procedure (SOP) for Aerial Spraying by aircraft/helicopter/drone is prepared by the Directorate of Plant Protection, Quarantine & Storage, Faridabad in line with FAO SOP on aerial spraying and considering requirement envisaged under the relevant provisions (Rule 43) of the Insecticides Act 1968 and Insecticides Rules 1971 for undertaking safe and effective control of desert locust by aerial spraying using aircraft/helicopter/drone. This SOP will render guidance to the locust officer/ pilot/operators while undertaking aerial control operations. This SOP is duly approved on 18th May, 2020.

(Rajesh Malik)

Plant Protection Adviser
Directorate of Plant Protection, Quarantine & Storage,

NH-IV, Faridabad-121001 India

Standard Operating Procedures (SOP) for Desert Locust Control

Objective

The objective of the Standard Operating Procedures (SOP) for Desert Locust Aerial Control is to give concise instructions for effective and safe control operations against the Desert Locust using aircraft/helicopter/drone. Considering the legal requirements envisaged under the relevant provisions of the Insecticides Act 1968 and Insecticides Rules 1971 for undertaking safe and effective control of desert locust by aerial spraying using aircraft/helicopter/drone these instructions required to be followed. These instructions are intended for use by the field staffs who are involved in Desert Locust aerial operations (including Locust Officers and pilots/ operators) to help them avoid dangerous, ineffective or wasteful operations. They are based on the **FAO Desert Locust Guidelines for aerial spraying** where more detailed information and references are available.

The instructions focus on:

- Aerial survey operations
- Aerial spraying of insecticides
- Aerial spray equipment's
- Techniques for safe and efficient operations

1. Control process

A series of steps need to be followed before, during and after aerial survey and control operations.

Aircraft/Helicopter are best for spraying large areas (5000+ ha) the smallest area an aircraft can spray is 100 ha and inaccessible through ground vehicles. Use of drone is useful for spot applications of pesticides such as long trees and in the area where vehicle entry is inaccessible such as acacia plantation, sand dune etc.

PREPARATIONS before control operations

- Determine what type and number of aircraft/helicopter/drone are required for control operations
- Select competent control teams and provide them with training or refresher training. The pilots/Operator shall undergo specialization training including clinical effects of the insecticides by the experienced locust officer as well as medical toxicologist authorized by the Plant Protection Adviser.
- Check and service of aircraft/helicopter/drone

- Check and test the spray system on the aircraft/helicopter/drone, that commonly needed spare parts are available and aircraft/helicopter/drone are equipped with a GPS-based track guidance system
- Distribute the required quantity and type of insecticides, protective clothing, aviation fuel and pumps to the likely spray sites
- Make sure that aircraft/helicopter/drone is available in the country and can be contracted by the MOA&FW for control operations. Check that airstrips have been maintained

BEFORE aerial control operations

- Determine if aerial control operations are required.
- If so, choose appropriate aircraft/helicopter/drone types, insecticide and spray coverage pattern (barrier or full cover).
- Calibrate the spray system on the aircraft/helicopter/drone in order to assure the correct amount of insecticide is applied in the right way and in the right place.
- All aerial operations shall be notified to the public not less than twenty-four hours in advance through competent authorities
- Animals and persons not connected with the operations shall be prevented from entering such areas for a specific period
- Ensure that local inhabitants are informed about the date, time and location of control operations, so that they can move their livestock, beehives and families to safety.
- Find the wind direction in order to establish a spray direction at right angles to it and demarcate the infested area.
- Make sure that temperature, wind and rainfall conditions are suitable and safe for the aerial control operations.
- Marking of the area shall be the responsibility of the operators
- Washing, decontamination and first-aid facilities shall be provided by the operators

DURING aerial control operations

- Make sure that all staff who are handling or applying insecticide use full protective clothing
- The operators shall use only approved insecticides and their formulations at approved concentration and height
- All other non-spraying personnel, vehicles and equipment are away from the target area to avoid contamination by the sprayed insecticide

- Start at the downwind edge of the target area and spray cross wind (at right angles to the wind direction), moving upwind after each spray pass, making sure to measure the correct track spacing using DGPS, flagmen or other means
- Make an extra spray pass upwind of the target area to prevent under-dosing at the upwind edge
- Stop spraying if the wind drops (less than 1 m/s) or becomes very strong (more than 10 m/s) and wait for the right conditions
- Stop spraying if it starts to rain or seems likely to rain soon
- Stop spraying if the wind direction changes by more than 45 degrees, adjust your new spray line and spray the remaining area

AFTER aerial control operations

- Monitor and record all relevant details on the *FAO Spray Monitoring Form* (Annexure-I).
- Empty any insecticide remaining in the aircraft/helicopter/drone spray tank back into the original insecticide container. Clean and maintain the spray system on the aircraft/helicopter/drone, and store the insecticide and the empty containers in safe places**.**
- Wash yourself and the protective clothing as soon as possible.

2. Ground support team and field equipment

Support Team: one locust officer, two drivers and two vehicles, plus support staff such as assistants and skilled laborers.

Equipment: to be available in each team

- Hand-held GPS *(1)*
- Maps, compass
- FAO forms *(2)*
- Clipboard, paper and pen
- Anemometer
- Hygrometer
- Flags
- Oil sensitive paper to sample ULV droplets
- Bucket and plastic measuring cylinder or jug

- VHF or UHF walkie-talkies for short range ground-to- air communication
- Vibrating tachometer
- Stop watch
- Hand lens (x10)
- Sweep net
- Plastic bags
- Tool kit, first aid kit
- HF radio
- Cages for mortality assessment
- Water and soap for washing
- Sets of protective clothing for all staff handling insecticides

(1) extra batteries, cigarette lighter adapter, remote antenna

(2) Survey & Control Forms and Spray Monitoring Forms

3. Principles of ULV application

Ultra low volume (ULV) spraying uses small amounts of concentrated insecticide. In locust control, about 1.0 litre/hectare is applied. The insecticide is not mixed with water or solvent. It is oil-based to prevent evaporation and is usually applied ready to spray.

Droplets of spray are carried by the wind. In full coverage treatments, the insecticide is sprayed as overlapping swaths onto the control target so that a uniform deposit is achieved and the locusts receive enough insecticide. Remember:

- Do **not** spray during the hottest part of the day (1100-1600 hr) when convection may occur and carry the spray up into the sky instead of down onto the locusts
- Do **not** spray at low wind speeds less than 1 m/s
- Do **not** spray at high wind speeds more than 10 m/s

4. ULV aerial spray system

A good ULV sprayer uses rotary atomizers (spinning discs or rotating cages) to produce droplets in a small size range (50-100 *u*m). If droplets are too large or too small, control will be poor and insecticide wasted. For aerial spraying, use the following:

- Volume median diameter (VMD): 75-100 *u*m
- Blade angle: 35° (AU4000), 40° (AU5000) *(1)*
- Emission height: 5-10 meters, depending on wind *(2)*

- Aircraft/helicopter speed: 140-160 km/h in consultation with pilot
  (1) at air speed of 160 km/h, 7000 rpm (AU4000), 8000 rpm (AU5000)
  (2) higher for milling and flying swarms and, possibly, barrier control
- Drone height and speed should coincide with effective and safe locust control operation.

5. Calibrating ULV spray system

The aerial spray system on the aircraft/helicopter/drone must be calibrated before the actual spraying takes place.

What is calibration?

The aerial spray equipment needs to be adjusted in order to apply the recommended amount of insecticide, in the right size spray droplets, to the right place.

Before setting flow rates for the first time, consult the manufacturer's manual to get a rough estimate of the required flow rate. On aircraft/helicopter/drone, flow rate is checked by recording the time spent spraying and the amount of insecticide used. Accordingly, the flow rate should be measured and reset if necessary.

Calibration should always be carried out by using the actual insecticide that will be applied

When do you calibrate spray equipment?

- When the aerial spray equipment is new
- When the insecticide formulation or concentration is changed
- When the volume application rate (VAR), track spacing or forward speed is changed
- Before the beginning of the campaign and at weekly intervals during it

How to calibrate a sprayer

Step 1. Find the recommended dose of the insecticide (g a.i./ha), from the drum label, FAO Guidelines, etc. If it is given as litres/hectare, go to step 3.

Step 2. Calculate the required Volume Application Rate (VAR).

$$\text{VAR (l/ha)} = \frac{\text{Recommended dose (g a.i./ha)}}{\textbf{Formulation concentration (g/l)}}$$

Example: If the recommended dose for chlorpyrifos is 250 g a.i./ha and its concentration is 450 g/l what is the VAR?

$$\text{VAR (l/ha)} = \frac{250}{\textbf{450}} = 0.55 \text{ l/ha}$$

If the formulation concentration expressed as a percentage of weight to volume (% w/v) convert the concentration to g a.i/l by using the formula:

$$\text{Concentration (g a.i./l)} = \frac{\text{Concentration given} \times 1000}{100}$$

Example: If the concentration given for Malathion is 96%, then this must be converted by using the formula:

$$\text{Concentration in g a.i./l} = \frac{96 \times 1000}{100} = 960 \text{ g a.i/l}$$

In short, multiply the given percentage concentration by 10.

Step 3. Calculate the Flow Rate (FR).

$$\text{FR (l/min)} = \frac{\text{VAR (l/ha)} \times \text{speed (km/h)} \times \text{track spacing (m)}}{600}$$

Example: What flow rate is required from an aircraft flying at 140 km/h using a 100m track spacing in order to apply 960 g a.i./ha of Malathion 96% ULV?

$$\text{FR (l/min)} = \frac{1 \text{ (l/ha)} \times 140 \text{ (km/h)} \times 1000 \text{ (m)}}{600} = 23.33 \text{ l/min}$$

It is important to remember that if one of the parameters (flow rate, track spacing or forward speed) is altered, then one or more of the others have to be changed in order to maintain the correct Volume Application Rate and Dose.

- If flow rate increases VAR increases (and vice versa)
- If track spacing increases VAR decreases (and vice versa)
- If forward speed increases VAR decreases (and vice versa)

Example: If the wind becomes stronger, it might be possible to increase the track spacing to allow a faster work rate. In order to maintain the correct VAR and dose, either the spray forward speed must be decreased or the flow rate must be increased. In order to achieve a faster work rate from the wider track spacing, the flow rate must be increased, rather than the forward speed being decreased.

How to measure the flow rate of aerial spray systems Electronic pesticide

pumps (collection technique):

Step 1. Calculate the required flow rate for each atomizer.

Step 2. Make sure that the aircraft engine is running so that the correct voltage is being supplied to the pump.

Step 3. Set the approximate flow rate based on tables in the user's handbook.

Step 4. Position a bucket under each atomizer. To prevent insecticide from squirting outside the collecting bucket, fasten plastic bags with a hole in the bottom over the atomizers.

Step 5. Put about 50 litres of insecticide into the spray tank in order to prime the pipework. Ensure that the pipes are full by pumping insecticide through the atomizers until air bubbles disappear (the pipework in an aircraft spray system can contain up to 30 litres of liquid). Return the collected insecticide to the sprayer tank.

Step 6. Put the buckets back under each atomizer, turn on the pump (*but not the atomizers*) and measure the volume of insecticide collected using a measuring cylinder. **Step 7.** Adjust the flow rate to bring it closer to the required rate calculated previously.

Repeat step 6 until this rate has been achieved to within about 5% error.

Step 8. When the required flow rate has been achieved, recheck it two more times to ensure that it is correct.

Windmill-driven pesticide pumps (loss technique while in flight):

Step 1. Calculate the desired flow rate (see page 12).

Step 2. Set the approximate flow rate based on tables in the user's handbook.

Step 3. Position a bucket under each atomizer. To prevent insecticide squirting outside the collecting bucket, fasten a plastic bag with a hole in the bottom over each atomizer. Put about 50 litres of insecticide into the spray tank in order to prime the pipework. Ensure that the pipes are full by pumping insecticide through the atomizers until air bubbles disappear (the pipework in an aircraft spray system can contain up to 30 litres of liquid). Return collected insecticide to the sprayer tank.

Step 4. Fill the spray tank to a known level with insecticide (either complete full or to a marked level).

Step 5. Take off and spray over the target area using normal spraying techniques for a specific number of minutes **(M)**.

Step 6. After landing, use a measuring cylinder to measure the amount of insecticide required to refill the spray tank to its original level. This is the number of litres emitted **(E)**.

Step 7. Calculate: **Flow rate (l/min) =** $\frac{\mathbf{E(l)}}{\text{M (mins)}}$

Step 8. Adjust the flow rate to bring it closer to the required rate calculated previously. Repeat steps 4-7 until this rate has been achieved to within about 5% error.

Step 9. When the required flow rate has been achieved, recheck it two more times to ensure that it is correct.

How to estimate work rate

A rough estimate of the work rate can be calculated from the formula:

$$\text{Work rate (ha/h)} = \frac{\text{Forward speed (km/h) x track spacing (m)}}{\mathbf{10}}$$

Note: this formula does not take into account the time required for turning at the end of each spray pass, which can be considerable for aircraft.

Typical track spacing in aerial control

A track spacing of **100 m** is generally used when spraying hopper bands, blocks of bands or settled swarms, milling swarms at roost and stratiform swarms using aircraft/helicopter.

6. Recording and reporting

Monitoring is very important in order to document the activities and to allow later analysis of the successes and failures of any campaigns. Most of the information concerning the control operations and their efficacy and the efficiency of the campaign are covered in the *FAO Spray Monitoring Form* (Annexure-I).

The form should be completed in order to include details on the location, rainfall, ecology and locusts. Duly filled forms should be returned to the National Locust Unit headquarters as soon as possible for review. Any problems (lack of protective clothing, overdosing, poor efficacy, non-target effects, etc.) can be noted on the form so they can be addressed later.

General flight report and job details produced by the DGPS, track guidance system and any flow control systems on board the aircraft/helicopter should be submitted to the Locust Control Unit Headquarters. Field staff recording the details of each control

Field staff recording the details of each control operation should use these forms

7. Cleaning, storing and disposal

Spray equipment should always be clean and ready to use. Properly dispose empty containers.

Always wear protective clothing while handling insecticides

Aerial spray system

- Drain unused insecticide back into the original containers
- To clean the sprayer, put some kerosene or diesel into it and spray it over the target area or waste ground, away from water bodies or supplies used by people or livestock; never dump this liquid in one place such as a pit
- Carry out any repair or required maintenance

- Wash the outside of the spray system with a cloth soaked in diesel or kerosene
- Cover the spray system (atomizer, variable restrictor unit and blades) with suitable protective covering to avoid any contamination (e.g. dust)

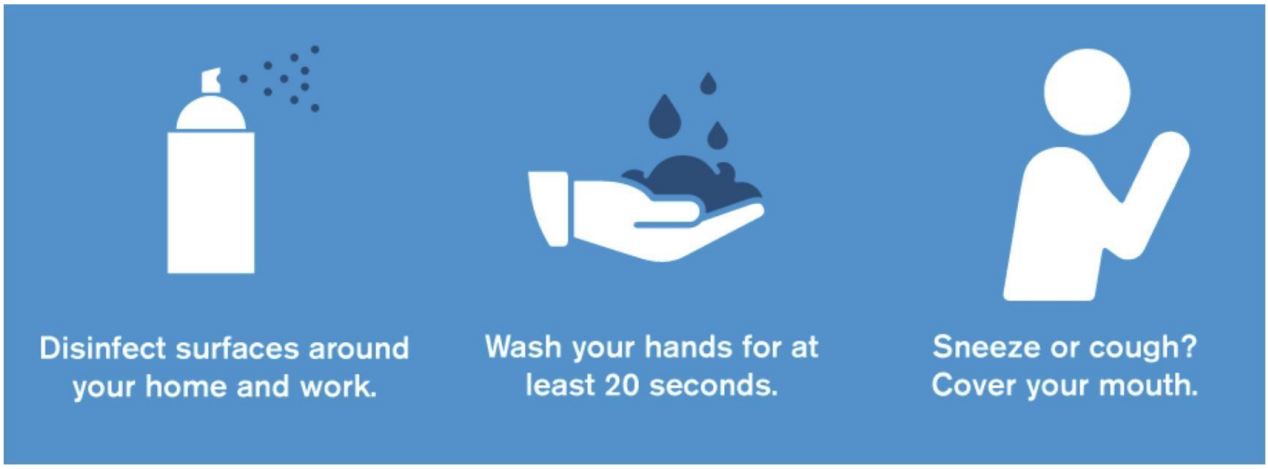
Insecticide storage

- Keep insecticide in original containers in a cool locked store to reduce deterioration caused by high temperatures
- Use older insecticides first (first-in-first-out system)

Disposal of empty insecticide containers

- Follow relevant provisions envisaged under the Insecticides Act 1968 and Rule 1971
- Clean empty insecticide containers three times inside and out with diesel or kerosene
- Collect the small volume of washings and dispose of by adding them to the insecticide in sprayer tanks during the next control operations or, if it is the end of the season, pour them into insecticide containers that are not full
- Never use empty containers for any other purpose than insecticides
- If they are to be recycled, they should be transported back to manufacturer
- Containers for disposal should be punctured, crushed and sent back to relevant authorities for appropriate disposal

Coronavirus: Safety Tips



Follow these steps to help keep you and others safe:

Download Arogya Setu App in your Mobile Phone

Stay home if you can and avoid any non-essential travel. Avoid social gatherings of more than 5 people.

Practice social distancing by keeping at least 6 feet — about two arm lengths — away from others if you must go out in public. Stay connected with loved ones through video and phone calls, texts and social media. Avoid close contact with people who are sick.

Wash your hands often with soap and water for at least 20 seconds, especially after being in a public place, or after blowing your nose, coughing or sneezing. If soap and water are not readily available, use a hand sanitizer with at least 60% alcohol.

Avoid touching your eyes, nose and mouth with unwashed hands.

Clean and disinfect household surfaces daily and high-touch surfaces frequently throughout the day. High-touch surfaces include phones, remote controls, counters, tabletops, doorknobs, bathroom fixtures, toilets, keyboards, tablets and bedside tables.

Cover your coughs and sneezes. Use a tissue to cover your nose and mouth and throw used tissues in a lined trash can. If a tissue isn't available, cough or sneeze into your elbow — not your hands. Wash your hands immediately.

Annexure-I

FAO SPRAY MONITORING FORM

Attach this form to the DL Survey and Control Form and submit both to the National Locust Unit in your country whenever control operations are carried out

(indicate appropriate information as required)

1	**CONTROL LOCATION**	**1**		**2**		**3**		**4**		**5**		**6**	
1-1	date												
1-2	name *(from DL Survey Form)*												
2	**VEGETATION DATA**												
2-1	vegetation type (Grass, Bushes, Trees, Crop)	G B T C		G B T C		G B T C		G B T C		G B T C		G B T C	
2-2	height (m)												
2-3	crop names and damage (%)												
3	**INSECTICIDE DATA**												
3-1	trade name												
3-2	concentration (g a.i./l or %)												
3-3	formulation (EC, ULV, Dust)	E U D		E U D		E U D		E U D		E U D		E U D	
3-4	expiry date												
3-5	is insecticide mixed with water or solvent?	Y N		Y N		Y N		Y N		Y N		Y N	
3-6	if yes, what solvent and mixing ratio												
4	**WEATHER CONDITIONS**												
	start and end of control operations:	start	end	start	end	start	end	start	end	start	end	start	end
4-1	time												
4-2	temperature (°C)												
4-3	relative humidity (%)												
4-4	wind speed (m/s)												
4-5	wind direction (degrees from N)												
4-6	spray direction (degrees from N)												
5	**SPRAY APPLICATION**												
5-1	sprayer type (Rotary, Airblast, ENS, Hydraulic, Other)	R A E H O		R A E H O		R A E H O		R A E H O		R A E H O		R A E H O	
5-2	sprayer operator (Pilot, Driver, Locust officer, Hired, Other)	P D L H O		P D L H O		P D L H O		P D L H O		P D L H O		P D L H O	
5-3	sprayer manufacturer												
5-4	sprayer model												
5-5	sprayer platform (Aerial, Vehicle, Handheld)	A V H		A V H		A V H		A V H		A V H		A V H	
5-6	date of last calibration												
5-7	atomizer height above ground (m)												
5-8	ROTARY SPRAYERS: speed setting (blade angle, pulley setting, no. batteries)												
5-9	speed of atomizer (rpm)												
5-10	flow rate setting (which nozzle or restrictor used)												
5-11	flow rate/atomizer (l/min)												
5-12	number of atomizers												
5-13	track spacing (m)												
5-14	BARRIERS ONLY: width and spacing (m)												
5-15	forward speed (km/h)												
5-16	AERIAL SPRAYING: support supplied	GP = ground party available RC = radio communication with aircraft TG = DGPS track guidance GP RC TG		GP RC TG		GP RC TG		GP RC TG		GP RC TG		GP RC TG	
5-17	ground marking (GPS, Flag, Mirror, Smoke, Vehicle, None)	G F M S V N		G F M S V N		G F M S V N		G F M S V N		G F M S V N		G F M S V N	
6	**CONTROL EFFICACY**												
6-1	locust mortality (% dead)												
6-2	time after treatment (hours)												
6-3	method of mortality estimation (Quadrats, Target size, Visual, Cages, Other)	Q T V C O		Q T V C O		Q T V C O		Q T V C O		Q T V C O		Q T V C O	
7	**SAFETY AND ENVIRONMENT**												
7-1	protective clothing: what did the operator wear?	G = goggles M = mask L = gloves O = overalls B = boots G M L O B		G M L O B		G M L O B		G M L O B		G M L O B		G M L O B	
7-2	was soap and water available?	Y N		Y N		Y N		Y N		Y N		Y N	
7-3	who was informed of spraying? (Farmer, Nomad, Villager, Official, Beekeeper)	F N V O B		F N V O B		F N V O B		F N V O B		F N V O B		F N V O B	
7-4	effect on non-target organisms	Y N		Y N		Y N		Y N		Y N		Y N	
7-5	if yes, what												
7-6	details of anyone who felt unwell or if other problems were encountered:												

Annexure-3.2.1

Templates/Proforma for the scrutiny of the applications of registration of pesticides in various category.

Content

1. **Chemistry**
   - **1.1. Template for 9(3) Applications**
     - **1.1.1. TI……2-11**
     - **1.1.2. FI……12-20**
     - **1.1.3. TIM……21-30**
     - **1.1.4. FIM……31-40**
   - **1.2. Template for 9(4) Applications……41-47**
   - **1.3. Template for Biopesticides……48-55**
2. **Bio efficacy (Herbicide unit)**
   - **2.1. Template for 9 (3) herbicide application**
     - **2.1.1. TI (NS)……57-69**
     - **2.1.2. TIM vs TI One Crop……70-76**
     - **2.1.3. TI / TIM ……77-80**
     - **2.1.4. FIM……81-94**
     - **2.1.5. PGR ……95-105**
     - **2.1.6. Template for Endorsement Application……106-117**
3. **Toxicology**
   - **3.1. Template for 9(3b) Chemical pesticides Applications**
     - **3.1.1. TI……119-141**
     - **3.1.2. TIM……142-164**
     - **3.1.3. FI……165-171**
     - **3.1.4. FIM……172-178**
   - **3.2. Template for 9(3) Chemical pesticides Applications**
     - **3.2.1. TI-New molecule……179-210**
     - **3.2.2. TI-New source……211-224**
     - **3.2.3. TIM New Molecule……225-255**
     - **3.2.4. TIM vs TI……256-262**
     - **3.2.5. TIM vs FI-WRT……263-265**
     - **3.2.6. FI……266-272**
     - **3.2.7. FIM……273-279**
   - **3.3. Template for 9(3b) Botanical pesticides Applications**

3.3.1. TI / TIM……280-286

3.3.2. FI/FIM……287-293

3.4. Template for 9(3) botanical pesticides Applications

3.4.1. TI /TIM……294-320

3.4.2. FI/FIM……321-327

3.5. Template for Biopesticide Applications……328-334

3.6. Template for Pheromones/ Semio-chemical pesticides Applications……335-380

3.7. Template for 9(4) TIM chemical pesticides applications……381-383

3.8. Template for Export pesticides applications……384-385

3.9. Template for other categories (MUP/ Household/ Public Health)……386-389

4. Packaging…… ……391-417

CHEMISTRY

COMPREHENSIVE TECHNICAL DOSSIER TEMPLATE

FOR SCRUTINY, CHEMICAL VERIFICATION & REGISTRATION

UNDER SECTION 9(3)

(Chemistry Division)

(Technical Import/ Technical Import (New Source))

COVER PAGE

DATA ON TECHNICAL

Particulars	Details
Category of Application	☐ 9(3) TI ☐ 9(3) TI (New source)
Name of Product	[Insert Product Name]
Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Number	[Insert]
Applicant Company	[Insert]
Date of Submission	[DD/MM/YYYY]

#Note:- In case of 9(3) TI (New source) application, the data shall be submitted as per extant RC guideline.

SECTION 1. GENERAL INFORMATION

Parameter	Details
Name of Applicant	[Insert]
Address of Applicant	[Insert]
Status of Insecticide in Schedule	[Insert]
Applicable BIS Specifications	[Insert BIS Number if available]
Type of Product	[Insert]
Details of source of supply of Technical with geo tagging details	**[Insert]**
Manufacturing/factory address of the manufacturer with geo tagging details	**[Insert]**
Office address of the manufacturer with geo tagging details	**[Insert]**
Manufacturing license details	**[Insert]**
Registration Certificate No. from Designated National Authority and validity	**[Insert]**
In case the supplies are made through a supplier, Mention supplier Name	**[Insert]**

Address of the supplier with geo tagging details, if proposed	**[Insert]**
Consent letter details from Manufacturer/source and supplier if any	**[Insert]**
RTT permit Details	**[Insert]**

In case of TI (New source), the following documents is required

Documents such as registration certificate/ certificate of DNA giving details CC, Manufacturing license or any other approval under any Govt. regulation will be acceptable to support that manufacturer is actual producer.

- For proof of chemical Composition (Where Chemical Composition is not published on the certificate of registration) the documents may be accepted: Such as apostille/notarized copy of application form containing Chemical Composition submitted to the regulatory authority in the exporting country with a undertaking by the source that their source can be cancelled without any notice, if their Chemical Composition is found different from the registered one at the time of registration.

SECTION 2.

CHEMICAL IDENTITY OF ACTIVE INGREDIENT

2.1 Identity Details

Insert more row or column, if required

Parameter	Details
ISO Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Name	[Insert]
CAS Number	[Insert]
Molecular Formula	[Insert]
Molecular Weight	[Insert]
Structural Formula	[Insert Structure]
Physical Appearance	[Insert]
Odour	[Insert]
Other parameters (If any)	**[Insert]**

SECTION 3.

CHEMICAL COMPOSITION

3.1 Composition of Technical Product

Insert more row or column, if required

S. No.	Name	Content [Insert % or w/w or v/v or w/v]
1.	[Insert name]	[insert xyz]
2.	[Insert name]	[insert abc]
3.	[Insert name]	[insert jkl]
4.	[Insert name]	[insert rst]
5.	[Insert name]	[insert def]
6.	Total	100.00 as per unit

SECTION 4.

IMPURITY PROFILE

4.1 Details of Impurities:

All identifiable impurities & unidentifiable impurities with details.

Insert more row or column, if required

Impurity Name	Chemical Name	CAS No.	Structural Formula	Molecular weight	Origin	Maximum Limit	Method of Analysis
[Insert]	[Insert]	[Insert]	[Insert]		[Insert]	[Insert]	[Insert]

SECTION 5. RAW MATERIAL DETAILS

5.1 Raw Material Specifications

Insert more row or column, if required

S. No.	Name of Raw Material	IUPAC Name	CAS No.	Purity (%)	Molecular Formula	Molecular Weight	Source of Supply	BIS/Analytical Specification
1	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

Attach supporting documents such as COA, SDS/MSDS, and technical specification sheets wherever applicable.

SECTION 6.

MANUFACTURING PROCESS

6.1 Manufacturing Chemistry

Provide detailed manufacturing process including:

6.2 Stepwise Manufacturing Process

Insert more row or column, if required

Step	Name of the raw material/reactants used with their respective purity	Weight of the Raw material (gm)	Reaction conditions (Temperature)	Approximate Time required (Hrs.)	Intermediate/substance/Product name	Chemical formula of Intermediate/substance/Product name	Chemical Name of the Impurities formed during the reaction	Yield (gm)
Step-I	[Insert name with purity]	[Insert value]			[Insert name]		[Insert name]	
	[Insert name with purity]	[Insert value]					[Insert name]	
	[Insert name with purity]	[Insert value]					[Insert name]	
Step-II	[Insert name with purity]	[Insert value]			[Insert name]		[Insert name]	
	[Insert name with purity]	[Insert value]					[Insert name]	
	[Insert name with purity]	[Insert value]					[Insert name]	
Final step	[Insert name with purity]	[Insert value]			[Insert name] [Insert name]	[Insert name] [Insert name]	[Insert name]	
	[Insert name with purity]	[Insert value]					[Insert name]	

6.3 Chemical Reaction Scheme

Provide:

6.4 Effluent Treatment Details

Parameter	Details
Type of Effluent	[Insert]
Treatment Method	[Insert]
Neutralization Method	[Insert]
Solvent Recovery System	[Insert]
Disposal Method	[Insert]

Insert more row or column, if required

SECTION 7. FIVE BATCH ANALYSIS DATA

Five representative commercial-scale batches shall be analysed.

Notes. (i) Insert data in the same order as the table of names, codes and structures. (ii) Insert additional rows as required, insert additional columns for additional batch data. (iii) Do not insert "not detected", insert < LOD[1].]

Code or name	Batch 1 [insert date] g/kg or w/w or w/v or g/L	Batch 2 [insert date] g/kg or w/w or w/v or g/L	Batch 3 [insert date] g/kg or w/w or w/v or g/L	Batch 4 [insert date] g/kg or w/w or w/v or g/L	Batch 5 [insert date] g/kg or w/w or w/v or g/L	Mean g/kg or w/w or w/v or g/L
[insert name AI]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]
[insert name impurity]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]
[insert name impurity]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]
[Insert name "unknowns", if appropriate]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]
Total, g/kg or w/w or w/v or g/L	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	-

[1] If an impurity is analysed but not detected (< LOD), the value of the LOD (limit of detection) has to be provided.

SECTION 8.

PHYSICO-CHEMICAL PROPERTIES

8.1 Technical Grade Active Ingredient

Parameter	Specification	Method
Appearance	[Insert]	BIS/CIPAC no. [Insert]
Melting Point	[Insert]	BIS no. [Insert]
Boiling Point	[Insert]	BIS no. [Insert]
Density	[Insert]	BIS no. [Insert]
Vapour Pressure	[Insert]	BIS no. [Insert]
Solubility in Water	[Insert]	BIS no. [Insert]
Solubility in Organic Solvents	[Insert]	BIS no. [Insert]
pH	[Insert]	BIS no. [Insert]
Flash Point	[Insert]	BIS no. [Insert]
Any other parameter (if any)	[Insert]	

Insert more row or column, if required

SECTION 9.

PRODUCT SPECIFICATION (In BIS FORMAT)

Technical Specification

Parameter	BIS Requirement	Observation	Method	Result
Active Ingredient	[Insert]	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]	[Insert]
Acidity/Alkalinity	[Insert]	[Insert]	[Insert]	[Insert]
Melting point range	[Insert]	[Insert]	[Insert]	[Insert]
Acetone/other organic solvent/water insoluble	[Insert]	[Insert]	[Insert]	[Insert]
Any other parameter (if any)	[Insert]	[Insert]	[Insert]	[Insert]

Insert more row or column, if required

SECTION 10. METHOD OF ANALYSIS

10.1 Active Ingredient Analysis

Type of Method of Analysis-Instrumental or Conventional.

If Instrumental, then provide details of Chromatographic Parameters-

HPLC Conditions

Parameter	Details
Column	[Insert]
Mobile Phase	[Insert]
Flow Rate	[Insert]
Injection Volume	[Insert]
Detector	[Insert]
Detection Wavelength	[Insert]
Run Time	[Insert]
Retention Time	[Insert]
Any other parameter (if applicable)	[Insert]

Insert more row or column, if required

Or GLC Conditions

Parameter	Details
Column Type	[Insert]
Carrier Gas	[Insert]
Detector	[Insert]
Injector Temperature	[Insert]
Detector Temperature	[Insert]
Any other parameter (if applicable)	[Insert]

Insert more row or column, if required

Insert any other method with details (If applicable)

SECTION 11.

Analytical Test Report and Identification & Quantification of Identifiable Impurities:

The study has been carried out by **M/s** _____
(Name of Laboratory/Institute) under the following details:

Particulars	Details
Product Name	
Active Ingredient(s)	
Batch Number(s)	
Manufacturing Date	
Date of expiry	
Proposed Shelf Life	

Study Details

Particulars	Details
Study Title	
Study Number	

Study Initiation Date	
Study Completion Date	
NABL Accreditation No.	
GLP Certification No., (if applicable)	
Whether NABL/GLP Accredited During the Period of Study	Yes / No
Validity of Accreditation During Study Period	

Replicates	**Weight of R/Std in mg**	**Weight of sample in mg**	**Area of the Active in R/Std. solution**	**Area of the Active in sample solution**	**% purity of the Ref. standard**	**Active content % by mass**	**Average content % by mass**	**Mean Avg. content % by mass**
A1B1	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
A2B2	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	

The complete ATR with all supporting instrumental analytical data including chromatograms, IR, UV, and NMR spectra etc.

Insert more row or column, if required

SECTION 12.

SHELF-LIFE & STORAGE STABILITY

12.1 Shelf-life Claim

[Insert shelf-life]-

[Regular/provisional]-

The study has been carried out by **M/s** _____
(Name of Laboratory/Institution) under the following conditions and details:

Product Details

Particulars	**Details**
Product Name	
Active Ingredient(s)	
Concentration(result)	
Batch Number(s)	
Manufacturing Date	
Packaging Type	

Proposed Shelf Life	
Others	

Study Details

Particulars	Details
Study Title	Shelf-Life / Stability Study
Study No.	
Study Start Date	
Study Completion Date	
Laboratory Name & Address	
NABL Accreditation Status	Yes / No
NABL/GLP Accredited During Study Period	Yes / No
Accreditation Validity Period	
Others	

Storage Conditions

A. Real-Time Stability Conditions

Parameter	Condition
Duration	[Insert]
Sampling Intervals	[Insert]
Three different Agro-Climatic Zones	[Insert]
Meteorological Data Authentic Meteorological data for corresponding study period from the three study locations.	

B. Accelerated Stability Conditions

Parameter	Condition
Temperature	[Insert]
Relative Humidity	[Insert]
Duration	[Insert]
Sampling Intervals	[Insert]

12.2 Storage Stability Data Observations-

Parameter	Initial	Final	Acceptance Criteria
Active Content	[Insert]	[Insert]	[Insert]
pH	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]
Melting point range	[Insert]	[Insert]	[Insert]
Acetone/other organic solvent/water insoluble	[Insert]	[Insert]	[Insert]
Physical Stability	[Insert]	[Insert]	[Insert]
Other parameters (as per submitted Product specification/ BIS)	[Insert]	[Insert]	[Insert]

Insert more row or column, if required

12.3 Storage Stability Study Details

Duration	Locations	Weight of R/Std in mg	Weight of sample in mg	Area of the active in R/Std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
[insert zero month]	Location-I	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-II	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-III	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[insert month wise details]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

In case of Accelerated Storage Stability data submitted: -

Day	Temperature	Relative Humidity (RH)
01	[Insert]	[Insert]
[Insert] days	[Insert]	[Insert]

Insert more row or column, if required

Duration	Weight of R/Std in mg	Weight of sample in mg	Area of the active in R/Std solution	Area of the active in sample solution	% purity of the R/standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
[at day 01]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[at the day of study completion]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

Insert more row or column, if required

COMPREHENSIVE TECHNICAL DOSSIER TEMPLATE

FOR SCRUTINY, CHEMICAL VERIFICATION & REGISTRATION

UNDER SECTION 9(3)

(Chemistry Division)

(Formulation Import)

COVER PAGE

DATA ON FORMULATION

Particulars	**Details**
Category of Application	☐ 9(3) FI
Name of Product	[Insert Product Name]
Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Number	[Insert]
Formulation Type	[Insert BIS/CIPAC code]
Applicant Company	[Insert]
Manufacturing Site	[Insert]
Date of Submission	[DD/MM/YYYY]

$ In case of 9(3B) FI application, the data shall be submitted as per extant RC guideline.

SECTION 1. GENERAL INFORMATION

Parameter	**Details**
Name of Applicant	[Insert]
Address of Applicant	[Insert]
Status of Insecticide in Schedule	[Insert]
Applicable BIS Specifications	[Insert BIS Number if available]
Applicable RC Guidelines	[Insert]
Type of Product	[Insert]
Details of source of supply of Technical	**[Insert]**
Registration Certificate No. (If technical is registered)	**[Insert]**
Registration Certificate No. from Designated National Authority and validity	**[Insert]**
In case the supplies are made through a supplier, Mention supplier Name	**[Insert]**
Consent letter details from Manufacturer/source and supplier if any	**[Insert]**

RTT permit Details	[Insert]

* Duly signed copy of Certificate of registration/notarised consent letter from the Source of supply of Technical shall be attached.

SECTION 2. CHEMICAL IDENTITY OF ACTIVE INGREDIENT

2.1 Identity Details

Parameter	Details
ISO Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Name	[Insert]
CAS Number	[Insert]
Molecular Formula	[Insert]
Molecular Weight	[Insert]
Structural Formula	[Insert Structure]
Chemical Class	[Insert]
Physical Appearance	[Insert]
Odour	[Insert]

SECTION 3. CHEMICAL COMPOSITION

3.1 Composition of Formulation (Applicable for FI)

S.No.	Ingredient	Function	Concentration (% w/w or w/v or g/L)
1	Active Ingredient	[Insert]	[Insert]
2	Solvent	[Insert]	[Insert]
3	Emulsifier	[Insert]	[Insert]
4	Stabilizer	[Insert]	[Insert]
5	Adjuvant	[Insert]	[Insert]
6	Others	[Insert]	[Insert]
	Total		[Insert]
Density (If applicable) :			

Insert more row or column, if required

Note: Adjuvants shall be disclosed by complete chemical identity and not by trade code

SECTION 4. RAW MATERIAL DETAILS

4.1 Raw Material Specifications

Insert more row or column, if required

S. No.	Name of Raw Material	IUPAC Name	CAS No.	Purity (%)	Molecular Formula	Molecular Weight	Source of Supply	BIS/Analytical Specification
1	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

SECTION 5. MANUFACTURING PROCESS

5.1 Manufacturing Chemistry

Provide detailed manufacturing process including:

5.2 Stepwise Manufacturing Process

Insert more row or column, if required

Step	Name of the raw material/reactants used with their respective purity	Weight of the Raw material (gm)	Equipment used	Reaction conditions (Temperature)	Approximate Time required (Hrs.)	Intermediate/substance/Product name	Chemical formula of Intermediate/substance/Product name	Yield (gm)
Step-I	**[Insert name with purity]**	**[Insert value]**	**[Insert**			**[Insert name]**		
	[Insert name with purity]	**[Insert value]**	**[Insert**					
	[Insert name with purity]	**[Insert value]**	**[Insert**					
Step-II	**[Insert name with purity]**	**[Insert value]**	**[Insert**			**[Insert name]**		
	[Insert name with purity]	**[Insert value]**	**[Insert**					
	[Insert name with purity]	**[Insert value]**	**[Insert**					
Final step	**[Insert name with purity]**	**[Insert value]**	**[Insert**			**[Insert name] [Insert name]**	**[Insert name] [Insert name]**	
	[Insert name with purity]	**[Insert**	**[Insert**					

		valu e]						

5.3 Chemical Reaction Scheme

5.4 Effluent Treatment Details

Parameter	Details
Type of Effluent	[Insert]
Treatment Method	[Insert]
Neutralization Method	[Insert]
Solvent Recovery System	[Insert]
Disposal Method	[Insert]

Insert more row or column, if required

SECTION 8. PHYSICO-CHEMICAL PROPERTIES OF PRODUCT ACTIVE INGREDIENT AND ITS ADJUVANTS

8.1 Physico-Chemical Properties of Active Ingredient: -

Parameter	Specification	Method
Appearance	[Insert]	BIS/CIPAC no. [Insert]
Melting Point	[Insert]	BIS no. [Insert]
Boiling Point	[Insert]	BIS no. [Insert]
Density	[Insert]	BIS no. [Insert]
Vapour Pressure	[Insert]	BIS no. [Insert]
Solubility in Water	[Insert]	BIS no. [Insert]
Solubility in Organic Solvents	[Insert]	BIS no. [Insert]
pH	[Insert]	BIS no. [Insert]
Flash Point	[Insert]	BIS no. [Insert]
Any other parameter (if applicable)	[Insert]	

Insert more row/column/table, if required

Adjuvants / Inert Ingredients

Physico-Chemical Properties of Each Adjuvant -

Name of Adjuvant: [Insert Name]

Name of Adjuvant 1 : [Insert Name]		
Parameter	**Description**	**Method**
Appearance	[Insert]	[Insert]
Colour	[Insert]	[Insert]
Odour	[Insert]	[Insert]
Density / Specific Gravity	[Insert]	[Insert]
Solubility	[Insert]	[Insert]

pH	[Insert]	[Insert]
CAS No.	[Insert]	[Insert]
Function in Formulation	[Insert]	[Insert]
Molecular Formula	[Insert]	[Insert]
Chemical Name / IUPAC Name	[Insert]	[Insert]
Molecular Weight	[Insert]	[Insert]
Purity (%)	[Insert]	[Insert]
Any Other required Parameter (if any)	[Insert]	[Insert]

 Add additional rows as required for all adjuvants present in the formulation.

 Attach supporting documents such as COA, SDS/MSDS, and technical specification sheets wherever applicable.

SECTION 9. PRODUCT SPECIFICATION (In BIS FORMAT)

Formulation Specification

Parameter	BIS Requirement	Method
Active Ingredient	[Insert]	[Insert]
Moisture	[Insert]	[Insert]
pH	[Insert]	[Insert]
Persistent foam	[Insert]	[Insert]
Wet sieve	[Insert]	[Insert]
Any other required parameter (if any)	[Insert]	[Insert]

SECTION 10. Method of Analysis

10.1 Active Ingredient Analysis-

Type of Method of Analysis-Instrumental or Conventional.

If Instrumental, then provide details of Chromatographic Parameters-

HPLC Conditions-

Parameter	Details
Column	[Insert]
Mobile Phase	[Insert]
Flow Rate	[Insert]
Injection Volume	[Insert]
Detector	[Insert]
Detection Wavelength	[Insert]
Run Time	[Insert]
Retention Time	[Insert]
Any other parameter (if applicable)	[Insert]

Or GLC Conditions-

Parameter	Details
Column Type	[Insert]
Carrier Gas	[Insert]
Detector	[Insert]
Injector Temperature	[Insert]
Detector Temperature	[Insert]
Any other parameter (if applicable)	[Insert]

Insert more row or column, if required

Insert any other method with details (If applicable)

SECTION 11. ANALYTICAL TEST REPORT

The study has been carried out by **M/s** _____
(Name of Laboratory/Institute) under the following details:

Particulars	Details
Product Name	
Active Ingredient(s)	
Batch Number(s)	
Manufacturing Date	
Date of expiry	

Study Details

Particulars	Details
Study Title	
Study Number	
Study Initiation Date	
Study Completion Date	
NABL Accreditation No.	
GLP Certification No., (if applicable)	
Whether NABL/GLP Accredited During the Period of Study	Yes / No
Validity of Accreditation During Study Period	

Replicates	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
A1B1	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**

A2B2	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	

Parameter	BIS Requirement	Method	Result/ Observations in ATR
Active Ingredient	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]
pH	[Insert]	[Insert]	[Insert]
Persistent foam	[Insert]	[Insert]	[Insert]
Wet sieve	[Insert]	[Insert]	[Insert]
Any other parameter (if applicable)	[Insert]	[Insert]	[Insert]

SECTION 12. SHELF-LIFE & STORAGE STABILITY

12.1 Shelf-life Claim

[Insert shelf-life]

The study has been carried out by **M/s** _____
(Name of Laboratory/Institution) under the following conditions and details:

Product Details

Particulars	Details
Product Name	
Active Ingredient(s)	
Concentration (Results)	
Batch Number(s)	
Manufacturing Date	
Packaging Type	
Proposed Shelf Life	
Others	

Study Details

Particulars	Details
Study Title	Shelf-Life / Stability Study
Study No.	
Study Start Date	
Study Completion Date	
Laboratory Name & Address	
NABL Accreditation Status	Yes / No
NABL/GLP Accredited During Study Period	Yes / No
Accreditation Validity Period	
Others	

Storage Conditions

A. Real-Time Stability Conditions

Parameter	Condition
Duration	[Insert]
Sampling Intervals	[Insert]
Three different Agro-Climatic Zones	[Insert]
Meteorological Data Authentic Meteorological data for corresponding study period from the three study locations.	

B. Accelerated Stability Conditions

Parameter	Condition
Temperature	[Insert]
Relative Humidity	[Insert]
Duration	[Insert]
Sampling Intervals	[Insert]

12.2 Storage Stability Data

Parameter	Initial	Final	Acceptance Criteria
Active Content	[Insert]	[Insert]	[Insert]
pH	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]
Physical Stability	[Insert]	[Insert]	[Insert]
Other parameters (as per submitted Product specification/ BIS)	[Insert]	[Insert]	[Insert]

#(Insert columns for more parameters)

12.3 Storage Stability Study Details

Duration	Locations	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
[insert zero month]	**Location-I**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**

	Location-II	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-III	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[insert month]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

In case of Accelerated Storage Stability data submitted:-

Day	Temperature	Relative Humidity (RH)
01	[Insert]	[Insert]
[Insert] days	[Insert]	[Insert]

Duration	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
[at day 01]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[at the day of study completion]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	

COMPREHENSIVE TECHNICAL DOSSIER TEMPLATE

FOR SCRUTINY, CHEMICAL VERIFICATION & REGISTRATION

UNDER SECTION 9(3)

(Chemistry Division)

(Technical Indigenous Manufacture)

DATA ON TECHNICAL

Particulars	Details
Category of Application	☐ 9(3) TIM ☐ 9(3b) TIM# ☐ 9(3) TIM vs TI ☐ 9(3) TIM vs FI ☐9(3) TIM vs FIM
Name of Product	[Insert Product Name]
Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Number	[Insert]
Applicant Company	[Insert]
Date of Submission	[DD/MM/YYYY]

#Note:- In case of 9(3B) TIM application, the data shall be submitted as per extant RC guideline.

SECTION 1. GENERAL INFORMATION

Parameter	Details
Name of Applicant	[Insert]
Address of Applicant	[Insert]
Address of Manufacturing Site	[Insert]
Status of Insecticide in Schedule	[Insert]
Applicable BIS Specifications	[Insert BIS Number if available]
Applicable RC Guidelines	[Insert]
Type of Product	[Insert]
Establishment of chemical equivalence with the earlier registrant with the reference of Registration committee meeting, in case of ☐ 9(3) TIM vs TI ☐ 9(3) TIM vs FI ☐9(3) TIM vs FIM	[Insert earlier 9(3) registrant name] [Insert reference of RC meeting]

SECTION 2. CHEMICAL IDENTITY OF ACTIVE INGREDIENT

2.1 Identity Details

Insert more row or column, if required

Parameter	Details
ISO Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Name	[Insert]
CAS Number	[Insert]
Molecular Formula	[Insert]
Molecular Weight	[Insert]
Structural Formula	[Insert Structure]
Chemical Class	[Insert]
Physical Appearance	[Insert]
Odour	[Insert]
Other parameters (If applicable)	

SECTION 3. CHEMICAL COMPOSITION

3.1 Composition of Technical Product

Insert more row or column, if required

S. No.	Component Name	Content [Insert % or w/w or v/v or w/v or g/L]
7.	[Insert name]	[insert xyz]
8.	[Insert name]	[insert abc]
9.	[Insert name]	[insert jkl]
10.	[Insert name]	[insert rst]
11.	[Insert name]	[insert def]
12.	Total	100.00 or (1000.00, if g/L)
Density (If applicable) :		

SECTION 4. IMPURITY PROFILE (Particularly for TIM)

4.1 Details of Impurities:

All impurities ≥1 g/kg and toxicologically relevant impurities below 1 g/kg shall be identified.

Insert more row or column, if required

Impurity Name	Chemical Name	CAS No.	Structural Formula	Molecular weight	Origin	Maximum Limit	Method of Analysis

[Insert]	[Insert]	[Insert]	[Insert]		[Insert]	[Insert]	[Insert]

SECTION 5. RAW MATERIAL DETAILS

5.1 Raw Material Specifications

Insert more row or column, if required

S.No.	Name of Raw Material	IUPAC Name	CAS No.	Purity (%)	Molecular Formula	Molecular Weight	Source of Supply	BIS/Analytical Specification
1	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

Attach supporting documents such as COA, SDS/MSDS, and technical specification sheets wherever applicable.

SECTION 6. MANUFACTURING PROCESS

6.1 Manufacturing Chemistry

Provide detailed manufacturing process including:

- Reaction chemistry
- Stoichiometry
- Intermediate formation
- Purification steps
- Recovery process
- Solvent recovery
- Yield optimization
- Process controls

6.2 Stepwise Manufacturing Process

Insert more row or column, if required

Step	Name of the raw material/reactants used with their respective purity	Weight of the Raw material (gm)	Reaction conditions (Temperature)	Approximate Time required (Hrs.)	Intermediate/substance/Product name	Chemical formula of Intermediate/substance/Product name	Chemical Name of the Impurities formed during the reaction	Yield (gm)
Step-I	**[Insert name with purity]**	**[Insert**			**[Insert name]**		**[Insert name]**	

		value]						
	[Insert name with purity]	**[Insert value]**					**[Insert name]**	
	[Insert name with purity]	**[Insert value]**					**[Insert name]**	
Step-II	**[Insert name with purity]**	**[Insert value]**			**[Insert name]**		**[Insert name]**	
	[Insert name with purity]	**[Insert value]**					**[Insert name]**	
	[Insert name with purity]	**[Insert value]**					**[Insert name]**	
Final step	**[Insert name with purity]**	**[Insert value]**			**[Insert name] [Insert name]**	**[Insert name] [Insert name]**	**[Insert name]**	
	[Insert name with purity]	**[Insert value]**					**[Insert name]**	

6.3 Chemical Reaction Scheme

Provide:

- Balanced chemical equations
- Structural reaction pathway
- Side reactions
- By-product formation
- Degradation pathway

Mandatory Attachments:
☐ Reaction mechanism
☐ Structural formulae
☐ Flow sheet diagram

6.4 Effluent Treatment Details

Parameter	Details
Type of Effluent	[Insert]
Treatment Method	[Insert]
Neutralization Method	[Insert]
Solvent Recovery System	[Insert]
Disposal Method	[Insert]

Insert more row or column, if required

SECTION 7. FIVE BATCH ANALYSIS DATA

Five representative commercial-scale batches shall be analyzed.

Notes. (i) Insert data in the same order as the table of names, codes and structures. (ii) Insert additional rows as required, insert additional columns for additional batch data. (iii) Do not insert "not detected", insert < LOD[2].]

Code or name	Batch 1 [insert date] g/kg or w/w or w/v or g/L	Batch 2 [insert date] g/kg or w/w or w/v or g/L	Batch 3 [insert date] g/kg or w/w or w/v or g/L	Batch 4 [insert date] g/kg or w/w or w/v or g/L	Batch 5 [insert date] g/kg or w/w or w/v or g/L	Mean g/kg or w/w or w/v or g/L	Standard deviation g/kg) or w/w or w/v or g/L	Mean - 3 SD (for a.i.) Mean + 3 SD (for impurities)	Manufacturing QC limit (technical specification), g/kg or w/w or w/v or g/L
[insert name AI]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]
[insert name impurity]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]
[insert name impurity]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]
[Insert name "unknowns", if appropriate]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]
Total, g/kg or w/w or w/v or g/L	[insert value]	[insert value]	[insert value]	[insert value]	[insert value]	-	-	-	-

[1] If an impurity is analysed but not detected (< LOD), the value of the LOD (limit of detection) has to be provided.

SECTION 8. PHYSICO-CHEMICAL PROPERTIES

8.1 Technical Grade Active Ingredient

Parameter	Specification	Method
Appearance	[Insert]	BIS/CIPAC no. [Insert]
Melting Point	[Insert]	BIS no. [Insert]
Boiling Point	[Insert]	BIS no. [Insert]
Density	[Insert]	BIS no. [Insert]

Vapour Pressure	[Insert]	BIS no. [Insert]
Solubility in Water	[Insert]	BIS no. [Insert]
Solubility in Organic Solvents	[Insert]	BIS no. [Insert]
pH	[Insert]	BIS no. [Insert]
Flash Point	[Insert]	BIS no. [Insert]
Any other parameter (if applicable)	[Insert]	

Insert more row or column, if required

SECTION 9. PRODUCT SPECIFICATION (BIS FORMAT)

Technical Specification

Parameter	BIS Requirement	Observation	Method	Result
Active Ingredient	[Insert]	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]	[Insert]
Acidity/Alkalinity	[Insert]	[Insert]	[Insert]	[Insert]
Any other parameter (if applicable)	[Insert]	[Insert]	[Insert]	[Insert]

Insert more row or column, if required

SECTION 10. METHOD OF ANALYSIS

10.1 Active Ingredient Analysis

HPLC Conditions

Parameter	Details
Column	[Insert]
Mobile Phase	[Insert]
Flow Rate	[Insert]
Injection Volume	[Insert]
Detector	[Insert]
Detection Wavelength	[Insert]
Run Time	[Insert]
Retention Time	[Insert]
Any other parameter (if applicable)	[Insert]

Insert more row or column, if required

GC Conditions

Parameter	Details
Column Type	[Insert]
Carrier Gas	[Insert]
Detector	[Insert]

Injector Temperature	[Insert]
Detector Temperature	[Insert]
Any other parameter (if applicable)	[Insert]

Insert more row or column, if required

Insert any other method (If applicable)

SECTION 11. Analytical Test Report and Identification & Quantification of Identifiable Impurities:

The study has been carried out by **M/s** _____ **(Name of Laboratory/Institute)** under the following details:

Particulars	Details
Product Name	
Active Ingredient(s)	
Batch Number(s)	
Manufacturing Date	
Date of expiry	
Proposed Shelf Life	

Study Details

Particulars	Details
Study Title	
Study Number	
Study Initiation Date	
Study Completion Date	
NABL Accreditation No.	
GLP Certification No., (if applicable)	
Whether NABL/GLP Accredited During the Period of Study	Yes / No
Validity of Accreditation During Study Period	

Replicates	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
A1B1	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**
A2B2	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**

The complete ATR with all supporting instrumental analytical data including chromatograms, IR, UV, and NMR spectra forms an integral part of this report.

Insert more row or column, if required

SECTION 12. SHELF-LIFE & STORAGE STABILITY

12.1 Shelf-life Claim

[Insert shelf-life]-

[complete/provisional]-

The study has been carried out by **M/s** _____
(Name of Laboratory/Institution) under the following conditions and details:

Product Details

Particulars	Details
Product Name	
Active Ingredient(s)	
Concentration	
Batch Number(s)	
Manufacturing Date	
Packaging Type	
Proposed Shelf Life	
Others	

Study Details

Particulars	Details
Study Title	Shelf-Life / Stability Study
Study No.	
Study Start Date	
Study Completion Date	
Laboratory Name & Address	
NABL Accreditation Status	Yes / No
NABL/GLP Accredited During Study Period	Yes / No
Accreditation Validity Period	
Others	

Storage Conditions

A. Real-Time Stability Conditions

Parameter	Condition
Duration	[Insert]
Sampling Intervals	[Insert]
Three different Agro-Climatic Zones	[Insert]
Meteorological Data Duly signed Meteorological data of the study locations during the study period is required to attached for reference.	

B. Accelerated Stability Conditions

Parameter	Condition
Temperature	[Insert]
Relative Humidity	[Insert]
Duration	[Insert]
Sampling Intervals	[Insert]

12.2 Storage Stability Data

Parameter	Initial	Final	Acceptance Criteria
Active Content	[Insert]	[Insert]	[Insert]
pH	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]
Physical Stability	[Insert]	[Insert]	[Insert]
Other parameters (as per submitted Product specification/ BIS)	[Insert]	[Insert]	[Insert]

Insert more row or column, if required

12.3 Storage Stability Study Details

Duration	Locations	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
[insert zero month]	Location-I	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-II	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-III	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[insert month]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

In case of Accelerated Storage Stability data submitted:-

Day	Temperature	Relative Humidity (RH)
01	[Insert]	[Insert]
[Insert] days	[Insert]	[Insert]

Insert more row or column, if required

Duration	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
[at day 01]	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**
[at the day of study completion]	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**

Insert more row or column, if required

SECTION 13. DECLARATION

We hereby declare that:

1. The information submitted is complete and authentic.
2. The manufacturing process disclosed represents actual commercial production.
3. All impurities known to the applicant have been disclosed.
4. The analytical methods are scientifically validated.
5. Studies were conducted in NABL/GLP compliant laboratories wherever applicable.

AUTHORIZED SIGNATORY

Parameter	Details
Name	[Insert]
Designation	[Insert]
Company	[Insert]
Signature	[Insert]
Date	[Insert]
Company Seal	[Insert]

COMPREHENSIVE TECHNICAL DOSSIER TEMPLATE

FOR SCRUTINY, CHEMICAL VERIFICATION & REGISTRATION

UNDER SECTION 9(3)

(Chemistry Division)

(Formulation Indigenous Manufacture)

DATA ON FORMULATION

Particulars	Details
Category of Application	☐ 9(3) FIM ☐ 9(3b) FIM $
Name of Product	[Insert Product Name]
Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Number	[Insert]
Formulation Type	[Insert BIS/CIPAC code]
Applicant Company	[Insert]
Manufacturing Site	[Insert]
Date of Submission	[DD/MM/YYYY]

$ In case of 9(3B) FIM application, the data shall be submitted as per extant RC guideline.

SECTION 1. GENERAL INFORMATION

Parameter	Details
Name of Applicant	[Insert]
Address of Applicant	[Insert]
Address of Manufacturing Site	[Insert]
Status of Insecticide in Schedule	[Insert]
Applicable BIS Specifications	[Insert BIS Number if available]
Applicable RC Guidelines	[Insert]
Type of Product	[Insert]
Details of Source of supply of Technical*	[Insert]

* Duly signed copy of Certificate of registration/notarised consent letter from the Source of supply of Technical shall be attached.

SECTION 2. CHEMICAL IDENTITY OF ACTIVE INGREDIENT

2.1 Identity Details

Parameter	Details
ISO Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]

CAS Name	[Insert]
CAS Number	[Insert]
Molecular Formula	[Insert]
Molecular Weight	[Insert]
Structural Formula	[Insert Structure]
Chemical Class	[Insert]
Physical Appearance	[Insert]
Odour	[Insert]

SECTION 3. CHEMICAL COMPOSITION

3.1 Composition of Formulation (Applicable for FIM)

S.No.	Ingredient	Function	Concentration (% w/w or w/v or g/L)
1	Active Ingredient	[Insert]	[Insert]
2	Solvent	[Insert]	[Insert]
3	Emulsifier	[Insert]	[Insert]
4	Stabilizer	[Insert]	[Insert]
5	Adjuvant	[Insert]	[Insert]
6	Others	[Insert]	[Insert]
	Total		[Insert]
Density (If applicable) :			

Insert more row or column, if required

Note: Adjuvants shall be disclosed by complete chemical identity and not by trade code alone.

3.3 Establishment of chemical equivalence:

Particulars	Details
Name of the Registrant	[Insert]
Reference of Registraton Committee Meeting	[Insert]
Category	[Insert]
Other details (if applicable)	[Insert]

Insert more row or column, if required

SECTION 4. RAW MATERIAL DETAILS

4.1 Raw Material Specifications

Insert more row or column, if required

S.No.	Name of Raw Material	IUPAC Name	CAS No.	Purity (%)	Molecular Formula	Molecular Weight	Source of Supply	BIS/Analytical Specification
1	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

SECTION 5. MANUFACTURING PROCESS

5.1 Manufacturing Chemistry

Provide detailed manufacturing process including:

- Reaction chemistry
- Stoichiometry
- Intermediate formation
- Purification steps
- Recovery process
- Solvent recovery
- Yield optimization
- Process controls

5.2 Stepwise Manufacturing Process

Insert more row or column, if required

Step	Name of the raw material/reactants used with their respective purity	Weight of the Raw material (gm)	Equipment used	Reaction conditions (Temperature)	Approximate Time required (Hrs.)	Intermediate/substance/Product name	Chemical formula of Intermediate/substance/Product name	Yield (gm)
Step-I	**[Insert name with purity]**	**[Insert value]**	**[Insert**			**[Insert name]**		
	[Insert name with purity]	**[Insert value]**	**[Insert**					
	[Insert name with purity]	**[Insert value]**	**[Insert**					
Step-II	**[Insert name with purity]**	**[Insert value]**	**[Insert**			**[Insert name]**		
	[Insert name with purity]	**[Insert value]**	**[Insert**					
	[Insert name with purity]	**[Insert**	**[Insert**					

		value]						
Final step	**[Insert name with purity]**	**[Insert value]**	**[Insert**			**[Insert name] [Insert name]**	**[Insert name] [Insert name]**	
	[Insert name with purity]	**[Insert value]**	**[Insert**					

5.3 Chemical Reaction Scheme

Provide:

- Balanced chemical equations
- Structural reaction pathway
- Side reactions
- By-product formation
- Degradation pathway

Mandatory Attachments:
☐ Reaction mechanism
☐ Structural formulae
☐ Flow sheet diagram

5.4 Effluent Treatment Details

Parameter	Details
Type of Effluent	[Insert]
Treatment Method	[Insert]
Neutralization Method	[Insert]
Solvent Recovery System	[Insert]
Disposal Method	[Insert]

Insert more row or column, if required

SECTION 8. PHYSICO-CHEMICAL PROPERTIES OF PRODUCT ACTIVE INGREDIENT AND ITS ADJUVANTS

8.1 Physico-Chemical Properties of Active Ingredient:-

Parameter	Specification	Method
Appearance	[Insert]	BIS/CIPAC no. [Insert]
Melting Point	[Insert]	BIS no. [Insert]
Boiling Point	[Insert]	BIS no. [Insert]
Density	[Insert]	BIS no. [Insert]
Vapour Pressure	[Insert]	BIS no. [Insert]
Solubility in Water	[Insert]	BIS no. [Insert]

Solubility in Organic Solvents	[Insert]	BIS no. [Insert]
pH	[Insert]	BIS no. [Insert]
Flash Point	[Insert]	BIS no. [Insert]
Any other parameter (if applicable)	[Insert]	

Insert more row/column/table, if required

Adjuvants / Inert Ingredients

Physico-Chemical Properties of Each Adjuvant (if required separately)

Physico-Chemical Properties of Adjuvant – 1

Name of Adjuvant: [Insert Name]

Name of Adjuvant 1 : [Insert Name]		
Parameter	**Description**	**Method**
Appearance	[Insert]	[Insert]
Colour	[Insert]	[Insert]
Odour	[Insert]	[Insert]
Density / Specific Gravity	[Insert]	[Insert]
Solubility	[Insert]	[Insert]
pH	[Insert]	[Insert]
CAS No.	[Insert]	[Insert]
Function in Formulation	[Insert]	[Insert]
Molecular Formula	[Insert]	[Insert]
Chemical Name / IUPAC Name	[Insert]	[Insert]
Molecular Weight	[Insert]	[Insert]
Purity (%)	[Insert]	[Insert]
Any Other Parameter (if applicable)	[Insert]	[Insert]
Name of Adjuvant 2 : [Insert Name]		
Appearance	[Insert]	[Insert]
Colour	[Insert]	[Insert]
Odour	[Insert]	[Insert]
Density / Specific Gravity	[Insert]	[Insert]
Solubility	[Insert]	[Insert]
pH	[Insert]	[Insert]
Any Other Parameter (if applicable)	[Insert]	[Insert]

 Add additional rows as required for all adjuvants present in the formulation.

 Attach supporting documents such as COA, SDS/MSDS, and technical specification sheets wherever applicable.

SECTION 9. PRODUCT SPECIFICATION (BIS FORMAT)

Formulation Specification

Parameter	BIS Requirement	Method
Active Ingredient	[Insert]	[Insert]
Moisture	[Insert]	[Insert]
pH	[Insert]	[Insert]
Persistent foam	[Insert]	[Insert]
Wet sieve	[Insert]	[Insert]
Any other parameter (if applicable)	[Insert]	[Insert]

SECTION 10. Method of Analysis

10.1 Active Ingredient Analysis

HPLC Conditions

Parameter	Details
Column	[Insert]
Mobile Phase	[Insert]
Flow Rate	[Insert]
Injection Volume	[Insert]
Detector	[Insert]
Detection Wavelength	[Insert]
Run Time	[Insert]
Retention Time	[Insert]
Any other parameter (if applicable)	[Insert]

GC Conditions

Parameter	Details
Column Type	[Insert]
Carrier Gas	[Insert]
Detector	[Insert]
Injector Temperature	[Insert]
Detector Temperature	[Insert]
Any other parameter (if applicable)	[Insert]

SECTION 11. ANALYTICAL TEST REPORT

The study has been carried out by **M/s** _____
(Name of Laboratory/Institute) under the following details:

Particulars	Details
Product Name	

Active Ingredient(s)	
Batch Number(s)	
Manufacturing Date	
Date of expiry	

Study Details

Particulars	Details
Study Title	
Study Number	
Study Initiation Date	
Study Completion Date	
NABL Accreditation No.	
GLP Certification No., (if applicable)	
Whether NABL/GLP Accredited During the Period of Study	Yes / No
Validity of Accreditation During Study Period	

Replicates	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
A1B1	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**
A2B2	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**

Parameter	BIS Requirement	Method	Result/ Observations in ATR
Active Ingredient	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]
pH	[Insert]	[Insert]	[Insert]
Persistent foam	[Insert]	[Insert]	[Insert]
Wet sieve	[Insert]	[Insert]	[Insert]
Any other parameter (if applicable)	[Insert]	[Insert]	[Insert]

SECTION 12. SHELF-LIFE & STORAGE STABILITY

12.1 Shelf-life Claim

[Insert shelf-life]

The study has been carried out by **M/s** _____
(Name of Laboratory/Institution) under the following conditions and details:

Product Details

Particulars	Details
Product Name	
Active Ingredient(s)	
Concentration	
Batch Number(s)	
Manufacturing Date	
Packaging Type	
Proposed Shelf Life	
Others	

Study Details

Particulars	Details
Study Title	Shelf-Life / Stability Study
Study No.	
Study Start Date	
Study Completion Date	
Laboratory Name & Address	
NABL Accreditation Status	Yes / No
NABL/GLP Accredited During Study Period	Yes / No
Accreditation Validity Period	
Others	

Storage Conditions

A. Real-Time Stability Conditions

Parameter	Condition
Duration	[Insert]
Sampling Intervals	[Insert]
Three different Agro-Climatic Zones	[Insert]
Meteorological Data Meteorological data of the study locations during the study period is required to attached for reference.	

B. Accelerated Stability Conditions

Parameter	Condition
Temperature	[Insert]
Relative Humidity	[Insert]
Duration	[Insert]
Sampling Intervals	[Insert]

12.2 Storage Stability Data

Parameter	Initial	Final	Acceptance Criteria
Active Content	[Insert]	[Insert]	[Insert]
pH	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]
Physical Stability	[Insert]	[Insert]	[Insert]
Other parameters (as per submitted Product specification/ BIS)	[Insert]	[Insert]	[Insert]

#(Insert columns for more parameters)

12.3 Storage Stability Study Details

Duration	Locations	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass
[insert zero month]	Location-I	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-II	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-III	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[insert month]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

In case of Accelerated Storage Stability data submitted:-

Day	Temperature	Relative Humidity (RH)
01	[Insert]	[Insert]
[Insert] days	[Insert]	[Insert]

Duration	Weight of std in mg	Weight of sample in mg	Area of the active in std solution	Area of the active in sample solution	% purity of the standard	Active content % by mass	Average content % by mass	Mean Avg. content % by mass

[at day 01]	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**
[at the day of study completion]	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**

SECTION 13. DECLARATION

We hereby declare that:

6. The information submitted is complete and authentic.
7. The manufacturing process disclosed represents actual commercial production.
8. All adjuvants and raw material known to the applicant have been disclosed.
9. The analytical methods are scientifically validated.
10. Studies were conducted in NABL/GLP compliant laboratories wherever applicable.

AUTHORIZED SIGNATORY

Parameter	Details
Name	[Insert]
Designation	[Insert]
Company	[Insert]
Signature	[Insert]
Date	[Insert]
Company Seal	[Insert]

COMPREHENSIVE TECHNICAL DOSSIER TEMPLATE

FOR SCRUTINY, CHEMICAL VERIFICATION & REGISTRATION

UNDER SECTION 9(4) TIM

(Chemistry Division)

(Technical Indigenous Manufacture)

COVER PAGE

DATA ON TECHNICAL

GENERAL INFORMATION

Detail of applicant:

Parameter	**Details**
Category of Application	9(4) TIM
Name & Address of the Applicant	[Insert]
Address of Manufacturing Site / R&D centre	[Insert]
Location of in-process sample drawl for PRV purpose	[Insert]

Detail of product:

Particulars	**Details**
Name of Product with purity	[Insert Product Name]
Type of Product	[Insert]
Date of Submission	[DD/MM/YYYY]
Establishment of Chemical equivalence	[Insert]

SECTION 1. CHEMICAL COMPOSITION

1.1 Chemical Composition of Technical, as per claimed 9(3) registrants

S. No.	**Component Name**	**Content in w/w %**
13.	**[Name of A.I]**	[insert in w/w %]
	Details of Associated Impurities	
14.	**[Insert name of Impurity-1]**	[insert in w/w %]
15.	**[Insert name of Impurity -2]**	[insert in w/w %]

16.	**[Insert name of Impurity -3]**	[insert in w/w %]
17.	**[Insert name of Impurity-4]**	[insert in w/w %]
18.	**Total**	**100.00 % w/w**

Insert more rows or column, if required

1.2 Details of Impurities

Impurity Name	**Chemical Name**	**CAS No.**	**Structural Formula**	**Molecular weight**	**Maximum acceptable Limit in % w/w**	**Method of Analysis**
[Insert name of Impurity-1]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[Insert name of Impurity-2]	[Insert	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[Insert name of Impurity-3]	[Insert	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

Insert more rows or column, if required

SECTION 2. CHEMICAL IDENTITY OF ACTIVE INGREDIENT

2.1 Identity Details

Parameter	**Details**
Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Name	[Insert]
CAS Number	[Insert]
Molecular Formula	[Insert]
Molecular Weight	[Insert]
Structural Formula	[Insert Structure]
Chemical Class	[Insert]
Physical Appearance	[Insert]
Other parameters (If applicable)	

Insert more rows or column, if required

SECTION 3. PHYSICO-CHEMICAL PROPERTIES

8.1 Technical Grade Active Ingredient

Parameter	Specification
Appearance	[Insert]
Melting Point	[Insert]
Boiling Point	[Insert]
Density	[Insert]
Vapour Pressure	[Insert]
Solubility in Water	[Insert]
Solubility in Organic Solvents	[Insert]
pH	[Insert]
Flash Point	[Insert]
Any other parameter (if applicable)	[Insert]

Insert more row or column, if required

SECTION 3. SPECIFICATION (As per BIS if BIS specification is not available)

Sr.No	Characteristics	Requirements	Observation	Test Method Reference
1	Physical appearance	[Insert]	[Insert]	[Insert]
2	a.i content	[Insert]	[Insert]	[Insert]
3	pH	[Insert]	[Insert]	[Insert]
4	Water	[Insert]	[Insert]	[Insert]
5	Melting point	[Insert]	[Insert]	[Insert]
6	Acidity (as H2SO4) Percent by mass; max. OR Alkalinity (as NaOH) Percent by mass; max.	[Insert]	[Insert]	[Insert]
7	Material insoluble in acetone	[Insert]	[Insert]	[Insert]
8	Any other parameter (if applicable)	[Insert]	[Insert]	[Insert]

Insert more row or column, if required

SECTION 4. METHOD OF ANALYSIS

Active Ingredient Analysis / Impurity analysis (if different from a.i determination) / Solvent Determination

<u>Outline Method:</u>

Weight of Standard taken (g) –

Weight of Sample taken (g) –

Dilution of standard and sample:

HPLC Conditions

Parameter	Details
Column	[Insert]
Mobile Phase or Gradient Profile	[Insert]
Flow Rate	[Insert]
Injection Volume	[Insert]
Detector	[Insert]
Detection Wavelength	[Insert]
Run Time	[Insert]
Any other parameter (if applicable)	[Insert]

Insert more rows or column, if required

GC Conditions

Parameter	Details
Column Type	[Insert]
Carrier Gas	[Insert]
Detector	[Insert]
Injector Temperature	[Insert]
Detector Temperature	[Insert]
Column Programme / Temperature	[Insert]
Any other parameter (if applicable)	[Insert]

Insert more rows or column, if required

SECTION 5. ANALYTICAL TEST REPORT

5.1 The study has been carried out by M/s._____
(Name of Laboratory/Institute) under the following details:

Particulars	Details
Product Name	[Insert]
Active Ingredient(s)	[Insert]
Batch Number(s)	[Insert]
Manufacturing Date	[Insert]
Date of expiry	[Insert]

5.2 Study Details

Particulars	Details
Study Title	[Insert]
Study Number	[Insert]

Study Initiation Date	[Insert]
Study Completion Date	[Insert]
Whether NABL/GLP Accredited During the Period of Study	**Yes / No**

5.3 For a.i content

Replicates	Weight of Standard	Weight of Sample	RT of Standard	RT of Internal Standard	Active content % by mass	Mean Avg. content % by mass
A1B1	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
A2B2	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

Insert more rows or column, if required

5.4 For impurity:

Name of impurity and active ingredient	RT of standard	RT of impurity in Concentrated Sample	Acceptable Limit	Observed value
[Insert a.i name]	NA	[Insert RT of a.i]	NA	NA
[Insert Impurity -1]	[Insert RT of Impurity-1]	[Insert RT of Impurity-1]	[Insert]	[Insert]
[Insert Impurity -2]	[Insert RT of Impurity-2]	[Insert RT of Impurity-2]	[Insert]	[Insert]

Insert more rows or column, if required

5.5 Details of Molecular Identity Test reports:

☐ UV-Visible Spectrum.

☐ FT-IR Spectrum.

☐ NMR Spectrum.

☐ Mass Spectrum.

SECTION 6. SHELF-LIFE

Shelf-life Claimed - _____ Years.

SECTION 7; MANUFACTURING PROCESS:

7.1. Manufacturing Process

Step	Name of the raw material/reactants	Weight of the Raw material (gm/kg)	Reaction conditions (Temperature/ pressure/solvent /catalyst etc.,)	Approximate Time required (Hrs.)	Intermediate/substance/Product name	Chemical formula of Intermediate/substance/Product name	Chemical Name of the Impurities formed during the reaction	Yield (gm/kg)
Step-I	[Insert name with purity]	[Insert value]			[Insert name]		[Insert name]	
	[Insert name with purity]	[Insert value]					[Insert name]	
	[Insert name with purity]	[Insert value]					[Insert name]	
Final step	[Insert name with purity]	[Insert value]			[Insert name] [Insert name]	[Insert name] [Insert name]	[Insert name]	
	[Insert name with purity]	[Insert value]					[Insert name]	

Insert more rows or column, if required

7.2. Step-wise Chemical Equation- Submitted / Not submitted.

7.3 Flow sheet diagram as per manufacturing process- Submitted / Not submitted.

SECTION 8. RAW MATERIAL DETAILS

8.1 Raw Material Specifications

S.No	Name of Raw Material	IUPAC Name	CAS No.	Purity (%)	Molecular Formula	Molecular Weight
1	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]

Insert more rows or column, if required.

Attach supporting documents such as COA, SDS/MSDS, and technical specification sheets wherever applicable.

8.2 Raw Material Source of Supply

S.No	Name of Raw Material	Source of Supply	Details of purchase of raw material
1	[Insert]	[Insert]	[Insert]

Insert more rows or column, if required.

SECTION 9; Effluent Treatment Details

Parameter	Details
Type of Effluent	[Insert]
Treatment Method	[Insert]
Neutralization Method	[Insert]
Solvent Recovery System	[Insert]
Disposal Method	[Insert]

Insert more row or column, if required

SECTION 10. DECLARATION

We hereby declare that:

11. The information submitted is complete and authentic.
12. The manufacturing process disclosed represents actual commercial production.
13. All impurities known to the applicant have been disclosed.
14. The analytical methods are scientifically validated.
15. Studies were conducted in NABL/GLP compliant laboratories wherever applicable.

AUTHORIZED SIGNATORY

Parameter	Details
Name	[Insert]
Designation	[Insert]
Company	[Insert]
Signature	[Insert]
Date	[Insert]
Company Seal	[Insert]

TEMPLATES 9(3) BIO-PESTICIDES OF ALREADY REGISTERED STRAIN & REGISTERED FORMULATION

COMPREHENSIVE DOSSIER TEMPLATE

FOR SCRUTINY & REGISTRATION

UNDER SECTION 9(3)

(Chemistry Division)

COVER PAGE

DATA ON Bio-pesticides

Particulars	Details
Category of Application	☐ 9(3) B/F (Already registered strain)
Name of Product	[Insert Product Name]
Strain and accession no.	[Insert]
Common Name	[Insert]
Systemic name(genus and species)	[Insert]
Applicant Company	[Insert]

SECTION 1. GENERAL INFORMATION

Parameter	Details
Status of Insecticide in Schedule	[Insert]
Type of Product such as WP/AS/LF	[Insert]
Consent letter details from Manufacturer/ original inventor /source or Self declaration as applicable	[Insert]
Authorization letter from the source from inventor of strain or undertaking by the applicant about the name of inventor/source of strain as per annexure	
Source of Origin	[Insert]
Details of sample submission in NBAIM (Fee receipt of sample submission for DNA finger printing at NBAIM, Mau	
Receipt of sample submission for PRV at CIL	
Any other item	

SECTION2.

CHEMICAL COMPOSITION

3.1 Composition of Product

Insert more row or column, if required

S. No.	Name	Content [Insert % or w/w or v/v or w/v]
1.	[Insert name]	[insert xyz]
2.	[Insert name]	[insert abc]
3.	[Insert name]	[insert jkl]
4.	[Insert name]	[insert rst]
5.	[Insert name]	[insert def]
6.	Total	100.00 as per unit

TEMPLATES 9(3) BIO-PESTICIDES OF NEW STRAIN

COMPREHENSIVE DOSSIER TEMPLATE

FOR SCRUTINY & REGISTRATION

UNDER SECTION 9(3b) /9(3)

(Chemistry Division)

COVER PAGE

DATA ON Bio-pesticides

Particulars	Details
Category of Application	☐ 9(3b) B/F (New strain) ☐ 9(3) B/F (New strain)
Name of Product	[Insert Product Name]
Strain and accession no.	[Insert]
Common Name	[Insert]
Systemic name (genus and species)	[Insert]
Applicant Company	[Insert]

SECTION 1. GENERAL INFORMATION

Parameter	Details
Address of Applicant	[Insert]
Status of Insecticide in Schedule	[Insert]
Type of Product such as WP/AS/LF etc	[Insert]
Consent letter details from Manufacturer/ original inventor /source and if any	[Insert]
Source of Origin	[Insert]
Fee receipt of sample submission for DNA finger printing at NBAIM, Mau	
Receipt of sample submission for PRV at CIL	
Any other	

SECTION 2.

CHEMICAL IDENTITY OF BIO PESTICIDES

2.1 Identity Details

Insert more row or column, if required

Parameter	Details

Common Name	[Insert]
Formulation Type	[Insert]
Physical Appearance	[Insert]
Odour	[Insert]
Other parameters (If any)	**[Insert]**

SECTION 3.

CHEMICAL COMPOSITION

3.1 Composition of Product

Insert more row or column, if required

S. No.	Name	Content [Insert % or w/w or v/v or w/v]
1.	[Insert name]	[insert xyz]
2.	[Insert name]	[insert abc]
3.	[Insert name]	[insert jkl]
4.	[Insert name]	[insert rst]
5.	[Insert name]	[insert def]
6.	Total	100.00 as per unit

SECTION 4. PHYSICO-CHEMICAL PROPERTIES

Parameter	Specification	Method
Physical and Morphological Appearance and habitat	[Insert]	[Insert]
pH	[Insert]	BIS no. [Insert]
Particle size	Insert]	
Moisture	Insert]	
Any other parameter (if any)	[Insert]	

Insert more row or column, if required

SECTION 4.A. SCIENTIFIC INFORMATION ON RAW MATERIAL

Parameter	Details
ISO Common Name	[Insert]
Chemical Name (IUPAC)	[Insert]
CAS Name	[Insert]
CAS Number	[Insert]
Molecular Formula	[Insert]
Molecular Weight	[Insert]

Structural Formula	[Insert Structure]
Physical Appearance	[Insert]
Odour	[Insert]

SECTION 5. PRODUCT SPECIFICATION

Parameter	Requirement	Method
CFU Count/Active Ingredient	[Insert]	[Insert]
Moisture	[Insert]	[Insert]
Ph	[Insert]	[Insert]
Pathogenic contaminants (such as gram-negative bacteria, salmonella, shigella, vibrio, etc	[Insert]	[Insert]
Other contaminants (chemical and botanical contaminants)	[Insert]	[Insert]
Potency if required	[Insert]	[Insert]
Any other parameter (if any)	[Insert]	[Insert]

Insert more row or column, if required

SECTION 6. RAW MATERIAL DETAILS

6.1 Raw Material

Insert more row or column, if required

S. No.	Name of Raw Material
1	[Insert]
2.	
3.	
	Any other (if any)

SECTION 7. MANUFACTURING PROCESS/Isolation of strain

7.1 Manufacturing Chemistry

Provide detailed manufacturing process/isolation of strain:

7.2 Stepwise Manufacturing Process (if applicable)

Insert row or column, if required

SECTION 8. METHOD OF ANALYSIS

Parameter	Method Details
CFU Count/Active Ingredient	[Insert]
Moisture	[Insert]

Pathogenic contaminants (such as gram-negative bacteria, salmonella, shigella, vibrio, etc	[Insert]
Other contaminants (chemical and botanical contaminants)	[Insert]
Potency if required	[Insert]
Any other parameter (if any)	[Insert]

Insert more row or column, if required

SECTION 9. Quantitative analysis Report:

The study has been carried out by **M/s** _____
(Name of Laboratory/Institute) under the following details:

9.1 Study Details

Particulars	Details
Study Title	
Study Number	
Product Name	
Formulation type	
Batch Number(s)	
Manufacturing Date	
Date of expiry	
Proposed Shelf Life	

9.2 Study results

Parameter	Requirement	Result	Method
CFU Count/Active Ingredient	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]
Ph	[Insert]	[Insert]	[Insert]
Pathogenic contaminants (such as gram-negative bacteria, salmonella, shigella, vibrio, etc	[Insert]	[Insert]	[Insert]
Other contaminants (chemical and botanical contaminants)	[Insert]	[Insert]	[Insert]
Potency if required	[Insert]	[Insert]	[Insert]
Plating for contaminants on specific media if required	[Insert]	[Insert]	[Insert]
Entomopathogenic capability on target insects by Bioassy if required	[Insert]	[Insert]	[Insert]
Any other parameter (if any)	[Insert]	[Insert]	[Insert]

The complete ATR with all supporting documents

Insert more row or column, if required

SECTION 10. SHELF-LIFE & STORAGE STABILITY

10.1 Shelf-life Claim

[Insert shelf-life]-

10.2 Shelf-life data study- NR for 9(3b)

The study has been carried out by **M/s** _____
(Name of Laboratory/Institution) under the following conditions and details:

Product Details

Particulars	Details
Study Title	
Study Number	
Product Name	
Formulation type	
Batch Number(s)	
Manufacturing Date	
Date of expiry	
Proposed Shelf Life	
Any Others	

Study Details

A. Real-Time Stability Conditions

Parameter	Details
Duration	[Insert]
Sampling Intervals	[Insert]
Different Agro-Climatic Zones	[Insert]
Any other parameters	
Meteorological Data Authentic Meteorological data for corresponding study period from the three study locations.	

10.3 Storage Stability Data Observations-

Parameter	Requirement	Result	Method
CFU Count/Active Ingredient	[Insert]	[Insert]	[Insert]
Moisture	[Insert]	[Insert]	[Insert]
Ph	[Insert]	[Insert]	[Insert]
Pathogenic contaminants (such as gram-negative bacteria, salmonella, shigella, vibrio, etc	[Insert]	[Insert]	[Insert]
Other contaminants (chemical and botanical contaminants)	[Insert]	[Insert]	[Insert]
Plating for contaminants on specific media if required	[Insert]	[Insert]	[Insert]

Entomopathogenic capability on target insects by Bioassyif required	[Insert]	[Insert]	[Insert]
Potency if required	[Insert]	[Insert]	[Insert]
Any other parameter (if any)	[Insert]	[Insert]	[Insert]

Insert more row or column, if required

10.4 Storage Stability Study Results-NR for 9(3b)

Duration	**Locations**	**Result of CFU Count (with requirement and result**	**Moisture (with requirement and result)**	**Ph (With requirement and result**	**Pathogenic contaminants (such as gram-negative bacteria, salmonella, shigella, vibrio, etc with requirement and result**	**Other contaminants (chemical and botanical contaminants) With requirement and result**	**Any other parameter (if any)**
[insert zero month]	Location-I	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-II	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-III	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
[insert month wise details]	Location-1	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]	[Insert]
	Location-II						
	Location-III						
[insert month wise details]							

BIOEFFICACY

(HERBICIDE UNIT)

Format for TI (NS) Application Scrutiny

Summary of Bioefficacy, Phytotoxicity, Effect on Succeeding Crop, Residue data

The applicant **M/s** _____ has submitted bioefficacy data in **Application No. (CSN)** _____ which is as follows:

Name of the Chemical to be Registered:

Category of the product:

Crops on which bio-efficacy trials conducted:

I. …….. Crop

Location 1:

Bio-efficacy data:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing/ transplanting : **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				

T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								

T5								
T6								

Yield parameters and yield:

Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:

Season 1

Treatments	Observation frequency (Days After Application)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1:							
T2:							
T3:							

Result:

<u>Effect on succeeding crop</u>: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:

Season 1

Treatments	Observation frequency (Days After sowing)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: bio effective dose							
T3: higher dose / dose for phytotoxicity							

Yield parameters and yield of succeeding crop:

Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				
T2				
T3				
T4				
T5				
T6				

Result:

***** *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Location 2:

<u>Bio-efficacy data</u>:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing/ transplanting : **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	**Treatments**	**Test dosages/hectare**		**Water volume (l/ha)**
		a.i. g/ha	**Formulation**	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield:

Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:

Season 1

Treatments	Observation frequency (Days After Application)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1:							
T2:							
T3:							

Result:

<u>Effect on succeeding crop</u>: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:

Season 1

Treatments	Observation frequency (Days After sowing)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: bio effective dose							
T3: higher dose / dose for phytotoxicity							

Yield parameters and yield of succeeding crop:

Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				

T2				
T3				
T4				
T5				
T6				

Result: * *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Data on Physico-Chemical & Biological Properties:

Institute of field studies from where samples were taken:

Location 1:

Location 2:

Location 3:

Institute where residue study carried out:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Dosage	Physical properties			Chemical properties			Biological properties		
Location-1; Season-1									
Before Application									
T1									
T2									
T3									

T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-1; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-2; Season-1									
Before Application									
T1									
T2									
T3									

T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-2; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-3; Season-1									
Before Application									
T1									
T2									
T3									

T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-3; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									

Result:

<u>Data on Residue in Plant and Soil (Multi-location studies)</u>:

Institute of field studies from where samples were taken:

Location-1:

Location-2:

Location-3:

Institute where study was carried out:

Instrument used for analysis:

Date of sampling:

Date of analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

S. No.	**Treatments**	**Test dosages/hectare**	
		a.i. g/ha	**Formulation**
T1 : untreated control			
T2 : Bioeffective dose			
T3: Higher dose (1.25x/ 2 X dose)			

Average recovery (%):

Substrate	**Fortification level (mg/kg)**	**Average Recovery (%)**

Mean Residue:

Substrate	**Location-1**			**Location-2**			**Location-3**		
	T1	**T2**	**T3**	**T1**	**T2**	**T3**	**T1**	**T2**	**T3**

Season-I									
Soil									
Season-II									
Soil									

Proposed PHI (days):

Name of the chemical	**Crop**	**Dosage**	**MRL Values (µg/g)**		**Proposed PHI (days)**
			WHO/FAO/Others	**Others (GOI*)**	

Result:

Copy of RTT Permit:

The applicant **M/s** _____ has submitted bioefficacy data in **Application No. (CSN)** _____ which is as follows:

Name of the chemical to be registered:

Category of the product:

Crops on which bio-efficacy trials conducted:

II. …….. Crop

Location 1:

<u>Bio-efficacy data</u>:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1:

Crop:

Variety:

Plot size:

Observation frequency:

Date of crop sowing or transplanting: **S1**:

Date of application: **S1:**

Date of harvest: **S1:**

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals: **Season 1**

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								

T6								

Weed dry weight at different intervals: **Season 1**

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals: **Season 1**

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield: Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (DAA)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: UTC							
T2: Bio-effective Dose							
T3: Higher dose							

Result:

<u>Effect on succeeding crop</u>: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (DAA)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: UTC							
T2: Bio-effective Dose							
T3: Higher dose							

Yield parameters and yield:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				
T2				
T3				
T4				
T5				
T6				

Result:

Location 2:

<u>Bio-efficacy data</u>:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1:

Crop:

Variety:

Plot size:

Observation frequency:

Date of crop sowing or transplanting: **S1**:

Date of application: **S1:**

Date of harvest: **S1:**

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	**Treatments**	**Test dosages/hectare**		**Water volume (l/ha)**
		a.i. g/ha	**Formulation**	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals: **Season 1**

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals: **Season 1**

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals: **Season 1**

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield: Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (DAA)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: UTC							
T2: Bio-effective Dose							
T3: Higher dose							

Result:

Effect on succeeding crop: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (DAA)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: UTC							
T2: Bio-effective Dose							
T3: Higher dose							

Yield parameters and yield:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				
T2				
T3				
T4				
T5				
T6				

Result:

Data on Physico-Chemical & Biological Properties:

Institute of field studies from where samples were taken:

Location 1:

Location 2:

Institute where residue study carried out:

Treatments:

S. No.	**Treatments**	**Test dosages/hectare**		**Water volume (l/ha)**
		a.i. g/ha	**Formulation**	
T1				
T2				
T3				
T4				
T5				
T6				

Dosage	**Physical properties**			**Chemical properties**			**Biological properties**		
Location-1; Season-1									
Before Application									
T1									

T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-2; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									

Result:

Data on Residue in Plant and soil:

Institute of field studies from where samples were taken:

Location-1:

Institute where study was carried out:

Instrument used for analysis:

Date of sampling:

Date of analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

S. No.	Treatments	Test dosages/hectare	
		a.i. g/ha	Formulation
T1: UTC			
T2: Bio-effective Dose			
T3: Higher dose (1.25 X/ 2X dose)			

Average recovery (%):

Substrate	Fortification level (mg/kg)	Average Recovery (%)

Mean Residue:

Substrate	Location-1		
	T1	T2	T3
Season-I			
Soil			

Result:

Format for 9(3) TI / TIM Application Scrutiny

Summary of Translocation, Metabolism and Persistence data

The applicant **M/s** _____ has submitted bioefficacy data in **Application No. (CSN)** _____ which is as follows:

Name of the chemical to be registered:

Category of the product:

Translocation in Plant:

Institute where study was carried out:

Institute where analysis was carried out:

Report Forwarding Authority:

Report Signatory Authority:

Instrument used for analysis:

Result:

Metabolism in Soil:

Institute where study was carried out:

Institute where analysis was carried out:

Report Forwarding Authority:

Report Signatory Authority:

Instrument used for analysis:

Result:

Metabolism in Water:

Institute where study was carried out:

Institute where analysis was carried out:

Report Forwarding Authority:

Report Signatory Authority:

Instrument used for analysis:

Result:

Metabolism in Plant:

Institute where study was carried out:

Institute where analysis was carried out:

Report Forwarding Authority:

Report Signatory Authority:

Instrument used for analysis:

Result:

Persistence in Plant:

Institute where study carried out:

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Treatments	Test dosages/hectare	
	a.i. g/ha	Formulation
T0:		
T1:		
T2:		

Sampling occasions and dates:

Sample Occasions	Dates

Average Recovery (%):

Fortification (mg/kg)	Substrate	% Recovery

Half life (days):

Crop	T1	T2

Residue study:

Sampling occasions (Days)	Residue (mg/kg)		
	T0	T1	T2

Result:

Persistence in Soil:

Institute where study carried out;

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Treatments	**Dose**
T0:	
T1:	
T2:	

Sampling occasions and dates:

Sampling occasions	**Dates**

Average recovery (%):

Fortification Level (mg/kg)	**Soil type**	**% Recovery**

Half life period (days):

Chemical name	**Soil type**	**Half-life (Days)**	
		T1	**T2**

Result:

Persistence in Water:

Institute where study carried out;

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Compound	Dose

Sampling occasions and dates:

Sampling occasions	Dates

Average recovery (%):

Fortification Level (mg/kg)	Type of water	% Recovery
	Acidic Water	
	Neutral Water	
	Alkaline Water	
	Acidic Water	
	Neutral Water	
	Alkaline Water	
	Acidic Water	
	Neutral Water	
	Alkaline Water	

Half life period (days):

Chemical name	Type of water	Half-life (Days)	
		T1	T2
	Acidic water		
	Neutral water		
	Basic water		

Result:

Registration status in foreign countries:

** If the applicant has submitted data for more than one crop and soil, submit that data also.*

***

Format for Formulation Application Scrutiny

Summary of Bioefficacy, Phytotoxicity, Effect on Succeeding Crop, Residue and Persistence data

The applicant **M/s** _____ has submitted bioefficacy data in **Application No. (CSN)** _____ which is as follows:

Name of the Chemical to be Registered:

Category of the product:

Crops on which bio-efficacy trials conducted:

III. …….. Crop

Location 1:

Bio-efficacy data:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing or transplanting: **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of application:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield:

Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:

Season 1

Treatments	Observation frequency (Days after Application)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: Bioeffective dose							
T3: higher dose (1.2X/ 2 X)							

Result:

<u>Effect on succeeding crop</u>: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days After sowing)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: Bioeffective dose							
T3: higher dose (1.2X/ 2 X)							

Yield parameters and yield:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				
T2				
T3				
T4				
T5				
T6				

Result:

***** *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Location 2:

<u>Bio-efficacy data</u>:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing or transplanting: **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of application:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield:
Season 1

Tr.	Y1	Y2	Y3	Y4	Y5	Y6	Y7
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days after Application)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: Bioeffective dose							
T3: higher dose (1.2X/ 2 X)							

Result:

<u>Effect on succeeding crop</u>: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days After sowing)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: Bioeffective dose							
T3: higher dose (1.2X/ 2 X)							

Yield parameters and yield:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				
T2				
T3				
T4				
T5				
T6				

Result:

***** *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Location 3:

<u>Bio-efficacy data</u>:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing or transplanting: **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of application:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8

T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield:
Season 1

Tr.	Y1	Y2	Y3	Y4	Y5	Y6	Y7
T1							
T2							
T3							
T4							
T5							
T6							

Result:

Phytotoxicity data:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days after Application)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: Bioeffective dose							
T3: higher dose (1.2X/ 2 X)							

Result:

Effect on succeeding crop: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days After sowing)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: Bioeffective dose							

T3: higher dose (1.2X/ 2 X)							

Yield parameters and yield:
Season 1

Tr.	Y1	Y2	Y3	Y4
T1				
T2				
T3				
T4				
T5				
T6				

Result:

***** *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

<u>Data on Physico-Chemical & Biological Properties</u>:

Institute of field studies from where samples were taken:

Location 1:

Location 2:

Location 3:

Institute where residue study carried out:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Dosage	Physical properties			Chemical properties			Biological properties		
Location-1; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-1; Season-2									
Before Application									

T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-2; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-2; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-3; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									

T6									
Location-3; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									

Result:

<u>Data on Residue in Plant and Soil (Multi-location studies)</u>:

Institute of field studies from where samples were taken:

Location-1:

Location-2:

Location-3:

Institute where study was carried out:

Instrument used for analysis:

Date of sampling:

Date of analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

S. No.	**Treatments**	**Test dosages/hectare**	
		a.i. g/ha	**Formulation**
T1 : untreated control			
T2: Bioeffective dose			
T3: higher dose			

Average recovery (%):

Substrate	**Fortification level (mg/kg)**	**Average Recovery (%)**

Mean Residue:

Substrate	Location-1			Location-2			Location-3		
	T1	T2	T3	T1	T2	T3	T1	T2	T3
Season-I									
Soil									
Season-II									
Soil									

Proposed PHI (days):

Name of the chemical	Crop	Dosage	MRL Values (µg/g)		Proposed PHI (days)
			WHO/FAO/Others	Others (GOI*)	

Result:

<u>Persistence in Plant</u>:

Institute where study carried out:

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Treatments	Test dosages/hectare	
	a.i. g/ha	Formulation
T0: untreated control		
T1: Bioeffective dose		
T2: higher dose		

Sampling occasions and dates:

Sample Occasions	Dates

Average Recovery (%):

Fortification (mg/kg)	Substrate	% Recovery

Half life (days):

Crop	T1	T2

Residue study:

Sampling occasions (Days)	Residue (mg/kg)		
	T0	T1	T2

Result:

** Submit the above mentioned data for other crops also, if the applicant has submitted data for more than one crop.*

<u>Persistence in Soil</u>:

Institute where study carried out;

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Treatments	Dose
T0: untreated control	
T1: Bioeffective dose	
T2: higher dose	

Sampling occasions and dates:

Sampling occasions	Dates

Average recovery (%):

Fortification Level (mg/kg)	Soil type	% Recovery

Half life period (days):

Chemical name	Soil type	Half-life (Days)	
		T1	T2

Result:

Persistence in Water:

Institute where study carried out;

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Compound	Dose
T0: untreated control	
T1: Bioeffective dose	
T2: higher dose	

Sampling occasions and dates:

Sampling occasions	Dates

Average recovery (%):

Fortification Level (mg/kg)	Type of water	% Recovery
	Acidic Water	
	Neutral Water	
	Alkaline Water	
	Acidic Water	
	Neutral Water	
	Alkaline Water	
	Acidic Water	
	Neutral Water	
	Alkaline Water	

Half life period (days):

Chemical name	Type of water	Half-life (Days)	
		T1	T2
	Acidic water		
	Neutral water		
	Basic water		

Result:

** Check the units of submitted data accordingly.*

Data on Compatibility with other chemicals:

Registration status in foreign countries:

Residue tolerance limits Fixed by the other Countries:

MRL Proforma:

RTT:

Label claims Sought/Proposed by the applicant:

Crop	Name of the weeds	Dosage/ha			Waiting period from last spray to harvest (days)
		a.i. (g)	Formulation (ml)	Water volume (L)	

***

Format for PGR Application Scrutiny

Summary of Bioefficacy, Phytotoxicity, Effect on Succeeding Crop, Residue and Persistence data For PGRs

The applicant **M/s** _____ has submitted bioefficacy data in **Application No. (CSN)** _____ which is as follows:

Name of the Chemical to be Registered:

Category of the product:

Crops on which bio-efficacy trials conducted:

IV…….. Crop

Location 1:

Bio-efficacy data:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing: **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Yield parameters and yield:

Season 1

Tr.	Y1	Y2	Y3	Y4	Y5	Y6	Y7
T1							
T2							

T3							
T4							
T5							
T6							

Result:

Phytotoxicity data:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (DAA)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: UTC							
T2: bioeffective Dose							
T3: higher dose (2X Dose)							

Result:

Location 2:

Bio-efficacy data:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing: **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				

T4				
T5				
T6				

Yield parameters and yield:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (DAA)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: UTC							
T2: bioeffective Dose							
T3: higher dose (2X Dose)							

Result:

***** *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Location 3:

<u>Bio-efficacy data</u>:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing: **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Yield parameters and yield:
Season 1

Tr.	Y1	Y2	Y3	Y4	Y5	Y6	Y7
T1							
T2							
T3							
T4							
T5							
T6							

Result:

Phytotoxicity data:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (DAA)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: UTC							
T2: bioeffective Dose							
T3: higher dose (2X Dose)							

Result:

***** *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Data on Physico-Chemical & Biological Properties (In case of soil applied PGR):

Institute of field studies from where samples were taken:

Location 1:

Location 2:

Location 3:

Institute where residue study carried out:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Dosage	Physical properties			Chemical properties			Biological properties		
Location-1; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-1; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-2; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									

Location-2; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-3; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-3; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									

Result:

<u>Data on Residue in Plant and Soil (Multi-location studies)</u>:

Institute of field studies from where samples were taken:

Location-1:

Location-2:

Location-3:

Location-4:

Institute where study was carried out:

Instrument used for analysis:

Date of sampling:

Date of analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

S. No.	Treatments	Test dosages/hectare	
		a.i. g/ha	Formulation
T1: UTC			
T2: Bioeffective Dose			
T3: Higher dose (1.25X Dose)			

Average recovery (%):

Substrate	Fortification level (mg/kg)	Average Recovery (%)

Mean Residue:

Substrate	Location-1			Location-2			Location-3			Location-4		
	T1	T2	T3	T1	T2	T3	T1	T2	T3	T1	T2	T3
Season-I												
Soil												
Season-II												
Soil												

Proposed PHI (days):

Name of the chemical	Crop	Dosage	MRL Values (µg/g)		Proposed PHI (days)
			WHO/FAO/Others	Others (GOI*)	

Result:

<u>Persistence in Plant</u>:

Institute where study carried out:

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Treatments	**Test dosages/hectare**	
	a.i. g/ha	**Formulation**
T0: UTC		
T1: Bioeffective Dose		
T2: Higher dose (1.25X Dose)		

Sampling occasions and dates:

Sample Occasions	**Dates**

Average Recovery (%):

Fortification (mg/kg)	**Substrate**	**% Recovery**

Half life (days):

Crop	**T1**	**T2**

Residue study:

Sampling occasions (Days)	**Residue (mg/kg)**		
	T0	**T1**	**T2**

Result:

** Submit the above mentioned data for other crops also, if the applicant has submitted data for more than one crop.*

<u>Persistence in Soil</u>:

Institute where study carried out;

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Treatments	**Dose**
T0: UTC	
T1: Bioeffective Dose	
T2: Higher dose (1.25X Dose)	

Sampling occasions and dates:

Sampling occasions	**Dates**

Average recovery (%):

Fortification Level (mg/kg)	**Soil type**	**% Recovery**

Half life period (days):

Chemical name	**Soil type**	**Half-life (Days)**	
		T1	**T2**

Result:

Persistence in Water:

Institute where study carried out;

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

T0: UTC	**Dose**
T1: Bioeffective Dose	
T2: Higher dose (1.25X Dose)	
T0: UTC	

Sampling occasions and dates:

Sampling occasions	**Dates**

Average recovery (%):

Fortification Level (mg/kg)	**Type of water**	**% Recovery**
	Acidic Water	
	Neutral Water	
	Alkaline Water	
	Acidic Water	
	Neutral Water	
	Alkaline Water	
	Acidic Water	
	Neutral Water	
	Alkaline Water	

Half life period (days):

Chemical name	**Type of water**	**Half-life (Days)**	
		T1	**T2**
	Acidic water		
	Neutral water		
	Basic water		

Result:

** Check the units of submitted data accordingly.*

Data on Compatibility with other chemicals:

Registration status in foreign countries:

Residue tolerance limits Fixed by the other Countries:

MRL Proforma:

RTT:

Label claims Sought/Proposed by the applicant:

Crop	Purpose	Dosage/ha			Waiting period from last spray to harvest (days)
		a.i. (g)	Formulation (ml)	Water volume (L)	

***

Format for Endorsement Application Scrutiny

Summary of Bioefficacy, Phytotoxicity, Effect on Succeeding Crop, Residue and Persistence data

The applicant **M/s** _____ has submitted bioefficacy data in **Application No. (CSN)** _____ which is as follows:

Name of the Chemical to be Registered:

Category of the product:

Crops on which bio-efficacy trials conducted:

V. …….. Crop

Location 1:

Bio-efficacy data:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing/ transplanting : **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8

T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals:

Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield:

Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:

Season 1

Treatments	Observation frequency (Days After Application)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1:							
T2:							
T3:							

Result:

<u>Effect on succeeding crop</u>: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days After sowing)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: bio effective dose							
T3: higher dose / dose for phytotoxicity							

Yield parameters and yield of succeeding crop:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				
T2				
T3				
T4				
T5				
T6				

Result:

***** *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Location 2:

<u>Bio-efficacy data</u>:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing/ transplanting : **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							
T6							

Result:

<u>Phytotoxicity data</u>:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days After Application)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1:							
T2:							
T3:							

Result:

<u>Effect on succeeding crop</u>: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days After sowing)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: bio effective dose							
T3: higher dose / dose for phytotoxicity							

Yield parameters and yield of succeeding crop:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				
T2				
T3				
T4				
T5				
T6				

Result: * *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Location 3:

<u>Bio-efficacy data</u>:

Report Forwarding Authority:

Report Signatory Authority:

Season and year 1: **Season and year 2**:

Crop:

Variety:

Plot size:

Observation frequency;

Date of crop sowing/ transplanting : **S1**: **S2**:

Date of application: **S1:** **S2**:

Date of harvest: **S1:** **S2**:

Method of spraying:

Number of Treatments:

Number of Replications:

Bio-effective dose:

Spray Volume:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Weeds on which the herbicide is effective:

Weed population at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed dry weight at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Weed Control Efficiency at different intervals:
Season 1

Tr.	W1	W2	W3	W4	W5	W6	W7	W8
T1								
T2								
T3								
T4								
T5								
T6								

Yield parameters and yield:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
T1							
T2							
T3							
T4							
T5							

T6							

Result:

Phytotoxicity data:

Observation frequency:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days After Application)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: bio effective dose							
T3: higher dose / dose for phytotoxicity							

Result:

Effect on succeeding crop: -

Name of succeeding crop:

Variety:

Date of sowing:

Date of harvest:

Phytotoxicity data at different intervals:
Season 1

Treatments	Observation frequency (Days after sowing)						
	Yellowing	Stunning	Leaf injury	Vein clearing	Wilting	Epinasty	Hyponasty
T1: untreated control							
T2: bio effective dose							
T3: higher dose / dose for phytotoxicity							

Yield parameters and yield of succeeding crop:
Season 1

Tr.	**Y1**	**Y2**	**Y3**	**Y4**
T1				
T2				
T3				
T4				
T5				
T6				

Result:

***** *Submit the bio-efficacy, phytotoxicity and succeeding crop data for both the seasons/years.*

Data on Physico-Chemical & Biological Properties:

Institute of field studies from where samples were taken:

Location 1:

Location 2:

Location 3:

Institute where residue study carried out:

Treatments:

S. No.	Treatments	Test dosages/hectare		Water volume (l/ha)
		a.i. g/ha	Formulation	
T1				
T2				
T3				
T4				
T5				
T6				

Dosage	Physical properties			Chemical properties			Biological properties		
Location-1; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-1; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-2; Season-1									
Before Application									
T1									
T2									
T3									
T4									

T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-2; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-3; Season-1									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									
Location-3; Season-2									
Before Application									
T1									
T2									
T3									
T4									
T5									
T6									
At Harvest									
T1									
T2									
T3									
T4									
T5									
T6									

Result:

Data on Residue in Plant and Soil (Multi-location studies):

Institute of field studies from where samples were taken:

Location-1:

Location-2:

Location-3:

Institute where study was carried out:

Instrument used for analysis:

Date of sampling:

Date of analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

S. No.	Treatments	Test dosages/hectare	
		a.i. g/ha	Formulation
T1 : untreated control			
T2 : Bioeffective dose			
T3: Higher dose (1.25x/ 2 X dose)			

Average recovery (%):

Substrate	Fortification level (mg/kg)	Average Recovery (%)

Mean Residue:

Substrate	Location-1			Location-2			Location-3		
	T1	T2	T3	T1	T2	T3	T1	T2	T3
Season-I									
Soil									
Season-II									
Soil									

Proposed PHI (days):

	Crop	Dosage	MRL Values (µg/g)	

Name of the chemical			WHO/FAO/Others	Others (GOI*)	Proposed PHI (days)

Result:

Persistence in Plant:

Institute where study carried out:

Instrument used for analysis:

Limit of detection (LOD):

Limit of quantification (LOQ):

Treatments:

Treatments	Test dosages/hectare	
	a.i. g/ha	Formulation
T1 : untreated control		
T2 : Bioeffective dose		
T3: Higher dose (1.25x/ 2 X dose)		

Sampling occasions and dates:

Sample Occasions	Dates

Average Recovery (%):

Fortification (mg/kg)	Substrate	% Recovery

Half life (days):

Crop	T1	T2

Residue study:

Sampling occasions (Days)	Residue (mg/kg)		
	T0	T1	T2

Result:

** Submit the above mentioned data for other crops also, if the applicant has submitted data for more than one crop.*

Data on Compatibility with other chemicals:

Registration status in foreign countries:

Residue tolerance limits Fixed by the other Countries:

MRL Proforma:

RTT:

Label claims Sought/Proposed by the applicant:

Crop	Name of the weeds	Dosage/ha			Waiting period from last spray to harvest (days)
		a.i. (g)	Formulation (ml)	Water volume (L)	

***

TOXICOLOGY

TOXICOLOGY TEMPLATE
9(3b) TI

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Manufacturing address		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. RTT Permit

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

3. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							

Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							

Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and							

histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

SHORT-TERM REPEATED DOSE TOXICITY STUDIES

1. Test study Information:

		Repeated dose 28-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Non rodent	Repeated dose 21 / 28-day dermal toxicity	Repeated dose 28-day inhalational toxicity
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						

OECD Guideline or others						

2. Study Design/Methodology

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *cross reference from study report and page no. to be provided*						
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						

Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

*** In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.**

3. Observations

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
NOAEL/ NOAEC/LOAEC						
Food and water consumption changes						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis**						

(cross reference from study report and page no. to be provided)						
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*						
Organs examined						
organ weight changes and organ/body weight ratios						
Statistical tool used for calculation						

4. Result Summary

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity- Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Number of animals dosed, survived and died						
Study Conclusion						

NEUROTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						

Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction)						

(cross reference from study report and page no. to be provided)						
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		Acute Neurotoxicity	Repeated dose Neurotoxicity	Delayed neurotoxicity -OP compound (acute exposure)	Delayed neurotoxicity-OP compound (Repeated administration)	Developmental Neurotoxicity study
NOAEL						
Food and water consumption changes (if any)						
Clinical observations schedule with respect to doses						

Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Neuropathological assessment report *(cross reference from study report and page no. to be provided)*						
Neurochemical assessment report *(cross reference from study report and page no. to be provided)*						
Electrophysiological assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report*						

and page no. to be provided)						
Necropsy findings and histopathological findings of brain *(cross reference from study report and page no. to be provided)*						
Statistical tool if used						

4. Result Summary

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Number of animals dosed, survived and died						
Study Conclusion						

LONG-TERM TOXICITY STUDIES

1. Test study Information:

Particulars		**Carcinogenicity- Mice**	**Combined Chronic toxicity and carcinogenicity-Rat**
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			
CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			

Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

Particulars		**Carcinogenicity- Mice**	**Combined Chronic toxicity and carcinogenicity-Rat**
Species/Strain			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			
Fasting conditions (Duration) pre-dose			
Housing conditions: Temperature, Humidity, Light cycle			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule, Duration of the study			
Rationale for dose selection			

Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Carcinogenicity- Mice**	**Combined Chronic toxicity and carcinogenicity-Rat**
NOAEL/LOAEL			
Food and water consumption changes			
Clinical observations schedule with respect to doses			
Nature and severity of the toxicity signs (if any)			
Time course of onset and Duration of the toxicity effect (if any)			
Date and time of death, if before scheduled sacrifice			
Total No. of Mortality per dose			
Body weight changes			
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*			
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*			
Necropsy findings • **Terminal body weight** • **Organ weights and their ratios, if applicable** • **Necropsy findings**			

• **Incidence and severity of abnormalities** *(cross reference from study report and page no. to be provided)*			
Histopathological findings • **Non neoplastic histopathological findings** • **Neoplastic histopathological findings** • **Correlation between gross and microscopic findings** • **Detailed description of all treatment-related histopathological findings including severity gradings** • **Report of any peer review of slides;** *(cross reference from study report and page no. to be provided)*			
Organs examined			
Statistical tool used for calculation			
Dose-Response relationship			

4. Result Summary

Particulars		**Carcinogenicity- Mice**	**Combined Chronic toxicity and carcinogenicity-Rat**
Number of animals dosed, survived and died			
Study Conclusion			

PHARMACOKINETICS STUDIES

1. Test study Information:

Particulars		**Pharmacokinetics and Metabolism in Rat**	**Feeding study including Metabolism in Livestock (Goat / Cow)**	**Feeding study including Metabolism in Livestock [Poultry (Hen)]**
Study Report No.				

Sponsor (Name and Address)				
Source (Name and Address)				
Study initiation Date				
Study completion Date				
Batch No.				
Purity				
CoA attached (Yes/No)				
IAEC No. (if available)				
GLP Statement (yes/no)				
Whether the study is done within GLP certificate validity date (yes/no)				
OECD Guideline or others				

Feeding study including Metabolism in Livestock (Goat / Cow) and Poultry – not applicable (NA) on case-to-case basis.

2. Study Design/Methodology

Particulars		**Pharmacokinetics and Metabolism in Rat**	**Feeding study including Metabolism in Livestock (Goat / Cow)**	**Feeding study including Metabolism in Livestock [Poultry (Hen)]**
Species/Strain				
No. and sex of animals per dose				
Age and Body weight range at dosing				
Dose of test chemical				
Vehicle used				
Route of administration				
Fasting conditions (Duration) pre-dose				
Housing conditions: Temperature, Humidity, Light cycle				
Total radioactivity per animal				
Analytical methods used for separation, quantitation and identification of metabolites				

Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*				
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*				
Acclimatization period and grouping				
Dosing Schedule/ Single Duration of the study				
Rationale for dose selection				
Any other special consideration				
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.				

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Pharmacokinetics and Metabolism in Rat**	**Feeding study including Metabolism in Livestock (Goat / Cow)**	**Feeding study including Metabolism in Livestock [Poultry (Hen)]**
NOAEL/NOAEC/LOAEC				
Food and water consumption changes				
Clinical observations schedule with respect to doses				
Absorption Data: • **Rate and extent of absorption by each route of exposure.**				

(additionally cross reference from study report and page no. to be provided)				
Tissue Distribution: • **Report as % of administered dose and as µg-equivalents per g of tissue.** • **Include tissue-to-blood and tissue-to-plasma ratios.** *(additionally cross reference from study report and page no. to be provided)*				
Plasma Concentrations & TK Parameters: • **Bioavailability, AUC, Cmax, Tmax, clearance, and half-life for relevant routes.** *(additionally cross reference from study report and page no. to be provided)*				
Excretion Data: • **Amounts of parent compound and metabolites in excreta (% of administered dose).** *(additionally cross reference from study report and page no. to be provided)*				
Metabolic pathway *(cross reference from study report and page no. to be provided)*				
Statistical tool used for calculation				

4. Result Summary

Particulars		Pharmacokinetics and Metabolism in Rat	Feeding study including Metabolism in Livestock (Goat / Cow)	Feeding study including Metabolism in Livestock [Poultry (Hen)]
Correlation with observed toxicity, dose proportionality, systemic exposure				
Study Conclusion				

IMMUNOTOXICITY STUDY

1. Test study Information:

Particulars		Immunotoxicity test
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

Particulars		Immunotoxicity test
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		
Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		

Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Immunotoxicity test**
NOAEL/BMC/NOAEC/LOAEC/BALF		
Food and water consumption changes (if any)		
Clinical observations schedule with respect to doses		
Nature and severity of the toxicity signs (if any)		
Time course of onset and Duration of the toxicity effect (if any)		
Body weight changes		
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*		
Necropsy findings and histopathological findings		

(cross reference from study report and page no. to be provided)		
Lymphoid Organs examined and weights *(cross reference from study report and page no. to be provided)*		
Phenotyping *(cross reference from study report and page no. to be provided)*		
Primary Functional test (TDAR) *(cross reference from study report and page no. to be provided)*		
Secondary functional Test Report *(cross reference from study report and page no. to be provided)*		
Statistical tool used for calculation		

4. Result Summary

Particulars		**Immunotoxicity test**
Number of animals dosed, survived and died		
Study Conclusion		

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							

Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose where applicable							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*							
Feed analysis statement /							

certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and							

histopathological findings *(cross reference from study report and page no. to be provided)*							
Organs examined							
Food and feed consumption data *(cross reference from study report and page no. to be provided)*							
Statistical tool if used							

4. Result Summary

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**
2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**
3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**
4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
9 (3b) TIM

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Manufacturing address		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							

CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							

Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

SHORT-TERM REPEATED DOSE TOXICITY STUDIES

1. Test study Information:

		Repeated dose 28-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Non rodent	Repeated dose 21 / 28-day dermal toxicity	Repeated dose 28-day inhalational toxicity
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

		Repeated dose 28-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Non rodent	Repeated dose 21 / 28-day dermal toxicity	Repeated dose 28-day inhalational toxicity
Species/Strain						
No. and sex of animals per dose						

Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *cross reference from study report and page no. to be provided*						
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

*** In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.**

3. Observations

		Repeated dose 28-day oral toxicity-Rodent	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
NOAEL/ NOAEC/LOAEC						
Food and water consumption changes						
Clinical observations schedule with respect to doses						

Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*						
Organs examined						
organ weight changes and organ/body weight ratios						
Statistical tool used for calculation						

4. Result Summary

		Repeated dose 28-day oral toxicity-Rodent	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**

Number of animals dosed, survived and died						
Study Conclusion						

NEUROTOXICITY STUDIES

1. Test study Information:

		Acute Neurotoxicity	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

		Acute Neurotoxicity	Repeated dose Neurotoxicity	Delayed neurotoxicity-OP compound (acute exposure)	Delayed neurotoxicity-OP compound (Repeated administration)	Developmental Neurotoxicity study
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*						
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*						
Acclimatization period and grouping						
Dosing Schedule,						

Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Acute Neurotoxicity	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity -OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
NOAEL						
Food and water consumption changes (if any)						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						

Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Neuropathological assessment report *(cross reference from study report and page no. to be provided)*						
Neurochemical assessment report *(cross reference from study report and page no. to be provided)*						
Electrophysiological assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings of brain *(cross reference from study report and page no. to be provided)*						
Statistical tool if used						

4. Result Summary

		Acute Neurotoxicity	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Number of animals						

dosed, survived and died						
Study Conclusion						

LONG-TERM TOXICITY STUDIES

1. Test study Information:

		Carcinogenicity- Mice	**Combined Chronic toxicity and carcinogenicity-Rat**
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			
CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

		Carcinogenicity- Mice	**Combined Chronic toxicity and carcinogenicity-Rat**
Species/Strain			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			

Fasting conditions (Duration) pre-dose			
Housing conditions: Temperature, Humidity, Light cycle			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule, Duration of the study			
Rationale for dose selection			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Carcinogenicity- Mice	**Combined Chronic toxicity and carcinogenicity-Rat**
NOAEL/LOAEL			
Food and water consumption changes			
Clinical observations schedule with respect to doses			
Nature and severity of the toxicity signs (if any)			

Time course of onset and Duration of the toxicity effect (if any)			
Date and time of death, if before scheduled sacrifice			
Total No. of Mortality per dose			
Body weight changes			
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*			
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*			
Necropsy findings • **Terminal body weight** • **Organ weights and their ratios, if applicable** • **Necropsy findings** • **Incidence and severity of abnormalities** *(cross reference from study report and page no. to be provided)*			
Histopathological findings • **Non neoplastic histopathological findings** • **Neoplastic histopathological findings** • **Correlation between gross and microscopic findings** • **Detailed description of all treatment-related histopathological findings including severity gradings**			

• **Report of any peer review of slides;** *(cross reference from study report and page no. to be provided)*			
Organs examined			
Statistical tool used for calculation			
Dose-Response relationship			

4. Result Summary

		Carcinogenicity- Mice	**Combined Chronic toxicity and carcinogenicity-Rat**
Number of animals dosed, survived and died			
Study Conclusion			

PHARMACOKINETICS STUDIES

1. Test study Information:

		Pharmacokinetics and Metabolism in Rat	**Feeding study including Metabolism in Livestock (Goat / Cow)**	**Feeding study including Metabolism in Livestock [Poultry (Hen)]**
Study Report No.				
Sponsor (Name and Address)				
Source (Name and Address)				
Study initiation Date				
Study completion Date				
Batch No.				
Purity				
CoA attached (Yes/No)				
IAEC No. (if available)				
GLP Statement (yes/no)				
Whether the study is done within GLP certificate validity date (yes/no)				

OECD Guideline or others				

Feeding study including Metabolism in Livestock (Goat / Cow) and Poultry – not applicable (NA) on case-to-case basis.

2. Study Design/Methodology

		Pharmacokinetics and Metabolism in Rat	**Feeding study including Metabolism in Livestock (Goat / Cow)**	**Feeding study including Metabolism in Livestock [Poultry (Hen)]**
Species/Strain				
No. and sex of animals per dose				
Age and Body weight range at dosing				
Dose of test chemical				
Vehicle used				
Route of administration				
Fasting conditions (Duration) pre-dose				
Housing conditions: Temperature, Humidity, Light cycle				
Total radioactivity per animal				
Analytical methods used for separation, quantitation and identification of metabolites				
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*				
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*				
Acclimatization period and grouping				
Dosing Schedule/ Single Duration of the study				
Rationale for dose selection				

Any other special consideration				
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.				

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Pharmacokinetics and Metabolism in Rat	**Feeding study including Metabolism in Livestock (Goat / Cow)**	**Feeding study including Metabolism in Livestock [Poultry (Hen)]**
NOAEL/NOAEC/LOAEC				
Food and water consumption changes				
Clinical observations schedule with respect to doses				
Absorption Data: • **Rate and extent of absorption by each route of exposure.** *(additionally cross reference from study report and page no. to be provided)*				
Tissue Distribution: • **Report as % of administered dose and as µg-equivalents per g of tissue.** • **Include tissue-to-blood and tissue-to-plasma ratios.** *(additionally cross reference from study report and page no. to be provided)*				
Plasma Concentrations & TK Parameters: • **Bioavailability, AUC, Cmax, Tmax, clearance,**				

and half-life for relevant routes. *(additionally cross reference from study report and page no. to be provided)*				
Excretion Data: • **Amounts of parent compound and metabolites in excreta (% of administered dose).** *(additionally cross reference from study report and page no. to be provided)*				
Metabolic pathway *(cross reference from study report and page no. to be provided)*				
Statistical tool used for calculation				

4. Result Summary

		Pharmacokinetics and Metabolism in Rat	**Feeding study including Metabolism in Livestock (Goat / Cow)**	**Feeding study including Metabolism in Livestock [Poultry (Hen)]**
Correlation with observed toxicity, dose proportionality, systemic exposure				
Study Conclusion				

IMMUNOTOXICITY STUDY

1. Test study Information:

		Immunotoxicity test
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		

Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

		Immunotoxicity test
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		
Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report*		

and page no. to be provided)		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Immunotoxicity test
NOAEL/BMC/NOAEC/LOAEC/BALF		
Food and water consumption changes (if any)		
Clinical observations schedule with respect to doses		
Nature and severity of the toxicity signs (if any)		
Time course of onset and Duration of the toxicity effect (if any)		
Body weight changes		
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*		
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*		
Lymphoid Organs examined and weights *(cross reference from study report and page no. to be provided)*		

Phenotyping *(cross reference from study report and page no. to be provided)*		
Primary Functional test (TDAR) *(cross reference from study report and page no. to be provided)*		
Secondary functional Test Report *(cross reference from study report and page no. to be provided)*		
Statistical tool used for calculation		

4. Result Summary

		Immunotoxicity test
Number of animals dosed, survived and died		
Study Conclusion		

ECOTOXICITY STUDIES

1. Test study Information:

		Acute Avian toxicity	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							

Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

		Acute Avian toxicity	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose where applicable							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*							
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*							

Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

		Acute Avian toxicity	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and histopathological findings *(cross reference from study report*							

and page no. to be provided)							
Organs examined							
Food and feed consumption data *(cross reference from study report and page no. to be provided)*							
Statistical tool if used							

4. Result Summary

		Acute Avian toxicity	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**
2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**
3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**
4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
9(3b) FI

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Type of Formulation		
In case of formulation, status of technical registration		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. RTT Permit

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

3. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							

Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							

Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

2. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							

Purity							
CoA attached (Yes/No)							
Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose where applicable							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report*							

and page no. to be provided)							
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							

Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*							
Organs examined							
Food and feed consumption data *(cross reference from study report and page no. to be provided)*							
Statistical tool if used							

4. Result Summary

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

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4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
9(3b) FIM

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Type of Formulation		
In case of formulation, status of technical registration		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							

Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							

Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

2. Result Summary

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		Acute Avian toxicity	Repeated dose avian toxicity	Avian reproductive toxicity	Acute toxicity-fish	Acute toxicity-Honeybee (oral and contact)	Acute toxicity-Earthworm
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		Acute Avian toxicity	Repeated dose avian toxicity	Avian reproductive toxicity	Acute toxicity-fish	Acute toxicity-Honeybee (oral and contact)	Acute toxicity-Earthworm
Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose where applicable							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*							
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							

Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*							
Organs examined							
Food and feed consumption data *(cross reference from study report and*							

page no. to be provided)							
Statistical tool if used							

4. Result Summary

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**
2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**
3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**
4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
9(3)-TI-New Molecule

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Manufacturing address		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. RTT Permit

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

1. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							

Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							

Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

SHORT-TERM REPEATED DOSE TOXICITY STUDIES

1. Test study Information:

		Repeated dose 28-day oral toxicity-Rodent	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						

Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *cross reference from study report and page no. to be provided*						
Feed analysis statement / certificate (as applicable) (optional)						

cross reference from study report and page no. to be provided						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

*** In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.**

3. Observations

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
NOAEL/ NOAEC/LOAEC						
Food and water consumption changes						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report						

(cross reference from study report and page no. to be provided)						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*						
Organs examined						
organ weight changes and organ/body weight ratios						
Statistical tool used for calculation						

4. Result Summary

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Number of animals dosed, survived and died						
Study Conclusion						

NEUROTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						

Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions						

(Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*						
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity -OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**

NOAEL						
Food and water consumption changes (if any)						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Neuropathological assessment report *(cross reference from study report and page no. to be provided)*						
Neurochemical assessment report *(cross reference from study report and page no. to be provided)*						
Electrophysiological assessment report *(cross reference from study report and page no. to be provided)*						

Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings of brain *(cross reference from study report and page no. to be provided)*						
Statistical tool if used						

4. Result Summary

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Number of animals dosed, survived and died						
Study Conclusion						

LONG-TERM TOXICITY STUDIES

1. Test study Information:

Particulars		**Carcinogenicity- Mice**	**Combined Chronic toxicity and carcinogenicity-Rat**
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			

CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

Particulars		**Carcinogenicity- Mice**	**Combined Chronic toxicity and carcinogenicity-Rat**
Species/Strain			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			
Fasting conditions (Duration) pre-dose			
Housing conditions: Temperature, Humidity, Light cycle			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule, Duration of the study			

Rationale for dose selection			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Carcinogenicity- Mice**	**Combined Chronic toxicity and carcinogenicity-Rat**
NOAEL/LOAEL			
Food and water consumption changes			
Clinical observations schedule with respect to doses			
Nature and severity of the toxicity signs (if any)			
Time course of onset and Duration of the toxicity effect (if any)			
Date and time of death, if before scheduled sacrifice			
Total No. of Mortality per dose			
Body weight changes			
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*			
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*			
Necropsy findings • **Terminal body weight** • **Organ weights and their ratios, if applicable** • **Necropsy findings**			

• **Incidence and severity of abnormalities** *(cross reference from study report and page no. to be provided)*			
Histopathological findings • **Non neoplastic histopathological findings** • **Neoplastic histopathological findings** • **Correlation between gross and microscopic findings** • **Detailed description of all treatment-related histopathological findings including severity gradings** • **Report of any peer review of slides;** *(cross reference from study report and page no. to be provided)*			
Organs examined			
Statistical tool used for calculation			
Dose-Response relationship			

4. Result Summary

Particulars		Carcinogenicity- Mice	Combined Chronic toxicity and carcinogenicity-Rat
Number of animals dosed, survived and died			
Study Conclusion			

DEVELOPMENTAL & REPRODUCTION TOXICITY STUDIES

1. Test study Information:

Particulars		Developmental toxicity study Rat	Developmental toxicity study - Rabbit	Two generational reproduction toxicity
Study Report No.				
Sponsor (Name and Address)				

Source (Name and Address)				
Study initiation Date				
Study completion Date				
Batch No.				
Purity				
CoA attached (Yes/No)				
IAEC No. (if available)				
GLP Statement (yes/no)				
Whether the study is done within GLP certificate validity date (yes/no)				
OECD Guideline or others				

2. Study Design/Methodology

Particulars		**Developmental toxicity study -Rat**	**Developmental toxicity study - Rabbit**	**Two generational reproduction toxicity**
Species/Strain				
No. and sex of animals per dose				
Age and Body weight range at dosing				
Dose of test chemical				
Vehicle used				
Route of administration				
Fasting conditions (Duration) pre-dose				
Housing conditions: Temperature, Humidity, Light cycle				
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*				
Feed analysis statement / certificate (as applicable) (optional)*				

(cross reference from study report and page no. to be provided)				
Acclimatization period and grouping				
Dosing Schedule, Duration of the study				
Rationale for dose selection				
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.				

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		Developmental toxicity study - Rat	Developmental toxicity study - Rabbit	Two generational reproduction toxicity
NOAEL/LOAEL				
Food, water and test material consumption changes (if any)				
Number of animals at study start, number surviving to termination.				
Number of confirmed pregnancies, abortions, and early deliveries.				
Day and cause of death (if any); survival status to scheduled sacrifice.				
Data from non-surviving animals reported separately (excluded from inter-group statistics). *(cross reference from study report and page no. to be provided)*				
Day of onset, duration, and progression of abnormal clinical signs.				

Daily clinical observations *(cross reference from study report and page no. to be provided)*		**Maternal toxicity (posture, respiration, activity, appearance).**	**Maternal toxicity (posture, respiration, activity, appearance).**	**Nature, Severity, Duration of Clinical Signs and Mortality**
Body weight changes and growth *(cross reference from study report and page no. to be provided)*		**body weight change, and gravid uterine weight; Optionally, body weight change corrected for gravid uterine weight.**	**body weight change, and gravid uterine weight; Optionally, body weight change corrected for gravid uterine weight.**	**Body weight data and changes including litter and pub at birth and during lactation.** **Body weight at sacrifice**
Consumption and Hormonal Data *(cross reference from study report and page no. to be provided)*		**T4, T3, and TSH levels (and other hormones if measured).** **Details of hormone kits/antibodies used, assay method, and laboratory historical control data (mean ± SD).**	**T4, T3, and TSH levels (and other hormones if measured).** **Details of hormone kits/antibodies used, assay method, and laboratory historical control data (mean ± SD).**	
Reproductive Indices/ Reproductive Performance and Fertility Indices *(additionally cross reference from study report and page no. to be provided)*		• **Number of corpora lutea (ovulation count).** • **Number of implantation sites per dam.** • **Number and percentage of live and dead fetuses.** • **Number and percentage of resorptions (early and late).** • **Pre-implantation loss (%) = (Corpora lutea – Implantations) / Corpora lutea × 100** • **Post-implantation loss (%) = (Implantations – Live fetuses) / Implantations × 100**	• **Number of corpora lutea (ovulation count).** • **Number of implantation sites per dam.** • **Number and percentage of live and dead fetuses.** • **Number and percentage of resorptions (early and late).** • **Pre-implantation loss (%) = (Corpora lutea – Implantations) / Corpora lutea × 100** • **Post-implantation loss (%) = (Implantations – Live fetuses) /**	• **Toxic Responses by Sex and Dose** • **Number of Animals Used** • **Toxic or Other Effects on Reproduction** • **Time-to-Mating** • **Gestation Length** • **Estrous Cycle Data** • **Sperm Parameters: Total cauda epididymal sperm count, Percent progressively motile sperm, Percent morphologically normal sperm, Percent of sperm with identified abnormalities**

			Implantations × 100	
Fetal Viability and Growth • **Number and percentage of live fetuses per litter.** • **Sex ratio (males/females per litter).** • **Fetal body weight, reported by sex and combined mean.** *(additionally cross reference from study report and page no. to be provided)*				
Abnormalities and Categorisation/ Developmental Landmarks *(additionally cross reference from study report and page no. to be provided)*		• **Criteria for classification of findings (malformations vs. variations) clearly described.** • **Record total number and percent of fetuses and litters showing any external, visceral, or skeletal alteration and specific anomalies with their incidence and type.**	• **Criteria for classification of findings (malformations vs. variations) clearly described.** • **Record total number and percent of fetuses and litters showing any external, visceral, or skeletal alteration and specific anomalies with their incidence and type.**	• **Physical Development in Pups: Eye opening, ear unfolding, tooth eruption, etc. (Justify selected landmarks).** • **Functional Observations: Motor, sensory, or behavioral assessments in pups and adults.**
Other critical observations *(additionally cross reference from study report and page no. to be provided)*		**Morphometric and Anatomical Observations** • **Anogenital distance (AGD) for all fetuses: Measured and statistically evaluated by sex, corrected**	**Morphometric and Anatomical Observations** • **Anogenital distance (AGD) for all fetuses: Measured and statistically evaluated by sex, corrected**	**Implantation and Offspring Data** 2. **Number of Implantations and Corpora Lutea** 3. **Litter Size and Live Births** 4. **Post-implantation Loss**

		for fetal body weight. • **External, visceral (soft tissue), and skeletal examinations performed for malformations or variations.** • **Sex-specific Findings- Indicate cases of incomplete testicular descent (cryptorchidism) or other sex-related developmental effects in male fetuses.**	**for fetal body weight.** • **External, visceral (soft tissue), and skeletal examinations performed for malformations or variations.** **Sex-specific Findings** **Indicate cases of incomplete testicular descent (cryptorchidism) or other sex-related developmental effects in male fetuses.**	**5. Grossly Visible Abnormalities in Pups** **6. Runts: Number if determined.**
Necropsy findings *(cross reference from study report and page no. to be provided)*				
Histopathological findings *(cross reference from study report and page no. to be provided)*				
Organs examined				
Statistical tool used for calculation				
Dose-Response relationship				

4. Result Summary

Particulars		**Developmental toxicity study - Rat**	**Developmental toxicity study - Rabbit**	**Two generational reproduction toxicity**
Number of animals dosed, survived and died				
Study Conclusion				

PHARMACOKINETICS STUDIES

1. Test study Information:

Particulars		Pharmacokinetics and Metabolism in Rat	Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			
CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

Particulars		Pharmacokinetics and Metabolism in Rat	Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)
Species/Strain			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			
Fasting conditions (Duration) pre-dose			
Housing conditions: Temperature, Humidity, Light cycle			

Total radioactivity per animal			
Analytical methods used for separation, quantitation and identification of metabolites			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule/ Single Duration of the study			
Rationale for dose selection			
Any other special consideration			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Pharmacokinetics and Metabolism in Rat**	**Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)**
NOAEL/NOAEC/LOAEC			
Food and water consumption changes			
Clinical observations schedule with respect to doses			

Absorption Data: • **Rate and extent of absorption by each route of exposure.** *(additionally cross reference from study report and page no. to be provided)*			
Tissue Distribution: • **Report as % of administered dose and as µg-equivalents per g of tissue.** • **Include tissue-to-blood and tissue-to-plasma ratios.** *(additionally cross reference from study report and page no. to be provided)*			
Plasma Concentrations & TK Parameters: • **Bioavailability, AUC, Cmax, Tmax, clearance, and half-life for relevant routes.** *(additionally cross reference from study report and page no. to be provided)*			
Excretion Data: • **Amounts of parent compound and metabolites in excreta (% of administered dose).** *(additionally cross reference from study report and page no. to be provided)*			
Metabolic pathway *(cross reference from study report and page no. to be provided)*			
Statistical tool used for calculation			

4. Result Summary

Particulars		Pharmacokinetics and Metabolism in Rat	Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)
Correlation with observed toxicity, dose proportionality, systemic exposure			
Study Conclusion			

IMMUNOTOXICITY STUDY

1. Test study Information:

Particulars		Immunotoxicity test
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

Particulars		Immunotoxicity test
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		
Dose of test chemical		

Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		Immunotoxicity test
NOAEL/BMC/NOAEC/LOAEC/BALF		

Food and water consumption changes (if any)		
Clinical observations schedule with respect to doses		
Nature and severity of the toxicity signs (if any)		
Time course of onset and Duration of the toxicity effect (if any)		
Body weight changes		
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*		
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*		
Lymphoid Organs examined and weights *(cross reference from study report and page no. to be provided)*		
Phenotyping *(cross reference from study report and page no. to be provided)*		
Primary Functional test (TDAR) *(cross reference from study report and page no. to be provided)*		
Secondary functional Test Report *(cross reference from study report and page no. to be provided)*		
Statistical tool used for calculation		

4. Result Summary

Particulars		**Immunotoxicity test**
Number of animals dosed, survived and died		
Study Conclusion		

MUTAGENECITY STUDIES

1. Test study Information:

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Report No.					

Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					
IAEC No. (in case of in vivo studies) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Species/Strain					
No. and sex of animals per dose (in case of in vivo)					
Age and Body weight range at dosing (in case of in vivo)					
Concentration/ Dose of test chemical					
Vehicle used					
Route of administration (in case of in vivo)					
Fasting conditions (Duration) pre-dose (in case of in vivo)					
Housing conditions: Temperature, Humidity, Light cycle (in case of in vivo)					
Feed report (source, restriction)					

(cross reference from study report and page no. to be provided)					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Food and water consumption changes (in vivo)					
Clinical observations schedule with respect to doses (in vivo)					
Body weight changes (in vivo)					
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*					
Scoring (in vivo)					
Primary endpoints (in vitro) frequency of micronucleated erythrocytes (PCE/NCE or					

reticulocyte MN) per animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome aberrations per metaphase in bone marrow/ spermatogonial chromosomal aberration frequency in testicular metaphases/ others *(cross reference from study report and page no. to be provided)*					
Primary endpoints (in vitro) revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others *(cross reference from study report and page no. to be provided)*					
Statistical tool used for calculation					

4. Result Summary

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Conclusion					

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee**	**Acute toxicity-Earthworm**

						(oral and contact)	
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions							

(Duration) pre-dose where applicable							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*							
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations							

schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*							
Organs examined							
Food and feed consumption data *(cross reference from study report and page no. to be provided)*							
Statistical tool if used							

4. Result Summary

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

5. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**

6. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**

7. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**

8. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
9(3)-TI-New Source

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Manufacturing address		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. RTT Permit

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

5. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

6. Study Design/Methodology

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							

Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

7. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							

Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

8. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

SHORT-TERM REPEATED DOSE TOXICITY STUDIES

1. Test study Information:

		Repeated dose 28-day oral toxicity-Rodent	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						

Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *cross reference from study report and page no. to be provided*						
Feed analysis statement / certificate (as applicable) (optional)						

cross reference from study report and page no. to be provided						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

*** In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.**

5. Observations

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
NOAEL/ NOAEC/LOAEC						
Food and water consumption changes						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report						

(cross reference from study report and page no. to be provided)						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*						
Organs examined						
organ weight changes and organ/body weight ratios						
Statistical tool used for calculation						

6. Result Summary

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Number of animals dosed, survived and died						
Study Conclusion						

<u>MUTAGENECITY STUDIES</u>

1. Test study Information:

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					

Batch No.					
Purity					
CoA attached (Yes/No)					
IAEC No. (in case of in vivo studies) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

5. Study Design/Methodology

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Species/Strain					
No. and sex of animals per dose (in case of in vivo)					
Age and Body weight range at dosing (in case of in vivo)					
Concentration/ Dose of test chemical					
Vehicle used					
Route of administration (in case of in vivo)					
Fasting conditions (Duration) pre-dose (in case of in vivo)					
Housing conditions: Temperature, Humidity, Light cycle (in case of in vivo)					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as					

applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

6. Observations

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Food and water consumption changes (in vivo)					
Clinical observations schedule with respect to doses (in vivo)					
Body weight changes (in vivo)					
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*					
Scoring (in vivo)					
Primary endpoints (in vitro) frequency of micronucleated erythrocytes (PCE/NCE or reticulocyte MN) per animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome					

aberrations per metaphase in bone marrow/ spermatogonial chromosomal aberration frequency in testicular metaphases/ others *(cross reference from study report and page no. to be provided)*					
Primary endpoints (in vitro) revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others *(cross reference from study report and page no. to be provided)*					
Statistical tool used for calculation					

7. Result Summary

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Conclusion					

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.							
Sponsor (Name and Address)							

Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose where applicable							
Housing conditions:							

Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*							
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							

Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*							
Organs examined							
Food and feed consumption data *(cross reference from study report and page no. to be provided)*							
Statistical tool if used							

4. Result Summary

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**

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4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
9(3) TIM-New molecule

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Manufacturing address		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

3. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							

Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

3. Study Design/Methodology

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional)							

cross reference from study report and page no. to be provided							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

SHORT-TERM REPEATED DOSE TOXICITY STUDIES

1. Test study Information:

		Repeated dose 28-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Non rodent	Repeated dose 21 / 28-day dermal toxicity	Repeated dose 28-day inhalational toxicity
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

		Repeated dose 28-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Non rodent	Repeated dose 21 / 28-day dermal toxicity	Repeated dose 28-day inhalational toxicity
Species/Strain						
No. and sex of animals per dose						

Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *cross reference from study report and page no. to be provided*						
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

*** In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.**

3. Observations

		Repeated dose 28-day oral toxicity-Rodent	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
NOAEL/ NOAEC/LOAEC						
Food and water consumption changes						
Clinical observations schedule with respect to doses						

Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*						
Organs examined						
organ weight changes and organ/body weight ratios						
Statistical tool used for calculation						

4. Result Summary

		Repeated dose 28-day oral toxicity-Rodent	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**

Number of animals dosed, survived and died						
Study Conclusion						

NEUROTOXICITY STUDIES

1. Test study Information:

		Acute Neurotoxicity	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

		Acute Neurotoxicity	Repeated dose Neurotoxicity	Delayed neurotoxicity-OP compound (acute exposure)	Delayed neurotoxicity-OP compound (Repeated administration)	Developmental Neurotoxicity study
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*						
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*						
Acclimatization period and grouping						
Dosing Schedule,						

Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Acute Neurotoxicity	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity -OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
NOAEL						
Food and water consumption changes (if any)						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						

Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Neuropathological assessment report *(cross reference from study report and page no. to be provided)*						
Neurochemical assessment report *(cross reference from study report and page no. to be provided)*						
Electrophysiological assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings of brain *(cross reference from study report and page no. to be provided)*						
Statistical tool if used						

4. Result Summary

		Acute Neurotoxicity	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Number of animals						

dosed, survived and died						
Study Conclusion						

LONG-TERM TOXICITY STUDIES

1. Test study Information:

		Carcinogenicity- Mice	**Combined Chronic toxicity and carcinogenicity-Rat**
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			
CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

		Carcinogenicity- Mice	**Combined Chronic toxicity and carcinogenicity-Rat**
Species/Strain			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			
Fasting conditions (Duration) pre-dose			

Housing conditions: Temperature, Humidity, Light cycle			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule, Duration of the study			
Rationale for dose selection			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Carcinogenicity- Mice	**Combined Chronic toxicity and carcinogenicity-Rat**
NOAEL/LOAEL			
Food and water consumption changes			
Clinical observations schedule with respect to doses			
Nature and severity of the toxicity signs (if any)			
Time course of onset and Duration of the toxicity effect (if any)			
Date and time of death, if before scheduled sacrifice			
Total No. of Mortality per dose			

Body weight changes			
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*			
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*			
Necropsy findings • **Terminal body weight** • **Organ weights and their ratios, if applicable** • **Necropsy findings** • **Incidence and severity of abnormalities** *(cross reference from study report and page no. to be provided)*			
Histopathological findings • **Non neoplastic histopathological findings** • **Neoplastic histopathological findings** • **Correlation between gross and microscopic findings** • **Detailed description of all treatment-related histopathological findings including severity gradings** • **Report of any peer review of slides;** *(cross reference from study report and page no. to be provided)*			
Organs examined			

Statistical tool used for calculation			
Dose-Response relationship			

4. Result Summary

		Carcinogenicity- Mice	**Combined Chronic toxicity and carcinogenicity-Rat**
Number of animals dosed, survived and died			
Study Conclusion			

DEVELOPMENTAL & REPRODUCTION TOXICITY STUDIES

1. Test study Information:

		Developmental toxicity study Rat	**Developmental toxicity study - Rabbit**	**Two generational reproduction toxicity**
Study Report No.				
Sponsor (Name and Address)				
Source (Name and Address)				
Study initiation Date				
Study completion Date				
Batch No.				
Purity				
CoA attached (Yes/No)				
IAEC No. (if available)				
GLP Statement (yes/no)				
Whether the study is done within GLP certificate validity date (yes/no)				
OECD Guideline or others				

2. Study Design/Methodology

		Developmental toxicity study -Rat	**Developmental toxicity study - Rabbit**	**Two generational reproduction toxicity**
Species/Strain				
No. and sex of animals per dose				
Age and Body weight range at dosing				

Dose of test chemical				
Vehicle used				
Route of administration				
Fasting conditions (Duration) pre-dose				
Housing conditions: Temperature, Humidity, Light cycle				
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*				
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*				
Acclimatization period and grouping				
Dosing Schedule, Duration of the study				
Rationale for dose selection				
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.				

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Developmental toxicity study - Rat	**Developmental toxicity study - Rabbit**	**Two generational reproduction toxicity**
NOAEL/LOAEL				
Food, water and test material consumption changes (if any)				
Number of animals at study start, number surviving to termination.				

Number of confirmed pregnancies, abortions, and early deliveries.				
Day and cause of death (if any); survival status to scheduled sacrifice.				
Data from non-surviving animals reported separately (excluded from inter-group statistics). *(cross reference from study report and page no. to be provided)*				
Day of onset, duration, and progression of abnormal clinical signs.				
Daily clinical observations *(cross reference from study report and page no. to be provided)*		**Maternal toxicity (posture, respiration, activity, appearance).**	**Maternal toxicity (posture, respiration, activity, appearance).**	**Nature, Severity, Duration of Clinical Signs and Mortality**
Body weight changes and growth *(cross reference from study report and page no. to be provided)*		**body weight change, and gravid uterine weight; Optionally, body weight change corrected for gravid uterine weight.**	**body weight change, and gravid uterine weight; Optionally, body weight change corrected for gravid uterine weight.**	**Body weight data and changes including litter and pub at birth and during lactation.** **Body weight at sacrifice**
Consumption and Hormonal Data *(cross reference from study report and page no. to be provided)*		**T4, T3, and TSH levels (and other hormones if measured).** **Details of hormone kits/antibodies used, assay method, and laboratory historical control data (mean ± SD).**	**T4, T3, and TSH levels (and other hormones if measured).** **Details of hormone kits/antibodies used, assay method, and laboratory historical control data (mean ± SD).**	
Reproductive Indices/ Reproductive Performance and Fertility Indices *(additionally cross reference from study report and page no. to be provided)*		• **Number of corpora lutea (ovulation count).** • **Number of implantation sites per dam.** • **Number and percentage of live and dead fetuses.**	• **Number of corpora lutea (ovulation count).** • **Number of implantation sites per dam.** • **Number and percentage of live and dead fetuses.**	• **Toxic Responses by Sex and Dose** • **Number of Animals Used** • **Toxic or Other Effects on Reproduction** • **Time-to-Mating**

		• **Number and percentage of resorptions (early and late).** • **Pre-implantation loss (%) = (Corpora lutea – Implantations) / Corpora lutea × 100** • **Post-implantation loss (%) = (Implantations – Live fetuses) / Implantations × 100**	• **Number and percentage of resorptions (early and late).** • **Pre-implantation loss (%) = (Corpora lutea – Implantations) / Corpora lutea × 100** • **Post-implantation loss (%) = (Implantations – Live fetuses) / Implantations × 100**	• **Gestation Length** • **Estrous Cycle Data** • **Sperm Parameters: Total cauda epididymal sperm count, Percent progressively motile sperm, Percent morphologically normal sperm, Percent of sperm with identified abnormalities**
Fetal Viability and Growth • **Number and percentage of live fetuses per litter.** • **Sex ratio (males/females per litter).** • **Fetal body weight, reported by sex and combined mean.** *(additionally cross reference from study report and page no. to be provided)*				
Abnormalities and Categorisation/ Developmental Landmarks *(additionally cross reference from study report and page no. to be provided)*		• **Criteria for classification of findings (malformations vs. variations) clearly described.** • **Record total number and percent of fetuses and litters showing any external,**	• **Criteria for classification of findings (malformations vs. variations) clearly described.** • **Record total number and percent of fetuses and litters showing any external,**	• **Physical Development in Pups: Eye opening, ear unfolding, tooth eruption, etc. (Justify selected landmarks).** • **Functional Observations: Motor, sensory, or**

		visceral, or skeletal alteration and specific anomalies with their incidence and type.	**visceral, or skeletal alteration and specific anomalies with their incidence and type.**	**behavioral assessments in pups and adults.**
Other critical observations *(additionally cross reference from study report and page no. to be provided)*		**Morphometric and Anatomical Observations** • **Anogenital distance (AGD) for all fetuses: Measured and statistically evaluated by sex, corrected for fetal body weight.** • **External, visceral (soft tissue), and skeletal examinations performed for malformations or variations.** • **Sex-specific Findings- Indicate cases of incomplete testicular descent (cryptorchidism) or other sex-related developmental effects in male fetuses.**	**Morphometric and Anatomical Observations** • **Anogenital distance (AGD) for all fetuses: Measured and statistically evaluated by sex, corrected for fetal body weight.** • **External, visceral (soft tissue), and skeletal examinations performed for malformations or variations.** **Sex-specific Findings** **Indicate cases of incomplete testicular descent (cryptorchidism) or other sex-related developmental effects in male fetuses.**	**Implantation and Offspring Data** **7. Number of Implantations and Corpora Lutea** **8. Litter Size and Live Births** **9. Post-implantation Loss** **10. Grossly Visible Abnormalities in Pups** **11. Runts: Number if determined.**
Necropsy findings *(cross reference from study report and page no. to be provided)*				
Histopathological findings *(cross reference from study report and page no. to be provided)*				
Organs examined				
Statistical tool used for calculation				

Dose-Response relationship				

4. Result Summary

		Developmental toxicity study - Rat	**Developmental toxicity study - Rabbit**	**Two generational reproduction toxicity**
Number of animals dosed, survived and died				
Study Conclusion				

PHARMACOKINETICS STUDIES

1. Test study Information:

		Pharmacokinetics and Metabolism in Rat	**Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)**
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			
CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

		Pharmacokinetics and Metabolism in Rat	**Feeding study including Metabolism in Livestock (Goat / Cow)**
Species/Strain			
No. and sex of animals per dose			

Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			
Fasting conditions (Duration) pre-dose			
Housing conditions: Temperature, Humidity, Light cycle			
Total radioactivity per animal			
Analytical methods used for separation, quantitation and identification of metabolites			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule/ Single Duration of the study			
Rationale for dose selection			
Any other special consideration			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Pharmacokinetics and Metabolism in Rat	**Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)**
NOAEL/NOAEC/LOAEC			
Food and water consumption changes			
Clinical observations schedule with respect to doses			
Absorption Data: • **Rate and extent of absorption by each route of exposure.** *(additionally cross reference from study report and page no. to be provided)*			
Tissue Distribution: • **Report as % of administered dose and as µg-equivalents per g of tissue.** • **Include tissue-to-blood and tissue-to-plasma ratios.** *(additionally cross reference from study report and page no. to be provided)*			
Plasma Concentrations & TK Parameters: • **Bioavailability, AUC, Cmax, Tmax, clearance, and half-life for relevant routes.** *(additionally cross reference from study report and page no. to be provided)*			
Excretion Data: • **Amounts of parent compound and metabolites in**			

excreta (% of administered dose). *(additionally cross reference from study report and page no. to be provided)*			
Metabolic pathway *(cross reference from study report and page no. to be provided)*			
Statistical tool used for calculation			

4. Result Summary

		Pharmacokinetics and Metabolism in Rat	**Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)**
Correlation with observed toxicity, dose proportionality, systemic exposure			
Study Conclusion			

IMMUNOTOXICITY STUDY

1. Test study Information:

		Immunotoxicity test
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

		Immunotoxicity test
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		
Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments		

(Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Immunotoxicity test
NOAEL/BMC/NOAEC/LOAEC/BALF		
Food and water consumption changes (if any)		
Clinical observations schedule with respect to doses		
Nature and severity of the toxicity signs (if any)		
Time course of onset and Duration of the toxicity effect (if any)		
Body weight changes		
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*		
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*		
Lymphoid Organs examined and weights *(cross reference from study report and page no. to be provided)*		
Phenotyping *(cross reference from study report and page no. to be provided)*		
Primary Functional test (TDAR) *(cross reference from study report and page no. to be provided)*		
Secondary functional Test Report *(cross reference from study report and page no. to be provided)*		
Statistical tool used for calculation		

4. Result Summary

		Immunotoxicity test
Number of animals dosed, survived and died		
Study Conclusion		

MUTAGENECITY STUDIES

1. Test study Information:

		Mutagenicity test AMES TEST	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					
IAEC No. (in case of in vivo studies) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

		Mutagenicity test AMES TEST	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Species/Strain					
No. and sex of animals per dose (in case of in vivo)					
Age and Body weight range at dosing (in case of in vivo)					
Concentration/ Dose of test chemical					

Vehicle used					
Route of administration (in case of in vivo)					
Fasting conditions (Duration) pre-dose (in case of in vivo)					
Housing conditions: Temperature, Humidity, Light cycle (in case of in vivo)					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Mutagenicity test AMES TEST	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Food and water consumption changes (in vivo)					
Clinical observations schedule with respect to doses (in vivo)					

Body weight changes (in vivo)					
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*					
Scoring (in vivo)					
Primary endpoints (in vitro) frequency of micronucleated erythrocytes (PCE/NCE or reticulocyte MN) per animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome aberrations per metaphase in bone marrow/ spermatogonial chromosomal aberration frequency in testicular metaphases/ others *(cross reference from study report and page no. to be provided)*					
Primary endpoints (in vitro) revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others *(cross reference from study report and page no. to be provided)*					

Statistical tool used for calculation					

4. Result Summary

		Mutagenicity test AMES TEST	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Conclusion					

ECOTOXICITY STUDIES

1. Test study Information:

		Acute Avian toxicity	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

		Acute Avian toxicity	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**

Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose where applicable							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*							
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or							

cross reference of report.							

3. Observations

		Acute Avian toxicity	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*							
Organs examined							
Food and feed consumption data *(cross reference from study report and page no. to be provided)*							
Statistical tool if used							

4. Result Summary

		Acute Avian toxicity	Repeated dose avian toxicity	Avian reproductive toxicity	Acute toxicity-fish	Acute toxicity-Honeybee (oral and contact)	Acute toxicity-Earthworm
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**
2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**
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4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE

9(3) TIM vs TI

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers (if available)		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							

Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							

Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

MUTAGENECITY STUDIES

1. Test study Information:

		Mutagenicity test AMES TEST	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vivo)*
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					
IAEC No. (in case of in vivo studies) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

		Mutagenicity test AMES TEST	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vivo)*
Species/Strain					
No. and sex of animals per dose (in case of in vivo)					

Age and Body weight range at dosing (in case of in vivo)					
Concentration/ Dose of test chemical					
Vehicle used					
Route of administration (in case of in vivo)					
Fasting conditions (Duration) pre-dose (in case of in vivo)					
Housing conditions: Temperature, Humidity, Light cycle (in case of in vivo)					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Mutagenicity test AMES TEST	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Food and water consumption changes (in vivo)					

Clinical observations schedule with respect to doses (in vivo)					
Body weight changes (in vivo)					
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*					
Scoring (in vivo)					
Primary endpoints (in vitro) frequency of micronucleated erythrocytes (PCE/NCE or reticulocyte MN) per animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome aberrations per metaphase in bone marrow/ spermatogonial chromosomal aberration frequency in testicular metaphases/ others *(cross reference from study report and page no. to be provided)*					
Primary endpoints (in vitro) revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others					

(cross reference from study report and page no. to be provided)					
Statistical tool used for calculation					

4. Result Summary

		Mutagenicity test AMES TEST	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Conclusion					

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

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TOXICOLOGY TEMPLATE
9(3) TIM vs FI-WRT

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

MUTAGENECITY STUDIES

1. Test study Information:

		Mutagenicity test AMES TEST
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (in case of in vivo studies) (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

		Mutagenicity test AMES TEST
Species/Strain		
No. and sex of animals per dose (in case of in vivo)		
Age and Body weight range at dosing (in case of in vivo)		

Concentration/ Dose of test chemical		
Vehicle used		
Route of administration (in case of in vivo)		
Fasting conditions (Duration) pre-dose (in case of in vivo)		
Housing conditions: Temperature, Humidity, Light cycle (in case of in vivo)		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Mutagenicity test **AMES TEST**
Food and water consumption changes (in vivo)		
Clinical observations schedule with respect to doses (in vivo)		
Body weight changes (in vivo)		
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*		
Scoring (in vivo)		
Primary endpoints (in vitro) frequency of micronucleated erythrocytes (PCE/NCE or reticulocyte MN) per animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome aberrations per metaphase in bone marrow/ spermatogonial chromosomal		

aberration frequency in testicular metaphases/ others *(cross reference from study report and page no. to be provided)*		
Primary endpoints (in vitro) revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others *(cross reference from study report and page no. to be provided)*		
Statistical tool used for calculation		

4. Result Summary

		Mutagenicity test **AMES TEST**
Study Conclusion		

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

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TOXICOLOGY TEMPLATE
9(3) FI

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Type of Formulation		
In case of formulation, status of technical registration		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. RTT Permit

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

4. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature,							

Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							

Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							

Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose where applicable							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*							
Feed analysis statement / certificate (as applicable) (optional)*							

(cross reference from study report and page no. to be provided)							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and histopathological findings							

(cross reference from study report and page no. to be provided)							
Organs examined							
Food and feed consumption data *(cross reference from study report and page no. to be provided)*							
Statistical tool if used							

4. Result Summary

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**
2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**
3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**
4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
9(3) FIM

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Type of Formulation		
In case of formulation, status of technical registration		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							

Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional)							

cross reference from study report and page no. to be provided							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							

Organs examined							

4. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
Ethical consideration (NO.) (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (yes/no)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		Acute Avian toxicity	Repeated dose avian toxicity	Avian reproductive toxicity	Acute toxicity-fish	Acute toxicity-Honeybee (oral and contact)	Acute toxicity-Earthworm
Species							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Dose of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose where applicable							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*							
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							

Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Avian toxicity**	**Repeated dose avian toxicity**	**Avian reproductive toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes							
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*							
Organs examined							
Food and feed consumption data *(cross reference from study report and page no. to be provided)*							

Statistical tool if used							

4. Result Summary

Particulars		Acute Avian toxicity	Repeated dose avian toxicity	Avian reproductive toxicity	Acute toxicity-fish	Acute toxicity-Honeybee (oral and contact)	Acute toxicity-Earthworm
Number of animals dosed, survived and died							
LD_{50}							
Study Conclusion							

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

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4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE-Botanicals
9(3b) TI and TIM

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Manufacturing address		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. RTT Permit (If applicable)

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

5. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature,							

Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							

Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					

Ethical consideration (NO.) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species					
No. and sex of animals per dose					
Age and Body weight range at dosing					
Dose of test chemical					
Vehicle used					
Route of administration					
Fasting conditions (Duration) pre-dose where applicable					
Housing conditions: Temperature, Humidity, Light cycle					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					

Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

3. Observations

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50					
Clinical observations schedule with respect to doses					
Nature and severity of the toxicity signs (if any)					
Time course of onset and Duration of the toxicity effect (if any)					
Date and time of death, if before scheduled sacrifice					
Total No. of Mortality per dose					
Body weight schedule changes					
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*					
Organs examined					
Food and feed consumption data *(cross reference from study report and page no. to be provided)*					
Statistical tool if used					

4. Result Summary

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died					

LD_{50}					
Study Conclusion					

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

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4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE-Botanicals
9(3b) FIM

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Type of Formulation		
In case of formulation, status of technical registration		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							

Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional)							

cross reference from study report and page no. to be provided							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and*							

page no. to be provided							
Organs examined							

4. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					
Ethical consideration (NO.) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

Particulars		Acute Avian toxicity	Acute toxicity-fish	Acute toxicity-Honeybee (oral and contact)	Acute toxicity-Earthworm
Species					
No. and sex of animals per dose					
Age and Body weight range at dosing					
Dose of test chemical					
Vehicle used					
Route of administration					
Fasting conditions (Duration) pre-dose where applicable					
Housing conditions: Temperature, Humidity, Light cycle					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

3. Observations

Particulars		Acute Avian toxicity	Acute toxicity-fish	Acute toxicity-Honeybee (oral and contact)	Acute toxicity-Earthworm

Result LD50					
Clinical observations schedule with respect to doses					
Nature and severity of the toxicity signs (if any)					
Time course of onset and Duration of the toxicity effect (if any)					
Date and time of death, if before scheduled sacrifice					
Total No. of Mortality per dose					
Body weight schedule changes					
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*					
Organs examined					
Food and feed consumption data *(cross reference from study report and page no. to be provided)*					
Statistical tool if used					

4. Result Summary

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died					
LD_{50}					
Study Conclusion					

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

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4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE-Botanicals
9(3) TI/TIM

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Manufacturing address		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. RTT Permit (If applicable)

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

6. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature,							

Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							

Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

SHORT-TERM REPEATED DOSE TOXICITY STUDIES

1. Test study Information:

		Repeated dose 28-day oral toxicity-Rodent	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-dog**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						

OECD Guideline or others						

2. Study Design/Methodology

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-dog**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *cross reference from study report and page no. to be provided*						
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

*** In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.**

3. Observations

Particulars		Repeated dose 28-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-dog	Repeated dose 21 / 28-day dermal toxicity	Repeated dose 28-day inhalational toxicity
NOAEL/ NOAEC/LOAEC						
Food and water consumption changes						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*						
Organs examined						

organ weight changes and organ/body weight ratios						
Statistical tool used for calculation						

4. Result Summary

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-dog**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
Number of animals dosed, survived and died						
Study Conclusion						

NEUROTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						

Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*						
Feed analysis statement /						

certificate (as applicable) * *(cross reference from study report and page no. to be provided)*						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity -OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
NOAEL						
Food and water consumption changes (if any)						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						

Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Neuropathological assessment report *(cross reference from study report and page no. to be provided)*						
Neurochemical assessment report *(cross reference from study report and page no. to be provided)*						
Electrophysiological assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings of brain *(cross reference from study report and*						

page no. to be provided)						
Statistical tool if used						

4. Result Summary

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Number of animals dosed, survived and died						
Study Conclusion						

LONG-TERM TOXICITY STUDIES

1. Test study Information:

Particulars		**Carcinogenicity/chronic toxicity/combined toxicity study**
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

Particulars		Carcinogenicity/chronic toxicity/combined toxicity study
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		
Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or		

cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		Carcinogenicity/chronic toxicity/combined toxicity study
NOAEL/LOAEL		
Food and water consumption changes		
Clinical observations schedule with respect to doses		
Nature and severity of the toxicity signs (if any)		
Time course of onset and Duration of the toxicity effect (if any)		
Date and time of death, if before scheduled sacrifice		
Total No. of Mortality per dose		
Body weight changes		
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*		
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*		
Necropsy findings • **Terminal body weight** • **Organ weights and their ratios, if applicable** • **Necropsy findings** • **Incidence and severity of abnormalities** *(cross reference from study report and page no. to be provided)*		

Histopathological findings • **Non neoplastic histopathological findings** • **Neoplastic histopathological findings** • **Correlation between gross and microscopic findings** • **Detailed description of all treatment-related histopathological findings including severity gradings** • **Report of any peer review of slides;** *(cross reference from study report and page no. to be provided)*		
Organs examined		
Statistical tool used for calculation		
Dose-Response relationship		

4. Result Summary

Particulars		**Carcinogenicity/chronic toxicity/combined toxicity study**
Number of animals dosed, survived and died		
Study Conclusion		

DEVELOPMENTAL & REPRODUCTION TOXICITY STUDIES

1. Test study Information:

Particulars		**TERATOGENECITY STUDY**	**REPRODUCTIVE TOXICITY**
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			

Purity			
CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

Particulars		**TERATOGENECITY STUDY**	**REPRODUCTIVE TOXICITY**
Species/Strain			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			
Fasting conditions (Duration) pre-dose			
Housing conditions: Temperature, Humidity, Light cycle			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			

Dosing Schedule, Duration of the study			
Rationale for dose selection			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**TERATOGENECITY STUDY**	**REPRODUCTIVE TOXICITY**
NOAEL/LOAEL			
Food, water and test material consumption changes (if any)			
Number of animals at study start, number surviving to termination.			
Number of confirmed pregnancies, abortions, and early deliveries.			
Day and cause of death (if any); survival status to scheduled sacrifice.			
Data from non-surviving animals reported separately (excluded from inter-group statistics). *(cross reference from study report and page no. to be provided)*			
Day of onset, duration, and progression of abnormal clinical signs.			
Daily clinical observations *(cross reference from study report and page no. to be provided)*		**Maternal toxicity (posture, respiration, activity, appearance).**	**Nature, Severity, Duration of Clinical Signs and Mortality**
Body weight changes and growth		**body weight change, and gravid uterine weight; Optionally, body weight change corrected for gravid uterine weight.**	**Body weight data and changes including litter and pub at birth and during lactation.** **Body weight at sacrifice**

(cross reference from study report and page no. to be provided)			
Consumption and Hormonal Data *(cross reference from study report and page no. to be provided)*		**T4, T3, and TSH levels (and other hormones if measured).** **Details of hormone kits/antibodies used, assay method, and laboratory historical control data (mean ± SD).**	
Reproductive Indices/ Reproductive Performance and Fertility Indices *(additionally cross reference from study report and page no. to be provided)*		• **Number of corpora lutea (ovulation count).** • **Number of implantation sites per dam.** • **Number and percentage of live and dead fetuses.** • **Number and percentage of resorptions (early and late).** • **Pre-implantation loss (%) = (Corpora lutea – Implantations) / Corpora lutea × 100** • **Post-implantation loss (%) = (Implantations – Live fetuses) / Implantations × 100**	• **Toxic Responses by Sex and Dose** • **Number of Animals Used** • **Toxic or Other Effects on Reproduction** • **Time-to-Mating** • **Gestation Length** • **Estrous Cycle Data** • **Sperm Parameters: Total cauda epididymal sperm count, Percent progressively motile sperm, Percent morphologically normal sperm, Percent of sperm with identified abnormalities**
Fetal Viability and Growth • **Number and percentage of live fetuses per litter.** • **Sex ratio (males/females per litter).** • **Fetal body weight, reported by sex and combined mean.** *(additionally cross reference from study report and page no. to be provided)*			
Abnormalities and Categorisation/ Developmental Landmarks *(additionally cross reference from study report and page no. to be provided)*		• **Criteria for classification of findings (malformations vs. variations) clearly described.** • **Record total number and percent of fetuses and litters showing any external, visceral, or skeletal alteration and specific**	• **Physical Development in Pups: Eye opening, ear unfolding, tooth eruption, etc. (Justify selected landmarks).** • **Functional Observations: Motor, sensory, or behavioral**

		anomalies with their incidence and type.	**assessments in pups and adults.**
Other critical observations *(additionally cross reference from study report and page no. to be provided)*		**Morphometric and Anatomical Observations** • **Anogenital distance (AGD) for all fetuses: Measured and statistically evaluated by sex, corrected for fetal body weight.** • **External, visceral (soft tissue), and skeletal examinations performed for malformations or variations.** • **Sex-specific Findings- Indicate cases of incomplete testicular descent (cryptorchidism) or other sex-related developmental effects in male fetuses.**	**Implantation and Offspring Data** **12. Number of Implantations and Corpora Lutea** **13. Litter Size and Live Births** **14. Post-implantation Loss** **15. Grossly Visible Abnormalities in Pups** **16. Runts: Number if determined.**
Necropsy findings *(cross reference from study report and page no. to be provided)*			
Histopathological findings *(cross reference from study report and page no. to be provided)*			
Organs examined			
Statistical tool used for calculation			
Dose-Response relationship			

4. Result Summary

Particulars		**TERATOGENECITY STUDY**	**REPRODUCTIVE TOXICITY**
Number of animals dosed, survived and died			
Study Conclusion			

PHARMACOKINETICS STUDIES

1. Test study Information:

Particulars		**Pharmacokinetics and Metabolism in Rat**	**Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)**
Study Report No.			
Sponsor (Name and Address)			

Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			
CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

Particulars		**Pharmacokinetics and Metabolism in Rat**	**Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)**
Species/Strain			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			
Fasting conditions (Duration) pre-dose			
Housing conditions: Temperature, Humidity, Light cycle			
Total radioactivity per animal			
Analytical methods used for separation, quantitation and identification of metabolites			
Feed report (source, restriction)			

(cross reference from study report and page no. to be provided)			
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule/ Single Duration of the study			
Rationale for dose selection			
Any other special consideration			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Pharmacokinetics and Metabolism in Rat**	**Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)**
NOAEL/NOAEC/LOAEC			
Food and water consumption changes			
Clinical observations schedule with respect to doses			
Absorption Data: • **Rate and extent of absorption by each route of exposure.** *(additionally cross reference from study report*			

and page no. to be provided)			
Tissue Distribution: • **Report as % of administered dose and as µg-equivalents per g of tissue.** • **Include tissue-to-blood and tissue-to-plasma ratios.** *(additionally cross reference from study report and page no. to be provided)*			
Plasma Concentrations & TK Parameters: • **Bioavailability, AUC, Cmax, Tmax, clearance, and half-life for relevant routes.** *(additionally cross reference from study report and page no. to be provided)*			
Excretion Data: • **Amounts of parent compound and metabolites in excreta (% of administered dose).** *(additionally cross reference from study report and page no. to be provided)*			
Metabolic pathway *(cross reference from study report and page no. to be provided)*			
Statistical tool used for calculation			

4. Result Summary

Particulars		**Pharmacokinetics and Metabolism in Rat**	**Feeding study including Metabolism in Livestock (Goat / Cow)/ Poultry (Hen)**
Correlation with observed toxicity, dose proportionality, systemic exposure			

Study Conclusion			

MUTAGENECITY STUDIES

1. Test study Information:

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					
IAEC No. (in case of in vivo studies) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Species/Strain					
No. and sex of animals per dose (in case of in vivo)					
Age and Body weight range at dosing (in case of in vivo)					
Concentration/ Dose of test chemical					
Vehicle used					
Route of administration (in case of in vivo)					
Fasting conditions (Duration) pre-dose (in case of in vivo)					

Housing conditions: Temperature, Humidity, Light cycle (in case of in vivo)					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Food and water consumption changes (in vivo)					
Clinical observations schedule with respect to doses (in vivo)					
Body weight changes (in vivo)					
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*					
Scoring (in vivo)					
Primary endpoints (in vitro) frequency of micronucleated					

erythrocytes (PCE/NCE or reticulocyte MN) per animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome aberrations per metaphase in bone marrow/ spermatogonial chromosomal aberration frequency in testicular metaphases/ others *(cross reference from study report and page no. to be provided)*					
Primary endpoints (in vitro) revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others *(cross reference from study report and page no. to be provided)*					
Statistical tool used for calculation					

4. Result Summary

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Conclusion					

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					

Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					
Ethical consideration (NO.) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species					
No. and sex of animals per dose					
Age and Body weight range at dosing					
Dose of test chemical					
Vehicle used					
Route of administration					
Fasting conditions (Duration) pre-dose where applicable					
Housing conditions: Temperature, Humidity, Light cycle					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					

Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

3. Observations

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50					
Clinical observations schedule with respect to doses					
Nature and severity of the toxicity signs (if any)					
Time course of onset and Duration of the toxicity effect (if any)					
Date and time of death, if before scheduled sacrifice					
Total No. of Mortality per dose					
Body weight schedule changes					
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*					
Organs examined					
Food and feed consumption data *(cross reference from study report and page no. to be provided)*					
Statistical tool if used					

4. **Result Summary**

Particulars		Acute Avian toxicity	Acute toxicity-fish	Acute toxicity-Honeybee (oral and contact)	Acute toxicity-Earthworm
Number of animals dosed, survived and died					
LD_{50}					
Study Conclusion					

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**
2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**
3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**
4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE-Botanicals
9(3) FI/FIM

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Type of Formulation		
In case of formulation, status of technical registration		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Residue tolerance limit fixed by foreign countries		
Medical data *cross reference from study report and page no. to be provided*		
Summary Report Derived from Health Records of Industrial Workers		
Summary Report Derived from International report on Carcinogenicity / Genotoxicity (if available)		

2. RTT Permit (if applicable)

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

3. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							

Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							

Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					

Ethical consideration (NO.) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Species					
No. and sex of animals per dose					
Age and Body weight range at dosing					
Dose of test chemical					
Vehicle used					
Route of administration					
Fasting conditions (Duration) pre-dose where applicable					
Housing conditions: Temperature, Humidity, Light cycle					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					

Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

3. Observations

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Result LD50					
Clinical observations schedule with respect to doses					
Nature and severity of the toxicity signs (if any)					
Time course of onset and Duration of the toxicity effect (if any)					
Date and time of death, if before scheduled sacrifice					
Total No. of Mortality per dose					
Body weight schedule changes					
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*					
Organs examined					
Food and feed consumption data *(cross reference from study report and page no. to be provided)*					
Statistical tool if used					

4. Result Summary

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Acute toxicity-Honeybee (oral and contact)**	**Acute toxicity-Earthworm**
Number of animals dosed, survived and died					

LD_{50}					
Study Conclusion					

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

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4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE

Registration of Biopesticides u/s 9(3)

1. General Information:

Test chemical Name/ strain/Accession number/Application no.		
Purpose		
Specify the status of the chemical listed in the schedule		
Global status with respect to Ban/restriction/withdrawn and reasons thereof		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**SAFETY PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars	**Single dose oral rat**	**Single dose Dermal rabbit**	**Acute inhalational**	**Single dose pulmonary rat**	**Single dose intraperitoneal rat**	**Single dose intravenous**	**Primary skin irritation-Rabbit**	**Primary eye irritation-Rabbit**	**Skin sensitization-Guinea Pig**
Study Report No.									
Sponsor (Name and Address)									
Source (Name and Address)									
Study initiation Date									
Study completion Date									
Batch No.									
Purity									

CoA attached (Yes/No)									
ETHICAL CONSIDERATION (NO.) (if available)									
OECD Guideline or others									

3. Study Design/Methodology:

Particulars	**Single dose oral rat**	**Single dose Dermal rabbit**	**Acute inhalational**	**Single dose pulmonary rat**	**Single dose intraperitoneal rat**	**Single dose intravenous**	**Primary skin irritation-Rabbit**	**Primary eye irritation-Rabbit**	**Skin sensitization-Guinea Pig**
Species/Strain									
No. and sex of animals per dose									
Age and Body weight range at dosing									
Dose of test chemical									
Vehicle used									
Route of administration									
Fasting conditions (Duration) pre-dose									
Housing conditions: Temperature, Humidity, Light cycle									
Feed report									

(source, restriction) *cross reference from study report and page no. to be provided*									
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*									
Acclimatization period and grouping									
Dosing Schedule, Duration of the study									
Rationale for dose selection									
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.									

4. Observations:

Particulars	Single dose oral rat	Single dose Dermal rabbit	Acute inhalational	Single dose pulmonary rat	Single dose intraperitoneal rat	Single dose intravenous	Primary skin irritation-Rabbit	Primary eye irritation-Rabbit	Skin sensitization-Guinea Pig
Result LD_{50}									
Clinical observations schedule with respect to doses									
Nature and severity of the toxicity signs (if any)									
Time course of onset and Duration of the toxicity effect (if any)									
Date and time of death, if before scheduled sacrifice									
Total No. of Mortality per dose									
Body weight schedule changes									
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*									
Organs examined									

5. Result Summary:

Particulars	Single dose oral rat	Single dose Dermal rabbit	Acute inhalational	Single dose pulmonary rat	Single dose intraperitoneal rat	Single dose intravenous	Primary skin irritation-Rabbit	Primary eye irritation-Rabbit	Skin sensitization-Guinea Pig
Study Conclusion									

ECOTOXICITY STUDIES

1. Test study Information:

	Acute Avian toxicity	Acute toxicity-fish	Acute oral toxicity-Honeybee	Acute contact toxicity-Honeybee	Toxicity to silkworm	Acute toxicity-Earthworm
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
Ethical consideration (NO.)						
GLP certificate validity date						
OECD Guideline or others						

2. Study Design/Methodology

	Acute Avian toxicity	Acute toxicity-fish	Acute oral toxicity-Honeybee	Acute contact toxicity-Honeybee	Toxicity to silkworm	Acute toxicity-Earthworm
Species						

No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose where applicable						
Housing conditions: Temperature, Humidity, Light cycle						
Diet and water report (source, restriction)						
Feed analysis certificate						
Acclimatization period and grouping						
Dosing Schedule/ Single or repeated, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments						

3. Observations

	Acute Avian toxicity	**Acute toxicity-fish**	**Acute oral toxicity-Honeybee**	**Acute contact toxicity-Honeybee**	**Toxicity to silkworm**	**Acute toxicity-Earthworm**
Result LD_{50}						
Clinical observations schedule with respect to doses						

Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight schedule changes						
Necropsy findings and histopathological findings						
Organs examined						
Food and feed consumption data						
Statistical tool if used						

4. Result Summary

	Acute Avian toxicity	**Acute toxicity-fish**	**Acute oral toxicity-Honeybee**	**Acute contact toxicity-Honeybee**	**Toxicity to silkworm**	**Acute toxicity-Earthworm**
Study Conclusion						

REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

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3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**
4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE- Pheromones/ Semio-chemicals

9(3b)

1. General Information:

Test chemical Name		
Specify the status of the chemical listed in the schedule		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		

2. RTT Permit (If applicable)

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

3. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							

IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							

Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

TERATOGENECITY STUDY

1. Test study Information:

Particulars		TERATOGENECITY STUDY
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

Particulars		TERATOGENECITY STUDY
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		

Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**TERATOGENECITY STUDY**

NOAEL/LOAEL		
Food, water and test material consumption changes (if any)		
Number of animals at study start, number surviving to termination.		
Number of confirmed pregnancies, abortions, and early deliveries.		
Day and cause of death (if any); survival status to scheduled sacrifice.		
Data from non-surviving animals reported separately (excluded from inter-group statistics). *(cross reference from study report and page no. to be provided)*		
Day of onset, duration, and progression of abnormal clinical signs.		
Daily clinical observations *(cross reference from study report and page no. to be provided)*		**Maternal toxicity (posture, respiration, activity, appearance).**
Body weight changes and growth *(cross reference from study report and page no. to be provided)*		**body weight change, and gravid uterine weight; Optionally, body weight change corrected for gravid uterine weight.**
Consumption and Hormonal Data *(cross reference from study report and page no. to be provided)*		**T4, T3, and TSH levels (and other hormones if measured).** **Details of hormone kits/antibodies used, assay method, and laboratory historical control data (mean ± SD).**
Reproductive Indices/ Reproductive Performance and Fertility Indices *(additionally cross reference from study report and page no. to be provided)*		• **Number of corpora lutea (ovulation count).** • **Number of implantation sites per dam.** • **Number and percentage of live and dead fetuses.** • **Number and percentage of resorptions (early and late).** • **Pre-implantation loss (%) = (Corpora lutea − Implantations) / Corpora lutea × 100** • **Post-implantation loss (%) = (Implantations − Live fetuses) / Implantations × 100**
Fetal Viability and Growth		

• **Number and percentage of live fetuses per litter.** • **Sex ratio (males/females per litter).** • **Fetal body weight, reported by sex and combined mean.** *(additionally cross reference from study report and page no. to be provided)*		
Abnormalities and Categorisation/ Developmental Landmarks *(additionally cross reference from study report and page no. to be provided)*		• **Criteria for classification of findings (malformations vs. variations) clearly described.** • **Record total number and percent of fetuses and litters showing any external, visceral, or skeletal alteration and specific anomalies with their incidence and type.**
Other critical observations *(additionally cross reference from study report and page no. to be provided)*		**Morphometric and Anatomical Observations** • **Anogenital distance (AGD) for all fetuses: Measured and statistically evaluated by sex, corrected for fetal body weight.** • **External, visceral (soft tissue), and skeletal examinations performed for malformations or variations.** • **Sex-specific Findings-** **Indicate cases of incomplete testicular descent (cryptorchidism) or other sex-related developmental effects in male fetuses.**
Necropsy findings *(cross reference from study report and page no. to be provided)*		
Histopathological findings *(cross reference from study report and page no. to be provided)*		
Organs examined		
Statistical tool used for calculation		
Dose-Response relationship		

4. Result Summary

Particulars		**TERATOGENECITY STUDY**
Number of animals dosed, survived and died		
Study Conclusion		

MUTAGENECITY STUDIES

1. Test study Information:

Particulars		Mutagenicity test AMES TEST	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vivo)*
Study Report No.					
Sponsor (Name and Address)					
Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					
IAEC No. (in case of in vivo studies) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

Particulars		Mutagenicity test AMES TEST	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vivo)*
Species/Strain					
No. and sex of animals per dose (in case of in vivo)					
Age and Body weight range at dosing (in case of in vivo)					
Concentration/ Dose of test chemical					
Vehicle used					
Route of administration (in case of in vivo)					
Fasting conditions (Duration) pre-dose (in case of in vivo)					
Housing conditions: Temperature, Humidity,					

Light cycle (in case of in vivo)					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Food and water consumption changes (in vivo)					
Clinical observations schedule with respect to doses (in vivo)					
Body weight changes (in vivo)					
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*					
Scoring (in vivo)					
Primary endpoints (in vitro) frequency of micronucleated erythrocytes (PCE/NCE or reticulocyte MN) per					

animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome aberrations per metaphase in bone marrow/ spermatogonial chromosomal aberration frequency in testicular metaphases/ others *(cross reference from study report and page no. to be provided)*					
Primary endpoints (in vitro) revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others *(cross reference from study report and page no. to be provided)*					
Statistical tool used for calculation					

4. Result Summary

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Conclusion					

IMMUNOTOXICITY STUDY

1. Test study Information:

Particulars		**Cellular immune response**
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		

Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

Particulars		**Cellular immune response**
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		
Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report*		

and page no. to be provided)		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Cellular immune response**
NOAEL/BMC/NOAEC/LOAEC/BALF		
Food and water consumption changes (if any)		
Clinical observations schedule with respect to doses		
Nature and severity of the toxicity signs (if any)		
Time course of onset and Duration of the toxicity effect (if any)		
Body weight changes		
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*		
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*		
Lymphoid Organs examined and weights *(cross reference from study report and page no. to be provided)*		

Phenotyping *(cross reference from study report and page no. to be provided)*		
Primary Functional test (TDAR) *(cross reference from study report and page no. to be provided)*		
Secondary functional Test Report *(cross reference from study report and page no. to be provided)*		
Statistical tool used for calculation		

4. Result Summary

Particulars		Cellular immune response
Number of animals dosed, survived and died		
Study Conclusion		

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		Acute Avian toxicity	Acute toxicity-fish
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			
CoA attached (Yes/No)			
Ethical consideration (NO.) (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

Particulars		Acute Avian toxicity	Acute toxicity-fish
Species			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			
Fasting conditions (Duration) pre-dose where applicable			
Housing conditions: Temperature, Humidity, Light cycle			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule, Duration of the study			
Rationale for dose selection			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

3. Observations

Particulars		Acute Avian toxicity	Acute toxicity-fish
Result LD50			
Clinical observations schedule with respect to doses			
Nature and severity of the toxicity signs (if any)			
Time course of onset and Duration of the toxicity effect (if any)			
Date and time of death, if before scheduled sacrifice			
Total No. of Mortality per dose			
Body weight schedule changes			
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*			
Organs examined			
Food and feed consumption data *(cross reference from study report and page no. to be provided)*			
Statistical tool if used			

4. Result Summary

Particulars		Acute Avian toxicity	Acute toxicity-fish
Number of animals dosed, survived and died			
LD_{50}			
Study Conclusion			

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

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TOXICOLOGY TEMPLATE-Pheromones/ Semio-chemicals

9(3)

1. General Information:

Test chemical Name		
Specify the status of the chemical listed in the schedule		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		

2. RTT Permit (If applicable)

RTT PERMIT DETAILS						
PERMIT NUMBER	**NAME OF THE INSECTICIDE/CHEMICAL**	**QUANTITY**	**NAME OF THE IMPORTER**	**NAME OF THE MANUFACTURER**	**SOURCE OF PROCUREMENT**	**PURPOSE OF THE IMPORT**

4. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							

IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

3. Study Design/Methodology

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							

Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.							

3. Observations

Particulars		**Acute Oral**	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

Particulars		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

SHORT-TERM REPEATED DOSE TOXICITY STUDIES (IF APPLICABLE)

1. Test study Information:

		Repeated dose 28-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Non rodent	Repeated dose 21 / 28-day dermal toxicity	Repeated dose 28-day inhalational toxicity
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

Particulars		Repeated dose 28-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Rodent	Repeated dose 90-day oral toxicity-Non rodent	Repeated dose 21 / 28-day dermal toxicity	Repeated dose 28-day inhalational toxicity
Species/Strain						

No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *cross reference from study report and page no. to be provided*						
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*						
Acclimatization period and grouping						
Dosing Schedule, Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

*** In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.**

3. Observations

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**
NOAEL/ NOAEC/LOAEC						
Food and water consumption changes						

Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						
Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*						
Organs examined						
organ weight changes and organ/body weight ratios						
Statistical tool used for calculation						

4. Result Summary

Particulars		**Repeated dose 28-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Rodent**	**Repeated dose 90-day oral toxicity-Non rodent**	**Repeated dose 21 / 28-day dermal toxicity**	**Repeated dose 28-day inhalational toxicity**

Number of animals dosed, survived and died						
Study Conclusion						

NEUROTOXICITY STUDIES (IF APPLICABLE)

1. Test study Information:

Particulars		Acute Neurotoxicity	Repeated dose Neurotoxicity	Delayed neurotoxicity-OP compound (acute exposure)	Delayed neurotoxicity-OP compound (Repeated administration)	Developmental Neurotoxicity study
Study Report No.						
Sponsor (Name and Address)						
Source (Name and Address)						
Study initiation Date						
Study completion Date						
Batch No.						
Purity						
CoA attached (Yes/No)						
IAEC No. (if available)						
GLP Statement (yes/no)						
Whether the study is done within GLP certificate validity date (yes/no)						
OECD Guideline or others						

2. Study Design/Methodology

Particulars		Acute Neurotoxicity	Repeated dose Neurotoxicity	Delayed neurotoxicity-OP compound (acute exposure)	Delayed neurotoxicity-OP compound (Repeated administration)	Developmental Neurotoxicity study
Species/Strain						
No. and sex of animals per dose						
Age and Body weight range at dosing						
Dose of test chemical						
Vehicle used						
Route of administration						
Fasting conditions (Duration) pre-dose						
Housing conditions: Temperature, Humidity, Light cycle						
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*						
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*						
Acclimatization period and grouping						
Dosing Schedule,						

Duration of the study						
Rationale for dose selection						
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.						

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity -OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
NOAEL						
Food and water consumption changes (if any)						
Clinical observations schedule with respect to doses						
Nature and severity of the toxicity signs (if any)						
Time course of onset and Duration of the toxicity effect (if any)						
Date and time of death, if before scheduled sacrifice						
Total No. of Mortality per dose						
Body weight changes						
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*						

Neurobehavioral assessment report *(cross reference from study report and page no. to be provided)*						
Neuropathological assessment report *(cross reference from study report and page no. to be provided)*						
Neurochemical assessment report *(cross reference from study report and page no. to be provided)*						
Electrophysiological assessment report *(cross reference from study report and page no. to be provided)*						
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*						
Necropsy findings and histopathological findings of brain *(cross reference from study report and page no. to be provided)*						
Statistical tool if used						

4. Result Summary

Particulars		**Acute Neurotoxicity**	**Repeated dose Neurotoxicity**	**Delayed neurotoxicity-OP compound (acute exposure)**	**Delayed neurotoxicity-OP compound (Repeated administration)**	**Developmental Neurotoxicity study**
Number of animals						

dosed, survived and died						
Study Conclusion						

DEVELOPMENTAL & REPRODUCTION TOXICITY STUDIES (IF APPLICABLE)

1. Test study Information:

Particulars		TERATOGENECITY STUDY	REPRODUCTIVE TOXICITY If applicable
Study Report No.			
Sponsor (Name and Address)			
Source (Name and Address)			
Study initiation Date			
Study completion Date			
Batch No.			
Purity			
CoA attached (Yes/No)			
IAEC No. (if available)			
GLP Statement (yes/no)			
Whether the study is done within GLP certificate validity date (yes/no)			
OECD Guideline or others			

2. Study Design/Methodology

Particulars		TERATOGENECITY STUDY	REPRODUCTIVE TOXICITY If applicable
Species/Strain			
No. and sex of animals per dose			
Age and Body weight range at dosing			
Dose of test chemical			
Vehicle used			
Route of administration			

Fasting conditions (Duration) pre-dose			
Housing conditions: Temperature, Humidity, Light cycle			
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*			
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*			
Acclimatization period and grouping			
Dosing Schedule, Duration of the study			
Rationale for dose selection			
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.			

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**TERATOGENECITY STUDY**	**REPRODUCTIVE TOXICITY If applicable**
NOAEL/LOAEL			
Food, water and test material consumption changes (if any)			
Number of animals at study start, number surviving to termination.			
Number of confirmed pregnancies, abortions, and early deliveries.			

Day and cause of death (if any); survival status to scheduled sacrifice.			
Data from non-surviving animals reported separately (excluded from inter-group statistics). *(cross reference from study report and page no. to be provided)*			
Day of onset, duration, and progression of abnormal clinical signs.			
Daily clinical observations *(cross reference from study report and page no. to be provided)*		**Maternal toxicity (posture, respiration, activity, appearance).**	**Nature, Severity, Duration of Clinical Signs and Mortality**
Body weight changes and growth *(cross reference from study report and page no. to be provided)*		**body weight change, and gravid uterine weight; Optionally, body weight change corrected for gravid uterine weight.**	**Body weight data and changes including litter and pub at birth and during lactation.** **Body weight at sacrifice**
Consumption and Hormonal Data *(cross reference from study report and page no. to be provided)*		**T4, T3, and TSH levels (and other hormones if measured).** **Details of hormone kits/antibodies used, assay method, and laboratory historical control data (mean ± SD).**	
Reproductive Indices/ Reproductive Performance and Fertility Indices *(additionally cross reference from study report and page no. to be provided)*		• **Number of corpora lutea (ovulation count).** • **Number of implantation sites per dam.** • **Number and percentage of live and dead fetuses.** • **Number and percentage of resorptions (early and late).** • **Pre-implantation loss (%) = (Corpora lutea − Implantations) / Corpora lutea × 100** • **Post-implantation loss (%) = (Implantations − Live fetuses) / Implantations × 100**	• **Toxic Responses by Sex and Dose** • **Number of Animals Used** • **Toxic or Other Effects on Reproduction** • **Time-to-Mating** • **Gestation Length** • **Estrous Cycle Data** • **Sperm Parameters: Total cauda epididymal sperm count, Percent progressively motile sperm, Percent morphologically normal sperm, Percent of sperm with identified abnormalities**
Fetal Viability and Growth			

• **Number and percentage of live fetuses per litter.** • **Sex ratio (males/females per litter).** • **Fetal body weight, reported by sex and combined mean.** *(additionally cross reference from study report and page no. to be provided)*			
Abnormalities and Categorisation/ Developmental Landmarks *(additionally cross reference from study report and page no. to be provided)*		• **Criteria for classification of findings (malformations vs. variations) clearly described.** • **Record total number and percent of fetuses and litters showing any external, visceral, or skeletal alteration and specific anomalies with their incidence and type.**	• **Physical Development in Pups: Eye opening, ear unfolding, tooth eruption, etc. (Justify selected landmarks).** • **Functional Observations: Motor, sensory, or behavioral assessments in pups and adults.**
Other critical observations *(additionally cross reference from study report and page no. to be provided)*		**Morphometric and Anatomical Observations** • **Anogenital distance (AGD) for all fetuses: Measured and statistically evaluated by sex, corrected for fetal body weight.** • **External, visceral (soft tissue), and skeletal examinations performed for malformations or variations.** • **Sex-specific Findings- Indicate cases of incomplete testicular descent (cryptorchidism) or other sex-related developmental effects in male fetuses.**	**Implantation and Offspring Data** 17. **Number of Implantations and Corpora Lutea** 18. **Litter Size and Live Births** 19. **Post-implantation Loss** 20. **Grossly Visible Abnormalities in Pups** 21. **Runts: Number if determined.**
Necropsy findings *(cross reference from study report and page no. to be provided)*			
Histopathological findings *(cross reference from study report and page no. to be provided)*			
Organs examined			

Statistical tool used for calculation			
Dose-Response relationship			

4. Result Summary

Particulars		**TERATOGENECITY STUDY**	**REPRODUCTIVE TOXICITY If applicable**
Number of animals dosed, survived and died			
Study Conclusion			

LONG-TERM TOXICITY STUDIES (IF APPLICABLE)

1. Test study Information:

Particulars		**Carcinogenicity/chronic toxicity/combined toxicity study**
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

Particulars		**Carcinogenicity/chronic toxicity/combined toxicity study**
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		

Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) * *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		Carcinogenicity/chronic toxicity/combined toxicity study
NOAEL/LOAEL		

Food and water consumption changes		
Clinical observations schedule with respect to doses		
Nature and severity of the toxicity signs (if any)		
Time course of onset and Duration of the toxicity effect (if any)		
Date and time of death, if before scheduled sacrifice		
Total No. of Mortality per dose		
Body weight changes		
Ophthalmological examination report *(cross reference from study report and page no. to be provided)*		
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*		
Necropsy findings • **Terminal body weight** • **Organ weights and their ratios, if applicable** • **Necropsy findings** • **Incidence and severity of abnormalities** *(cross reference from study report and page no. to be provided)*		
Histopathological findings • **Non neoplastic histopathological findings** • **Neoplastic histopathological findings** • **Correlation between gross and microscopic findings**		

• **Detailed description of all treatment-related histopathological findings including severity gradings** • **Report of any peer review of slides;** *(cross reference from study report and page no. to be provided)*		
Organs examined		
Statistical tool used for calculation		
Dose-Response relationship		

4. Result Summary

Particulars		**Carcinogenicity/chronic toxicity/combined toxicity study**
Number of animals dosed, survived and died		
Study Conclusion		

PHARMACOKINETICS STUDIES (IF APPLICABLE)

1. Test study Information:

Particulars		**Metabolism**
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		

Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

Particulars		**Metabolism**
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		
Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Total radioactivity per animal		
Analytical methods used for separation, quantitation and identification of metabolites		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report*		

and page no. to be provided)		
Acclimatization period and grouping		
Dosing Schedule/ Single Duration of the study		
Rationale for dose selection		
Any other special consideration		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Metabolism**
NOAEL/NOAEC/LOAEC		
Food and water consumption changes		
Clinical observations schedule with respect to doses		
Absorption Data: • **Rate and extent of absorption by each route of exposure.** *(additionally cross reference from study report and page no. to be provided)*		
Tissue Distribution: • **Report as % of administered dose and as µg-equivalents per g of tissue.** • **Include tissue-to-blood and tissue-to-plasma ratios.**		

(additionally cross reference from study report and page no. to be provided)		
Plasma Concentrations & TK Parameters: • **Bioavailability, AUC, Cmax, Tmax, clearance, and half-life for relevant routes.** *(additionally cross reference from study report and page no. to be provided)*		
Excretion Data: • **Amounts of parent compound and metabolites in excreta (% of administered dose).** *(additionally cross reference from study report and page no. to be provided)*		
Metabolic pathway *(cross reference from study report and page no. to be provided)*		
Statistical tool used for calculation		

4. Result Summary

Particulars		Metabolism
Correlation with observed toxicity, dose proportionality, systemic exposure		
Study Conclusion		

MUTAGENECITY STUDIES

1. Test study Information:

Particulars		Mutagenicity test AMES TEST	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vitro)*	Mutagenicity test *(In Vivo)*
Study Report No.					
Sponsor (Name and Address)					

Source (Name and Address)					
Study initiation Date					
Study completion Date					
Batch No.					
Purity					
CoA attached (Yes/No)					
IAEC No. (in case of in vivo studies) (if available)					
GLP Statement (yes/no)					
Whether the study is done within GLP certificate validity date (yes/no)					
OECD Guideline or others					

2. Study Design/Methodology

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Species/Strain					
No. and sex of animals per dose (in case of in vivo)					
Age and Body weight range at dosing (in case of in vivo)					
Concentration/ Dose of test chemical					
Vehicle used					
Route of administration (in case of in vivo)					
Fasting conditions (Duration) pre-dose (in case of in vivo)					
Housing conditions: Temperature, Humidity, Light cycle (in case of in vivo)					
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*					
Feed analysis statement / certificate (as applicable) (optional)*					

(cross reference from study report and page no. to be provided)					
Acclimatization period and grouping					
Dosing Schedule, Duration of the study					
Rationale for dose selection					
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.					

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Food and water consumption changes (in vivo)					
Clinical observations schedule with respect to doses (in vivo)					
Body weight changes (in vivo)					
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*					
Scoring (in vivo)					
Primary endpoints (in vitro) frequency of micronucleated erythrocytes (PCE/NCE or reticulocyte MN) per animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome aberrations per metaphase in bone marrow/ spermatogonial chromosomal aberration frequency in testicular metaphases/ others					

(cross reference from study report and page no. to be provided)					
Primary endpoints (in vitro) **revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others** *(cross reference from study report and page no. to be provided)*					
Statistical tool used for calculation					

4. Result Summary

Particulars		**Mutagenicity test AMES TEST**	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vitro)***	**Mutagenicity test *(In Vivo)***
Study Conclusion					

IMMUNOTOXICITY STUDY

1. Test study Information:

Particulars		**Cellular immune response**
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (if available)		
GLP Statement (yes/no)		

Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

Particulars		**Cellular immune response**
Species/Strain		
No. and sex of animals per dose		
Age and Body weight range at dosing		
Dose of test chemical		
Vehicle used		
Route of administration		
Fasting conditions (Duration) pre-dose		
Housing conditions: Temperature, Humidity, Light cycle		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		

Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

Particulars		**Cellular immune response**
NOAEL/BMC/NOAEC/LOAEC/BALF		
Food and water consumption changes (if any)		
Clinical observations schedule with respect to doses		
Nature and severity of the toxicity signs (if any)		
Time course of onset and Duration of the toxicity effect (if any)		
Body weight changes		
Clinical pathology • **Hematology** • **Biochemistry** • **Urinanalysis** *(cross reference from study report and page no. to be provided)*		
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*		
Lymphoid Organs examined and weights *(cross reference from study report and page no. to be provided)*		
Phenotyping *(cross reference from study report and page no. to be provided)*		
Primary Functional test (TDAR) *(cross reference from study report and page no. to be provided)*		
Secondary functional Test Report *(cross reference from study report and page no. to be provided)*		

Statistical tool used for calculation		

4. Result Summary

Particulars		Cellular immune response
Number of animals dosed, survived and died		
Study Conclusion		

ECOTOXICITY STUDIES

1. Test study Information:

Particulars		Acute Avian toxicity	Acute toxicity-fish	Toxicity to Honeybees (oral and contact) If applicable
Study Report No.				
Sponsor (Name and Address)				
Source (Name and Address)				
Study initiation Date				
Study completion Date				
Batch No.				
Purity				
CoA attached (Yes/No)				
Ethical consideration (NO.) (if available)				
GLP Statement (yes/no)				
Whether the study is done within GLP certificate validity date (yes/no)				
OECD Guideline or others				

2. Study Design/Methodology

Particulars		Acute Avian toxicity	Acute toxicity-fish	Toxicity to Honeybees (oral and contact) If applicable
Species				
No. and sex of animals per dose				

Age and Body weight range at dosing				
Dose of test chemical				
Vehicle used				
Route of administration				
Fasting conditions (Duration) pre-dose where applicable				
Housing conditions: Temperature, Humidity, Light cycle				
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*				
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*				
Acclimatization period and grouping				
Dosing Schedule, Duration of the study				
Rationale for dose selection				
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.				

3. Observations

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Toxicity to Honeybees (oral and contact) If applicable**
Result LD50				

Clinical observations schedule with respect to doses				
Nature and severity of the toxicity signs (if any)				
Time course of onset and Duration of the toxicity effect (if any)				
Date and time of death, if before scheduled sacrifice				
Total No. of Mortality per dose				
Body weight schedule changes				
Necropsy findings and histopathological findings *(cross reference from study report and page no. to be provided)*				
Organs examined				
Food and feed consumption data *(cross reference from study report and page no. to be provided)*				
Statistical tool if used				

4. Result Summary

Particulars		**Acute Avian toxicity**	**Acute toxicity-fish**	**Toxicity to Honeybees (oral and contact) If applicable**
Number of animals dosed, survived and died				
LD_{50}				
Study Conclusion				

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**

2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**

3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**

4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
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1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		

5. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

MUTAGENECITY STUDIES

1. Test study Information:

		Mutagenicity test AMES TEST
Study Report No.		
Sponsor (Name and Address)		
Source (Name and Address)		
Study initiation Date		
Study completion Date		
Batch No.		
Purity		
CoA attached (Yes/No)		
IAEC No. (in case of in vivo studies) (if available)		
GLP Statement (yes/no)		
Whether the study is done within GLP certificate validity date (yes/no)		
OECD Guideline or others		

2. Study Design/Methodology

		Mutagenicity test AMES TEST
Species/Strain		
No. and sex of animals per dose (in case of in vivo)		

Age and Body weight range at dosing (in case of in vivo)		
Concentration/ Dose of test chemical		
Vehicle used		
Route of administration (in case of in vivo)		
Fasting conditions (Duration) pre-dose (in case of in vivo)		
Housing conditions: Temperature, Humidity, Light cycle (in case of in vivo)		
Feed report (source, restriction) *(cross reference from study report and page no. to be provided)*		
Feed analysis statement / certificate (as applicable) (optional)* *(cross reference from study report and page no. to be provided)*		
Acclimatization period and grouping		
Dosing Schedule, Duration of the study		
Rationale for dose selection		
Any deviation from protocol or amendments (Yes/No). If yes provide justification or cross reference of report.		

* In case of repeated dose studies, feed analysis report shall be provided from feed supplier / in-house / third party.

3. Observations

		Mutagenicity test **AMES TEST**
Food and water consumption changes (in vivo)		
Clinical observations schedule with respect to doses (in vivo)		
Body weight changes (in vivo)		
Clinical pathology if any (in vivo) *(cross reference from study report and page no. to be provided)*		
Scoring (in vivo)		
Primary endpoints (in vitro) frequency of micronucleated erythrocytes (PCE/NCE or reticulocyte MN) per animal; PCE:NCE ratio for marrow toxicity/ frequency of structural chromosome aberrations		

per metaphase in bone marrow/ spermatogonial chromosomal aberration frequency in testicular metaphases/ others *(cross reference from study report and page no. to be provided)*		
Primary endpoints (in vitro) revertant colony counts/ number (%) of cells with structural aberrations/ mutant frequency, clonogenicity, expression time, viability/ % micronucleated cells, CBPI or MN frequency, cytotoxicity metric/ others *(cross reference from study report and page no. to be provided)*		
Statistical tool used for calculation		

4. Result Summary

		Mutagenicity test **AMES TEST**
Study Conclusion		

NOTE- REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**
2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**
3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**
4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

TOXICOLOGY TEMPLATE
EXPORT

1. General Information:

Test chemical Name		
Type of chemical (Insecticide/ Fungicide/ Herbicide/ Other)		
Specify the status of the chemical listed in the schedule		
Global status of pesticide molecules with respect to Ban/restriction/withdrawn and reasons thereof		
Summary Report Derived from International report on Carcinogenicity & Genotoxicity (if available)		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

Study Type	**Results**	**Toxic Category**	**Source of information**
Acute Oral Toxicity			
Acute Dermal Toxicity			
Acute Inhalation Toxicity			
Acute Eye Irritation			
Acute Dermal Irritation			
Skin Sensitization			
Effect on Reproduction			
Developmental Toxicity			
Carcinogenicity			

Mutagenicity			
Neurotoxicity			

NOTE: The template is required for first time registration of molecule for export

TOXICOLOGY TEMPLATE

1. General Information:

Test chemical Name/ strain/Accession number/Application no.		
Purpose		
Specify the status of the chemical listed in the schedule		
Global status with respect to Ban/restriction/withdrawn and reasons thereof		

2. Label and Leaflet

LABEL AND LEAFLET							
CHEMICAL COMPOSITION	**PURPOSE OF USE**	**SAFETY PRECAUTIONS**	**SYMPTOMS OF POISONING**	**CAUTIONARY STATEMENT**	**FIRST AID**	**ANTIDOTE**	**TOXICITY TRIANGLE**

ACUTE TOXICITY STUDIES

1. Test study Information:

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Study Report No.							
Sponsor (Name and Address)							
Source (Name and Address)							
Study initiation Date							
Study completion Date							
Batch No.							
Purity							
CoA attached (Yes/No)							
IAEC No. (if available)							
GLP Statement (yes/no)							
Whether the study is done within GLP certificate validity date (Yes/No)							
OECD Guideline or others							

2. Study Design/Methodology

		Acute Oral	Acute dermal	Acute inhalational	Primary skin irritation	Acute eye irritation	Skin sensitization
Species/Strain							
No. and sex of animals per dose							
Age and Body weight range at dosing							
Doses of test chemical							
Vehicle used							
Route of administration							
Fasting conditions (Duration) pre-dose							
Housing conditions: Temperature, Humidity, Light cycle							
Feed report (source, restriction) *cross reference from study report and page no. to be provided*							
Feed analysis statement / certificate (as applicable) (optional) *cross reference from study report and page no. to be provided*							
Acclimatization period and grouping							
Dosing Schedule, Duration of the study							
Rationale for dose selection							
Any deviation from protocol or amendments (Yes/No). If yes provide justification							

or cross reference of report.							

3. Observations

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Result LD_{50} / LC_{50}							
Clinical observations schedule with respect to doses							
Nature and severity of the toxicity signs (if any)							
Time course of onset and Duration of the toxicity effect (if any)							
Date and time of death, if before scheduled sacrifice							
Total No. of Mortality per dose							
Body weight schedule changes (If any)							
Necropsy findings and histopathological findings *cross reference from study report and page no. to be provided*							
Organs examined							

4. Result Summary

		Acute Oral	**Acute dermal**	**Acute inhalational**	**Primary skin irritation**	**Acute eye irritation**	**Skin sensitization**
Number of animals dosed, survived and died							
LD_{50} / LC_{50} (GHS Classification)							
Study Conclusion							

REFER TO THE GUIDANCE DOCUMENT ON TOXICOLOGY FOR THE REGISTRATION OF CHEMICAL PESTICIDES IN INDIA.

NOTE:

1. **Hyperlink(s), relevant cross-reference(s), and page number(s) of the study report shall be provided wherever feasible for ease of scrutiny and verification.**
2. **This template is dynamic in nature and may be revised/updated from time to time in view of regulatory requirements, scientific developments, or related considerations, as deemed necessary by the competent authority.**
3. **Submission of information in the present template shall not preclude seeking any additional information, clarification, or supporting document(s), wherever considered necessary during evaluation of the application.**
4. **In case of any discrepancy, the contents of the original study report/dossier shall prevail.**

PACKAGING

Template for Packaging Discipline for Applications U/s 9(3)-TI

S. No.	Parameters	Reply/Response of applicant
1.	Form I	
2.	Labels and Leaflets per IR-1971, all fields (as applicable) and as amended from time to time	
3.	Manner of labeling and Leaflet	
4.	Type of packaging (Ultra small, small or Big whichever is applicable)	
5.	Manner of packaging Manner of Import packaging with details UN IMDG Code, packaging group, Hazard class, (a)Primary Packing (b) Secondary Packing (c) Transport Packing	
6.	Specification for primary, Secondary and Transport packages (whichever is applicable) Specification shall include: 1) capacity, 2) thickness, 3) shape, 4) size (LxWxH), 5) brimful capacity, 6) sealing system, 7) Material of construction, Grade, 8) Closure system, 9) Gross/net weight of primary, secondary and transport packing,	
7.	Details of packaging material and its compatibility with content: Container Content Compatibility (CCC) test from NABL accrediated labs as per standard test protocols/BIS specifications. a) Name of the Laboratory : b) NABL Status (Validity of NABL Certificate): c) Period of the Study : From ____ to _____ d) BIS specification/protocol followed : e) Number of Samplesset taken for Test : f) Temperature maintained during the test : g) Primary Packing taken for test : h) Test result of AI before the test : i) Test result of AI after the test : j) Test result of Packaging material before test : k) Test result of Packaging material after test : l) Final Conclusion of the Study :	
8.	Performance of container with content during storage stability test(Shelf life Study) at three different agro climate conditions for	

	30 months/ ASS Data for a.i. content and physico chemical properties of the product and test parameter on container:-- 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - Shelf life Study / ASS: _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) Three Agro-climatic locations selected for Shelf life Study: Location 01 :_____ Location 02 :_____ Location 03:_____ 7) BIS specification followed : 8) Primary Packing taken for test : 9) Test result of AI before the test : 10) Test result of AI after the test (for ASS) : 11) Zero Month Test result of AI after the test : Location 12) 06 Months Test result of AI after the test : 13) 12 Months Test result of AI after the test : 14) 18 Months Test result of AI after the test : 15) 24 Months Test result of AI after the test : 16) 30 Months Test result of AI after the test : Final Conclusion of Test results of AI after 30 Months : 17) Test result of Packaging Material before the test : 18) Test result of Packaging Material after the test(for ASS) : 19) Zero Month Test result of Packaging Material after the test : 20) 06 Months Test result of Packaging Material after the test : 21) 12 Months Test result of Packaging Materialafter the test : 22) 18 Months Test result of Packaging Material after the test : 23) 24 Months Test result of Packaging Material after the test : 24) 30 Months Test result of Packaging Material after the test : Final Conclusion of Test results of Packaging Material after 30 Months : 25) Metrological data of 3 locations: Name of the Institute/Organization of Metrological data: a) Location 01 :_____ b) Location 02 :_____ c) Location 03:_____ 26) Average Temperature range during the entire Shelf Life Study: (i) Location 01 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃	

	24 Months : _____ to _____℃ 30 Months : _____ to _____℃ (ii) Location 02 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ (iii) Location 03 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃	
9.	Transport worthiness test (TWT) for proposed transport packing from NABL accrediated lab. 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) BIS specification followed : 7) Primary Packing taken for test : 8) Secondary Packaging Taken for Test : 9) Transport Packing Taken for test : 10) Whether the primary,secondary and transport packing taken for tests are same as proposed in Manner of Packing : Yes/No 11) Number of Samples taken for Test : !2) Details of the Tests Conducted :(to cover all type of proposed packing combinations -(Primary+Secondary+Transport) **A) Vertical Drop Test** a) BIS No. : b) Scheme for Dropping Test : c) Test results : **B) Stack Load Test :** a) BIS No. : b)Scheme for Stack Load Test : c) Test results : C) **Vibration Leak Test :** a) BIS No. : b) Scheme for Vibration Test :	

	c) Test results : D) **Rolling Test:** a) BIS No. : b) Scheme for Rolling Test : c) Test results : **E) Inclined Impact Test :** a) BIS No. : b) Scheme for Rolling Test : c) Test results : F) Any other Packaging Test under TWT as per IMDG / BIS /RC requirements : a) BIS No. : b) Test results :	

Template for Packaging Discipline for Applications U/s 9(3)-FI

S. No.	Parameters	Reply/Response of applicant
1.	Form I	
2.	Labels and Leaflets per IR-1971, all fields (as applicable) and as amended from time to time	
3.	Manner of labelling and Leaflet	
4.	Type of packaging (Ultra small, small or Big whichever is applicable)	
5.	Manner of packaging: Manner of Import packaging with details UN IMDG Code, packaging group, Hazard class, (a)Primary Packing (b) Secondary Packing (c) Transport Packing Manner of Re-packing: (a)Primary Packing (b) Secondary Packing (c) Transport Packing	
6.	Specification for primary, Secondary and Transport packages (whichever is applicable) Specification shall include: 1) capacity, 2) thickness, 3) shape, 4) size (LxWxH), 5) brimful capacity, 6) sealing system, 7) Material of construction, Grade, 8) Closure system, 9) Gross/net weight of primary, secondary and transport packing,	
7.	Details of packaging material and its compatibility with content: Container Content Compatibility (CCC) test from NABL accrediated labs as per standard test protocols/BIS specifications. a) Name of the Laboratory : b) NABL Status (Validity of NABL Certificate): c) Period of the Study : From ____ to _____ d) BIS specification/protocol followed : e) Number of Samplesset taken for Test : f) Temperature maintained during the test : g) Primary Packing taken for test : h) Test result of AI before the test : i) Test result of AI after the test : j) Test result of Packaging material before test :	

	k) Test result of Packaging material after test : l) Final Conclusion of the Study :	
8.	Performance of container with content during storage stability test(Shelf life Study) at three different agro climate conditions for 30 months/ ASS Data for a.i. content and physico chemical properties of the product and test parameter on container:-- 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - Shelf life Study / ASS: _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) Three Agro-climatic locations selected for Shelf life Study: 7) Location 01 :_____ 8) Location 02 :_____ 9) Location 03:_____ 10) BIS specification followed : 11) Primary Packing taken for test : 12) Test result of AI before the test : 13) Test result of AI after the test (for ASS) : 14) Zero Month Test result of AI after the test : 15) Location 16) 06 Months Test result of AI after the test : 17) 12 Months Test result of AI after the test : 18) 18 Months Test result of AI after the test : 19) 24 Months Test result of AI after the test : 20) 30 Months Test result of AI after the test : 21) Final Conclusion of Test results of AI after 30 Months : 22) Test result of Packaging Material before the test : 23) Test result of Packaging Material after the test(for ASS) : 24) Zero Month Test result of Packaging Material after the test : 25) 06 Months Test result of Packaging Material after the test : 26) 12 Months Test result of Packaging Materialafter the test : 27) 18 Months Test result of Packaging Material after the test : 28) 24 Months Test result of Packaging Material after the test : 29) 30 Months Test result of Packaging Material after the test : 30) Final Conclusion of Test results of Packaging Material after 30 Months :	

	31) Metrological data of 3 locations: Name of the Institute/Organization of Metrological data: Location 01 :_____ Location 02 :_____ Location 03:_____ 32) Average Temperature range during the entire Shelf Life Study: (iv) Location 01 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ (v) Location 02 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ (vi) Location 03 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃	
9.	Transport worthiness test (TWT) for proposed transport packing from NABL accrediated lab. 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) BIS specification followed : 7) Primary Packing taken for test : 8) Secondary Packaging Taken for Test : 9) Transport Packing Taken for test : 10) Whether the primary,secondary and transport packing taken for tests are same as proposed in Manner of Packing : Yes/No 11) Number of Samples taken for Test :	

	12) Details of the Tests Conducted :(to cover all type of proposed packing combinations -(Primary+Secondary+Transport) **A. Vertical Drop Test** BIS No. : Scheme for Dropping Test : Test results : **B. Stack Load Test :** a) BIS No. : b)Scheme for Stack Load Test : c) Test results : C. **Vibration Leak Test :** a) BIS No. : b) Scheme for Vibration Test : c) Test results : D. **Rolling Test:** a) BIS No. : b) Scheme for Rolling Test : c) Test results : **E. Inclined Impact Test :** a) BIS No. : b) Scheme for Rolling Test : c) Test results : F. Any other Packaging Test under TWT as per IMDG / BIS /RC requirements : a) BIS No. : b) Test results :	

Template for Packaging Discipline for Applications U/s 9(3)-TIM

S. No.	Parameters	Reply/Response of applicant
1.	Form I	
2.	Labels and Leaflets per IR-1971, all fields (as applicable) and as amended from time to time	
3.	Manner of labeling and Leaflet	
4.	Type of packaging (Ultra small, small or Big whichever is applicable)	
5.	Manner of packaging: (a)Primary Packing (b) Secondary Packing (c) Transport Packing	
6.	Specification for primary, Secondary and Transport packages (whichever is applicable) Specification shall include: a) capacity, b) thickness, c) shape, d) size (LxWxH), e) brimful capacity, f) sealing system, g) Material of construction, Grade, h) Closure system, i) Gross/net weight of primary, secondary and transport packing,	
7.	Details of packaging material and its compatibility with content: Container Content Compatibility (CCC) test from NABL accrediated labs as per standard test protocols/BIS specifications. 1) Name of the Laboratory : 2) NABL Status (Validity of NABL Certificate): 3) Period of the Study : From ____ to _____ 4) BIS specification/protocol followed : 5) Number of Samplesset taken for Test : 6) Temperature maintained during the test : 7) Primary Packing taken for test : 8) Test result of AI before the test : 9) Test result of AI after the test : 10) Test result of Packaging material before test : 11) Test result of Packaging material after test : 12) Final Conclusion of the Study :	
8.	Performance of container with content during storage stability test(Shelf life Study) at three different agro climate conditions for 30 months/ ASS Data for a.i. content	

	and physico chemical properties of the product and test parameter on container:-- 1. Name of the Laboratory : 2. NABL/GLP Status (Validity of NABL/GLP Certificate): 3. Name of the Study - Shelf life Study / ASS: _____ 4. Period of the Study : From ____ to _____ 5. Number of Months/Days : 6. Three Agro-climatic locations selected for Shelf life Study: 7. Location 01 :_____ 8. Location 02 :_____ 9. Location 03:_____ 10. BIS specification followed : 11. Primary Packing taken for test : 12. Test result of AI before the test : 13. Test result of AI after the test (for ASS) : 14. Zero Month Test result of AI after the test : 15. Location 16. 06 Months Test result of AI after the test : 17. 12 Months Test result of AI after the test : 18. 18 Months Test result of AI after the test : 19. 24 Months Test result of AI after the test : 20. 30 Months Test result of AI after the test : 21. Final Conclusion of Test results of AI after 30 Months : 22. Test result of Packaging Material before the test : 23. Test result of Packaging Material after the test(for ASS) : 24. Zero Month Test result of Packaging Material after the test : 25. 06 Months Test result of Packaging Material after the test : 26. 12 Months Test result of Packaging Materialafter the test : 27. 18 Months Test result of Packaging Material after the test : 28. 24 Months Test result of Packaging Material after the test : 29. 30 Months Test result of Packaging Material after the test : 30. Final Conclusion of Test results of Packaging Material after 30 Months : 31. Metrological data of 3 locations: Name of the Institute/Organization of Metrological data: Location 01 :_____ Location 02 :_____ Location 03:_____	

	32. Average Temperature range during the entire Shelf Life Study: Location 01 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ Location 02 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ Location 03 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃	
9.	Transport worthiness test (TWT) for proposed transport packing from NABL accrediated lab. 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) BIS specification followed : 7) Primary Packing taken for test : 8) Secondary Packaging Taken for Test : 9) Transport Packing Taken for test : 10) Whether the primary,secondary and transport packing taken for tests are same as proposed in Manner of Packing : Yes/No 11) Number of Samples taken for Test : 12) Details of the Tests Conducted :(to cover all type of proposed packing combinations -(Primary+Secondary+Transport) **A. Vertical Drop Test** BIS No. :	

	Scheme for Dropping Test : Test results : **B.Stack Load Test :** a) BIS No. : b)Scheme for Stack Load Test : c) Test results : **C.Vibration Leak Test :** a) BIS No. : b) Scheme for Vibration Test : c) Test results : **D.Rolling Test:** a) BIS No. : b) Scheme for Rolling Test : c) Test results : **E.Inclined Impact Test :** a) BIS No. : b) Scheme for Rolling Test : c) Test results : F.Any other Packaging Test under TWT as per IMDG / BIS /RC requirements : a) BIS No. : b) Test results :	

Template for Packaging Discipline for Applications U/s 9(3)-FIM

S. No.	Parameters	Reply/Response of applicant
1.	Form I	
2.	Labels and Leaflets per IR-1971, all fields (as applicable) and as amended from time to time	
3.	Manner of labeling and Leaflet	
4.	Type of packaging (Ultra small, small or Big whichever is applicable)	
5.	Manner of packaging: (a)Primary Packing (b) Secondary Packing (c) Transport Packing	
6.	Specification for primary, Secondary and Transport packages (whichever is applicable) Specification shall include: a) capacity, b) thickness, c) shape, d) size (LxWxH), e) brimful capacity, f) sealing system, g) Material of construction, Grade, h) Closure system, i) Gross/net weight of primary, secondary and transport packing,	
7.	Details of packaging material and its compatibility with content: Container Content Compatibility (CCC) test from NABL accrediated labs as per standard test protocols/BIS specifications. 1) Name of the Laboratory : 2) NABL Status (Validity of NABL Certificate): 3) Period of the Study : From ____ to _____ 4) BIS specification/protocol followed : 5) Number of Samplesset taken for Test : 6) Temperature maintained during the test : 7) Primary Packing taken for test : 8) Test result of AI before the test : 9) Test result of AI after the test : 10) Test result of Packaging material before test : 11) Test result of Packaging material after test : 12) Final Conclusion of the Study :	
8.	Performance of container with content during storage stability test(Shelf life Study) at three different agro	

	climate conditions for 30 months/ ASS Data for a.i. content and physico chemical properties of the product and test parameter on container:-- 1. Name of the Laboratory : 2. NABL/GLP Status (Validity of NABL/GLP Certificate): 3. Name of the Study - Shelf life Study / ASS: _____ 4. Period of the Study : From ____ to _____ 5. Number of Months/Days : 6. Three Agro-climatic locations selected for Shelf life Study: 7. Location 01 :_____ 8. Location 02 :_____ 9. Location 03:_____ 10. BIS specification followed : 11. Primary Packing taken for test : 12. Test result of AI before the test : 13. Test result of AI after the test (for ASS) : 14. Zero Month Test result of AI after the test : 15. Location 16. 06 Months Test result of AI after the test : 17. 12 Months Test result of AI after the test : 18. 18 Months Test result of AI after the test : 19. 24 Months Test result of AI after the test : 20. 30 Months Test result of AI after the test : 21. Final Conclusion of Test results of AI after 30 Months : 22. Test result of Packaging Material before the test : 23. Test result of Packaging Material after the test(for ASS) : 24. Zero Month Test result of Packaging Material after the test : 25. 06 Months Test result of Packaging Material after the test : 26. 12 Months Test result of Packaging Material after the test : 27. 18 Months Test result of Packaging Material after the test : 28. 24 Months Test result of Packaging Material after the test : 29. 30 Months Test result of Packaging Material after the test : 30. Final Conclusion of Test results of Packaging Material after 30 Months : 31. Metrological data of 3 locations: Name of the Institute/Organization of Metrological data: Location 01 :_____ Location 02 :_____ Location 03:_____	

	32. Average Temperature range during the entire Shelf Life Study: Location 01 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ Location 02 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ Location 03 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃	
9.	Transport worthiness test (TWT) for proposed transport packing from NABL accrediated lab. 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) BIS specification followed : 7) Primary Packing taken for test : 8) Secondary Packaging Taken for Test : 9) Transport Packing Taken for test : 10) Whether the primary,secondary and transport packing taken for tests are same as proposed in Manner of Packing : Yes/No 11) Number of Samples taken for Test : 12) Details of the Tests Conducted :(to cover all type of proposed packing combinations -(Primary+Secondary+Transport) **A.Vertical Drop Test**	

	BIS No. : Scheme for Dropping Test : Test results : **B.Stack Load Test :** a) BIS No. : b)Scheme for Stack Load Test : c) Test results : **C.Vibration Leak Test :** a) BIS No. : b) Scheme for Vibration Test : c) Test results : **D.Rolling Test:** a) BIS No. : b) Scheme for Rolling Test : c) Test results : **E.Inclined Impact Test :** a) BIS No. : b) Scheme for Rolling Test : c) Test results : F.Any other Packaging Test under TWT as per IMDG / BIS /RC requirements : a) BIS No. : b) Test results :	

Template for Packaging Discipline for Applications U/s 9(3)-Bio-Pesticides: Primary culture/mother culture & Formulated product

S. No.	Parameters	Reply/Response of applicant
1.	Form I	
2.	Labels and Leaflets per IR-1971, all fields (as applicable) and as amended from time to time	
3.	Manner of labeling and Leaflet	
4.	Type of packaging (Ultra small, small or Big whichever is applicable)	
5.	Manner of packaging: (a)Primary Packing (b) Secondary Packing (c) Transport Packing	
6.	Specification for primary, Secondary and Transport packages (whichever is applicable) Specification shall include: 1) capacity, 2) thickness, 3) shape, 4) size (LxWxH), 5) brimful capacity, 6) sealing system, 7) Material of construction, Grade, 8) Closure system, 9) Gross/net weight of primary, secondary and transport packing,	
7.	Details of packaging material and its compatibility with content: Container Content Compatibility (CCC) test from NABL accrediated labs as per standard test protocols/BIS specifications. a) Name of the Laboratory : b) NABL Status (Validity of NABL Certificate): c) Period of the Study : From ____ to _____ d) BIS specification/protocol followed : e) Number of Samplesset taken for Test : f) Temperature maintained during the test : g) Primary Packing taken for test : h) Test result of AI before the test : i) Test result of AI after the test : j) Test result of Packaging material before test : k) Test result of Packaging material after test : l) Final Conclusion of the Study :	

8.	Performance of container with content during storage stability test(Shelf life Study) at three different agro climate conditions for 30 months/ ASS Data for a.i. content and physico chemical properties of the product and test parameter on container:-- 1. Name of the Laboratory : 2. NABL/GLP Status (Validity of NABL/GLP Certificate): 3. Name of the Study - Shelf life Study / ASS: _____ 4. Period of the Study : From ____ to _____ 5. Number of Months/Days : 6. Three Agro-climatic locations selected for Shelf life Study: 7. Location 01 :_____ 8. Location 02 :_____ 9. Location 03:_____ 10. BIS specification followed : 11. Primary Packing taken for test : 12. Test result of AI before the test : 13. Test result of AI after the test (for ASS) : 14. Zero Month Test result of AI after the test : 15. Location 16. 06 Months Test result of AI after the test : 17. 12 Months Test result of AI after the test : 18. 18 Months Test result of AI after the test : 19. 24 Months Test result of AI after the test : 20. 30 Months Test result of AI after the test : 21. Final Conclusion of Test results of AI after 30 Months : 22. Test result of Packaging Material before the test : 23. Test result of Packaging Material after the test(for ASS) : 24. Zero Month Test result of Packaging Material after the test : 25. 06 Months Test result of Packaging Material after the test : 26. 12 Months Test result of Packaging Materialafter the test : 27. 18 Months Test result of Packaging Material after the test : 28. 24 Months Test result of Packaging Material after the test : 29. 30 Months Test result of Packaging Material after the test : 30. Final Conclusion of Test results of Packaging Material after 30 Months : 31. Metrological data of 3 locations: Name of the Institute/Organization of Metrological data: Location 01 :_____	

	Location 02 :_____ Location 03:_____ 32. Average Temperature range during the entire Shelf Life Study: Location 01 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ Location 02 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ Location 03 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃	
9.	Transport worthiness test (TWT) for proposed transport packing from NABL accrediated lab. 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) BIS specification followed : 7) Primary Packing taken for test : 8) Secondary Packaging Taken for Test : 9) Transport Packing Taken for test : 10) Whether the primary,secondary and transport packing taken for tests are same as proposed in Manner of Packing : Yes/No 11) Number of Samples taken for Test : 12) Details of the Tests Conducted :(to cover all type of proposed packing combinations -(Primary+Secondary+Transport)	

	A.Vertical Drop Test BIS No. : Scheme for Dropping Test : Test results : **B.Stack Load Test :** a) BIS No. : b)Scheme for Stack Load Test : c) Test results : **C.Vibration Leak Test :** a) BIS No. : b) Scheme for Vibration Test : c) Test results : **D.Rolling Test:** a) BIS No. : b) Scheme for Rolling Test : c) Test results : **E.Inclined Impact Test :** a) BIS No. : b) Scheme for Rolling Test : c) Test results : F.Any other Packaging Test under TWT as per IMDG / BIS /RC requirements : a) BIS No. : b) Test results :	

Template for Packaging Discipline for Applications U/s 9(4)-TIM

S. No.	Parameters	Reply/Response of applicant
1.	Form I	
2.	Manner of labeling and Leaflet	
3.	Type of packaging (Ultra small, small or Big whichever is applicable)	
4.	Manner of packaging: (a)Primary Packing (b) Secondary Packing (c) Transport Packing	
5.	Specification for primary, Secondary and Transport packages (whichever is applicable) Specification shall include: a) capacity, b) thickness, c) shape, d) size (LxWxH), e) brimful capacity, f) sealing system, g) Material of construction, Grade, h) Closure system, i) Gross/net weight of primary, secondary and transport packing,	
6.	Details of packaging material and its compatibility with content: Container Content Compatibility (CCC) test from NABL accrediated labs as per standard test protocols/BIS specifications. 1) Name of the Laboratory : 2) NABL Status (Validity of NABL Certificate): 3) Period of the Study : From ____ to _____ 4) BIS specification/protocol followed : 5) Number of Samplesset taken for Test : 6) Temperature maintained during the test : 7) Primary Packing taken for test : 8) Test result of AI before the test : 9) Test result of AI after the test : 10) Test result of Packaging material before test : 11) Test result of Packaging material after test : 12) Final Conclusion of the Study :	

Template for Endorsement Applications of New Packaging Material U/s 9(3)/ 9(4):

S. No.	Parameters	Reply/Response of applicant
1.	Duly signed Endorsement application	
2.	Notarized copy of Certificate of Registration	
3.	Type of packaging (Ultra small, small or Big whichever is applicable)	
4.	Manner of packaging: (a)Primary Packing (b) Secondary Packing (c) Transport Packing	
5.	Specification for primary, Secondary and Transport packages (whichever is applicable) Specification shall include: 1) capacity, 2) thickness, 3) shape, 4) size (LxWxH), 5) brimful capacity, 6) sealing system, 7) Material of construction, Grade, 8) Closure system, 9) Gross/net weight of primary, secondary and transport packing,	
6.	Details of packaging material and its compatibility with content: Container Content Compatibility (CCC) test from NABL accrediated labs as per standard test protocols/BIS specifications. a) Name of the Laboratory : b) NABL Status (Validity of NABL Certificate): c) Period of the Study : From ____ to _____ d) BIS specification/protocol followed : e) Number of Samplesset taken for Test : f) Temperature maintained during the test : g) Primary Packing taken for test : h) Test result of AI before the test : i) Test result of AI after the test : j) Test result of Packaging material before test : k) Test result of Packaging material after test : l) Final Conclusion of the Study :	
7.	Performance of container with content during storage stability test(Shelf life Study) at three different agro climate conditions for 30 months/ ASS Data for a.i. content	

	and physico chemical properties of the product and test parameter on container:-- 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - Shelf life Study / ASS: _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) Three Agro-climatic locations selected for Shelf life Study: 7) Location 01 :_____ 8) Location 02 :_____ 9) Location 03:_____ 10) BIS specification followed : 11) Primary Packing taken for test : 12) Test result of AI before the test : 13) Test result of AI after the test (for ASS) : 14) Zero Month Test result of AI after the test : 15) Location 16) 06 Months Test result of AI after the test : 17) 12 Months Test result of AI after the test : 18) 18 Months Test result of AI after the test : 19) 24 Months Test result of AI after the test : 20) 30 Months Test result of AI after the test : 21) Final Conclusion of Test results of AI after 30 Months : 22) Test result of Packaging Material before the test : 23) Test result of Packaging Material after the test(for ASS) : 24) Zero Month Test result of Packaging Material after the test : 25) 06 Months Test result of Packaging Material after the test : 26) 12 Months Test result of Packaging Materialafter the test : 27) 18 Months Test result of Packaging Material after the test : 28) 24 Months Test result of Packaging Material after the test : 29) 30 Months Test result of Packaging Material after the test : 30) Final Conclusion of Test results of Packaging Material after 30 Months : 31) Metrological data of 3 locations: Name of the Institute/Organization of Metrological data: Location 01 :_____ Location 02 :_____ Location 03:_____	

	32) Average Temperature range during the entire Shelf Life Study: Location 01 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ Location 02 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃ Location 03 :_____ Zero Month : _____ to _____℃ 06 Months : _____ to _____℃ 12 Months : _____ to _____℃ 18 Months: _____ to _____℃ 24 Months : _____ to _____℃ 30 Months : _____ to _____℃	
8.	Transport worthiness test (TWT) for proposed transport packing from NABL accrediated lab. 1. Name of the Laboratory : 2. NABL/GLP Status (Validity of NABL/GLP Certificate): 3. Name of the Study - _____ 4. Period of the Study : From ____ to _____ 5. Number of Months/Days : 6. BIS specification followed : 7. Primary Packing taken for test : 8. Secondary Packaging Taken for Test : 9. Transport Packing Taken for test : 10. Whether the primary,secondary and transport packing taken for tests are same as proposed in Manner of Packing : Yes/No 11. Number of Samples taken for Test : 12. Details of the Tests Conducted :(to cover all type of proposed packing combinations -(Primary+Secondary+Transport) **A.Vertical Drop Test** BIS No. : Scheme for Dropping Test :	

	Test results : **B.Stack Load Test :** a) BIS No. : b)Scheme for Stack Load Test : c) Test results : **C.Vibration Leak Test :** a) BIS No. : b) Scheme for Vibration Test : c) Test results : **D.Rolling Test:** a) BIS No. : b) Scheme for Rolling Test : c) Test results : **E.Inclined Impact Test :** a) BIS No. : b) Scheme for Rolling Test : c) Test results : F.Any other Packaging Test under TWT as per IMDG / BIS /RC requirements : a) BIS No. : b) Test results :	
9.	Justification for change	
10.	Affidavit duly notarized regarding endorsement made earlier in respect of registration certificate	
11.	Label for New pack size (if applicable)	

Template for Endorsement Applications of New Pack Size with already approved packing material U/s 9(3)/ 9(4)

S. No.	Parameters	Reply/Response of applicant
1.	Duly signed Endorsement application	
2.	Notarized copy of Certificate of Registration	
3.	Type of packaging (Ultra small, small or Big whichever is applicable)	
4.	Manner of packaging: (a)Primary Packing (b) Secondary Packing (c) Transport Packing	
5.	Specification for primary, Secondary and Transport packages (whichever is applicable) Specification shall include: a) capacity, b) thickness, c) shape, d) size (LxWxH), e) brimful capacity, f) sealing system, g) Material of construction, Grade, h) Closure system, i) Gross/net weight of primary, secondary and transport packing,	
6.	Transport worthiness test (TWT) for proposed transport packing from NABL accrediated lab. 1) Name of the Laboratory : 2) NABL/GLP Status (Validity of NABL/GLP Certificate): 3) Name of the Study - _____ 4) Period of the Study : From ____ to _____ 5) Number of Months/Days : 6) BIS specification followed : 7) Primary Packing taken for test : 8) Secondary Packaging Taken for Test : 9) Transport Packing Taken for test : 10) Whether the primary,secondary and transport packing taken for tests are same as proposed in Manner of Packing : Yes/No 11) Number of Samples taken for Test : 12) Details of the Tests Conducted :(to cover all type of proposed packing combinations -(Primary+Secondary+Transport) **A.Vertical Drop Test** BIS No. :	

	Scheme for Dropping Test : Test results : **B.Stack Load Test :** a) BIS No. : b)Scheme for Stack Load Test : c) Test results : **C.Vibration Leak Test :** a) BIS No. : b) Scheme for Vibration Test : c) Test results : **D.Rolling Test:** a) BIS No. : b) Scheme for Rolling Test : c) Test results : **E.Inclined Impact Test :** a) BIS No. : b) Scheme for Rolling Test : c) Test results : F.Any other Packaging Test under TWT as per IMDG / BIS /RC requirements : a) BIS No. : b) Test results :	
7.	Justification for change	
8.	Affidavit duly notarized regarding endorsement made earlier in respect of registration certificate	
9.	Label for New pack size (if applicable)	

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